Chapter 6

Algebraic Geometry

I develop theory of varieties over a non-algebraically closed field k which more closely resembles "Faisceaux algébriques cohérents" [Ser55] than say EGA, and therefore is a bit more elementary because the structure sheaf consists of functions into a single ambient field $\overline{k}$ (and perhaps psychologically because the underlying set consists of solutions to equations rather than Galois orbits of solutions). By dealing with irreducible subsets rather than just points then we can also avoid the use of generic points or a large ambient field of infinite transcendence degree. I summarise the features of this exposition versus other popular expositions from which I sourced the material / inspiration

Author	Book	Non-Algebraically Closed Field	Abstract Variety	Real Functions	Arithmetic Case
J.P. Serre	Faisceaux algébriques cohérents [Ser55]	No	Yes	Yes	No
J.S. Milne	Algebraic Geometry [Mil17]	No	Yes	No	No
R. Hartshorne	Algebraic Geometry [Har13] (Chapter I)	No	No	Yes	No
George R. Kempf	Algebraic Varieties	No	Yes	Yes	No
A. Grothendieck	Éléments de géométrie algébrique	Yes	Yes	No	Yes
	These Notes	Yes	Yes	Yes	No

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6.1 Affine Varieties

In order to generalise the usual notions to non-algebraically closed field, and develop a "coordinate-free" approach, we introduce the following concept

Definition 6.1.1
Let A be a finitely generated k-algebra and $\mathfrak{m}$ a maximal ideal. Recall (3.34.21) that $k(\mathfrak{m})/k$ is an algebraic (indeed finite) field extension, and we call it the ***residue field*** *for $\mathfrak{m}$.*

Now we may introduce the Zero-Loci and study relationships to ideals

Proposition 6.1.2 (Correspondence between Zero-Loci and Ideals)
Let $A = k[X_1, ..., X_n]$ be the polynomial ring in n-variables over a field k. For a set $S \subseteq k[X_1, ..., X_n]$ and an algebraic extension K/k define the ***zero-locus***

$$V_K(S) \coloneqq \{\alpha \in K^n \mid f(\alpha) = 0 \quad \forall f \in S\}.$$

Similarly for a subset $Y \subseteq K^n$ define

$$I_k(Y) \coloneqq \{f \in A \mid f(y) = 0 \quad \forall y \in Y\}$$

The pair of maps V_K, I_k constitute a Galois Connection

$$\mathcal{I}(k[X_1, ..., X_n]) \underset{I_k}{\overset{V_K}{\rightleftharpoons}} \mathcal{P}(K^n)$$

namely they satisfy

a) V_K and I_k are order-reversing
b) $S \subseteq I_k(V_K(S))$
c) $Y \subseteq V_K(I_k(Y))$

Furthermore (omitting the subscripts)

d) $VIV = V$ and $IVI = I$
e) $I(Y)$ is a radical ideal and $\sqrt{\langle S\rangle} \subseteq I(V(S))$
f) $V(S) = V(\langle S\rangle) = V\left(\sqrt{\langle S\rangle}\right)$ and $V(\mathfrak{a}) = V\left(\sqrt{\mathfrak{a}}\right)$
g) $\bigcap_i V(S_i) = V\left(\bigcup_i S_i\right)$ and $\bigcap_i V(\mathfrak{a}_i) = V\left(\sum_i \mathfrak{a}_i\right)$
h) $\bigcap_i I(W_i) = I\left(\bigcup_i W_i\right)$
i) $V(\mathfrak{a}) \cup V(\mathfrak{b}) = V(\mathfrak{a} \cap \mathfrak{b}) = V(\mathfrak{a}\mathfrak{b})$
j) $V((0)) = K^n$ and $V(A) = \emptyset$

The sets of the form $V_K(\mathfrak{a})$ constitute the closed sets of a topology on K^n, denoted by $\mathrm{Zar}_k(K^n)$. In this case we have the following form for the topological closure

$$V_K(I_k(Y)) = \overline{Y}$$

Furthermore

$$I_k(Y) = I_k\left(\overline{Y}\right)$$

Proof. We make use of general results on Galois connections (Foundations, Set Theory), though many results may be shown more directly. The fact it's a Galois connection follows from (2.1.54).

1-3. These follow (2.1.50)

4. This follows from (2.1.52).

5. It's clear that $I(Y)$ is an ideal. It is radical because K is reduced (...). The second statement is straightforward.

6. This follows from (2.1.53) by considering the closure operators $\sqrt{\langle - \rangle}$ and $\langle - \rangle$.

7. The first equality follows from (2.1.55). The second equality follows from (3.3.29).

8. This follows from (2.1.55).

9. Observe that $\mathfrak{m}_x$ is prime (because K is an integral domain) and $x \in V(\mathfrak{a}) \iff \mathfrak{a} \subseteq \mathfrak{m}_x$. the result follows from (3.3.38) because $\mathfrak{a} \subseteq \mathfrak{m}_x \vee \mathfrak{b} \subseteq \mathfrak{m}_x \iff \mathfrak{a}\mathfrak{b} \subseteq \mathfrak{m}_x \iff \mathfrak{a} \cap \mathfrak{b} \subseteq \mathfrak{m}_x$

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The family of sets $\mathrm{Zar}_k(K^n) := \mathrm{Im}(V_K)$ constitute the closed sets of a topology precisely because they are closed under arbitrary intersections and finite unions. Furthermore by (2.1.52) $V_K \circ I_k$ is a closure operator with image precisely the closed sets. Therefore by (2.1.41)

$$(V_K \circ I_k)(Y) = \bigcap_{Y \subseteq Z \in \mathrm{Zar}_k(K^n)} Z$$

which is the definition of the topological closure. The last statement follows by applying I_k and using **4.** ∎

For a fixed ideal $\mathfrak{a} \lhd k[X_1, ..., X_n]$ we may vary the field K to obtain families of solutions in different fields.

Definition 6.1.3 (Algebraic Set and L-valued points)
*Let $\mathfrak{a} \lhd k[X_1, ..., X_n]$ be a **radical** ideal. For every k-algebra R define the R-valued points by*

$$V(\mathfrak{a})(R) := \{r \in R^n \mid f(r_1, ..., r_n) \quad \forall f \in \mathfrak{a}\}$$

Definition 6.1.4 (Affine Variety)
*Let $\mathfrak{a} \lhd k[X_1, ..., X_n]$ be a **radical** ideal. The family $X(-) := V(\mathfrak{a})(-)$ is called an **affine variety** to which we associate the **coordinate ring** $k[X] := k[X_1, ..., X_n]/\mathfrak{a}$, which is a reduced f.g. k-algebra. Note the elements of $k[X]$ may be regarded as R-valued functions on the R-valued points $X(R)$.*

Note we may regard this as a functor, suppose we have compatible maps $i_{jk} : K_j/k \to K_k/k$ then these induce compatible maps

$$X(i_{jk}) : X(K_j) \to X(X_k)$$

Definition 6.1.5 (Residue Field)
*Let $X = V(\mathfrak{a})$ be an affine variety and L/k a field extension. For a point $x \in X(L)$ define the **residue field** to be*

$$k(x) := k(x_1, ..., x_n) \subset L/k$$

and the degree of x to be

$$\deg(x) := [k(x) : k]$$

Furthermore we define

$$\mathfrak{M}_{X,x} := I_k(\{(x)\})$$

for $x \in X(L)$. When x is algebraic then $\deg(x)$ is finite and $\mathfrak{M}_{X,x} = \mathfrak{m}_x/\mathfrak{a}$ is maximal (3.34.21).

Remark 6.1.6
$\mathbb{A}^n_k := V((0))$ is an affine variety and $\mathbb{A}^n_k(L) = L^n$.

Proposition 6.1.7 (Rational Points are Algebra Homomorphisms)
Let $X = V(\mathfrak{a})$ be an affine variety and R a k-algebra. Then there is a bijection

$$\{x \in X(R)\} \longleftrightarrow \mathrm{AlgHom}_k(k[X], R)$$

Proof. Given $x \in X(R)$ then the evaluation map $\mathrm{ev}_x : k[X_1, ..., X_n] \to R$ contains $\mathfrak{a}$ by definition and therefore (3.10.3) yields a unique map $\varphi_x : k[X] \to R$ such that $\varphi_x\left(\overline{X}_i\right) = x_i$.

Similarly given an algebra homomorphism $\varphi : k[X] \to R$ we may define $x = \left(\varphi\left(\overline{X}_1\right), ..., \varphi\left(\overline{X}_n\right)\right)$. Then by uniqueness these are mutual inverses. ∎

For completeness we also consider affine subvarieties in the same way as before

Proposition 6.1.8 (Affine subvarieties)
*Let $X = V(\mathfrak{a}) \subset \mathbb{A}^n_k$ be an affine variety and $A := k[X]$ the coordinate ring. For any ideal $\mathfrak{b} \lhd A$ and field extension K/k define the **zero-locus** by*

$$V_K(\mathfrak{b}) := \{(x) \in X(K) \mid f(x) = 0 \quad \forall f \in \mathfrak{b}\}$$

Similarly for a subset $Y \subset X(K)$ define

$$I_k(Y) := \{f \in k[X] \mid f(x) = 0 \quad \forall x \in Y\}$$

Suppose $\pi : k[X_1, ..., X_n] \to k[X]$ is the quotient map. Then we have the following properties

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$$V_K(\mathfrak{b}) = V_K(\pi^{-1}(\mathfrak{b})) \cap X(K)$$
$$I_k(Y \cap X(K)) = \pi(I_k(Y)) \quad Y \subset \mathbb{A}_k^n(K)$$
$$I_k(V_K(\mathfrak{b})) = \pi(I_k(V_K(\pi^{-1}(\mathfrak{b}))))$$

The pair (V_K, I_k) satisfies the same properties as (...). The image of V_K forms the closed sets of a topology on $X(K)$ which coincides with the subspace topology from $\mathbb{A}_k^n(K)$.

We may now generalize the Weak Nullstellensatz to arbitrary affine varieties and non-algebraically closed fields K/k.

Proposition 6.1.9 (Galois Group Action)
Let $X = V(\mathfrak{a})$ be an affine variety and K/k a normal *algebraic field extension (e.g. $\overline{k}$). There is a well-defined group action*

$$\mathrm{Aut}(K/k) \times X(K) \to X(K)$$
$$(\sigma, (x_1, ..., x_n)) \to (\sigma(x_1), ..., \sigma(x_n))$$

For $x, y \in X(K)$ the following are equivalent

a) x and y are topologically indistinguishable
b) $x = \sigma(y)$ for some $\sigma \in \mathrm{Aut}(K/k)$
c) $x \in \overline{\{y\}}$
d) $y \in \overline{\{x\}}$
e) $\overline{\{x\}} = \overline{\{y\}}$
f) $\mathfrak{M}_{X,x} = \mathfrak{M}_{X,y}$

For every point x we have

$$V_K(\mathfrak{M}_{X,x}) = \overline{\{x\}} = \{\sigma(x) \mid \sigma \in \mathrm{Aut}(K/k)\}$$

and x is quasi-closed (4.1.90) and therefore $X(K)$ is symmetric. Denote the set of Galois Orbits by $X_0(K)$ then the quotient map

$$\pi : X(K) \to X_0(K)$$

is precisely the Kolmogorov Quotient (4.1.36) and in particular a quasi-homeomorphism.

Proof. Given $F \in \mathfrak{a}$ we have $\sigma(F(x)) = F(\sigma(x))$. This shows that $x \in X(K) \implies \sigma(x) \in X(K)$. Further it's clear that $\sigma(\tau(x)) = (\sigma \circ \tau)(x)$ so the group action is well-defined.

By (4.1.34) a) $\iff$ e) $\implies$ c), d). By (6.1.2) $V(\mathfrak{M}_{X,x}) = \overline{\{x\}}$ and $I(\overline{\{x\}}) = \mathfrak{M}_{X,x}$. So e) $\iff$ f). Suppose $\mathfrak{M}_{X,x} = \mathfrak{M}_{X,y}$ then this means there exists an isomorphism $k(x) \cong k(y) \subset K$ which lifts to an element of $\mathrm{Aut}(K/k)$ by (3.18.69), so f) $\implies$ b). Conversely b) $\implies$ f) is straightforward. If $x \in \overline{\{y\}} = V(\mathfrak{M}_{X,y})$ then $\mathfrak{M}_{X,y} \subset \mathfrak{M}_{X,x}$ whence they are equal by maximality, so c), d) $\implies$ f). ∎

Corollary 6.1.10
Let k be an algebraically closed field and X an affine variety. Then $X(k)$ is a Kolmogorov topological space.

Proposition 6.1.11 (Generalized Weak Nullstellensatz)
Let $X = V(\mathfrak{a})$ be an affine variety. There is a bijection between maximal ideals and Galois orbits of points

$$\mathrm{Specm}(k[X]) \longleftrightarrow X_0(\overline{k}) = X(\overline{k})/\mathrm{Aut}(\overline{k}/k)$$
$$\mathfrak{M} \to [V_K(\mathfrak{M})]$$
$$\mathfrak{M}_{X,x} \leftarrow [x]$$

Proof. We may deduce this from (3.34.24) by observing that $x \in X(\overline{k}) \iff \mathfrak{a} \subset \mathfrak{m}_x$, $\mathfrak{M}_{X,x} = \mathfrak{m}_x/\mathfrak{a}$ and using the correspondence of ideals of quotient ring (3.3.56)

$$\{\mathfrak{M} \in \mathrm{Specm}(k[X_1, ..., X_n]) \mid \mathfrak{a} \subset \mathfrak{M}\} \longleftrightarrow \mathrm{Specm}(k[X]).$$

∎

Lemma 6.1.12 (Rabinowitsch Trick)
Let $\mathfrak{a} \lhd A$ and $f \in A$. Consider the ring $B = A[Y]$. If $\mathfrak{a}B + (1 - Yf) = B$ then $f \in \sqrt{\mathfrak{a}}$.

Proof. The hypothesis implies

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$$1 = (1 - Yf)g(Y) + ah(Y)$$

for $a \in \mathfrak{a}$ and $h(Y) \in A[Y]$. Consider the quotient map $\bar{\cdot} : A \to A/\mathfrak{a}$ and the corresponding map $A[Y] \to (A/\mathfrak{a})[Y]$. Applying this to the above shows $1 - Y\overline{f}$ is invertible in $(A/\mathfrak{a})[Y]$. So by (3.7.5) $\overline{f}$ is nilpotent in $(A/\mathfrak{a})$ whence $f \in \sqrt{\mathfrak{a}}$. ∎

Proposition 6.1.13 (Strong Nullstellensatz)
Let X be an affine variety and $\mathfrak{a} \lhd k[X]$. Then

$$I_k V_{\overline{k}}(\mathfrak{a}) = \sqrt{\mathfrak{a}}^J = \sqrt{\mathfrak{a}}$$

In particular the $\overline{k}$-radical ideals are precisely the radical ideals. and there is a dual isomorphism

$$\mathrm{Rad}(k[X]) \longleftrightarrow \mathrm{Zar}_k\left(X\left(\overline{k}\right)\right)$$

Proof. Observe that $x \in V_{\overline{k}}(\mathfrak{a})$ if and only if $\mathfrak{a} \subseteq \mathfrak{M}_{X,x}$. Therefore

$$I_k(V_{\overline{k}}(\mathfrak{a})) = \bigcap_{x \in V_{\overline{k}}(\mathfrak{a})} I_k(\{x\}) = \bigcap_{\mathfrak{a} \subseteq \mathfrak{M}_{X,x}} \mathfrak{M}_{X,x}$$

Therefore by the correspondence in (6.1.11) this is equal to precisely $\sqrt{\mathfrak{a}}^J$.

For the second equality we consider first the case $X = \mathbb{A}^n_k$ and $k[X] = k[X_1, ..., X_n]$. Let $\mathfrak{a} \lhd k[X]$ and choose $f \in I_k V_{\overline{k}}(\mathfrak{a})$. Consider the ring $B \coloneqq k[X_1, ..., X_n, Y]$ and the ideal $\tilde{\mathfrak{a}} = \mathfrak{a}B + (1 - Yf)$. Clearly this has no zeros in $\overline{k}^{n+1}$, so by the Weak Nullstellensatz (3.34.24) it is not proper. By the (6.1.12) $f \in \sqrt{\mathfrak{a}}$ as required. The reverse inclusion is clear.

Now suppose that $X = V(\mathfrak{a})$, $k[X] = k[X_1, ..., X_n]/\mathfrak{a}$ and $\mathfrak{b} \lhd k[X]$ is proper. Using (6.1.8) and (3.3.50) together with the result just proven, shows

$$I_k(V_{\overline{k}}(\mathfrak{b})) = \pi(I_{\overline{k}} V_{\overline{k}}(\pi^{-1}\mathfrak{b})) = \pi\left(\sqrt{\pi^{-1}(\mathfrak{b})}\right) = \sqrt{\mathfrak{b}}$$

where $\pi : k[X_1, ..., X_n] \to k[X]$ is canonical surjective morphism. ∎

Corollary 6.1.14
Let $k[X]$ be any finitely generated reduced k-algebra, then $k[X]$ is a Jacobson ring, i.e.

$$\sqrt{\mathfrak{a}} = \sqrt{\mathfrak{a}}^J$$

In particular the intersection of all maximal ideals is zero

$$\bigcap_{\mathfrak{m}} \mathfrak{m} = 0$$

6.1.1 Topological Properties

Proposition 6.1.15 (Principal Open Sets)
Let $X = V(\mathfrak{a})$ be an affine variety over k. For every $f \in k[X]$ the sets

$$D(f) \coloneqq X\left(\overline{k}\right) \setminus V_{\overline{k}}((f)) = \left\{(x) \in X\left(\overline{k}\right) \;\middle|\; f(x) \neq 0\right\}$$

form a base for the Zariski topology $\mathrm{Zar}_k\left(X\left(\overline{k}\right)\right)$.

Proof. Let $U = X\left(\overline{k}\right) \setminus V(\mathfrak{b})\left(\overline{k}\right)$ be a non-empty open set. For every $x \in U$ there by definition exists $f \in \mathfrak{b}$ such that $f(x) \neq 0$. Therefore $D(f)$ is a neighbourhood of x in the Zariski topology. ∎

The topological notion of irreducibility is important, and may be reduced to a purely algebraic statement on the coordinate ring.

Proposition 6.1.16 (Criterion for Irreducibility)
Let X be an affine variety (e.g. $\mathbb{A}^n_k$) and $Y = V(\mathfrak{b})$ an affine subvariety corresponding to a radical ideal $\mathfrak{b} \lhd k[X]$. Then the following are equivalent

a) $Y\left(\overline{k}\right)$ is an irreducible subset of $X\left(\overline{k}\right)$
b) $Y_0\left(\overline{k}\right)$ is an irreducible subset of $X_0\left(\overline{k}\right)$
c) $\mathfrak{b}$ is prime
d) $k[Y]$ is an integral domain.

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Proof. Suppose X is not irreducible. Then we have $X \subseteq V_{\overline{k}}(\mathfrak{b}) \cup V_{\overline{k}}(\mathfrak{c})$ a non-trivial decomposition into closed subsets (and associated ideals). Then by the dual isomorphism we have also $\mathfrak{a} \subsetneq \mathfrak{b}$ and we may choose $f \in \mathfrak{b} \setminus \mathfrak{a}$ and similarly $g \in \mathfrak{c} \setminus \mathfrak{a}$. However fg vanishes on $X\left(\overline{k}\right)$ ands so we have $fg \in \mathfrak{a}$. Therefore $\mathfrak{a}$ is not prime.

Conversely suppose X is irreducible and $\mathfrak{bc} \subseteq \mathfrak{a}$. Then $X \subseteq V_{\overline{k}}(\mathfrak{b}) \cup V_{\overline{k}}(\mathfrak{c})$. By irreducibility we have $X \subseteq V_{\overline{k}}(\mathfrak{b})$, whence applying $I_{\overline{k}}(-)$ we see $\mathfrak{b} \subseteq I_{\overline{k}}V_{\overline{k}}(\mathfrak{b}) \subseteq \mathfrak{a}$ (since $\mathfrak{a}$ is radical). Therefore $\mathfrak{a}$ is prime. ∎

When $W \subset X$ is an irreducible subset then write $\mathfrak{M}_{X,W} \coloneqq I(W)$, by the previous result this equals $I\left(\overline{W}\right)$ and $\mathfrak{M}_{X,W}$ is a prime ideal.

Proposition 6.1.17
Let X be an affine variety, W, Z be subsets of $X\left(\overline{k}\right)$. Then

$$W \subset \overline{Z} \iff \mathfrak{M}_{X,Z} \subset \mathfrak{M}_{X,W}$$

and

$$\overline{W} = \overline{Z} \iff \mathfrak{M}_{X,Z} = \mathfrak{M}_{X,W}$$

Proof. Observe that $V\left(\mathfrak{M}_{X,W}\right) = \overline{W}$ by (6.1.2). Therefore the result follows immediately from the same result. ∎

Proposition 6.1.18 (Points are Closed)
Let $X = V(\mathfrak{a})$ be an affine variety and $x \in X\left(\overline{k}\right)$. Then

$$\overline{\{x\}} = V\left(\mathfrak{M}_{X,x}\right) = \left\{\sigma(x) \;\middle|\; \sigma \in \mathrm{Aut}\left(\overline{k}/k\right)\right\}$$

is irreducible and finite of order $[k(x) : k]_s$. Further X is a symmetric space.

Proof. The equalities are proven in (6.1.9). Further $\overline{\{x\}}$ is irreducible by (4.1.67). We may replace $\mathrm{Aut}\left(\overline{k}/k\right)$ with $\mathrm{Mor}_k\left(k(x), \overline{k}\right)$ to determine the order.

Clearly X is symmetric because it satisfies criteria (4.1.41) b). ∎

Proposition 6.1.19 (Closed sets $\leftrightarrow$ radical ideals)
Let $X = V(\mathfrak{a})$ be an affine variety. Recall (6.1.13) there is a dual lattice isomorphism between radical ideals and "Zariski"-closed subsets of $X\left(\overline{k}\right)$ (and indeed $X_0\left(\overline{k}\right)$).

$$\mathrm{Rad}(k[X]) \underset{I_k(-)}{\overset{V_{\overline{k}}(-)}{\rightleftarrows}} \mathrm{Zar}_k\left(X\left(\overline{k}\right)\right)$$

Under this isomorphism we have

- *maximal ideals correspond to $\mathrm{Aut}\left(\overline{k}/k\right)$-orbits of $X\left(\overline{k}\right)$ (or elements of $X_0\left(\overline{k}\right)$)*
- *prime ideals of $k[X]$ correspond to irreducible subsets of $X\left(\overline{k}\right)$ (and therefore of $X_0\left(\overline{k}\right)$)*
- *minimal prime ideals of $k[X]$ correspond to irreducible components of $X\left(\overline{k}\right)$ (and therefore of $X_0\left(\overline{k}\right)$)*

Recall that prime ideals are precisely the meet-prime radical ideals and irreducible subsets are precisely the join-prime closed subsets (see (4.1.61)). Therefore we have a dual isomorphism between the Krull Lattice of radical ideals of $k[X]$ and the Krull Latice of closed subsets of $X\left(\overline{k}\right)$.

Proof. The content of the Strong Nullstellensatz is precisely that $I_k(-) \circ V_{\overline{k}}(-) = \mathbf{1}$. The other direction was already proven so we have a dual order isomorphism. The statement about maximal ideals was already shown in (6.1.11) and prime ideals in (6.1.16). Then as irreducible components are precisely maximal irreducible subsets the final statement follows from the dual order isomorphism. ∎

Corollary 6.1.20
Let $X = V(\mathfrak{a})$ be an affine variety. There is a bijective correspondence between affine subvarieties of X and closed subsets of $X\left(\overline{k}\right)$.

Definition 6.1.21 (Irreducible Affine Variety)
*We say an affine variety $X = V(\mathfrak{a})$ is **integral** or **irreducible** if the topological space $X\left(\overline{k}\right)$ is irreducible.*

This is the case precisely when $\mathfrak{a}$ is prime, or when $k[X]$ is an integral domain by (6.1.19), or equivalently irreducible (3.3.67), since it is assumed to be reduced.

Note trivially that $\mathbb{A}^n_k$ is irreducible.

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Proposition 6.1.22 (Decomposition into Irreducible Components)
Let $X = V(\mathfrak{a})$ be an affine variety then the topological space $X\left(\overline{k}\right)$ is Noetherian. Furthermore it has finitely many irreducible components X_i and the decomposition

$$X\left(\overline{k}\right) = X_1 \cup ... \cup X_n$$

is the unique incomparable decomposition into irreducible closed subsets.

Proof. By Hilbert's Basis Theorem (3.14.8) $k[X]$ is a Noetherian ring, so by (6.1.19) $X\left(\overline{k}\right)$ is Noetherian. The result then follows from (4.1.76) ∎

Proposition 6.1.23 (Subspace Topology)
Let $X = V(\mathfrak{a})$ an affine variety and $Y = V(\mathfrak{b})$ an affine subvariety. Then there is a commutative diagram

$$\begin{array}{ccc} \{\mathfrak{c} \in \mathrm{Rad}(k[X]) \mid \mathfrak{b} \subseteq \mathfrak{c}\} & \underset{I}{\overset{V}{\rightleftarrows}} & \left\{Z \in \mathrm{Zar}_k\left(X\left(\overline{k}\right)\right) \;\middle|\; Z \subseteq Y\left(\overline{k}\right)\right\} \\ \pi^{-1} \uparrow \;\downarrow \pi & & \| \\ \mathrm{Rad}(k[X]/\mathfrak{b}) & \underset{I}{\overset{V}{\rightleftarrows}} & \mathrm{Zar}_k\left(Y\left(\overline{k}\right)\right) \end{array}$$

under which prime ideals correspond to irreducible subsets and the horizontal arrows induce dual isomorphisms.

In particular the subspace topology for $Y\left(\overline{k}\right)$ coincides with the Zariski topology.

Proof. The left hand arrows are mutual inverses by (3.3.56). The horizontal maps are dual isomorphisms by (6.1.19). The equality then follows from (6.1.8). ∎

Proposition 6.1.24
Let X be an affinve variety. The correspondence between points and ideals (6.1.11) restricts as follows

$$\mathrm{Specm}\left(k[X]_f\right) \longleftrightarrow \{\mathfrak{M} \in \mathrm{Specm}(k[X]) \mid f \notin \mathfrak{M}\} \longleftrightarrow D(f)_0 := D(f)/\operatorname{Aut}\left(\overline{k}/k\right)$$

More precisely a point $x \in D(f)$ corresponds to the maximal ideal $\left(\mathfrak{M}_{X,x}\right)_f$ of $k[X]_f$.

Proof. For the first correspondence consider the localisation map $i : k[X] \to k[X]_f$. By (3.5.27) the extension of a maximal ideal not containing f is maximal. Conversely i satisfies the hypotheses of (3.34.27) so the inverse image of a maximal ideal is maximal. This shows the inverse image of a maximal ideal is maximal, not containing f. Therefore the correspondence of prime ideals (3.5.36) restricts to maximal ideals.

The second correspondence is immediate from (6.1.11) because $f \notin \mathfrak{M}_{X,x} \iff f(x) \neq 0$. ∎

6.1.2 Structure Sheaf

Definition 6.1.25 (Sheaf of Regular Functions)
*Let X be an affine variety, $U \subset X\left(\overline{k}\right)$ an open subset and $f : U \to \overline{k}$. We say that f is **regular** if for every $x \in U$ there exists $g, h \in k[X]$ and an open neighbourhood V of x such that*

$$f(y) = \frac{g(y)}{h(y)} \quad \forall y \in V$$

*We denote the **structure sheaf** by*

$$\mathcal{O}_X(U) := \left\{f : U \to \overline{k} \;\middle|\; f \text{ regular}\right\}$$

which is a sheaf of k-algebras.

If $U \subset X$ is an open subset then we denote the structure sheaf by the restriction

$$\mathcal{O}_U := \mathcal{O}_X \mid_U$$

Proposition 6.1.26 (Sections are localisation of coordinate ring)
Let $X = V(\mathfrak{a})$ be an affine variety and $f \in k[X]$. There is a canonical isomorphism

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$$\rho_f : k[X]_f \xrightarrow{\sim} \mathcal{O}_X(D(f))$$
$$\frac{g}{f^n} \to \frac{g(-)}{f(-)^n}$$

Furthermore $D(g) \subseteq D(f) \iff S_f \subseteq \overline{S_g}$ *and we have the following commutative diagram*

$$\begin{array}{ccc} k[X]_f & \xrightarrow{\sim} & \mathcal{O}_X(D(f)) \\ \downarrow{\scriptstyle i_{S_f S_g}} & & \downarrow{\scriptstyle (-)|_{D(g)}} \\ k[X]_g & \xrightarrow{\sim} & \mathcal{O}_X(D(g)) \end{array}$$

In particular there is a canonical isomorphim

$$k[X] \xrightarrow{\sim} \mathcal{O}_X(X)$$

Proof. The map is trivially well-defined and injective. Consider a regular map $\sigma \in \mathcal{O}_X(D(f))$. Consider the ideal

$$\mathfrak{a} \coloneqq \left\{ g \in k[X] \mid g\sigma \in \mathrm{Im}\left(i_f\right) \right\}$$

It is enough to show $f \in \mathfrak{a}$. Suppose $f \notin \mathfrak{a}$ then it's contained in a maximal ideal which is of the form $\mathfrak{M}_{X,x}$ for some $x \in X\left(\overline{k}\right)$ by (6.1.11). By definition $x \in D(f)$ and there is an open neighbourhood $W \subseteq D(f)$ and elements $h_1, h_2 \in k[X]$ such that

$$\sigma(y) = h_1 \frac{y}{h_2}(y) \quad \forall y \in W$$

Choose $h_3 \in k[X]$ such that $x \in D(h_3) \subseteq W$ then clearly

$$\sigma(y) = (h_1 h_3)\frac{y}{h_2 h_3}(y) \quad \forall y \in D(h_3)$$

and in particular $(h_2 h_3 \sigma)(y) = (h_1 h_3)(y)$ for all $y \in D(f)$. Therefore $h_2 h_3 \in \mathfrak{a} \subseteq \mathfrak{m}_x$ which implies $h_2(x) = 0$ or $h_3(x) = 0$ a contradiction.

Therefore $f \in \mathfrak{a}$ and clearly $\sigma \in \mathrm{Im}\left(i_f\right)$ as required. ∎

Corollary 6.1.27
Suppose $f \in k[X]$ *is a regular function then* $f \neq 0 \implies D(f) \neq \emptyset$.

In what follows we may systematically identify $\mathcal{O}_X(X)$ and $k[X]$, and we formalise this in the following result.

Proposition 6.1.28 (Properties of the Structure Sheaf)
Let $X = V(\mathfrak{a})$ *be an affine variety and let* $\widehat{\cdot} : k[X] \xrightarrow{\sim} \mathcal{O}_X(X)$ *be the canonical isomorphism defined in (6.1.26). Then the following properties hold*

a) For all $f \in \mathcal{O}_X(X)$ *we have* $D(f) = \{x \in U \mid f(x) \neq 0\}$ *is open and* $f|_{D(f)}$ *is invertible.*
b) For an irreducible subset $W \subset X$ *define the prime ideal*

$$\mathfrak{M}_{X,W} \coloneqq \{f \in \mathcal{O}_X(X) \mid f(x) = 0 \quad \forall x \in W\}$$

c) For $x \in X$ *we have* $\mathfrak{M}_{X,x}$ *is maximal and every maximal ideal is of this form*
d) For $W, Z \subset X$ *be irreducible subsets then*

$$\mathfrak{M}_{X,Z} \subset \mathfrak{M}_{X,W} \iff W \subset \overline{Z}$$

and

$$\mathfrak{M}_{X,Z} = \mathfrak{M}_{X,W} \iff \overline{W} = \overline{Z}$$

In particular $\mathfrak{M}_{X,W} = \mathfrak{M}_{X,\overline{W}}$
e) For all $f \in \mathcal{O}_X(X)$ *we have a canonical isomorphism*

$$\begin{array}{ccc} \mathcal{O}_X(X) & & \\ \downarrow & \searrow^{\rho_{X D(f)}} & \\ \mathcal{O}_X(X)_f & \xrightarrow{\sim} & \mathcal{O}_X(D(f)) \end{array}$$

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which is the unique homomorphism making this diagram commute

For X an affine variety we denote by $\overline{X}_1, ..., \overline{X}_n$ the image of the coordinate functions under $\hat{\cdot}$.

6.1.3 Regular Morphisms of Affine Varieties

Definition 6.1.29 (Regular Morphism of Affine Varieties)
*Let $X \subset \mathbb{A}_k^n$, $Y \subset \mathbb{A}_k^m$ be affine varieties and $\phi : X\left(\overline{k}\right) \to Y\left(\overline{k}\right)$ a function. Then we say that ϕ is a **regular morphism** (at x) if $\pi_j \circ \phi$ is regular (at x) in the sense of (6.1.25) for $j = 1, ..., m$.*

A regular morphism is an isomorphism if it has a two-sided inverse which is also regular.

Note a regular function $X \to \overline{k}$ is completely equivalent to a regular morphism $X \to \mathbb{A}_k^1$.

Proposition 6.1.30 (Regular Morphisms of Affine Varieties)
Let $X = V(\mathfrak{a})$ and $Y = V(\mathfrak{b})$ be affine varieties. Then there is a bijection

$$\begin{aligned} \mathrm{AlgHom}_k(\mathcal{O}_Y(Y), \mathcal{O}_X(X)) &\longleftrightarrow \{\phi : X \to Y \mid \phi \text{ regular}\} \\ \phi^\sharp \to & \left(\phi^\sharp\left(\overline{Y}_1\right)(-), ..., \phi^\sharp\left(\overline{Y}_m\right)(-)\right) \\ \overline{G} \to G(\pi_1 \circ \phi, ..., \pi_m \circ \phi_m) \leftarrow & \phi \end{aligned}$$

Proof. Let $\phi^\sharp$ be a k-algebra homomorphism, then $\phi^\sharp\left(\overline{Y}_i\right) \in \mathcal{O}_Y(Y)$ is a regular function. Therefore by definition the first map is well-defined. Conversely $\pi_j \circ \phi$ is regular so corresponds to an element of $\phi_j \in \mathcal{O}_X(X)$ and the second map is well-defined. As ϕ is uniquely determined by $\phi^\sharp\left(Y_j\right)$ for $j = 1, ..., m$ we see that the maps are mutually inverse. ■

Lemma 6.1.31
Let $\phi : X \to Y$ be a regular morphism of affine varieties and $W \subset X$ an irreducible subset. Then

$$\left(\phi^\sharp\right)^{-1}\left(\mathfrak{M}_{X,W}\right) = \mathfrak{M}_{Y,\phi(W)} = \mathfrak{M}_{Y,\overline{\phi(W)}}$$

Proof. This follows immediately from the definitions and (6.1.28). ■

Proposition 6.1.32
Let $\phi : X \to Y$ be a regular morphism of affine varieties and $W \subset X$, $Z \subset Y$ irreducible. Then

$$Z \subset \overline{\phi(W)} \iff \left(\phi^\sharp\right)^{-1}\left(\mathfrak{M}_{X,W}\right) \subset \mathfrak{M}_{Y,Z}$$

and

$$\overline{Z} = \overline{\phi(W)} \iff \left(\phi^\sharp\right)^{-1}\left(\mathfrak{M}_{X,W}\right) = \mathfrak{M}_{Y,Z}$$

Proof. This follows from (6.1.28) and (6.1.31). ■

Proposition 6.1.33
Let $\phi : X \to Y$ be a regular morphism of affine varieties and $x \in X$, $y \in Y$ points. Then the following are equivalent

a) $y \in \overline{\{\phi(x)\}}$
b) $\phi(x) \in \overline{\{y\}}$
c) $\overline{\{\phi(x)\}} = \overline{\{y\}}$
d) $\left(\phi^\sharp\right)^{-1}\left(\mathfrak{M}_{X,x}\right) = \mathfrak{M}_{Y,y}$

When k is algebraically closed then this is equivalent to $\phi(x) = y$.

Proof. We have a)-c) are equivalent because Y is symmetric (6.1.9), (4.1.42), and then c) $\iff$ d) follows from (6.1.32). ■

Corollary 6.1.34
Let $\phi : X \to Y$ be a regular morphism of affine varieties. Then ϕ is continuous.

Proof. Observe that $\phi^{-1}(D(f)) = D(f \circ \phi) = D\left(\phi^\sharp(f)\right)$ so we are done by (6.1.15) and (4.1.27). ■

Corollary 6.1.35
Let $\phi : X \to Y$ and $\psi : Y \to Z$ be regular morphisms of affine varieties. Then $\psi \circ \phi$ is regular and

$$(\psi \circ \phi)^\sharp = \phi^\sharp \circ \psi^\sharp$$

Proposition 6.1.36 (Affine Varieties form a Category)
The collection of affine varieties together with regular morphisms form a category $\mathbf{AffVar}_k$. Furthermore there is an equivalence of categories

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$$\mathbf{AffVar}_k \xrightarrow{\sim} \mathbf{ReducedFgAlg}_k^{op}$$

Proof. By using the correspondence with k-algebra homomorphisms (6.1.30) we may show that the law of composition for regular morphisms also satisfy the axioms of a category. We have therefore shown that the assignment $X \to k[X]$ is full and faithful functor. To show it is essentially surjective consider a reduced finitely-generated k-algebra A. Then by definition $A \cong k[X_1, ..., X_n]/\mathfrak{a}$ for some ideal $\mathfrak{a}$. As A is reduced we see $\mathfrak{a}$ is radical and therefore A is the coordinate ring of the affine variety $X = V(\mathfrak{a})$ and by definition $k[X] \cong A$. ∎

Lemma 6.1.37
Let $\phi : X \to Y$ be a regular morphism of affine varieties. Suppose that $V \subset Y$ is open and $f : V \to \overline{k}$ is regular. Then $f \circ \phi : \phi^{-1}(V) \to \overline{k}$ is regular.

In other words there is a well-defined functor

$$\mathbf{AffVar}_k \to \mathbf{LocalSpaceFunctions}_k$$

which we show is full and faithful in (6.1.39).

Proposition 6.1.38 (Principal Open Sets are Affine)
Let $X = V(\mathfrak{a})$ be an affine variety and $f \in \mathcal{O}_X(X)$. Define $Y \coloneqq V(\mathfrak{a}^e + (1 - X_{n+1}F)) \subset \mathbb{A}_k^{n+1}$ where $f = \overline{F}$ and $F \in k[X_1, ..., X_n]$. Then there is an isomorphism of spaces with functions

$$\theta : \left(D(f), \mathcal{O}_X|_{D(f)}\right) \to \left(Y\left(\overline{k}\right), \mathcal{O}_Y\right)$$
$$(x_1, ..., x_n) \to \left(x_1, ..., x_n, \frac{1}{F(x_1, ..., x_n)}\right)$$

Proof. Observe $V(\mathfrak{a}^e + (1 - X_{n+1}F)) = V(\mathfrak{a}^e) \cap V(1 - X_{n+1}F)$. Therefore θ is well-defined. Conversely given $(x_1, ..., x_{n+1}) \in V(\mathfrak{a}^e + (1 - X_{n+1}F)) = V(\mathfrak{a}^e) \cap V(1 - X_{n+1}F)$, by definition $(x_1, ..., x_n) \in V(\mathfrak{a})$ and $F(x_1, ..., x_n)x_{n+1} = 1 \implies F(x_1, ..., x_n) \neq 0$. Therefore the inverse map is well-defined. Evidently this restricts to a bijection

$$D\left(\overline{G}\right) \cap D(F) \longleftrightarrow D(G) \cap V(\mathfrak{a}^e + (1 - X_{n+1}F))$$

By (6.1.24) and (4.1.27) θ is a homeomorphism.

Suppose that $V \subset Y$ is open and $g : V \to \overline{k}$ is regular, then for every $y \in V$ there is some neighbourhood W for which

$$g(z_1, ..., z_{n+1}) = \frac{P(z_1, ..., z_{n+1})}{Q(z_1, ..., z_{n+1})} \quad \forall z \in W$$

where $P, Q \in k[Y_1, ..., Y_{n+1}]$ and $Q(z) \neq 0$. Define the rational functions in $k(X_1, ..., X_n)$ as follows

$$R(X_1, ..., X_n) \coloneqq P\left(X_1, ..., X_n, F(X_1, ..., X_n)^{-1}\right)$$
$$S(X_1, ..., X_n) \coloneqq Q\left(X_1, ..., X_n, F(X_1, ..., X_n)^{-1}\right)$$

Let $U \coloneqq \theta^{-1}(W)$. Then evidently $R(x) = P(\theta(x))$ and $S(x) = Q(\theta(x))$ for all $x \in U$. In particular we deduce that $S(\theta^{-1}(y)) \neq 0 \implies S \neq 0$ and that $g \circ \theta$ is regular at $\theta^{-1}(y)$. The converse is straightforward so we deduce that θ is an isomorphism. ∎

When considering regular maps over an open subset of an affine variety, there are two plausible definitions which we show are equivalent.

Proposition 6.1.39 (Quasi-affine Morphisms)
Let $X \subset \mathbb{A}_k^n$, $Y \subset \mathbb{A}_k^m$ be affine varieties, $U \subset X\left(\overline{k}\right)$ an open subset and $\phi : U \to Y\left(\overline{k}\right)$ a function. Then the following are equivalent

a) $\pi_j \circ \phi$ is regular for all $j = 1, ..., m$.
b) ϕ determines a regular morphism $(U, \mathcal{O}_X|_U) \to (Y, \mathcal{O}_Y)$ of spaces with functions.

If $V \subset Y\left(\overline{k}\right)$ is an open subset and $\phi(U) \subset V$ then ϕ determines a regular morphism $(U, \mathcal{O}_X|_U) \to (V, \mathcal{O}_Y|_V)$.

Proof. b) $\implies$ a) follows trivially because $\pi_j : Y\left(\overline{k}\right) \to \overline{k}$ is regular.

a) $\implies$ b) Firstly we observe that for any $f \in \mathcal{O}_X(X)$ for which $D(f) \subset U$ there is a commutative diagram

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$$\begin{array}{ccc} D(f) & \xrightarrow{\phi|_{D(f)}} & Y\left(\overline{k}\right) \\ \sim\downarrow\theta & \nearrow_{\tilde{\phi}_f} & \\ V(\mathfrak{a}^e + (1 - X_{n+1}F)) & & \end{array}$$

where θ is defined in (6.1.38) and $\tilde{\phi}_f(x_1, ..., x_{n+1}) = \phi(x_1, ..., x_n)$. By definition $\tilde{\phi}_f$ is a regular morphism of affine varieties and therefore determines a regular morphism of spaces of functions (6.1.37). As θ is a regular isomorphism we find that $\phi|_{D(f)}$ is a regular morphism of spaces of functions. As the $D(f)$ form a base on the topology of U we may conclude from (4.3.27) that ϕ is a regular morphism of spaces with functions. ■

Proposition 6.1.40
Let X, Y, Z be affine varieties with $U \subset X$ and $V \subset Y$ open subsets. Suppose that $\phi : U \to V$ is regular and $\psi : V \to Z$ is regular, then $\psi \circ \phi : U \to Z$ is regular.

Corollary 6.1.41
Let X be an affine variety, $U \subset X$ an open subset and $f \in \mathcal{O}_X(U)$. Then the set

$$D(f) \coloneqq \{x \in U \mid f(x) \neq 0\}$$

is open and $f|_{D(f)}$ is invertible.

Proof. By (6.1.39) we may regard f as a regular morphism $(U, \mathcal{O}_X|_U) \to \mathbb{A}^1_k\left(\overline{k}\right)$, and it is in particular continuous. Then $\{0\} = V(X_1)$ is a closed subset of $\mathbb{A}^1_k\left(\overline{k}\right)$ and so $D(f) = U \setminus f^{-1}(\{0\})$ is an open subset. It is clear by the definition of the structure sheaf that $f|_{D(f)}$ is invertible. ■

Corollary 6.1.42 (Affine Mapping Property)
Let X, Y be affine varieties. Then there are bijections

$$\mathrm{Mor}(X, Y) \xrightarrow{\sim} \mathrm{Mor}((X, \mathcal{O}_X), (Y, \mathcal{O}_Y)) \xrightarrow{\sim} \mathrm{AlgHom}_k(\mathcal{O}_Y(Y), \mathcal{O}_X(X))$$

where the second map is given by $\phi^\sharp_Y$. Furthermore these are natural in either X or Y, and preserve isomorphisms.

Proof. The first map is well-defined by (6.1.39) and evidently injective. The composite map is bijective by (6.1.30), which shows the first map is surjective and hence bijective. We conclude trivially the second map is also a bijection. ■

Proposition 6.1.43 (Equaliser is Closed)
Let $\phi, \psi : X \to Y$ be regular morphisms of affine varieties. Then the equaliser

$$\left\{x \in X\left(\overline{k}\right) \;\middle|\; \phi(x) = \psi(x)\right\}$$

is closed.

Proof. By applying the projections we may reduce to the case $Y = \mathbb{A}^1_k$. Then $\phi - \psi$ is regular and the equaliser is the inverse image of $\{0\}$ which is evidently closed in $\mathbb{A}^1_k$. As regular maps are continuous then we are done. ■

Proposition 6.1.44
Let $Y = V(\mathfrak{a})$ be an affine variety and $Z \coloneqq V(\mathfrak{b})$ a closed subvariety for $\mathfrak{a} \lhd k[Y]$. Then the space of functions $(Z, \mathcal{O}_Z)$ coincides with the notion of a closed subspace (4.3.21).

Further the inclusion map $i : Z \hookrightarrow Y$ is regular and the induced map

$$i^\sharp_Y : \mathcal{O}_Y(Y) \longrightarrow \mathcal{O}_Z(Z)$$

is a surjective homomorphism with kernel precisely $\rho(\mathfrak{b})$ where $\rho : k[Y] \xrightarrow{\sim} \mathcal{O}_Y(Y)$ is the canonical isomorphism.

Proof. We have a surjective homomorphism $\pi : k[Y] \to k[Z] \coloneqq k[Y]/\mathfrak{b}$ with kernel precisely $\mathfrak{b}$. Suppose $f \in \mathcal{O}_Y(U)$ is a regular map, then we wish to show that $f|_{U \cap Z}$ is regular. But if f is given locally by a ratio $\frac{g}{h}$ then $f|_{U\cap Z} = i^\sharp_U(f)$ is given locally by the ratio $\frac{\pi(g)}{\pi(h)}$. And because π is surjective this means $i^\sharp_Y$ is surjective. Furthermore we have a commutative diagram

$$\begin{array}{ccc} \mathcal{O}_Y(Y) & \xrightarrow{i^\sharp} & \mathcal{O}_Z(Z) \\ \sim\uparrow\rho & & \sim\uparrow\rho \\ k[Y] & \xrightarrow{\pi} & k[Z] \end{array}$$

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from which the properties of $i_Y^\sharp$ follows immediately.

Recall that the Zariski topology on Z coincides with the subspace topology (4.3.21), and we have shown the notion of regular function coincides with the closed subspace definition (4.3.23). Therefore the topology on Z and structure sheaves $\mathcal{O}_Z$ coincide. ∎

Proposition 6.1.45
Let $\phi : X \to Y$ be a regular morphism of affine varieties. Then ϕ is a closed immersion (4.3.23) if and only if $\phi^\sharp$ is surjective.

Proof. Suppose that $\phi^\sharp$ is surjective and let $Z \coloneqq V\big(\ker\big(\phi^\sharp\big)\big)$ be a closed subset of Y. Then $i : Z \hookrightarrow Y$ is regular and we have a commutative diagram

$$\begin{array}{ccc} \mathcal{O}_Y(Y) & \xrightarrow{\phi^\sharp} & \mathcal{O}_X(X) \\ \downarrow{\scriptstyle i^\sharp} & \nearrow_{\tilde{\phi}^\sharp} & \\ \mathcal{O}_Z(Z) & & \end{array}$$

By (6.1.44) $\ker\big(i^\sharp\big) = \ker\big(\phi^\sharp\big)$, so $\tilde{\phi}^\sharp$ exists and is an isomorphism by (3.3.56). By (6.1.42) there exists a corresponding regular morphism $\tilde{\phi} : X \to Z$ which is an isomorphism which satisfies $i \circ \tilde{\phi} = \phi$. By the criteria in (4.3.23) this means ϕ is a closed immersion.

Conversely if ϕ is a closed immersion then by definition $\phi = i \circ \tilde{\phi}$ for $i : Z \hookrightarrow Y$ the inclusion of a closed subset where $\tilde{\phi}$ is a regular isomorphism. By (6.1.42) $\tilde{\phi}^\sharp \circ i^\sharp = \phi^\sharp$ and $\tilde{\phi}^\sharp$ is an isomorphism, which shows that $\phi^\sharp$ is surjective. ∎

6.1.4 Dominant Morphisms

Definition 6.1.46
Let $\phi : X \to Y$ be a regular morphism of affine varieties. Then we say it is

- ***finite*** *if $\phi_Y^\sharp : \mathcal{O}_Y(Y) \to \mathcal{O}_X(X)$ is module-finite (equivalently integral)*
- ***dominant*** *if $\phi\big(X\big(\overline{k}\big)\big)$ is dense in $Y\big(\overline{k}\big)$*

Proposition 6.1.47 (Criteria for dominant morphisms)
A regular morphism $\phi : X \to Y$ is dominant if and only if $\phi_Y^\sharp : \mathcal{O}_Y(Y) \to \mathcal{O}_X(X)$ is injective. In particular when both X and Y are integral then this induces a field extension

$$\phi_\star : k(Y) \hookrightarrow k(X)$$

Proof. Recall $\phi(X)$ is dense precisely when the closure $\overline{\phi(X)} = Y$. Then

$$\begin{aligned} \ker\big(\phi_Y^\sharp\big) = I_k(\phi(X)) &\implies V_{\overline{k}}\big(\ker\big(\phi^\sharp\big)\big) = V_{\overline{k}}(I_k(\phi(X))) = \overline{\phi(X)} \\ &\implies \sqrt{\ker(\phi^\sharp)} = I_k\big(\overline{\phi(X)}\big) \end{aligned}$$

by (6.1.2) and (6.1.13). If $\ker\big(\phi_Y^\sharp\big) = 0$ then clearly $\overline{\phi(X)} = Y$. Conversely if this holds then

$$\sqrt{\ker\big(\phi_Y^\sharp\big)} = I_k\big(Y\big(\overline{k}\big)\big) = I_k(V_{\overline{k}}(0)) = \sqrt{(0)} = (0)$$

as $\mathcal{O}_Y(Y)$ is reduced, whence $\ker\big(\phi_Y^\sharp\big) = 0$. ∎

6.1.5 Dimension

Definition 6.1.48 (Dimension)
Let X be an affine variety. Then we define the dimension to be

$$\dim X \coloneqq \dim X\big(\overline{k}\big) = \dim X_0\big(\overline{k}\big)$$

where this is the Krull Dimension (4.1.81) of the $\overline{k}$-rational points with the k-Zariski topology. This is the supremum of dimension over all irreducible components by (4.1.83).

We say X is of ***pure dimension*** *n if all irreducible components have the same dimension n. Note an integral variety is always of pure dimension.*

We say $X = V(\mathfrak{a})$ is an ***affine curve*** *if it is of pure dimension 1.*

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Proposition 6.1.49 (Dimension of subspace)
Let $X = V(\mathfrak{a})$ be an affine variety and $Y = V(\mathfrak{b})$ an affine subvariety Then

a) $\dim Y = \dim(\mathfrak{b}) = \dim k[X]/\mathfrak{b} = \dim k[Y]$
b) $\operatorname{codim}(Y, X) = \operatorname{ht}(\mathfrak{b})$

Further when $Y = V(\mathfrak{p})$ is integral then $\operatorname{codim}(Y, X) = \dim k[X]_{\mathfrak{p}}$.

Proof. a) follows from (6.1.23).

b) follows similarly by observing that the definition of ideal height (3.21.1) is dual to the topological definition of codimension (4.1.81). The last statement follows from (3.21.4). ■

Corollary 6.1.50
Let $X = V(\mathfrak{a})$ be an affine variety then $\dim X = \dim k[X]$. *In particular* $\dim \mathbb{A}_k^n = n$.

Proof. The first statement is a consequence of (6.1.49) in the case $\mathfrak{b} = (0)$. The dimension for $\mathbb{A}_k^n$ then follows (3.34.28). ■

As we showed when studying the Krull dimension of affine algebras, the lattice of irreducible subsets is particularly well behaved.

Proposition 6.1.51 (Biequidimensional Algebraic Sets)
Let $X = V(\mathfrak{a})$ be an affine variety. Then $X\left(\overline{k}\right)$ is quasi-biequidimensional.

Further X is equidimensional (e.g. integral) iff it is biequidimensional. In this case for every closed subset $Y = V(\mathfrak{b})$ the codimension formula is satisfied

$$\dim X = \dim Y + \operatorname{codim}(Y, X)$$

or in algebraic terms

$$\dim k[X] = \dim \mathfrak{b} + \operatorname{ht}(\mathfrak{b})$$

Proof. We've observed that the lattice of radical (resp. prime) ideals of $k[X]$ is isomorphic to the lattice of closed (resp. irreducible) subsets of $X\left(\overline{k}\right)$. Therefore by (3.34.35) $X\left(\overline{k}\right)$ is quasi-biequidimensional.

In the equidimensional case, the codimension formula follows from (4.1.87) and the algebraic version from (6.1.49).■

In the integral case we recover the "classical" field-theoretic version of dimension

Proposition 6.1.52 (Dimension of Function Field)
Let $X = V(\mathfrak{p})$ be an integral affine variety. Then

$$\dim X = \dim k[X] = \operatorname{trdeg}(\operatorname{Frac}(k[X])/k)$$

Proof. We have already shown that $\dim X = \dim k[X]$. The second equality follows from Noether Normalisation (3.34.28). ■

Proposition 6.1.53
Let X be an affine variety and $Z \subset X$ a closed subset. Then the following are equivalent

a) $\dim Z = 0$
b) Z *is finite*

More precisely

$$Z = \overline{\{x_1\}} \cup \ldots \cup \overline{\{x_n\}}$$

for a finite number of points $x_1, \ldots, x_n \in Z$.

Proof. This follows from (4.1.91) and (6.1.18). ■

Proposition 6.1.54 (Hypersurfaces)
Let $X = V(\mathfrak{a})$ be an affine variety and $f \in k[X]$. Define the ***hypersurface*** $Y \coloneqq V(f)$ *then* $\dim Y = n - 1$ *(and Y is equidimensional).*

Conversely if $k[X]$ is a UFD and $Y \subset X$ is an irreducible closed subset of dimension 1, then Y is a hypersurface.

Proof. **TODO.** ■

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6.1.6 Local Rings

Definition 6.1.55 (Local Ring)
*Let X be an affine variety and $W \subset X\left(\overline{k}\right)$ be an irreducible subset. Then we define the **local ring** at W to be*

$$\mathcal{O}_{X,W} \coloneqq \varinjlim_{U \cap W \neq \emptyset} \mathcal{O}_X(U)$$

It is a local ring with unique maximal ideal

$$\mathfrak{m}_{X,W} = \{(U,\sigma) \mid \sigma(x) = 0 \quad \forall x \in W\}.$$

In the case $W = \{(x)\}$ then we write it as $\left(\mathcal{O}_{X,x}, \mathfrak{m}_{X,x}\right)$.

Proof. First of all the family of open sets such that $U \cap W \neq \emptyset$ is directed by reverse inclusion precisely because W is irreducible, see (4.1.60).

Any section $\sigma \in \mathcal{O}_X(U)$ is continuous with respect to the cofinite topology on k. Therefore $D(\sigma) \coloneqq \sigma^{-1}(k \setminus \{0\}) \subset U$ is an open set and $\sigma \notin \mathfrak{m}_{X,W} \implies D(\sigma) \cap W \neq \emptyset$. Then evidently $[(D(\sigma), \sigma^{-1})]$ is a multiplicative inverse for (U,σ) and by (...) $\mathcal{O}_{X,W}$ is a local ring with unique maximal ideal $\mathfrak{m}_{X,W}$. ∎

Corollary 6.1.56 (Field of Rational Functions)
*Let X be an integral affine variety. Then $\mathcal{O}_{X,X} =: k(X)$ is a field, which we denote as **field of rational functions**.*

Proof. Evidently $\mathfrak{m}_{X,X} = (0)$. Explicitly (V,σ) has inverse $(D(\sigma), \sigma^{-1})$. ∎

Proposition 6.1.57 (Local Ring is Localization of Coordinate Ring)
Let $X = V(\mathfrak{a})$ be an affine variety and $W \subset X\left(\overline{k}\right)$ an irreducible subset with prime ideal $\mathfrak{M}_{X,W} \coloneqq I(W)$. For all such W we have a canonical local isomorphism

$$\begin{array}{ccc} \mathcal{O}_X(X) & \longrightarrow & \mathcal{O}_X(X)_{\mathfrak{M}_{X,W}} \\ & \searrow^{\rho_W} & \downarrow \sim \\ & & \mathcal{O}_{X,W} \end{array}$$

$$\frac{f}{g} \mapsto \left[\left(D(g), \frac{f}{g}\right)\right]$$

The inverse image of $\mathfrak{m}_{X,W}$ is $\mathfrak{M}_{X,W}$ and $\mathfrak{M}_{X,W}\mathcal{O}_X(X)_{\mathfrak{M}_{X,W}}$ respectively. This induces isomorphisms

$$\mathcal{O}_X(X)/\mathfrak{M}_{X,W} \xrightarrow{\sim} k\left(\mathfrak{M}_{X,W}\mathcal{O}_X(X)_{\mathfrak{M}_{X,W}}\right) \xrightarrow{\sim} k\left(\mathfrak{m}_{X,W}\right)$$

Further we have

$$\dim \mathcal{O}_{X,W} = \operatorname{ht}\left(\mathfrak{M}_{X,W}\right) = \operatorname{codim}(W,X) = \dim X - \dim W$$

In particular

$$\dim \mathcal{O}_{X,x} = \dim X$$

Proof. This is a formal consequence of generic facts regarding localization and direct limits

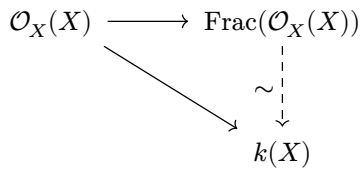
$$\mathcal{O}_X(X)_{\mathfrak{p}} \overset{(3.5.44)}{\cong} \varinjlim_{f \notin \mathfrak{p}} \mathcal{O}_X(X)_f \cong \varinjlim_{D(f) \cap W \neq \emptyset} \mathcal{O}_X(D(f)) \overset{(2.6.127)}{\cong} \varinjlim_{U \cap W \neq \emptyset} \mathcal{O}_X(U)$$

We may demonstrate this more directly. For it is surjective because the principal open sets form a basis and by (6.1.26), observing that $f \notin \mathfrak{p} \iff D(f) \cap V(\mathfrak{p}) \neq \emptyset$. Suppose $\frac{g}{f}$ and $\frac{g'}{f'}$ have the same image then there is some h such that $D(h) \subseteq D(ff') = D(f) \cap D(f')$ and $h \notin \mathfrak{p}$ such that $\frac{g}{f} = \frac{g'}{f'}$ are equal in $k[X]_h$ and a-fortiori in $k[X]_{\mathfrak{p}}$. Therefore the map is injective as required.

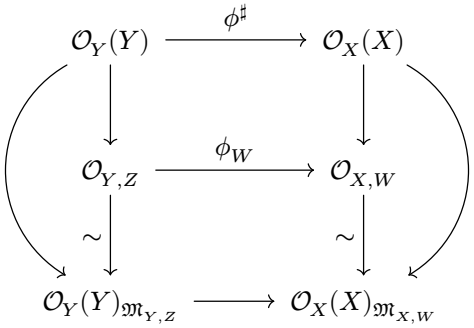
The first two dimension identities follow from (6.1.49) the third from (6.1.51). The final identity follows from (6.1.53).∎

Corollary 6.1.58
Let X be an irreducible variety then there is a canonical isomorphism with the field of rational functions

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Proposition 6.1.59
Let $\phi : X \to Y$ be a regular morphism of affine varieties. Suppose $W \subset X$ is irreducible and $Z \subset \overline{\phi(W)}$ for some irreducible $Z \subset Y$. Then the stalk map $\phi_W : \mathcal{O}_{Y,Z} \to \mathcal{O}_{X,Z}$ (see (4.3.5)) makes the following diagram commute

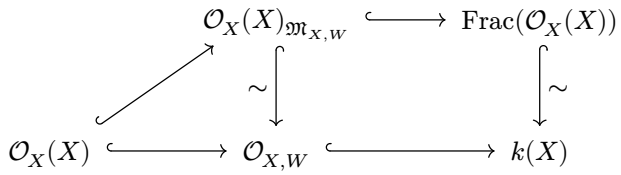


where the bottom arrow is the localisation map induced by $\phi^\sharp$ (3.5.22). In fact it is the unique map making either square commute.

Proposition 6.1.60
Let X be an integral affine variety, $W = V(\mathfrak{M}_{X,W})$ an irreducible subset. Then there are canonical injections

$$\mathcal{O}_X(X) \hookrightarrow \mathcal{O}_{X,W} \hookrightarrow k(X)$$

and the restriction maps are injective. Further there is a commutative diagram



which shows $k(X)$ is a quotient field for $\mathcal{O}_{X,W}$ and $\mathcal{O}_X(X)$.

Proof. We first show the restriction maps are injective. Suppose $\sigma|_V = 0$. Note that V is dense (4.1.59), σ is continuous (6.1.34) with respect to the k-Zariski topology on $\overline{k}$ so that $\sigma^{-1}(\{0\})$ is closed. By definition this contains V and therefore also $\mathrm{cl}_U(V) = \mathrm{cl}_X(V) \cap U = U$. We conclude $\sigma = 0$ and the restriction maps are injective. The same argument shows that the remaining maps are injective.

The vertical maps are isomorphisms by (6.1.57), and we may verify directly the diagram commutes. ∎

Therefore if it's convenient to do so we may identify $\mathcal{O}_X(X)$ and $\mathcal{O}_{X,W}$ as subrings of $k(X)$.

Proposition 6.1.61
Let X be an affine variety and $W \subset X$ an irreducible closed subset. Then the minimal primes of $\mathcal{O}_{X,W}$ are in bijection with the irreducible components of X containing W.

$\mathcal{O}_{X,W}$ is an integral domain if and only if W lies in a unique irreducible component.

Proof. By (6.1.57) and (3.5.42) the minimal primes of $\mathcal{O}_{X,W}$ correspond to minimal primes of $k[X]$ contained in $\mathfrak{M}_{X,W}$. These correspond to irreducible components of X containing $\overline{W}$ (and hence W) by (6.1.19).

As $\mathcal{O}_{X,W}$ is reduced, it is an integral domain iff it has a unique minimal prime ideal (3.3.67). ∎

Proposition 6.1.62
Let X be an affine variety, R an algebraic k-algebra (3.34.25) and $\phi : \mathcal{O}_{X,x} \to R$ a k-algebra homomorphism. Then $\phi^{-1}(\mathfrak{m}_R) = \mathfrak{m}_{X,x}$ and therefore factors through $k(\mathfrak{m}_{X,x})$

This in particular this includes the case R is an algebraic field extension.

Proof. See (3.34.27). ∎

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6.1.7 Rational Points

Proposition 6.1.63 (K-rational points)
Let $X = V(\mathfrak{a})$ be an affine variety and K an algebraic field extension. Then there are natural bijections

$$\begin{aligned} X(K) &\longleftrightarrow \mathrm{AlgHom}_k(\mathcal{O}_X(X), K) \\ &\longleftrightarrow \{(\mathfrak{M}, \phi) \mid \mathfrak{M} \in \mathrm{Specm}(\mathcal{O}_X(X)) \quad \phi : \mathcal{O}_X(X)/\mathfrak{M} \hookrightarrow K\} \\ &\longleftrightarrow \left\{(x, \phi) \;\middle|\; x \in X_0\left(\overline{k}\right) \quad \phi : k\left(\mathfrak{m}_{X,x}\right) \hookrightarrow K\right\} \\ &\longleftrightarrow \left\{(x, \phi) \;\middle|\; x \in X_0\left(\overline{k}\right) \quad \phi : \mathcal{O}_{X,x} \to K\right\} \end{aligned}$$

Proof. The first correspondence is simply (6.1.7) and (6.1.26).

For the second correspondence consider a k-algebra homomorphism $\phi : \mathcal{O}_X(X) \to K$. By (3.34.27) we have $\ker(\phi) =: \mathfrak{M}$ is maximal, which induces a homomorphism $\mathcal{O}_X(X)/\mathfrak{M} \hookrightarrow K$. Evidently this correspondence is injective and surjective.

By (6.1.11) there is a bijection $\mathrm{Specm}(\mathcal{O}_X(X)) \longleftrightarrow X_0\left(\overline{k}\right)$. Further there is an isomorphism $\mathcal{O}_X(X)/\mathfrak{M}_{X,x} \cong k\left(\mathfrak{m}_{X,x}\right)$ (6.1.57) from which the third correspondence is well-defined and bijective.

The final correspondence follows from (6.1.62) because all such homomorphisms factor through $k\left(\mathfrak{m}_{X,x}\right)$. ∎

6.1.8 Generic Points

When considering irreducible subsets $W \subset X$ there is another way of looking at these in terms of "generic points".

Definition 6.1.64 (Generic Point)
Let $X = V(\mathfrak{a})$ be an affine variety and $(\xi) \in X(\Omega)$ for some extension field Ω/k. Then we may define the irreducible subset of $X\left(\overline{k}\right)$

$$W_\xi := V_{\overline{k}}(\mathfrak{p}_\xi) = \left\{(x) \in X\left(\overline{k}\right) \;\middle|\; \forall f \in k[X](f(\xi) = 0 \implies f(x) = 0)\right\}$$

There are canonical isomorphisms

$$k\left(\mathfrak{m}_{X,W}\right) \xrightarrow{\sim} k[X]_{\mathfrak{p}}/\mathfrak{p}k[X]_{\mathfrak{p}} \xrightarrow{\sim} \mathrm{Frac}(k[X]/\mathfrak{p}) \xrightarrow{\sim} k(\xi)$$

*We say that (ξ) is a **generic point** corresponding to the irreducible closed subset W_ξ. Moreover every irreducible subset is of this form for we may simply consider $\Omega := \mathrm{Frac}(k[X]/\mathfrak{p})$ and $\xi = \left(\overline{X}_1, ..., \overline{X}_n\right)$.*

*If $(x) \in W_\xi$ then we say that (x) is a **specialization** of (ξ) and this induces the specialization homomorphism*

$$\begin{array}{ccc} k[X]/\mathfrak{p}_\xi & \longrightarrow & k[X]/\mathfrak{m}_x \\ \wr\wr & & \wr\wr \\ k[\xi] & \dashrightarrow & k[x] \end{array}$$

If $W_\xi = X$ then we say simply (ξ) is a generic point.

6.1.9 Tangent Space and Non-Singular Points

We propose two definitions for the tangent space and show that under mild technical conditions they are naturally isomorphic. The latter definition may be identified geometrically.

Definition 6.1.65
*Let X be an affine variety and let $W \subset X$ be an irreducible subset. We define the **cotangent space** at W to be the $k\left(\mathfrak{m}_{X,W}\right)$-vector space*

$$T^\star_W X := \mathfrak{m}_{X,W}/\mathfrak{m}^2_{X,W}$$

*and the **tangent space** to be the $k\left(\mathfrak{m}_{X,W}\right)$-vector space*

$$T_W X := \mathrm{Der}_k\left(\mathcal{O}_{X,W}, k\left(\mathfrak{m}_{X,W}\right)\right)$$

For points this has a concrete interpretation

Proposition 6.1.66 (Concrete interpretation of Tangent Space)
Let $X = V(\mathfrak{a}) \subset \mathbb{A}^n_k$ be an affine variety and $x \in X\left(\overline{k}\right)$. Then there are natural $k(x)$-module isomorphisms

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$$\begin{array}{lll} T_xX \xrightarrow{\sim} \mathrm{Der}_k(k[X], k(x)) \xrightarrow{\sim} \left\{ v \in k(x)^n \;\middle|\; \sum_{i=1}^n v_i \frac{\partial F}{\partial X_i}(x) = 0 \quad \forall F \in \mathfrak{a} \right\} \\ D \qquad\quad \to \qquad\qquad\qquad \left(D\left(\overline{X}_1\right), ..., D\left(\overline{X}_n\right) \right) \end{array}$$

where we have used the identification $k\left(\mathfrak{m}_{X,x}\right) \cong k(x)$ *and the* k*-algebra homomorphism* $k[X] \to k\left(\mathfrak{m}_{X,x}\right)$*. Suppose* $\mathfrak{a} = \langle F_1, ..., F_m \rangle$ *then the right hand side is the kernel of the following* $k(x)$*-module homomorphism*

$$\begin{pmatrix} \frac{\partial F_1}{\partial X_1}(x) & \cdots & \frac{\partial F_1}{\partial X_n}(x) \\ \vdots & \ddots & \vdots \\ \frac{\partial F_m}{\partial X_1}(x) & \cdots & \frac{\partial F_m}{\partial X_n}(x) \end{pmatrix} : k(x)^n \to k(x)^m$$

In particular

$$\dim_{k\left(\mathfrak{m}_{X,x}\right)} T_xX = n - \mathrm{rk}\left(\frac{\partial F_i}{\partial X_j}(x) \right)$$

Proof. Recall by (6.1.57) and (6.1.64) that $\mathcal{O}_{X,x} \cong k[X]_{\mathfrak{M}_{X,x}}$ and $k\left(\mathfrak{m}_{X,x}\right) \cong k(x)$ so we have isomorphisms

$$\mathrm{Der}_k\left(\mathcal{O}_{X,x}, k\left(\mathfrak{m}_{X,x}\right)\right) \cong \mathrm{Der}_k\left(k[X]_{\mathfrak{M}_{X,x}}, k(x)\right) \overset{(3.25.10)}{\cong} \mathrm{Der}_k(k[X], k(x))$$

and the remaining isomorphism is from (3.34.41) ∎

Proposition 6.1.67 (Dimension Formula of Tangent Space)
Let $X = V(\mathfrak{a}) \subset \mathbb{A}_k^n$ *be an* integral *affine variety,* $W \subset X$ *an irreducible subset. Suppose that* $\mathfrak{a} = \langle F_1, ..., F_m \rangle$ *and let* $\xi := \left(\overline{X}_1, ..., \overline{X}_n\right) \in k\left(\mathfrak{m}_{X,W}\right)^n$ *be the image of the coordinate functions. Then we have the dimension formula*

$$\dim T_WX = \dim \mathrm{Der}\left(k[X], k\left(\mathfrak{m}_{X,W}\right)\right) = n - \mathrm{rk}\left(\frac{\partial F_i}{\partial X_j}(\xi) \right)$$

Suppose that $Z \subset W$ *is an irreducible subset then we have the inequality*

$$\dim T_ZX \geq \dim T_WX$$

and

$$\dim T_WX \geq \dim T_XX = \mathrm{Der}(k(X))$$

which equals $\dim X$ *when* $k(X)$ *is separably generated (e.g.* k *is perfect).*

Proof. We have isomorphisms

$$\begin{aligned} \dim T_WX := \quad & \mathrm{Der}_k\left(\mathcal{O}_{X,W}, k\left(\mathfrak{m}_{X,W}\right)\right) \\ \overset{(6.1.57)}{\to} \quad & \mathrm{Der}_k\left(\mathcal{O}_X(X)_{\mathfrak{M}_{X,W}}, k\left(\mathfrak{m}_{X,W}\right)\right) \\ \overset{(3.25.10)}{\to} \quad & \mathrm{Der}_k\left(\mathcal{O}_X(X), k\left(\mathfrak{m}_{X,W}\right)\right) \\ \overset{(6.1.26)}{\to} \quad & \mathrm{Der}_k\left(k[X], k\left(\mathfrak{m}_{X,W}\right)\right) \end{aligned}$$

and the rank formula follows from (3.34.41) and (3.3.135).

Let ζ be the corresponding generic point of Z, then because $\mathfrak{M}_{X,W} \subset \mathfrak{M}_{X,Z}$ we may construct a k-algebra homomorphism $k[\xi] \to k[\zeta]$. So by the determinant criteria of rank (3.8.43) we deduce that

$$\mathrm{rk}\left(\frac{\partial F_i}{\partial X_j}(\zeta) \right) \leq \mathrm{rk}\left(\frac{\partial F_i}{\partial X_j}(\xi) \right)$$

from which the inequality follows.

Observe that $k\left(\mathfrak{m}_{X,X}\right) =: k(X)$ and $T_XX \xrightarrow{\sim} \mathrm{Der}_k(k[X], k(X)) \xrightarrow{\sim} \mathrm{Der}_k(k(X))$. When $k(X)$ is separably generated this equals $\dim X$ by (3.34.44). ∎

Proposition 6.1.68 (Duality between Tangent and Cotangent Space)
Let X *be an affine variety and* $W \subset X$ *be an irreducible subset. There is a canonical* $k\left(\mathfrak{m}_{X,W}\right)$*-module homomorphism*

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$$T_W X \to (T^\star_W X)^\vee$$

Under the condition that $k(\mathfrak{m}_{X,W})/k$ *is separably generated (e.g.* k *is perfect) then this map is an isomorphism. In particular* $\dim T_W X = \dim T^\star_W X$ *by (3.3.109).*

Proof. This follows directly from (3.34.50) by considering the local ring $A := \mathcal{O}_{X,W}$ with maximal ideal $\mathfrak{m}_{X,W}$. ∎

Lemma 6.1.69
Let A *be a reduced ring with minimal prime ideal* $\mathfrak{p}$ *and prime ideal* $\mathfrak{q} \supset \mathfrak{p}$ *containing no other minimal prime ideals. Suppose that*

$$\phi : A \to B$$

is a surjective ring homomorphism with $\ker(\phi) = \mathfrak{p}$*. Then* ϕ *induces an isomorphism of rings*

$$\tilde{\phi} : A_{\mathfrak{q}} \to B_{\phi(\mathfrak{q})}$$

which is the unique morphism commuting with ϕ*.*

Proof. The map $\tilde{\phi}$ exists and has kernel $\mathfrak{p}A_{\mathfrak{q}}$ by (3.5.5). We need simply to show that $\mathfrak{p}A_{\mathfrak{q}} = (0)$. By (3.5.42) it is the unique minimal prime ideal of $A_{\mathfrak{q}}$. As $A_{\mathfrak{q}}$ is reduced then we are done by (3.3.46). ∎

Proposition 6.1.70
Let X *be an affine variety,* W *an irreducible subset contained in precisely one irreducible component* X_α*. Then there is a canonical isomorphism*

$$\begin{aligned} \mathcal{O}_{X,W} &\xrightarrow{\sim} \mathcal{O}_{X_\alpha,W} \\ (U,\sigma) &\to \left(U \cap X_\alpha, \sigma|_{X_\alpha \cap U}\right) \end{aligned}$$

and in particular an isomorphism of tangent spaces

$$T_W X \xrightarrow{\sim} T_W X_\alpha$$

Proof. The regular map $i : X_\alpha \hookrightarrow X$ corresponds to a ring homomorphism $i^\sharp$ with kernel $\mathfrak{M}_{X,X_\alpha}$ by (6.1.44). By (...) we have a commutative diagram

$$\begin{array}{ccc} \mathcal{O}_X(X) & \xrightarrow{i^\sharp} & \mathcal{O}_{X_\alpha}(X_\alpha) \\ \downarrow & & \downarrow \\ \mathcal{O}_{X,W} & \xrightarrow{i_W} & \mathcal{O}_{X_\alpha,W} \\ \downarrow \sim & & \downarrow \sim \\ \mathcal{O}_X(X)_{\mathfrak{M}_{X,W}} & \longrightarrow & \mathcal{O}_{X_\alpha}(X_\alpha)_{\mathfrak{M}_{X_\alpha,W}} \end{array}$$

and by (6.1.69) we deduce that the stalk map is an isomorphism. ∎

Definition 6.1.71 (Non-Singular Points)
Let X *be an affine variety and* $W \subset X$ *be an irreducible subset. We say that* X *is* ***non-singular*** *at* W *if*

a) W *is contained in only one irreducible component* X_α*, and*
b) $\dim T_W X = \dim \mathrm{Der}_k(k(X_\alpha))$

Note that by (6.1.61) a) $\iff \mathcal{O}_{X,W}$ *is an integral domain.*

Recall by (6.1.67) that $\dim T_W X = \dim T_W X_\alpha \geq \dim \mathrm{Der}_k(k(X_\alpha))$ *and in the case of* k *perfect this equals* $\dim X_\alpha$*.*

We may now show the following important result

Proposition 6.1.72 (Non-Singular Point)
Let $X = V(\mathfrak{a}) \subset \mathbb{A}^n_k$ *be an affine variety then the non-singular points form an open dense subset of* X*.*

Proof. We may reduce to the case of X irreducible. Let $\Omega := k(X)$ and $\xi := \left(\overline{X}_1, ..., \overline{X}_n\right) \in X(\Omega)$ and suppose $\mathfrak{p} = \langle F_1, ..., F_m \rangle \lhd k[X_1, ..., X_n]$ is the ideal defining X. Then

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$$\mathrm{Der}(\Omega) \overset{(3.25.10)}{=} \mathrm{Der}(k[X], \Omega) \overset{(3.34.41)}{\cong} \ker\left(\frac{\partial F_i}{\partial X_j}(\xi)\right)$$

For $x \in X\left(\overline{k}\right)$ there is a k-algebra homomorphism $k[\xi] \to k[x]$ and so by the determinant criteria of rank (3.8.43) we see

$$\mathrm{rk}\left(\frac{\partial F_i}{\partial X_j}(x)\right) \leq \mathrm{rk}\left(\frac{\partial F_i}{\partial X_j}(\xi)\right)$$

Suppose the rank of the right hand side is r and consider the r-minors $g_1, ..., g_m \in k[X]$. Then by the same result at least one is non-zero and the set of non-singular points is given by

$$\bigcup_p D(g_p)$$

which is non-empty by (6.1.27). ∎

Proposition 6.1.73 (Non-Singular Variety)
Let X be an integral affine variety. Then the following are equivalent

a) X is non-singular at all points $x \in X\left(\overline{k}\right)$
b) X is non-singular at all irreducible subsets $W \subset X\left(\overline{k}\right)$

*In this case we say that X is a **non-singular affine variety**.*

Proof. Evidently $b) \implies a)$ because $T_x X = T_{\overline{\{x\}}} X$. Conversely given any $x \in W$ we have by (6.1.67)

$$T_x X \geq T_W X \geq \dim \mathrm{Der}(k(X))$$

whence they are both equal. ∎

Definition 6.1.74 (Regular Point)
*Let X be an affine variety and $W \subset X$ an irreducible subset. Then we say that W is **regular** if*

a) W is contained in precisely one irreducible component X_α, and
b) $\mathcal{O}_{X,W}$ is a regular local ring (i.e. $\dim T^\star_W X = \dim \mathcal{O}_{X,W}$)

*We say X is **regular** if it is regular at all irreducible subsets.*

Fortunately these concepts are typically equivalent.

Proposition 6.1.75 (Regular = Non-Singular in Perfect Case)
Let $X = V(\mathfrak{a}) \subset \mathbb{A}^n_k$ an affine variety with k perfect and $W \subset X$ an irreducible subset. Then

$$\dim T^\star_W X = \dim T_W X$$

and the following are equivalent

a) X is regular at W
b) X is non-singular at W
c) $\dim T_W X = \dim \mathcal{O}_{X,W} = \mathrm{codim}(W, X)$

When $\mathrm{codim}(W, X) = 1$ this is equivalent to $\mathcal{O}_{X,W}$ being a discrete valuation ring.

Proof. Recall by (6.1.57) that $\dim \mathcal{O}_{X,x} = \dim X$ and by definition $T^\star_x X = \mathfrak{m}_{X,x}/\mathfrak{m}^2_{X,x}$. Further by definition (x) is regular iff $\dim T^\star_x X = \dim \mathcal{O}_{X,x}$. Finally by (6.1.68) $\dim T^\star_x X = \dim T_x X$ so we see $a) \iff c)$.

Similarly $b) \iff c)$ is essentially by definition. The final statement follows from (3.30.2). ∎

Example 6.1.76 (Simple Points of a Plane Curve)
The canonical example is a plane curve $k[X] = k[X,Y]/(F(X,Y))$ which has dimension 1 (...). Given $(x) \in X(K)$ we see that

$$T_x X = \left\{ (\alpha, \beta) \in k(x)^2 \;\middle|\; 0 = \alpha \frac{\partial F}{\partial X}(x) + \beta \frac{\partial F}{\partial Y}(x) \right\}$$

*Clearly this has dimension 1 (in which case (x) is a **simple point**) unless both the partial derivatives vanish at (x) in which case it has dimension 2.*

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Clearly $F(X,Y) = Y^2 - X^3$ has a non-simple point at $(0,0)$ but $F(X,Y) = Y - X^2$ has simple points everywhere.

6.1.10 Zeta Function over Finite Fields

For this section let $k = \mathbb{F}_p$ with algebraic closure $\overline{k}$. Then for every $n \geq 1$ by (...) $\overline{k}$ has a unique subfield k_n of degree n, and order p^n. Furthermore we have the following properties

a) $k_m \subseteq k_n \iff [k_m : k] \mid [k_n : k] \iff m \mid n$
b) $[k_n : k_m] = \frac{n}{m}$
c) k_n/k_m is Galois with the group of automorphisms generated by ϕ^m for all integers $m \mid n$, and the Galois group has order $\frac{n}{m}$.
d) every finite subextension is of this form

Lemma 6.1.77
Let $x_1, ..., x_N \in \overline{k}$. Then

$$k(x_1, ..., x_N) = k_n$$

where

$$n \coloneqq \operatorname{lcm}_i [k(x_i) : k]$$

Proof. By definition $[k(x_i) : k] \mid n$ whence $k(x_i) \subseteq k_n$ by the previous observations, and therefore $k(x_1, ..., x_N) \subseteq k_n$.

Similarly as $k(x_i) \subset k(x_1, ..., x_N)$ we have $[k(x_i) : k] \mid [k(x_1, ..., x_N) : k]$. Then by definition $n \mid [k(x_1, ..., x_N) : k]$ which shows $k_n \subset k(x_1, ..., x_N)$. ∎

The following result shows that the number of rational points over a finite field may be characterized by algebraic properties of the coordinate ring $k[X]$. This is essentially because maximal ideals correspond to Galois orbits of rational points and we may count rational points by summing over Galois orbits.

Proposition 6.1.78 (Counting Rational Points)
Let $X = V(\mathfrak{a}) \subset \mathbb{A}_k^N$ be an affine variety. Then for all integers $n \geq 1$ we have the following relation

$$\#X(k_n) = \sum_{d \mid n} d \times \#\{\mathfrak{m} \mid \dim_k k(\mathfrak{m}) = d\}$$

Proof. For $(x) \in X\left(\overline{k}\right)$, then by the previous discussion

$$(x) \in X(k_n) \iff k(x) \subseteq k_n \iff \dim_k k(x) \mid n$$

This shows that

$$\begin{aligned} \#X(k_n) &= \sum_{d \mid n} \#\{(x) \in X(k_n) \mid \dim_k k(x) = d\} \\ &= \sum_{d \mid n} \#\{(x) \in X(k_d) \mid \dim_k k(x) = d\} \end{aligned} \tag{*}$$

By (6.1.11) we have a bijection

$$X(k_d)/\operatorname{Gal}(k_d/k) \xrightarrow{\sim} \{\mathfrak{m} \mid \dim_k k(\mathfrak{m}) \mid d\}$$

As we have an isomorphism $k(x) \cong k(\mathfrak{m}_x)$ this restricts to a bijection

$$\{(x) \in X(k_d) \mid \dim_k k(x) = d\}/\operatorname{Gal}(k_d/k) \xrightarrow{\sim} \{\mathfrak{m} \mid \dim_k k(\mathfrak{m}) = d\}$$

We claim that the action of $\operatorname{Gal}(k_d/k) = \langle \phi \rangle$ is free. For suppose $(x) \in X(k_d)$ such that $\dim_k k(x) = d$ and $\phi^r(x_i) = x_i$ for $i = 1, ..., N$ and $0 < r \leq d$. Then by (3.18.108) we have $x_i \in k_r$ whence $\dim_k k(x) \leq r$. This shows we must have $r = d$ and in particular only the identity automorphism fixes (x).

Therefore we see that

$$\#\{(x) \in X(k_d) \mid \dim_k k(x) = d\} = \#\operatorname{Gal}(k_d/k) \times \#\{\mathfrak{m} \mid \dim_k k(\mathfrak{m}) = d\}$$

From (3.18.111) we recall $\operatorname{Gal}(k_d/k)$ is cyclic of order d so we may combine this with (*) to obtain the required result. ∎

Proposition 6.1.79 (Zeta function of an affine variety over a finite field)
Formally as elements of the power series ring $\mathbb{Q}[[T]]$ we have

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$$Z(X,T) \coloneqq \prod_{\mathfrak{m}\in \operatorname{Specm}(k[X])} \left(1-T^{\deg(\mathfrak{m})}\right)^{-1} = \exp\left(\sum_{m=1}^{\infty} \#\frac{X(k_m)}{m} T^m\right)$$

Proof. Define

$$b_d \coloneqq \#\{\mathfrak{m} \mid \dim_k k(\mathfrak{m}) = d\}$$

Let $Z(X,T)$ be the right hand side then

$$\begin{aligned}
\log(Z(X,T)) &= \sum_{m=1}^{\infty} \#X(k_m)\frac{T^m}{m} \\
&= \sum_{m=1}^{\infty}\sum_{d|m}(d\times b_d)\frac{T^m}{m} \\
&= \sum_{d=1}^{\infty} b_d \sum_{r=1}^{\infty}\frac{T^{rd}}{r} \\
&= -\sum_{d=1}^{\infty} b_d \log\left(1-T^d\right)
\end{aligned}$$

■

Example 6.1.80
For $X(k) = k^n$ we have $\#X(k_m) = p^{mn}$. Then

$$Z(X,T) = \exp\left(\sum_{m=1}^{\infty}\frac{p^{mn}T^m}{m}\right) = \exp(-\log(1-p^nT)) = \frac{1}{1-p^nT}$$

6.1.11 Base Change

Definition 6.1.81 (Affine Base Change)
Let $X = V(\mathfrak{a}) \subset \mathbb{A}^n_k$ be an affine variety over k and K/k a field extension. Consider the canonical ring homomorphism

$$i_{kK} : k[X_1,...,X_n] \hookrightarrow K[X_1,...,X_n]$$

and let $\mathfrak{a}^e \coloneqq \mathfrak{a}K[X_1,...,X_n]$ denote the extension under this map.

*Further define the **base change** of X with respect to K/k to be the affine variety*

$$X_K \coloneqq V\left(\sqrt{\mathfrak{a}^e}\right) \subset \mathbb{A}^n_K$$

Proposition 6.1.82 (Coordinate Ring of Base Change)
Let $X = V(\mathfrak{a}) \subset \mathbb{A}^n_k$ be an affine variety over k and K/k a field extension. There is a canonical K-algebra isomorphism

$$\begin{aligned}
(k[X]\otimes_k K)_{\text{red}} &\xrightarrow{\sim} K[X_K] \\
\overline{F}\otimes\lambda &\to \lambda\cdot\overline{i_{kK}(F)}
\end{aligned}$$

Proof. By (3.4.42) we have K-algebra isomorphisms

$$k[X]\otimes_k K \xrightarrow{\sim} (k[X_1,...,X_n]\otimes_k K)/\mathfrak{a}^e \xrightarrow{\sim} K[X_1,...,X_n]/\mathfrak{a}^e$$

By (3.34.4) and (3.34.2) this determines a unique K-algebra isomorphism

$$(k[X]\otimes_k K)_{\text{red}} \xrightarrow{\sim} (K[X_1,...,X_n]/\mathfrak{a}^e)_{\text{red}} \xrightarrow{\sim} K[X_1,...,X_n]/\sqrt{\mathfrak{a}^e} =: K[X_K]$$

■

It may be interesting to understand the cases when $\mathfrak{a}^e$ remains radical under the base change.

Proposition 6.1.83
Let $X = V(\mathfrak{a})$ be an affine variety. Then the extension $\mathfrak{a}^e$ under $A \to A\otimes_k K$ remains radical under any of the following conditions

a) $k[X]$ is geometrically reduced
b) K/k is geometrically reduced (e.g. separably generated)

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c) k is perfect.

Proof. By (3.4.42) we have an isomorphism

$$K[X_1, ..., X_n]/\mathfrak{a}^e \xrightarrow{\sim} k[X] \otimes_k K$$

so that $\mathfrak{a}^e$ is radical iff $k[X] \otimes_k K$ is reduced. Then we may reduce to results on geometrically reduced algebras. ∎

We are particularly interested when the variety remains integral (equivalently irreducible) under base change.

Definition 6.1.84 (Geometrically Integral)
*Let $X = V(\mathfrak{a})$ be an affine variety over k. Then we say X is **geometrically integral** if X_K is integral for all field extensions K. By (...) this is equivalent to $k[X]$ being geometrically irreducible.*

Proposition 6.1.85 (Criteria to be Geometrically Integral)
Let $X = V(\mathfrak{a}) \subset \mathbb{A}^n_k$ be an integral affine variety. Then the following are equivalent

a) X is geometrically integral
b) $X_{\overline{k}}$ is integral
c) $X_{k'}$ is integral for every finite separable extension k'/k
d) $k[X]$ is geometrically irreducible.
e) $k[X] \otimes \overline{k}$ is irreducible
f) $k[X] \otimes k'$ is irreducible for every finite separable extension k'/k.
g) $k(X)/k$ is relatively separably closed

Proof. We have *a)* $\iff$ *d)*, *b)* $\iff$ *e)* and *c)* $\iff$ *f)* are straightforward since a ring A is irreducible iff A_{red} is integral.

For *d)* $\iff$ *e)* $\iff$ *f)* $\iff$ *g)* we may use (3.35.41). ∎

Proposition 6.1.86
Let $X = \mathbb{A}^n_k$. Then there is a canonical isomorphism $X_K \cong \mathbb{A}^n_K$. In particular X is geometrically integral.

Proposition 6.1.87
Let $X \subset \mathbb{A}^n_k$ be an affine variety, K/k a field extension and R a K-algebra. Then there is a bijection

$$\begin{array}{ccc} X\big(R_{(k)}\big) & \xrightarrow{\quad\sim\quad} & X_K(R) \\ \downarrow \sim & & \downarrow \sim \\ \mathrm{AlgHom}_k\big(k[X], R_{(k)}\big) & \xrightarrow{\sim} & \mathrm{AlgHom}_K(K[X_K], R) \\ \phi & \to & \big(\lambda \cdot \overline{F} \to \lambda \cdot \phi\big(\overline{i_{kK}(F)}\big)\big) \end{array}$$

for $F \in k[X_1, ..., X_n]$, where the vertical arrows are given by (...).

Corollary 6.1.88
Let X be an affine variety and $L/K/k$ a tower of extensions, then we may identify $X(L)$ with $X_K(L)$.

Proposition 6.1.89
Let $X = V(\mathfrak{a})$ an affine variety and $\overline{k}/K/k$ a tower of extensions. Then after identifying $X\big(\overline{k}\big)$ with $X_K\big(\overline{k}\big)$ we have the following properties

a) U open in $X\big(\overline{k}\big) \implies U$ open in $X_K\big(\overline{k}\big)$,
b) There is a well-defined K-algebra homomorphism

$$(\mathcal{O}_X(U) \otimes_k K)_{\text{red}} \to \mathcal{O}_{X_K}(U)$$

$$f \otimes \lambda \to \lambda \cdot f$$

which is an isomorphism when $U = X$,
c) sets of the form $D\big(\sum_{i=1}^n \lambda_i f_i\big)$ are open in $X_K\big(\overline{k}\big)$ and form a base for the Zariski topology.

Proof.

a) A closed subset of $X\big(\overline{k}\big)$ is of the form $Z = V(\mathfrak{b})$ for $\mathfrak{b} \subset \mathfrak{a}$. Then evidently $Z = V(\mathfrak{b}^e)$ and so it is closed as a subset of $X_K\big(\overline{k}\big)$
b) By definition we have an open cover $U = \bigcup_{i=1}^n U_i$ and polynomials $g_i, h_i \in k[X]$ such that $f|_{U_i} = \frac{g_i}{h_i}$. This shows that $\lambda \cdot f$ is also regular. The map is an isomorphism on global sections by (6.1.82).

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c) By (6.1.39) $f \coloneqq \sum_{i=1}^{n} \lambda_i f_i$ is continuous. As $\{0\}$ is a closed subset of $\overline{k}$ in the Zariski (co-finite) topology this shows that $D(f)$ is open. These form a base because $D(fg) = D(f) \cap D(g)$, and specifically for the Zariski topology by (6.1.15) and b).

■

6.1.12 Valuation Rings on the Function Field

Proposition 6.1.90
Let K/k be a field extension and R a valuation ring of K/k. Then there exists a valuation ring $R' \subset R$ such that $k(\mathfrak{m}_{R'})/k$ is algebraic and $\mathfrak{m}_{R'} = \mathfrak{m}_R \cap R'$.

Proof. By (3.24.11) the identity map $1 : k \to \overline{k}$ extends to a ring homomorphism $S \to \overline{k}$ where $(S, \mathfrak{m}_S)$ is a valuation ring of $k(\mathfrak{m}_R)/k$. Considering the quotient map $\pi : R \to k(\mathfrak{m}_R)$ let $R' \coloneqq \pi^{-1}(S)$. The restriction map

$$\pi' : R' \to S \to k(\mathfrak{m}_S)$$

is surjective, and so the kernel $\mathfrak{m}_{R'} \coloneqq \ker(\pi') = \mathfrak{m}_R \cap R'$ is a maximal ideal of R' and $k(\mathfrak{m}_{R'}) \cong k(\mathfrak{m}_S)$. Observe that $R' \subseteq R$ and $\mathfrak{m}_R \subseteq \mathfrak{m}_{R'}$. If $x \notin R$ then $x^{-1} \in \mathfrak{m}_R \implies x^{-1} \in R$ by (3.24.2). If $x \in R \setminus R'$ then $\pi(x) \notin S \implies \pi(x^{-1}) \in S \implies x^{-1} \in R'$. Therefore we conclude that R' is a valuation ring with maximal ideal $\mathfrak{m}_{R'}$ and algebraic residue field as required. ■

The following result was proven in the case of a finitely generated k-algebra (3.34.28), where equality holds. In general only the inequality holds by considering for example the case $\dim K = 0$. The proof below is taken from [Rey11].

Lemma 6.1.91
Let R be a subring of a finitely generated field extension K/k. Then $\dim R \leq \operatorname{trdeg}(K/k)$.

Proof. Consider a chain of prime ideals

$$\mathfrak{p}_0 \subsetneq \ldots \subsetneq \mathfrak{p}_n \subset R$$

Choose $x_i \in \mathfrak{p}_i \setminus \mathfrak{p}_{i-1}$ and define $A \coloneqq k[x_1, \ldots, x_n] \subset R$ with $K' \coloneqq \operatorname{Frac}(A) \subset K$. Then there is a chain of prime ideals of A

$$\mathfrak{p}_0 \cap A \subsetneq \ldots \subsetneq \mathfrak{p}_n \cap A$$

whence $n \leq \dim A \overset{(3.34.28)}{=} \operatorname{trdeg}(K'/k) \overset{(3.18.140)}{\leq} \operatorname{trdeg}(K/k)$. Taking supremum over n we reach the required result. ■

Lemma 6.1.92
Let $R \subseteq S$ be proper valuation rings of K with $\dim R < \infty$. Then $\dim S \leq \dim R$, with equality iff $R = S$.

Proof. Firstly we claim that $\mathfrak{m}_S \subseteq \mathfrak{m}_R$. For $x \in \mathfrak{m}_S \implies x^{-1} \notin S \implies x^{-1} \notin R \implies x \in \mathfrak{m}_R$.

Secondly if $R \subsetneq S$ then $\mathfrak{m}_S \subsetneq \mathfrak{m}_R$. For given $x \in S \setminus R$ we have $x^{-1} \in \mathfrak{m}_R \implies x^{-1} \in S \implies x^{-1} \in S^\star \implies x^{-1} \notin \mathfrak{m}_S$.

Let $\mathfrak{p}_0 \subsetneq \ldots \subsetneq \mathfrak{p}_n$ be a chain of prime ideals in S. As these are contained in $\mathfrak{m}_S$, they are also contained in $\mathfrak{m}_R$ and therefore R. Therefore this constitutes a chain of prime ideals in R from which we deduce the inequality. Further if $R \subsetneq S$ then we may always extend the chain by at least one prime ideal, namely $\mathfrak{m}_R$. ■

Proposition 6.1.93
Let $X = V(\mathfrak{a}) \subset \mathbb{A}^n_k$ be an integral affine variety, and $\mathcal{O}_X(X) \subset R \subset k(X)$ a valuation ring. Then there exists a point $x \in X\big(\overline{k}\big)$ such that R dominates $\mathcal{O}_{X,x}$. Further $\dim R \leq \dim X$ with equality iff $\mathcal{O}_{X,x} = R$ as subrings of $k(X)$.

Proof. By (6.1.90) we may assume that $k(\mathfrak{m}_R)/k$ is algebraic. Choose any embedding $k(\mathfrak{m}_R)/k \hookrightarrow \overline{k}/k$ (3.18.61). Then there is a composite k-algebra homomorphism

$$\mathcal{O}_X(X) \hookrightarrow R \to k(\mathfrak{m}_R) \hookrightarrow \overline{k}$$

Let x_i be the image of the affine coordinates $\overline{X}_i$ under this homomorphism. The image $k[x_1, \ldots, x_n]$ is a (finite) field extension of k by (3.18.44). Then under this map the image of f is $f(x)$, so the kernel is precisely $\mathfrak{M}_{X,x} = \mathfrak{m}_R \cap \mathcal{O}_X(X)$. Suppose that $f \notin \mathfrak{M}_{X,x}$ then $f \notin \mathfrak{m}_R \implies f^{-1} \in R$. In particular we conclude that $\mathcal{O}_{X,x} = \mathcal{O}_X(X)_{\mathfrak{M}_{X,x}} \subset R$. Further $\mathfrak{m}_{X,x} = \mathfrak{M}_{X,x}\mathcal{O}_{X,x} \subseteq \mathfrak{m}_R R \subseteq \mathfrak{m}_R$, so R dominates $\mathcal{O}_{X,x}$ by (3.24.4).

The final statement follows from (6.1.92) and $\dim X = \dim \mathcal{O}_{X,x}$ (6.1.57). ■

6.1.13 Products of Affine Varieties

In what follows we let A_{red} denote the reduced ring (resp k-algebra) $A/N(A)$.

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Proposition 6.1.94
There is a canonical isomorphism

$$\begin{aligned} \psi : k[X_1,...,X_n] \otimes_k k[Y_1,...,Y_m] &\xrightarrow{\sim} k[X_1,...,X_n,Y_1,...,Y_m] \\ F(X_1,...,X_n) \otimes G(Y_1,...,Y_m) &\to \quad F(X_1,...,X_n)G(Y_1,...,Y_m) \end{aligned}$$

Let $\mathfrak{a} \lhd k[X_1,...,X_n]$ *and* $\mathfrak{b} \lhd k[Y_1,...,Y_m]$ *be ideals. Then there is a commutative diagram*

$$\begin{array}{ccc} k[X_1,...,X_n] \otimes_k k[Y_1,...,Y_m] & \xrightarrow[\sim]{\psi} & k[X_1,...,X_n,Y_1,...,Y_m] \\ \downarrow{\scriptstyle \pi_1 \otimes \pi_2} & & \downarrow \\ \frac{k[X_1,...,X_n]}{\mathfrak{a}} \otimes_k \frac{k[Y_1,...,Y_m]}{\mathfrak{b}} & \xrightarrow[\sim]{\overline{\psi}} & \frac{k[X_1,...,X_n,Y_1,...,Y_m]}{\mathfrak{a}^e+\mathfrak{b}^e} \end{array}$$

$$\overline{F} \otimes \overline{G} \to \overline{F(X_1,...,X_n)G(Y_1,...,Y_m)}$$

where the bottom arrow is uniquely determined and an isomorphism. The nilradical of the right hand side is $\sqrt{\mathfrak{a}^e+\mathfrak{b}^e}/(\mathfrak{a}^e+\mathfrak{b}^e)$ *and so there is a corresponding isomorphism*

$$\left(\frac{k[X_1,...,X_n]}{\mathfrak{a}} \otimes_k \frac{k[Y_1,...,Y_m]}{\mathfrak{b}}\right)_{\text{red}} \xrightarrow{\sim} \frac{k[X_1,...,X_n,Y_1,...,Y_m]}{\sqrt{\mathfrak{a}^e+\mathfrak{b}^e}}$$

Proof. By (3.4.37) there is a unique k-algebra homomorphism ψ such that $\psi(F \otimes 1) = F$ and $\psi(1 \otimes G) = G$, whence $\psi(F \otimes G) = FG$. Similarly there is an inverse morphism ϕ such that $\phi(X_i) = X_i \otimes 1$ and $\phi(Y_j) = 1 \otimes Y_j$. Evidently $\phi(F(X_1,...,X_n)) = F \otimes 1$ and $\phi(G(Y_1,...,Y_m)) = 1 \otimes G$. This shows that $\phi \circ \psi = \mathbf{1}$. Similarly $(\psi \circ \phi)(X_i) = X_i$ and $(\psi \circ \phi)(Y_j) = Y_j$. This shows that $\psi \circ \phi = \mathbf{1}$ as required.

There is a commutative diagram

$$\begin{array}{ccc} k[X_1,...,X_n] & \xrightarrow{i} & k[X_1,...,X_n] \otimes_k k[Y_1,...,Y_m] \\ & \searrow & \wr \downarrow \psi \\ & & k[X_1,...,X_n,Y_1,...,Y_m] \end{array}$$

By (3.4.42) $\mathfrak{a} \otimes k[Y_1,...,Y_m]$ is the extension of $\mathfrak{a}$ under i. As the diagram commutes we may show that $\psi(\mathfrak{a} \otimes k[Y_1,...,Y_m]) = \mathfrak{a}^e$ and similarly $\psi(k[X_1,...,X_n] \otimes \mathfrak{b}) = \mathfrak{b}^e$. Therefore by (3.3.51)

$$\psi(\mathfrak{a} \otimes k[Y_1,...,Y_m] + k[X_1,...,X_n] \otimes \mathfrak{b}) = \mathfrak{a}^e + \mathfrak{b}^e$$

By (3.4.29) $\ker(\pi_1 \otimes \pi_2) = \mathfrak{a} \otimes k[Y_1,...,Y_m] + k[X_1,...,X_n] \otimes \mathfrak{b}$ therefore $\overline{\psi}$ exists and is an isomorphism by (...). ∎

Lemma 6.1.95
Let $X = V(\mathfrak{a}) \subset \mathbb{A}_k^n$ *and* $Y = V(\mathfrak{b}) \subset \mathbb{A}_k^m$*. Consider the affine variety* $\widetilde{X \times Y} := V\left(\sqrt{\mathfrak{a}^e+\mathfrak{b}^e}\right) \subset \mathbb{A}_k^{n+m}$*. The projection maps*

$$\begin{aligned} \pi_X &: \widetilde{X \times Y}\left(\overline{k}\right) \to X\left(\overline{k}\right) \\ \pi_Y &: \widetilde{X \times Y}\left(\overline{k}\right) \to Y\left(\overline{k}\right) \end{aligned}$$

are regular and there is an isomorphism

$$\begin{aligned} \left(\mathcal{O}_X(X) \otimes_k \mathcal{O}_Y(Y)\right)_{\text{red}} &\xrightarrow{\sim} \mathcal{O}_{\widetilde{X \times Y}}\left(\widetilde{X \times Y}\right) \\ f \otimes g \to \quad & (f \circ \pi_X) \cdot (g \circ \pi_Y) \end{aligned}$$

where $\pi_X : \widetilde{X \times Y}\left(\overline{k}\right) \to X\left(\overline{k}\right)$ *and* $\pi_Y : \widetilde{X \times Y}\left(\overline{k}\right) \to Y\left(\overline{k}\right)$ *are the obvious regular projection maps.*

Proof. It's straightforward to show that π_X and π_Y are regular using the definition (6.1.29). By (6.1.94) and (6.1.26) there is a unique morphism making the following diagram commute which must be an isomorphism, and we simply need to check it is of the required form.

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$$\begin{array}{ccc} (k[X] \otimes_k k[Y])_{\text{red}} & \xrightarrow{\sim} & k\left[\widetilde{X \times Y}\right] \\ \downarrow \sim & & \downarrow \sim \\ (\mathcal{O}_X(X) \otimes_k \mathcal{O}_Y(Y))_{\text{red}} & \dashrightarrow & \mathcal{O}_{\widetilde{X \times Y}}\left(\widetilde{X \times Y}\right) \end{array}$$

Given $f \in k[X]$ and $g \in k[Y]$ where $f = \overline{F}$ and $g = \overline{G}$. Then the image of $f \otimes g$ is simply $h := \overline{F(X_1, ..., X_n)G(Y_1, ..., Y_n)}$ and

$$h(x_1, ..., x_n, y_1, ..., y_m) = f(x_1, ..., x_n)g(y_1, ..., y_m)$$

from which the result follows. ∎

Proposition 6.1.96 (Affine Product)
Let $X = V(\mathfrak{a}) \subset \mathbb{A}^n_k$ and $Y = V(\mathfrak{b}) \subset \mathbb{A}^m_k$ be affine varieties. Then there is a bijection

$$\begin{aligned} X\left(\overline{k}\right) \times Y\left(\overline{k}\right) &\xrightarrow{\sim} \widetilde{X \times Y}\left(\overline{k}\right) \\ (x, y) &\to (x_1, ..., x_n, y_1, ..., y_m) \end{aligned}$$

Define $\left(X\left(\overline{k}\right) \times Y\left(\overline{k}\right), \mathcal{O}_{X \times Y}\right)$ as a local space of functions such that this bijection is a regular isomorphism.

The projection maps $\pi_X : X \times Y \to X$ and $\pi_Y : X \times Y \to Y$ are regular and for every affine variety Z we have a natural bijection

$$\begin{aligned} \text{Mor}((Z, \mathcal{O}_Z), (X \times Y, \mathcal{O}_{X \times Y})) &\xrightarrow{\sim} \text{Mor}((Z, \mathcal{O}_Z), (X, \mathcal{O}_X)) \times \text{Mor}((Z, \mathcal{O}_X), (Y, \mathcal{O}_Y)) \\ f &\to ((\pi_X \circ f), (\pi_Y \circ f)) \end{aligned}$$

Proof. Evidently it is enough to prove all the same properties for $\widetilde{X \times Y}$, as then they follow immediately for $X \times Y$ by composing with θ, which is by construction a regular isomorphism. We have natural isomorphisms

$$\begin{aligned} \text{Mor}\left(Z, \widetilde{X \times Y}\right) &\xrightarrow{\sim} \text{AlgHom}_k\left(\mathcal{O}_{\widetilde{X \times Y}}\left(\widetilde{X \times Y}\right), \mathcal{O}_Z(Z)\right) \\ &\xrightarrow{\sim} \text{AlgHom}_k\left((\mathcal{O}_X(X) \otimes_k \mathcal{O}_Y(Y))_{\text{red}}, \mathcal{O}_Z(Z)\right) \\ &\xrightarrow{\sim} \text{AlgHom}_k(\mathcal{O}_X(X) \otimes_k \mathcal{O}_Y(Y), \mathcal{O}_Z(Z)) \\ &\xrightarrow{\sim} \text{AlgHom}_k(\mathcal{O}_X(X), \mathcal{O}_Z(Z)) \times \text{AlgHom}_k(\mathcal{O}_Y(Y), \mathcal{O}_Z(Z)) \\ &\xrightarrow{\sim} \text{Mor}(Z, X) \times \text{Mor}(Z, Y) \end{aligned}$$

by (6.1.95) and (6.1.30). We may verify that the image of $1_{\widetilde{X \times Y}}$ is (π_X, π_Y) and so the result follows from naturality in Z. ∎

Proposition 6.1.97 (Affine Varieties are Separated)
Let $f : X \to Y$ be a regular map of affine varieties. Then the graph

$$\Gamma_f(K) := \{(x, f(x)) \mid x \in X(K)\}$$

is a closed subset of $(X \times Y)(K)$. In particular the diagonal

$$\Delta_X := \{(x, x) \mid x \in X(K)\}$$

is a closed subset of $(X \times X)(K)$.

Proof. Let $\overline{X}_1, ..., \overline{X}_n$ and $\overline{Y}_1, ..., \overline{Y}_m$ be the coordinate functions of $k[X \times Y]$. Further we may suppose that $f = \left(\overline{F}_1, ..., \overline{F}_m\right)$ for $F_j \in k[X_1, ..., X_n]$. Then Γ_f is the zero locus of the ideal

$$\left\langle \overline{Y}_1 - F_1\left(\overline{X}_1, ..., \overline{X}_n\right), ..., \overline{Y}_m - F_m\left(\overline{X}_1, ..., \overline{X}_n\right) \right\rangle$$

∎

Proposition 6.1.98 (Criteria for Product to be Integral)
Let X, Y be integral affine varieties such that either X or Y is geometrically *integral (e.g. if k is algebraically closed). Then $X \times Y$ is integral.*

Proof. See (3.35.42). ∎

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6.1.14 Affine Curves

Definition 6.1.99
*An **affine curve** is an integral affine variety of dimension* 1.

For example an irreducible polynomial $F(X,Y)$ yields an affine curve and is known as a plane curve. The singular points are precisely the common roots of

$$F, \frac{\partial F}{\partial X}, \frac{\partial F}{\partial Y}$$

in $\overline{k}^2$. If there are no such roots then the affine curve is non-singular.

Proposition 6.1.100 (Characterisation Regular Point on a curve)
Let $X = V(\mathfrak{a})$ be an affine curve and $x \in X\left(\overline{k}\right)$. Then the following are equivalent

a) x is regular
b) $\mathcal{O}_{X,x}$ is a discrete valuation ring
c) $\mathcal{O}_{X,x}$ is an integrally closed domain

When k is perfect then this is equivalent to x being a non-singular point.

Proof. Note that by assumption X is integral, so $\mathcal{O}_{X,x}$ is an integral domain by (6.1.61).

a)-c) then follows directly from (3.30.2) and the fact $\dim \mathcal{O}_{X,x} = \dim X = 1$ (6.1.57). The final statement follows from (6.1.75). ∎

Proposition 6.1.101
Let $X = V(\mathfrak{a})$ be a regular affine curve. Then there exists a bijection

$$X_0 \coloneqq X\left(\overline{k}\right) / \operatorname{Aut}\left(\overline{k}/k\right) \longleftrightarrow \{\mathcal{O}_X(X) \subset R \subsetneq k(X) \mid R \text{ valuation ring}\}$$

Proof. The map $x \to \mathcal{O}_{X,x}$ is well-defined by (6.1.100), (6.1.9) and (6.1.60). If $\mathcal{O}_{X,x} = \mathcal{O}_{X,y}$ then $\mathcal{O}^\star_{X,x} = \mathcal{O}^\star_{X,y} \implies \mathfrak{m}_{X,x} = \mathfrak{m}_{X,y} \implies \mathfrak{M}_{X,x} = \mathfrak{M}_{X,y} \implies x \sim y$ by (6.1.9) so the map is injective.

To show surjectivity, observe by (6.1.93) there exists $x \in X\left(\overline{k}\right)$ such that $\mathcal{O}_{X,x} \subseteq R$. Furthermore by repeated application of (6.1.92) we see

$$0 < \dim R \leq \dim \mathcal{O}_{X,x} \overset{(6.1.57)}{=} \dim X = 1$$

This forces $\dim R = \dim \mathcal{O}_{X,x} = 1$ and so $R = \mathcal{O}_{X,x}$ by the same result. ∎

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6.2 Abstract Varieties

In order to deal with affine, quasi-affine, projective and quasi-projective varieties uniformly it is convenient to define an abstract variety. Further some constructions are more natural in this setting (for example product variety). This is analogous to the notion of manifold, and much closer to the notion of scheme. Some differences

a) Sections of the structure sheaf are concrete $\overline{k}$-valued functions rather than abstract elements of a ring in the case of a scheme.
b) When k is not algebraically closed the underlying topological space of the corresponding scheme is the Kolmogorov Quotient (sober-ified to include all the possible generic points). When k is algebraically closed the only difference is the inclusion of generic points.
c) Varieties here are always Noetherian (i.e. there is a finite affine cover and the affines are themselves Noetherian).

The differences are primarily psychological and almost all the theory carries over *mutatis mutandis* to the scheme setting, with additional hypotheses where necessary.

Definition 6.2.1 (Abstract Variety)
*A **variety** over a field k is a space with functions $(X, \mathcal{O}_X)$ such that*

- *There exists a finite open cover $X = \bigcup_{i=1}^n U_i$ together with isomorphisms*
$$\theta_i : \left(U_i, \mathcal{O}_X|_{U_i}\right) \cong \left(X_i\left(\overline{k}\right), \mathcal{O}_{X_i}\right)$$
for some affine varieties $X_1, ..., X_n$ in the sense of (6.1.4).

*The sheaf $\mathcal{O}_X$ is called the **structure sheaf** or **sheaf of regular functions**. A regular morphism is then a morphism of space with functions.*

It is immediate that every affine variety is an abstract variety.

Proposition 6.2.2
A variety $(X, \mathcal{O}_X)$ is a local space with functions. That is for all $f \in \mathcal{O}_X(U)$, the set
$$D(f) := \{x \in U \mid f(x) \neq 0\}$$
is open and $f|_U$ is invertible.

Proof. We have an open affine cover $X = \bigcup_{i=1}^n U_i$. Then we have
$$D(f) = \bigcup_{i=1}^n D\left(f|_{U_i}\right)$$
By (6.1.41) $D\left(f|_{U_i}\right)$ is open and $f|_{U_i}$ is invertible. We deduce that $D(f)$ is open, and using the sheaf conditions that $f|_{D(f)}$ is invertible. ∎

Definition 6.2.3 (Affine (Abstract) Variety)
*Let X be a space of functions. We say it is an **affine variety** if it is isomorphic to $\left(Y\left(\overline{k}\right), \mathcal{O}_Y\right)$, where Y is an affine variety in the sense of (6.1.4) and $\mathcal{O}_Y$ is the corresponding sheaf of regular functions (6.1.25).*

Proposition 6.2.4
Let X be a variety and $f : X \to \overline{k}$ be a function. Then it is regular iff the corresponding map $f : X \to \mathbb{A}^1_k\left(\overline{k}\right)$ is a regular morphism of abstract varieties.

Proof. Suppose that $f : X \to \mathbb{A}^1_k\left(\overline{k}\right)$ is a regular morphism. Evidently the identity map $1 : \mathbb{A}^1_k\left(\overline{k}\right) \to \overline{k}$ is a regular function. Therefore by definition $f : X \to \overline{k}$ is a regular function.

Conversely if $f : X \to \overline{k}$ is regular then so is $f|_U$ for every affine open subset $U \subset X$. By (6.1.29) and (6.1.39) this a regular morphism of abstract varieties $U \to \mathbb{A}^1_k\left(\overline{k}\right)$. By (4.3.27) we have f is a regular morphism. ∎

6.2.1 Topological Properties

Definition 6.2.5 (Integral Variety)
*We say a variety X is **integral** (or irreducible) if it is irreducible as a topological space.*

Proposition 6.2.6 (Criteria for Integrality)
Let X be a variety. Then the following are equivalent

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a) X is integral
b) $\mathcal{O}_X(U)$ is integral for every open subset $U \subset X$

Proposition 6.2.7 (Varieties are Noetherian and Hereditarily Quasi-Compact)
Let X be a variety. Then X is Noetherian as a topological space and every open subset is quasi-compact (and Noetherian).

Proof. (4.1.78) and (4.1.114). ∎

Corollary 6.2.8
Let X be a variety. Then it may be expressed uniquely as a finite union of (closed) irreducible components.

$$X = X_1 \cup ... \cup X_n$$

Proof. See (4.1.76). ∎

Proposition 6.2.9
Suppose $(X, \mathcal{O}_X)$ is a space with functions such that $X = \bigcup_{i=1}^n U_i$ and $\left(U_i, \mathcal{O}_X|_{U_i}\right)$ is a variety. Then so is $(X, \mathcal{O}_X)$.

Proof. Each U_i is a finite union of affine varieties and therefore so is X. ∎

Proposition 6.2.10
Let X be a variety. The (affine) principal open subsets form a base for the topology on X.

Proposition 6.2.11 (Varieties are Symmetric)
Let X be a variety. Then X is symmetric as a topological space and $\overline{\{x\}}$ is finite for all $x \in X$. When k is algebraically closed then every point is closed and therefore X is Kolmogorov.

Furthermore for every open neighbourhood U of x we have $\overline{\{x\}} = \mathrm{cl}_U(\{x\})$.

Proof. That X is symmetric is from (4.1.44) and the affine case (6.1.18). The last statement follows from (4.1.43), then finiteness follows from the affine case. ∎

Proposition 6.2.12 (Closed Subsets of Dimension 0)
Let X be a variety and $Z \subset X$ a closed subset. Then the following are equivalent

a) $\dim Z = 0$
b) $Z = \overline{\{z_1\}} \cup ... \cup \overline{\{z_n\}}$ for some $z_1, ..., z_n \in Z$
c) Z is finite

Proof. Recall X is Noetherian (6.2.7) and therefore by (4.1.91) we have $a) \iff b)$. Then $b) \implies c)$ is (6.2.11) and conversely if Z is finite then $Z = \bigcup_{z \in Z} \overline{\{z\}}$ is a finite union. ∎

Definition 6.2.13 (Separated Variety)
*Let X be a variety. Then we say X is **separated** if for every affine variety Z and pair of morphisms $\phi_1, \phi_2 : Z \rightrightarrows X$ the equaliser*

$$\{z \in Z \mid \phi_1(z) = \phi_2(z)\}$$

is closed in Z.

By (6.1.43) every affine variety is separated.

Proposition 6.2.14
Let X be a separated variety and Z a variety. Then the equaliser of any two maps $\phi_1, \phi_2 : Z \rightrightarrows X$ is closed.

Proof. By definition $Z = \bigcup_{i=1}^n U_i$ where U_i is affine. By assumption the equaliser of $\phi_1|_{U_i}, \phi_2|_{U_i} : U_i \rightrightarrows X$ is closed. As a finite union of closed sets is closed, we are done. ∎

6.2.2 Local Ring

We recall the definition of local ring (4.3.3) for a variety X with irreducible subset W

$$\mathcal{O}_{X,W} := \varinjlim_{U \cap W \neq \emptyset} \mathcal{O}_X(U).$$

Note that if $\overline{W'} = \overline{W}$ then $\mathcal{O}_{X,W} = \mathcal{O}_{X,W'}$, so for example $\mathcal{O}_{X,W} = \mathcal{O}_{X,\overline{W}}$.

In the case $W = \{x\}$ we write $\mathcal{O}_{X,x}$ for the local ring and $\mathfrak{m}_{X,x}$ and $k(\mathfrak{m}_{X,x})$ for the maximal ideal and residue field respectively.

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In the case $X = W$ is irreducible then we recover the **field of rational functions** written $k(X)$. For a variety we can make stronger statement about the structure of $k(X)$

Proposition 6.2.15
Let X be an integral variety, $x \in X$ and U an open neighbourhood of x. Then there are canonical injections

$$\mathcal{O}_X(U) \hookrightarrow \mathcal{O}_{X,x} \hookrightarrow k(X)$$

which makes $k(X)$ the field of fractions for $\mathcal{O}_{X,x}$. When U is affine then $k(X)$ is also the field of fractions for $\mathcal{O}_X(U)$. These maps commute with the restriction maps on $\mathcal{O}_X$, which are also injective.

Proof. By restricting to an affine open cover we may check that the restriction maps ρ_{UV} are injective. This shows that the first map is injective because every stalk can be lifted to some open neighbourhood.

Let U be an affine open neighbourhood of x then by (4.3.16) $k(X) \xrightarrow{\sim} k(U)$. We may deduce from the affine case (6.1.60) that $k(X)$ is the field of fractions for $\mathcal{O}_{X,x}$ and $\mathcal{O}_X(U)$. ∎

As $\mathcal{O}_{X,x}$ is isomorphic to the local ring of an affine neighbourhood we can recover many statements from the affine case.

Proposition 6.2.16
Let X be a variety and $W \subset X$ an irreducible subset. Then the following are equivalent

a) W is contained in only one irreducible component
b) $\mathcal{O}_{X,W}$ is an integral domain

Proof. Firstly we may reduce to the case W is closed by considering $\overline{W}$. As X is a variety, $W \cap U \neq \emptyset$ for U affine. By (4.1.73) we may reduce to the case X is affine, which is (6.1.61). ∎

Proposition 6.2.17 (Dominant Morphisms induce Function Field Maps)
Let X, Y be integral varieties and $\phi : X \to Y$ a dominant regular morphism. Then there is a canonical injective k-algebra homomorphism

$$\begin{aligned} \phi_\star : k(Y) &\hookrightarrow k(X) \\ (V, \sigma) &\to (\phi^{-1}(V), \sigma \circ \phi) \end{aligned}$$

This is functorial in the sense that

$$(\phi \circ \psi)_\star = \psi_\star \circ \phi_\star$$

If ϕ is an open immersion then $\phi_\star$ is an isomorphism.

Proof. See (4.3.7) and (4.3.16). ∎

Proposition 6.2.18 (Function Field Maps are compatible)
Let X, Y be integral varieties, $\phi : X \to Y$ a dominant regular morphism and $y \in \overline{\{\phi(x)\}}$. Then for all neighbourhoods V of y we have a commutative diagram

$$\begin{array}{ccc} \mathcal{O}_Y(V) & \xrightarrow{\phi_V^\sharp} & \mathcal{O}_X(\phi^{-1}(V)) \\ \downarrow & & \downarrow \\ \mathcal{O}_{Y,\phi(x)} & \xrightarrow{\phi_x} & \mathcal{O}_{X,x} \\ \downarrow & & \downarrow \\ k(Y) & \xrightarrow{\phi_\star} & k(X) \end{array}$$

In particular when identified as subrings of $k(X)$ and $k(Y)$ we have the identities

$$\phi_\star^{-1}(\mathcal{O}_{X,x}) = \mathcal{O}_{Y,\phi(x)}$$

and

$$\phi_\star^{-1}(\mathcal{O}_X(\phi^{-1}(V))) = \mathcal{O}_Y(V)$$

Proof. See (4.3.9) and (6.2.15). ∎

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Lemma 6.2.19
For a variety X, R an algebraic k-algebra and $\phi : \mathcal{O}_{X,x} \to R$ a k-algebra homomorphism Then $\phi^{-1}(\mathfrak{m}) = \mathfrak{m}_{X,x}$ for every maximal ideal $\mathfrak{m} \lhd R$.

Proof. Restatement of (6.1.62), since $\mathcal{O}_{X,x}$ is isomorphic to local ring of affine neighbourhood. ∎

Proposition 6.2.20 (Equivalent Points)
Let X be a variety and $x, y \in X$. Then the following are equivalent

a) x, y are topologically indistinguishable
b) $x \in \overline{\{y\}}$
c) $y \in \overline{\{x\}}$
d) $\overline{\{x\}} = \overline{\{y\}}$
e) $[x] = [y]$ in the Kolmogorov Quotient X_0

When X is integral these are equivalent to $\mathcal{O}_{X,x} = \mathcal{O}_{X,y}$ as subrings of $k(X)$.

Proof. By (6.2.11) X is symmetric and the equivalent conditions a)-e) follow from (4.1.42).

If $x \sim y$ then by definition $\mathcal{O}_{X,x} = \mathcal{O}_{X,y}$.

Conversely suppose $f \in \mathcal{O}_{X,x} \setminus \mathcal{O}_{X,y}$ in $k(X)$. In some open neighbourhood U of x we have $f = \frac{g}{h}$ for $g, h \in \mathcal{O}_X(U)$ and $h(x) \neq 0$. By definition we must have $h(y) = 0$. Therefore $D(h)$ contains x but not y, and therefore x and y are not topologically indistinguishable. ∎

6.2.3 Affine Varieties

We restate various results relating to "abstract" affine varieties to avoid the need to switch notations later on.

Proposition 6.2.21 (Properties of the Structure Sheaf)
Let X be an abstract variety $U \subset X$ an affine open subset. Then the following properties hold

a) For all irreducible subsets $W \subset U$ we have $\mathfrak{M}_{U,W} \coloneqq \{f \in \mathcal{O}_X(U) \mid f|_W = 0\}$ is a prime ideal of $\mathcal{O}_X(U)$. Further we have a canonical local isomorphism

$$\begin{array}{ccc} \mathcal{O}_X(U) & \longrightarrow & \mathcal{O}_X(U)_{\mathfrak{M}_{U,W}} \\ & \searrow^{\rho_W} & \downarrow \sim \\ & & \mathcal{O}_{X,W} \end{array}$$

which is the unique homomorphism making this diagram commute.

The inverse image of $\mathfrak{m}_{X,W}$ is $\mathfrak{M}_{U,W}\mathcal{O}_X(U)_{\mathfrak{M}_{U,W}}$ and $\mathfrak{M}_{U,W}$ in $\mathcal{O}_X(U)_{\mathfrak{M}_{U,W}}$ and $\mathcal{O}_X(U)$ respectively.

When $W \subset \overline{\{x\}}$ then $\mathfrak{M}_{U,W} = \mathfrak{M}_{U,x}$ is a maximal ideal.

b) There is a well-defined bijection

$$\begin{aligned} U_0 &\longleftrightarrow \operatorname{Specm}(\mathcal{O}_X(U)) \\ x &\to \mathfrak{M}_{U,x} \end{aligned}$$

Further $\mathfrak{M}_{U,x} = \mathfrak{M}_{U,x'} \iff x \in \overline{\{x'\}} \iff x$ and x' are topologically indistinguishable.

c) For all $f \in \mathcal{O}_X(U)$ we have $\left(D(f), \mathcal{O}_X|_{D(f)}\right)$ is an affine variety. Further we have a canonical isomorphism

$$\begin{array}{ccc} \mathcal{O}_X(U) & \longrightarrow & \mathcal{O}_X(U)_f \\ & \searrow^{\rho_{XD(f)}} & \downarrow \sim \\ & & \mathcal{O}_X(D(f)) \end{array}$$

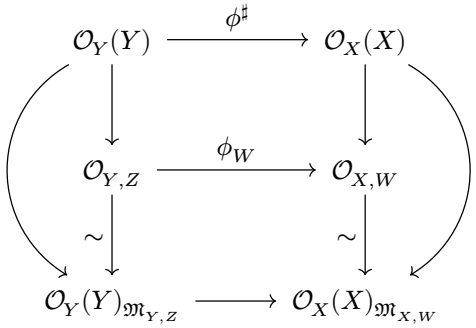
which is the unique homomorphism making this diagram commute. For irreducible $W \subset D(f)$, under this isomorphism the inverse image of $\mathfrak{M}_{D(f),W}$ is $\left(\mathfrak{M}_{U,W}\right)_f$. Similarly the following diagram commutes

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$$\begin{array}{ccc} \mathcal{O}_X(U)_f & \xrightarrow{\sim} & \mathcal{O}_X(D(f)) \\ \downarrow & & \downarrow \\ \mathcal{O}_X(U)_{\mathfrak{M}_{U,W}} & \xrightarrow{\sim} & \mathcal{O}_{X,W} \end{array}$$

and the inverse image of $\mathfrak{m}_{X,W}$ *is* $\mathfrak{M}_{D(f),W}$ *and* $\left(\mathfrak{M}_{U,W}\right)_f$ *in* $\mathcal{O}_X(D(f))$ *and* $\mathcal{O}_X(U)_f$ *respectively.*

d) Suppose X, Y *are affine varieties and* $W \subset X$ *is irreducible and* $Z \subset \overline{\phi(W)}$ *for some irreducible* $Z \subset Y$*. Let* $\phi : X \to Y$ *be a regular morphism. Then the stalk map* $\phi_W : \mathcal{O}_{Y,Z} \to \mathcal{O}_{X,W}$ *(see (4.3.5)) makes the following diagram commute*



where the bottom arrow is the localisation map induced by $\phi^\sharp$ *(3.5.22). In fact it is the unique map making either square commute.*

Proposition 6.2.22 (Affine Mapping Property)
Let Y *be an affine variety, and* X *a variety. Then there is a canonical bijection*

$$\begin{aligned} \mathrm{Mor}(X,Y) &\xrightarrow{\sim} \mathrm{AlgHom}_k(\mathcal{O}_Y(Y), \mathcal{O}_X(X)) \\ \phi &\to \ \phi^\sharp_Y \end{aligned}$$

When X *is affine and* $Z \subset Y$*,* $W \subset X$ *are irreducible subsets, then*

$$Z \subset \overline{\phi(W)} \iff \left(\phi^\sharp_Y\right)^{-1}\left(\mathfrak{M}_{X,W}\right) \subset \mathfrak{M}_{Y,Z}$$

Proof. When X is affine then this is just (6.1.30) and (6.1.32). In general suppose $X = \bigcup_{i=1}^n U_i$ is an affine covering then we have a commutative diagram

$$\begin{array}{ccc} \mathrm{Mor}(X,Y) & \xrightarrow{(-)^\sharp} & \mathrm{AlgHom}_k(\mathcal{O}_Y(Y), \mathcal{O}_X(X)) \\ \downarrow \alpha & & \downarrow \beta \\ \prod_{i=1}^n \mathrm{Mor}(U_i, Y) & \xrightarrow{\gamma} & \prod_{i=1}^n \mathrm{AlgHom}_k(\mathcal{O}_Y(Y), \mathcal{O}_X(U_i)) \end{array}$$

where $\alpha(\phi)_i = \phi|_{U_i}$ and $\beta(\psi)_i = \rho_{XU_i} \circ \psi$ and γ is the same definition as $(-)^\sharp$.

By the affine case γ is bijective, and evidently α is injective. Therefore $\beta \circ (-)^\sharp$ is injective, whence $(-)^\sharp$ is injective. To show surjectivity consider $\psi \in \mathrm{AlgHom}_k(\mathcal{O}_Y(Y), \mathcal{O}_X(X))$ then as γ is bijective we have maps $\phi_i : U_i \to X$ such that $(\phi_i)^\sharp = \rho_{YU_i} \circ \psi$. Suppose $V \subset U_i \cap U_j$ is affine then

$$(\phi_i|_V)^\sharp = \rho_{U_iV} \circ \rho_{XU_i} \circ \psi = \rho_{XV} \circ \psi$$

Therefore $(\phi_i|_V)^\sharp = \left(\phi_j|_V\right)^\sharp$ and from the affine case $\phi_i|_V = \phi_j|_V$. As the open affines from a base (6.2.10) we conclude $\phi_i|_{U_i \cap U_j} = \phi_j|_{U_i \cap U_j}$. Therefore there exists a morphism $\phi : X \to Y$ such that $\phi|_{U_i} = \phi_i$ and therefore $\beta\left(\phi^\sharp\right) = \beta(\psi)$. Evidently β is injective and so $\phi^\sharp = \psi$. ∎

Corollary 6.2.23
There is a full and faithful contravariant functor

$$\mathbf{AbstractAffVar}_k \longrightarrow \mathbf{ReducedFGAlgebras}_k$$

which in particular preserves and reflects isomorphisms

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Corollary 6.2.24
Let X be an affine variety, Y a variety and $f \in \mathcal{O}_X(X)$. Then there is a canonical bijection

$$\begin{aligned} \mathrm{Mor}(D(f), Y) &\xrightarrow{\sim} \mathrm{AlgHom}_k\big(\mathcal{O}_Y(Y), \mathcal{O}_X(X)_f\big) \\ \phi &\to \ \theta_f \circ \phi_Y^\sharp \end{aligned}$$

where $\theta_f : \mathcal{O}_X(D(f)) \xrightarrow{\sim} \mathcal{O}_X(X)_f$ is the isomorphism (6.2.21).c). The image of $i : D(f) \to X$ is simply the canonical localisation map.

Suppose Y is affine and $W \subset D(f)$, $Z \subset Y$ are irreducible subsets. Then $Z \subset \overline{\phi(W)} \iff \big(\theta_f \circ \phi_Y^\sharp\big)^{-1}\Big(\big(\mathfrak{M}_{X,W}\big)_f\Big) \subset \mathfrak{M}_{Y,Z}$.

Proof. We may apply (6.2.22). ∎

Corollary 6.2.25
Let X be an affine variety and $\phi, \psi : X \to Y$ regular morphisms, and $f \in \mathcal{O}_X(X)$. Then $\phi|_{D(f)} = \psi|_{D(f)}$ if and only if $\rho_f \circ \phi^\sharp = \rho_f \circ \psi^\sharp$, where $\rho_f : \mathcal{O}_X(X) \to \mathcal{O}_X(X)_f$ is the canonical localisation map.

Proof. Denote by $i : D(f) \hookrightarrow X$ the inclusion then by (6.2.22) $\phi|_{D(f)} = \psi|_{D(f)} \iff \phi \circ i = \psi \circ i \iff i^\sharp \circ \phi^\sharp = i^\sharp \circ \psi^\sharp$ where $i^\sharp : \mathcal{O}_X(X) \to \mathcal{O}_{D(f)}(D(f))$ is simply the restriction map. We may compose this with the isomorphism (6.2.21).c) to get the required result. ∎

There are some properties of affine varieties which generalise

Proposition 6.2.26
Let X be an affine variety and $f_1, ..., f_r \in \mathcal{O}_X(X)$. Then the following are equivalent

a) $X = \bigcup_{i=1}^r D(f_i)$
b) $(f_1, ..., f_r) = \mathcal{O}_X(X)$

Proof. We may reduce to the case of X a concrete affine variety. Then

$$X = \bigcup_{i=1}^r D(f_i) \iff \emptyset = \bigcap_{i=1}^r V(f_i) = V(f_1, ..., f_r) \iff k[X] = IV(f_1, ..., f_r) = \sqrt{(f_1, ..., f_r)} \iff k[X] = (f_1, ..., f_r)$$

where we have identified $\mathcal{O}_X(X)$ and $k[X]$. ∎

Proposition 6.2.27
Let X be a variety and $f \in \mathcal{O}_X(X)$. There is a canonical isomorphism

$$\mathcal{O}_X(X)_f \xrightarrow{\sim} \mathcal{O}_X(D(f))$$

Proof. The following is adapted from [Liu06] (Prop. 2.3.12).

Recall (6.2.2) that $D(f)$ is an open subset and $f|_{D(f)}$ is invertible. This shows that the homomorphism is well-defined, and we simply need to show it is bijective, with the case X affine already proven (6.1.28).

Let $X = \bigcup_{i=1}^n U_i$ be an affine open covering then $D(f) = \bigcup_{i=1}^n D\big(f|_{U_i}\big)$. By the affine case we have an isomorphism

$$\mathcal{O}_X(U_i)_{f|_{U_i}} \xrightarrow{\sim} \mathcal{O}_X\Big(D\big(f|_{U_i}\big)\Big)$$

By the sheaf conditions we have an exact sequence of $\mathcal{O}_X(X)$-modules

$$0 \longrightarrow \mathcal{O}_X(X) \longrightarrow \bigoplus_{i=1}^n \mathcal{O}_X(U_i) \longrightarrow \bigoplus_{i,j=1}^n \mathcal{O}_X(U_i \cap U_j)$$

Localising at $f \in \mathcal{O}_X(X)$ is exact (3.5.13) and commutes with direct sums (...) so we obtain the exact sequences

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{O}_X(X)_f & \longrightarrow & \bigoplus_{i=1}^n \mathcal{O}_X(U_i)_{f|_{U_i}} & \longrightarrow & \bigoplus_{i,j=1}^n \mathcal{O}_X(U_i \cap U_j)_{f|_{U_i \cap U_j}} \\ & & \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma \\ 0 & \longrightarrow & \mathcal{O}_X(D(f)) & \longrightarrow & \bigoplus_{i=1}^n \mathcal{O}_X\Big(D\big(f|_{U_i}\big)\Big) & \longrightarrow & \bigoplus_{i,j=1}^n \mathcal{O}_X\Big(D\big(f|_{U_i \cap U_j}\big)\Big) \end{array}$$

where α, β, γ are just the canonical maps already defined. By the affine case β is an isomorphism. We claim that diagram chasing shows α is injective, for suppose that $\alpha(x) = 0$ then $\alpha(x)|_{D(f|_{U_i})} = 0$ whence $x|_{U_i} = 0$, and by exactness $x = 0$. As $U_i \cap U_j$ is a variety, then we conclude γ is also injective.

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We further claim that α is surjective, for given $y \in \mathcal{O}_X(D(f))$ then there exists $x_i \in \mathcal{O}_X(U_i)_{f_i}$ such that $\beta(x_i) = y|_{D(f|_{U_i})}$. From injectivity of γ and exactness of the bottom sequence we conclude that $x_i|_{U_i \cap U_j} = x_j|_{U_i \cap U_j}$, and so by exactness of the top sequence there is $x \in \mathcal{O}_X(X)_f$ such that $x|_{U_i} = x_i$. Consequently $\alpha(x)|_{D(f|_{U_i})} = y|_{D(f|_{U_i})}$. Exactness of the bottom sequence shows that $\alpha(x) = y$ as required. ∎

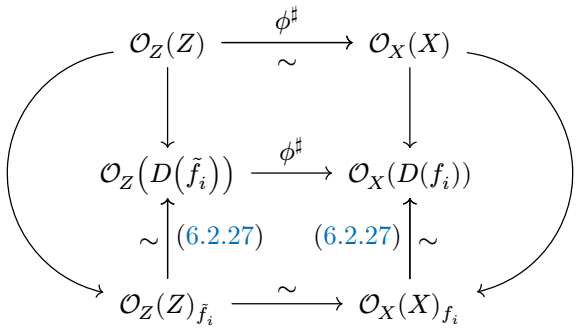
Proposition 6.2.28
Let X be a variety and suppose $f_1, ..., f_r \in \mathcal{O}_X(X)$ generate $\mathcal{O}_X(X)$. If $D(f_i)$ is an affine variety for all $i = 1...r$ then so is X. Furthermore

$$X = \bigcup_{i=1}^{n} D(f_i)$$

Proof. First we claim that $\mathcal{O}_X(X)$ is a reduced finitely generated k-algebra. Recall by (6.2.27) there is a k-algebra isomorphism $\mathcal{O}_X(X)_{f_i} \xrightarrow{\sim} \mathcal{O}_X(D(f_i))$. By assumption $D(f_i)$ is affine so $\mathcal{O}_X(X)_{f_i}$ is finitely-generated as a k-algebra. Therefore by (3.23.21) so is $\mathcal{O}_X(X)$.

By (6.1.36) we may find an affine variety Z together with an isomorphism $\phi^\sharp : \mathcal{O}_Z(Z) \xrightarrow{\sim} \mathcal{O}_X(X)$. By (6.2.22) there is a corresponding morphism of varieties $\phi : X \to Z$, which we wish to show is an isomorphism. Let $\tilde{f}_i := \left(\phi^\sharp\right)^{-1}(f_i)$ then by definition $\phi^{-1}\left(D\left(\tilde{f}_i\right)\right) = D(f_i)$.

We have a commutative diagram



whence the middle arrow is an isomorphism. As $D(f_i)$ is an affine variety we deduce from (6.2.23) that $\phi|_{D(f_i)} : D(f_i) \xrightarrow{\sim} D\left(\tilde{f}_i\right)$ is a regular isomorphism. Further $X = \phi^{-1}(Z)$ whence the $D(f_i)$ cover X and ϕ is a regular isomorphism as required. ∎

6.2.4 Lifting Stalk Maps

In the same way that stalks can be lifted to an open neighbourhood, for varieties we can lift stalk maps to an open neighbourhood. After reducing to the affine case this is essentially an algebraic statement about localisation.

Proposition 6.2.29 (Lifting Stalk Maps I)
Let $\phi, \psi : X \to Y$ be regular morphisms of varieties and $W \subset X$, $Z \subset Y$ non-empty irreducible subsets. Suppose

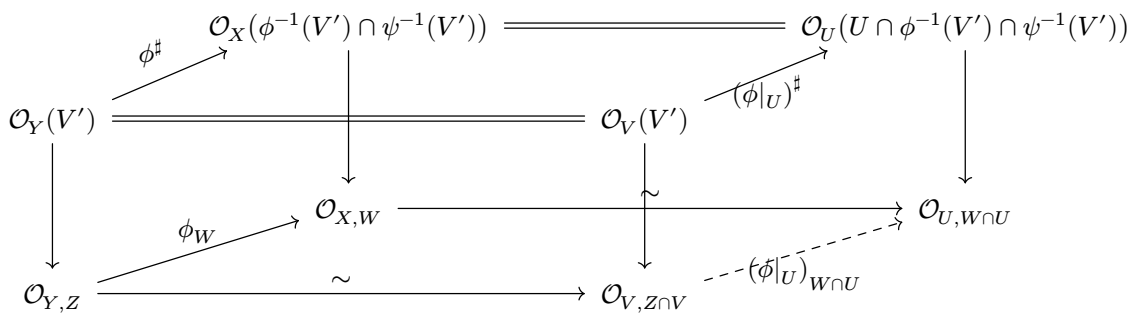
a) $Z \subset \overline{\phi(W)} = \overline{\psi(W)}$, and
b) the stalk maps $\phi_W, \psi_W : \mathcal{O}_{Y,Z} \to \mathcal{O}_{X,W}$ agree.

Then there exists an open subset $U \subset X$ for which $U \cap W \neq \emptyset$ and $\phi|_U = \psi|_U$.

Proof. For any open subset $V' \subset Y$ meeting Z (and therefore $\phi(W)$ (4.1.19)), then $\phi^{-1}(V')$ meets W. Similarly for ψ. So by irreducibility $\phi^{-1}(V') \cap \psi^{-1}(V')$ meets W (4.1.60).

Let $V \subset Y$ be an affine open subset meeting Z and choose an affine open subset $U \subset \phi^{-1}(V) \cap \psi^{-1}(V)$ meeting W.

From (4.3.19) we have a commutative cube for every open subset $V' \subset V$



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By uniqueness of the bottom square we conclude that $(\phi|_U)_{W\cap U} = (\psi|_U)_{W\cap U}$ and we may reduce to the case of X and Y affine.

By (6.2.25) it is sufficient to find $f \in \mathcal{O}_X(X)$ such that $\rho_f \circ \phi_Y^\sharp = \rho_f \circ \psi_Y^\sharp$. Suppose that $D(f) \cap W \neq \emptyset$ and consider the commutative diagram

$$\begin{array}{ccc} \mathcal{O}_Y(Y) & \overset{\phi^\sharp}{\underset{\psi^\sharp}{\longrightarrow}} & \mathcal{O}_X(X) \\ \downarrow & & \downarrow{\scriptstyle \rho_f} \\ & & \mathcal{O}_X(X)_f \\ & & \downarrow \\ \mathcal{O}_Y(Y)_{\mathfrak{m}_{Y,Z}} & \overset{\phi_W}{\underset{\psi_W}{\longrightarrow}} & \mathcal{O}_X(X)_{\mathfrak{m}_{X,W}} \end{array}$$

where we have used (6.2.21) to replace $\mathcal{O}_{Y,Z}$ with $\mathcal{O}_Y(Y)_{\mathfrak{m}_{Y,Z}}$ and $\mathcal{O}_{X,W}$ with $\mathcal{O}_X(X)_{\mathfrak{m}_{X,W}}$.

Let $y_1, ..., y_n$ be generators for $\mathcal{O}_Y$. Then by assumption there exists $t_i \notin \mathfrak{M}_{X,W}$ such that $t_i\phi^\sharp(y_i) = t_i\psi^\sharp(y_i)$. In particular $f = t_1 \cdots t_n$ works, since then $f\phi^\sharp(y_i) = f\psi^\sharp(y_i)$ for all i. ∎

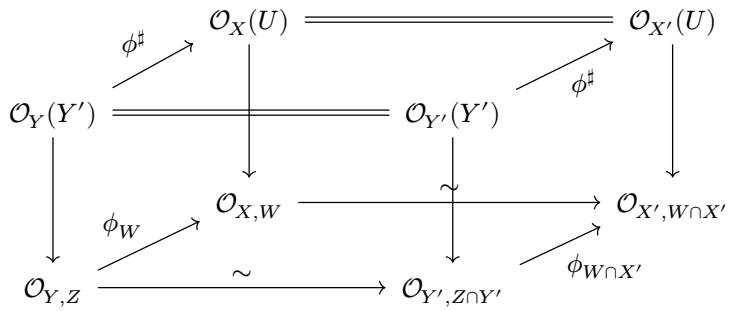
Proposition 6.2.30 (Lifting Stalk Maps II)
Let X, Y be varieties, $W \subset X$, $Z \subset Y$ irreducible subsets and suppose there exists a local k-algebra homomorphism $\underline{\phi_W : \mathcal{O}_{Y,Z} \to \mathcal{O}_{X,W}}$. Then there exists an open subset $U \subset X$ and a regular morphism $\phi : U \to Y$ such that $\overline{Z} = \overline{\phi(W \cap U)}$ and the following diagram commutes for all open subsets $V \subset Y$ meeting Z and $U' \subset \phi^{-1}(V)$ meeting W

$$\begin{array}{ccc} \mathcal{O}_Y(V) & \xrightarrow{\phi^\sharp} & \mathcal{O}_X(U') \\ \downarrow & & \downarrow \\ \mathcal{O}_{Y,Z} & \xrightarrow[\phi_W]{} & \mathcal{O}_{X,W} \end{array}$$

Proof. Let $Y' \subset Y$ be an open affine subset meeting Z, and $X' \subset \phi^{-1}(V)$ an affine open subset meeting W.

Then $W \cap X'$ is an irreducible subset of X' and similarly for $Z \cap Y'$ (...).

For every open subset $U \subset X'$ meeting W we have a commutative diagram



Therefore we may assume without loss of generality that X and Y are affine, because we may reduce to finding a regular morphism $\phi : U \to Y'$.

We wish to find $f \in \mathcal{O}_X(X)$ and $\phi_Y^\sharp$ which completes the diagram

$$\begin{array}{ccc} \mathcal{O}_Y(Y) & \overset{\phi_Y^\sharp}{\dashrightarrow} & \mathcal{O}_X(X)_f \\ \downarrow & & \downarrow \\ \mathcal{O}_{Y,Z} & \xrightarrow{\phi_W^\sharp} & \mathcal{O}_{X,W} \\ \downarrow \sim & & \downarrow \sim \\ \mathcal{O}_Y(Y)_{\mathfrak{m}_{Y,Z}} & \xrightarrow{\phi_W^\sharp} & \mathcal{O}_X(X)_{\mathfrak{m}_{X,W}} \end{array}$$

with i_Z and i_f the corresponding vertical maps.

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For then a regular morphism $\phi : D(f) \to Y$ exists by (6.2.24). and by uniqueness (6.2.21).d) it has the required stalk map. As (6.2.19) $\phi^\sharp_W$ is local we may chase the diagram to show that $\left(\phi^\sharp\right)_Y^{-1}\left(\left(\mathfrak{M}_{X,W}\right)_f\right) = \mathfrak{M}_{Y,Z}$ and therefore $\overline{Z} = \overline{\phi(W \cap D(f))}$ as required.

By inspection it is sufficient to find f such that i_f is injective and $\mathrm{Im}\left(\phi^\sharp_W \circ i_Z\right) \subset \mathrm{Im}\left(i_f\right)$. We may find g such that i_g is injective by (3.5.37). Suppose $\mathcal{O}_Y(Y)$ is generated by $y_1, ..., y_m$ then $\phi^\sharp_W(y_i/1) = x_i/g_i$ for some $g_i \notin \mathfrak{M}_{X,W}$. Then $f = gg_1 \cdots g_m$ satisfies both criteria. As the stalk map is uniquely defined such that the diagram commutes we find that the regular morphism ϕ has stalk map precisely $\phi^\sharp_W$. ∎

Corollary 6.2.31 (Lifting Stalk Maps I)
Let X, Y be varieties, $x \in X$ and $\phi, \psi : X \to Y$ regular morphisms for which

a) $\overline{\{\phi(x)\}} = \overline{\{\psi(x)\}}$
b) $\phi_x = \psi_x$

Then there exists an open neighbourhood U of x such that $\phi|_U = \psi|_U$.

Corollary 6.2.32 (Lifting Stalk Maps II)
Let X, Y be varieties and $\phi^\sharp_x : \mathcal{O}_{Y,y} \to \mathcal{O}_{X,x}$ a local k-algebra homomorphism. Then there exists an open subset $U \subset X$ and a regular morphism $\phi : U \to Y$ such that $y \in \overline{\{\phi(x)\}}$ and has stalk map precisely $\phi^\sharp_x$.

6.2.5 Open Subvarieties

Proposition 6.2.33 (Open Subset is Abstract Variety)
*Let X be a variety (resp. separated variety) and $U \subset X$ an open subset. Then $(U, \mathcal{O}_X|_U)$ is a variety (resp. separated variety), which we call an **open subvariety**. We may write $\mathcal{O}_U \coloneqq \mathcal{O}_X|_U$.*

The inclusion $i : U \hookrightarrow X$ is an open immersion and an open immersion $\phi : X \to Y$ induces an isomorphism between X and the open subvariety $\phi(X)$.

Proof. By definition $X = \bigcup_{i=1}^n U_i$ where U_i is affine. By (...) we have

$$U \cap U_i = \bigcup_{\alpha \in \mathcal{A}_i} D(f_\alpha)$$

where $f_\alpha \in \mathcal{O}_X(U_i)$. By (6.2.21) $\left(D(f_\alpha), \mathcal{O}_X|_{D(f_\alpha)}\right)$ is affine. Therefore we have an open cover of affine sets. By (6.2.7) U is quasi-compact and therefore there exists a finite affine cover. This shows that U is a variety.

Clearly the inclusion map $U \to X$ is injective and regular and so we see that U inherits separatedness. ∎

Proposition 6.2.34 (Quasi-Affine Varieties are Abstract Varieties)
Let X be a quasi-affine variety. Then $(X, \mathcal{O}_X)$ is a separated variety. More precisely there exists a full and faithful functor

$$\mathbf{QuasiAffVar}_k \longrightarrow \mathbf{Var}_k$$

Proof. This follows from (6.2.33) and (6.1.39) ∎

Proposition 6.2.35 (Morphisms into Open Subvarieties)
Let X, Y be varieties and $U \subset Y$ an open subvariety. Then a regular map $\phi : X \to U$ is regular when regarded as a map $\phi : X \to Y$.

Conversely a regular map $\phi : X \to Y$ such that $\phi(X) \subset U$ may be regarded as a regular map $\phi : X \to U$.

Proposition 6.2.36 (Morphisms from Open Subvarieties)
Let $\phi : X \to Y$ be a regular map of varieties and $U \subset X$ an open subvariety. Then $\phi|_U : U \to Y$ is a regular map.

Proposition 6.2.37 (Patching Varieties)
Suppose $X = \bigcup_{i=1}^n U_i$ is a finite union of sets, $\left(U_i, \mathcal{O}_{U_i}\right)$ is a variety, and suppose that for all $i, j = 1...n$

- *$U_i \cap U_j$ is open in U_j, and*
- *the identity map $1 : \left(U_i \cap U_j, \mathcal{O}_{U_i}|_{U_i \cap U_j}\right) \to \left(U_j \cap U_i, \mathcal{O}_{U_j}|_{U_i \cap U_j}\right)$ is a regular morphism of spaces of functions.*

Then there exists a unique structure of a variety on X such that

- *U_i is open in X, and*
- *$\mathcal{O}_{U_i} = \mathcal{O}_X|_{U_i}$.*

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Proposition 6.2.38 (Glueing Varieties)
Suppose $X = \bigcup_{i=1}^{n} U_i$ is a finite union of sets and there exists bijections $\theta_i : U_i \xrightarrow{\sim} W_i$, with W_i varieties such that for all $i, j = 1...n$

- *$\theta_i(U_i \cap U_j)$ is open in W_i, and*
- *$\theta_{ij} = \theta_j \circ \theta_i^{-1} : \theta_i(U_i \cap U_j) \to \theta_j(U_i \cap U_j)$ is regular.*

Then there exists a unique structure of a variety on X such that

- *U_i is open in X, and*
- *θ_i is a regular isomorphism.*

Proof. The topology exists by (4.3.26) and we may simply define

$$\mathcal{O}_X(U) := \left\{ f : U \to \overline{k} \;\middle|\; f \circ \theta_i^{-1}|_{\theta_i(U_i \cap U)} \text{ regular for all } i = 1...n \right\}$$

which gives X the structure of a space of functions. As each U_i is a finite union of affine varieties, so is X. ∎

6.2.6 Closed Subvarieties

Proposition 6.2.39 (Affine Closed Subvarieties)
Let $(X, \mathcal{O}_X)$ be an affine variety with explicit isomorphism $\theta : X \cong V(\mathfrak{a})$, and $Z \subset X$ a closed subset. Then the structure $(Z, \mathcal{O}_Z)$ defined in (4.3.21) is also affine with explicit isomorphism $\theta|_Z : Z \cong V(\mathfrak{b})$ where $\mathfrak{b} := I(\theta(Z))$.

Proof. This follows largely from (6.1.44). ∎

Proposition 6.2.40 (Closed Subvariety)
Let $(X, \mathcal{O}_X)$ be a variety with Z a closed subset. Then $(Z, \mathcal{O}_Z)$ with the structure defined in (4.3.21) is a variety.

Proof. By assumption $X = \bigcup_{i=1}^{n} U_i$ with $\left(U_i, \mathcal{O}_{U_i}\right)$ affine. Evidently $Z = \bigcup_{i=1}^{n} (U_i \cap Z)$ is an open cover. Further $U_i \cap Z$ is a closed subset of U_i. By (6.2.39) $\left(U_i \cap Z, \mathcal{O}_{U_i}|_{U_i \cap Z}\right) \overset{(6.2.47)}{=} \left(U_i \cap Z, \mathcal{O}_Z|_{U_i \cap Z}\right)$ is affine. Therefore Z is a variety. ∎

Definition 6.2.41 (Closed Immersion)
*Let $\phi : X \to Y$ be a regular morphism of varieties. Then it is a **closed immersion** if $\phi(X)$ is closed in Y and ϕ determines an isomorphism between X and the closed subvariety $\phi(X)$.*

Proposition 6.2.42 (Closed Subvariety is a Variety)
Let X be a variety and $Z \subset X$ a closed subset. Then the space with functions $(Z, \mathcal{O}_Z)$ is a variety and the map $j : (Z, \mathcal{O}_Z) \to (X, \mathcal{O}_X)$ is a closed immersion.

When X is affine, then so is Z and the map $j^\sharp : \mathcal{O}_X(X) \to \mathcal{O}_Z(Z)$ is surjective.

Proof. By definition $X = \bigcup_{i=1}^{n} U_i$ where U_i is affine. Then $Z \cap U_i$ is a closed subset of U_i and by (6.2.47) $\mathcal{O}_Z|_{Z \cap U_i} = \mathcal{O}_{U_i}|_{Z \cap U_i}$. Therefore by (6.2.39) $Z \cap U_i$ is affine, and open in Z. This shows that Z is a variety.

Suppose $f \in \mathcal{O}_X(U)$ then $f \circ j \in \mathcal{O}_Z(Z \cap U)$, which shows that j is a regular morphism.

The final statement follows from (6.1.45). ∎

Proposition 6.2.43
Let $\phi : X \to Y$ be a closed immersion and Y an affine variety. Then X is affine.

Proposition 6.2.44
Let $\phi : X \to Y$ be a regular map of varieties and $Z \subset X$ a closed subvariety. Then $\phi|_Z$ is regular.

Proposition 6.2.45
Let $\phi : X \to Y$ be a regular map of varieties and $Z \subset Y$ a closed (resp. open) subvariety. If $\phi(X) \subset Z$ then ϕ may be regarded as a regular map $X \to Z$.

Proposition 6.2.46
Let $\phi : X \to Y$ be a regular map of affine varieties. Then ϕ is a closed immersion if and only if the induced homomorphism $\phi_Y^\sharp : \mathcal{O}_Y(Y) \to \mathcal{O}_X(X)$ is surjective.

Proof. Restatement of (6.1.45). ∎

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6.2.7 Locally Closed Subvariety

Proposition 6.2.47 (Locally Closed Subvariety)
Let X be a variety, Z a closed subset and U an open subset. Then $Z \cap U$ is a variety (viewed as a closed subvariety of U or an open subvariety of Z).

6.2.8 Rational Points

For abstract varieties we use a more local definition of rational point which coincides with the affine case (6.1.63).

Definition 6.2.48 (K-Rational Points)
Let X be a variety and K/k an algebraic field extension. Define the K-rational points as follows

$$X(K) \coloneqq \left\{(x, \iota) \;\middle|\; x \in X_0, \quad \iota : k\left(\mathfrak{m}_{X,x}\right) \hookrightarrow K\right\}$$

where X_0 is the Kolmogorov Quotient (4.1.35) of X, and recalling that $\mathcal{O}_{X,x} = \mathcal{O}_{X,y}$ whenever x, y become equal in X_0 (6.2.11).

By (6.2.19) specifying ι is completely equivalent to specifying a k-algebra homomorphism $\mathcal{O}_{X,x} \to K$.

Proposition 6.2.49
Let $\phi : X \to Y$ be a regular morphism of varieties and K/k an algebraic field extension. Then the map

$$\begin{aligned} \phi_K : X(K) &\longrightarrow Y(K) \\ (x, \iota) &\to \quad (\phi(x), \iota \circ \phi_x) \end{aligned}$$

is well-defined and satisfies

$$\psi_K \circ \phi_K = (\psi \circ \phi)_K$$

In particular if ϕ is an isomorphism then ϕ_K is a bijection.

Proposition 6.2.50 (Points = Rational Points over $\overline{k}$)
Let X be a variety. Then there is a canonical bijection

$$\begin{aligned} X &\xrightarrow{\sim} X\left(\overline{k}\right) \\ x &\to \quad ([x], \mathrm{ev}_x) \end{aligned}$$

which is functorial in X. More precisely if $\phi : X \to Y$ is a regular morphism then

$$\mathrm{ev}_{\phi(x)} = \mathrm{ev}_x \circ \overline{\phi}_x$$

Further

$$y = \phi(x) \iff \phi(x) \in \overline{\{y\}} \text{ and } \mathrm{ev}_x \circ \overline{\phi}_x = \mathrm{ev}_y$$

Proof. Recall (4.1.34) that X may be written as a disjoint union

$$X \coloneqq \bigsqcup_{x \in \mathcal{I}} \overline{\{x\}}$$

for some representative subset $\mathcal{I} \subset X$ which is in bijection with X_0. By (6.2.11) X is symmetric and for all $z \in \overline{\{x\}}$ we have $[z] = [x]$ as elements of X_0. So for a given $x \in \mathcal{I}$ we need to show that the map

$$\begin{aligned} \overline{\{x\}} &\to \mathrm{AlgHom}_k\left(k\left(\mathfrak{m}_{X,x}\right), \overline{k}\right) \\ z &\to \mathrm{ev}_z \end{aligned}$$

is bijective. It is well-defined because $z \in \overline{\{x\}}$ implies $\mathcal{O}_{X,x} = \mathcal{O}_{X,z}$. Let U be an affine open neighbourhood of x then by (4.1.43) $\overline{\{x\}} = \mathrm{cl}_U(\{x\})$. Further by (4.3.15) $U \hookrightarrow X$ induces a local isomorphism on stalks and therefore residue fields $k\left(\mathfrak{m}_{X,x}\right) \cong k\left(\mathfrak{m}_{U,x}\right)$. By functoriality (4.3.10) we may then reduce to the case of X affine. The result then follows from the affine case (6.1.63).

There is a commutative diagram

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$$\begin{array}{ccc} X & \xrightarrow{\sim} & X\left(\overline{k}\right) \\ \downarrow{\scriptstyle \phi} & & \downarrow{\scriptstyle \phi_{\overline{k}}} \\ Y & \xrightarrow{\sim} & Y\left(\overline{k}\right) \end{array}$$

from which the final statement follows easily. ∎

Proposition 6.2.51
Let X be a variety. Then the evaluation map

$$\begin{aligned} \mathrm{ev}_X : X &\longrightarrow \mathrm{AlgHom}_k\left(\mathcal{O}_X(X), \overline{k}\right) \\ x \to \quad & f \to f(x) \end{aligned}$$

is natural in X.

When X is affine it is also a bijection.

Proof. More precisely when $\phi : X \to Y$ is a regular morphism of varieties we require to show that the following diagram commutes

$$\begin{array}{ccc} X & \xrightarrow{\mathrm{ev}} & \mathrm{AlgHom}_k\left(\mathcal{O}_X(X), \overline{k}\right) \\ \downarrow{\scriptstyle \phi} & & \downarrow{\scriptstyle - \circ \phi^\sharp} \\ Y & \xrightarrow{\mathrm{ev}} & \mathrm{AlgHom}_k\left(\mathcal{O}_Y(Y), \overline{k}\right) \end{array}$$

Suppose $g \in \mathcal{O}_Y(Y)$ and $x \in X$ then this amounts to

$$g(\phi(x)) = \phi^\sharp(g)(x)$$

which holds by definition.

Let $\theta : X \cong V(\mathfrak{a})$ be an isomorphism of varieties. Then $\phi^\sharp$ is an isomorphism of k-algebras, and so $- \circ \phi^\sharp$ is a bijection (since functors preserve isomorphisms). The evaluation map for $V(\mathfrak{a})$ is a bijection (6.1.63) and so we deduce it is also for X. ∎

Corollary 6.2.52
Let X be a variety and $U \subset X$ an affine open subset. Then the evaluation map

$$\mathrm{ev}_U : U \longrightarrow \mathrm{AlgHom}_k\left(\mathcal{O}_X(U), \overline{k}\right)$$

is bijective.

Suppose $\phi : X \to Y$ is a regular morphism, U is an affine neighbourhood of x and V is an affine neighbourhood of y such that $U \subset \phi^{-1}(V)$, then

$$\phi(x) = y \iff \mathrm{ev}_U(x) \circ \rho_{\phi^{-1}(V)U} \circ \phi^\sharp = \mathrm{ev}_V(y)$$

6.2.9 Dimension

Definition 6.2.53 (Dimension of a Variety)
*Let X be a variety. Then we define the **dimension** to be the Krull Dimension of X as a topological space.*

Proposition 6.2.54
Let X be an integral variety. Then it is biequidimensional and

$$\dim X = \mathrm{trdeg}_k(k(X))$$

The dimension equals $\dim U$ for every open subset.

Proof. Let U be an affine open subset then by (6.1.52) $\dim U = \mathrm{trdeg}_k(k(U))$ which equals $\mathrm{trdeg}_k(k(X))$ by (4.3.16). Further by (6.1.51) U is biequidimensional and so by (4.1.73) we conclude that X is biequidimensional with the same dimension as U.

Every open subset can be covered by affines so we are done by (4.1.88). ∎

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Corollary 6.2.55
Let X be a variety. Then it is quasi-biequidimensional. Moreover every closed subset $W \subset X$ satisfies the codimension formula

$$\dim X = \dim W + \operatorname{codim}(W, X)$$

If W is irreducible then we also have the identity

$$\dim \mathcal{O}_{X,W} = \operatorname{codim}(W, X) = \dim X - \dim W$$

Proof. Every irreducible component is an integral closed subvariety and therefore biequidimensional (6.2.54). So by definition X is quasi-biequidimensional. The codimension formula is (4.1.87).

Let U be an affine open subset of X meeting W. Then there is an isomorphism $\mathcal{O}_{X,W} \xrightarrow{\sim} \mathcal{O}_{U,W\cap U}$ (4.3.17). Further by (4.1.73) we have $\operatorname{codim}(W, X) = \operatorname{codim}(W \cap U, U)$. Therefore we may deduce from the affine case (6.1.57) that

$$\dim \mathcal{O}_{X,W} = \dim \mathcal{O}_{U,W\cap U} = \operatorname{codim}(W \cap U, U) = \operatorname{codim}(W, X)$$

as required. ∎

Proposition 6.2.56 (Irreducible Closed Subsets of Dimension 0)
Let X be a variety and $Z \subset X$ an irreducible closed subset. Then the following are equivalent

a) $\dim Z = 0$
b) $Z = \overline{\{z\}}$ *for some* $z \in Z$
c) Z *is finite*

In particular for every $x \in X$ we have $\dim \mathcal{O}_{X,x} = \dim X$.

Proof. Follows from (6.2.12). The dimension formula follows from (6.2.55). ∎

Proposition 6.2.57
Let X be a variety. Then there is a bijective correspondence

$$X_0 \longleftrightarrow \{W \subset Z \mid W \text{ closed and irreducible and } \dim W = 0\}$$

Proof. By (6.2.11) and (4.1.94). ∎

6.2.10 Galois Action

The notion of rational point allows us to define a Galois Action which coincides with concrete notions in both the affine and projective cases.

Definition 6.2.58 (Abstract Galois Action)
Let X be an abstract variety and K/k an algebraic field extension. Then there is a well-defined group action

$$\operatorname{Aut}(K/k) \times X(K) \longrightarrow X(K)$$

given by

$$\sigma, (x, \iota) \longrightarrow (x, \sigma \circ \iota)$$

When $K = \overline{k}$ then this naturally gives a well-defined group action

$$\operatorname{Aut}\left(\overline{k}/k\right) \times X \longrightarrow X$$

in light of the bijection $X \xrightarrow{\sim} X\left(\overline{k}\right)$ (6.2.50).

Proposition 6.2.59 (Topological Properties of Galois Action)
Let X be an abstract variety. For every $x \in X$ and $\sigma \in \operatorname{Aut}\left(\overline{k}/k\right)$ we have x and $\sigma(x)$ are topologically indistinguishable. Furthermore the orbit of x is simply $\overline{\{x\}}$, which has order $\left[k\left(\mathfrak{m}_{X,x}\right) : k\right]$.

In particular σ preserves every open set and is therefore a homeomorphism of X. Finally every σ restricts to a well-defined action on X_0 which is simply the identity.

For every affine open subset $U \subset X$ with explicit isomorphism $U \xrightarrow{\sim} V(\mathfrak{a})$. Then the action of $\operatorname{Aut}\left(\overline{k}/k\right)$ coincides with the usual one under this isomorphism.

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6.2.11 Product Variety

We construct the product in the category of varieties, but note that the topology of the product does *not* coincide with the usual product topology.

We first show that the product, if it exists, is unique in concrete sense.

Lemma 6.2.60 (Uniqueness of Product)
Let X, Y be varieties (resp. affine varieties), then there is at most one topology on $X \times Y$ and local space of functions $\mathcal{O}_{X\times Y}$ for which the projection maps are regular and the following map

$$\begin{aligned} \mathrm{Mor}(Z, X \times Y) &\xrightarrow{\sim} \mathrm{Mor}(Z, X) \times \mathrm{Mor}(Z, Y) \\ \phi &\to \ (\pi_X \circ \phi, \pi_Y \circ \phi) \end{aligned}$$

is a well-defined bijection for all varieties (resp. affine varieties) Z, natural in Z.

Proof. Suppose we have two such structures, denote them by $X \times Y$ and $\widetilde{X \times Y}$, then we have a natural bijection

$$\begin{aligned} \mathrm{Mor}(Z, X \times Y) &\to \mathrm{Mor}\big(Z, \widetilde{X \times Y}\big) \\ \phi &\to \phi \end{aligned}$$

which is the identity on the underlying set function because $\phi = \phi' \iff \pi_X \circ \phi = \pi_X \circ \phi'$ and $\pi_Y \circ \phi = \pi_Y \circ \phi'$. By the dual version of the Yoneda Lemma (2.6.58) the identity map $1 : X \times Y \to \widetilde{X \times Y}$ is a regular isomorphism. Therefore the topology is the same and so are the sheaf of functions. ∎

Proposition 6.2.61
Let X, Y be varieties, then the categorical product $X \times Y$ exists, and coincides with the set-theoretic product. More precisely

a) there exists a unique topology on $X \times Y$ and sheaf of functions $\mathcal{O}_{X\times Y}$ such that the projection maps $\pi_X : X \times Y \to X$ and $\pi_Y : X \times Y \to Y$ are regular and there is a natural bijection for all varieties Z

$$\begin{aligned} \mathrm{Mor}(Z, X \times Y) &\xrightarrow{\sim} \mathrm{Mor}(Z, X) \times \mathrm{Mor}(Z, Y) \\ \phi &\to \ (\pi_X \circ \phi, \pi_Y \circ \phi) \end{aligned}$$

b) If X and Y are affine (resp. separated), then so is $X \times Y$, and there is an isomorphism

$$\big(\mathcal{O}_X(X) \otimes_k \mathcal{O}_Y(Y)\big)_{\mathrm{red}} \xrightarrow{\sim} \mathcal{O}_{X\times Y}(X \times Y)$$

c) If $U \subset X$ and $V \subset Y$ are open then so is $U \times V$, and $\mathcal{O}_{U\times V} = \mathcal{O}_{X\times Y}|_{U\times V}$

d) If $T \subset X$ and $S \subset Y$ are closed then $T \times S$ is a closed subset of $X \times Y$

Proof. Suppose X, Y are both affine. Then $X \times Y$ is given the structure of an affine variety in (6.1.96). This means the following map

$$\begin{aligned} \mathrm{Mor}(Z, X \times Y) &\to \mathrm{Mor}(Z, X) \times \mathrm{Mor}(Z, Y) \\ \phi &\to (\pi_X \circ \phi, \pi_Y \circ \phi) \end{aligned}$$

is bijective when Z is affine. Consider the case $Z = \bigcup_{i=1}^n U_i$ is a variety. Injectivity is obvious, for surjectivity suppose $f : Z \to X$ and $g : Z \to Y$ are regular. Then $f|_{U_i}$ and $g|_{U_i}$ are regular, and by the affine case so is $f|_{U_i} \times g|_{U_i}$. By (4.3.27) $f \times g$ is also regular.

Consider the general case $X = \bigcup_{i=1}^n U_i$ and $Y = \bigcup_{j=1}^m V_j$ are varieties with given affine coverings. Define $W_{ij} \coloneqq U_i \times V_j$, then by the affine case W_{ij} may be given the structure of an (affine) variety with regular set-theoretic projections $\pi_X^{ij} : W_{ij} \to U_i$ and $\pi_Y^{ij} : W_{ij} \to V_j$ such that the following map is a bijection

$$\mathrm{Mor}\big(Z, W_{ij}\big) \xrightarrow{\sim} \mathrm{Mor}(Z, U_i) \times \mathrm{Mor}\big(Z, V_j\big)$$

for all varieties Z. Then consider $Z = W_{ij} \cap W_{kl} = \big(\pi_X^{kl}\big)^{-1}(U_i \cap U_k) \cap \big(\pi_Y^{kl}\big)^{-1}\big(V_j \cap V_k\big)$ regarded as an open subvariety of W_{kl}. With this structure the projection maps $\pi_X : Z \to U_i$ and $\pi_Y : Z \to V_j$ may be shown to be regular by

- changing the codomain of $\pi_X^{kl} : W_{kl} \to U_k$ (resp. π_Y) to X, (6.2.35)
- restricting the domain to Z (6.2.36),
- changing the codomain to U_i (resp. V_j) (6.2.35)

Consequently the identity map $W_{ij} \cap W_{kl} \to W_{ij}$ is regular and therefore so is $W_{ij} \cap W_{kl} \to W_{ij} \cap W_{kl}$ (6.2.35) when considered as open subvarieties of W_{kl} and W_{ij} respectively. By symmetry this is a regular isomorphism. Therefore

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the W_{ij} satisfy the criteria of (6.2.37) and there exists a unique structure of a variety on $X \times Y$ for which $\mathcal{O}_{X\times Y}|_{W_{ij}} = \mathcal{O}_{W_{ij}}$. By (...) the set-theoretic projections $\pi_X : X \times Y \to X$ and $\pi_Y : X \times Y \to Y$ are regular. As before we need to show the following map is bijective

$$\mathrm{Mor}(Z, X \times Y) \to \mathrm{Mor}(Z, X) \times \mathrm{Mor}(Z, Y)$$
$$\phi \to (\pi_X \circ \phi, \pi_Y \circ \phi)$$

with injectivity obvious. Suppose $f : Z \to X$ and $g : Z \to Y$ are regular. Define $T_{ij} \coloneqq f^{-1}(U_i) \cap g^{-1}(V_j)$. Then by (6.2.36) $f_{ij} \coloneqq f|_{T_{ij}}$ and $g_{ij} \coloneqq g|_{T_{ij}}$ are regular. Therefore by the universal property for W_{ij} the maps $f_{ij} \times g_{ij}$ are regular. Clearly they are compatible so these glue to give a regular map $f \times g : Z \to X \times Y$ (6.2.35) (4.3.27) as required. Further $X \times Y = \bigcup_{i,j} W_{ij}$ so it is indeed a variety. Uniqueness of the structure follows from (6.2.60).

For *b*) we already demonstrated that the affine product is also a product in the category of varieties, so it follows from uniqueness. For separatedness this is an easy exercise.

For *c*) we have $U \times V = \pi_X^{-1}(U) \cap \pi_Y^{-1}(V)$ is open, and the universal property naturally restricts to the subset $U \times V$. The identity then follows from uniqueness (6.2.60).

For *d*) we have $T \times S = \pi_X^{-1}(T) \cap \pi_Y^{-1}(S)$ is closed. ∎

Proposition 6.2.62
Let X be an integral variety and Y a geometrically integral affine variety (see (6.1.84)). Then $X \times Y$ is an integral variety.

Proof. By definition $X = \bigcup_{i=1}^n U_i$ for U_i affine. By (6.2.61) $X \times Y = \bigcup_{i=1}^n U_i \times Y$ and $U_i \times Y$ is affine. By (6.1.98) $U_i \times Y$ is integral. By (4.1.59) $U_i \cap U_j \neq \emptyset$ whence $(U_i \times Y) \cap (U_j \times Y) \neq \emptyset$. Therefore by (4.1.74) we conclude that $X \times Y$ is integral. ∎

6.2.12 Separated Varieties

Proposition 6.2.63
Let $\phi_1, \phi_2 : X \rightrightarrows Y$ be regular morphisms of varieties with Y separated. Then the equaliser

$$\{x \in X \mid \phi_1(x) = \phi_2(x)\}$$

is a closed subset of X.

Proof. Let $X = \bigcup_{i=1}^n U_i$ where U_i are affine. Then the equaliser is the finite union of

$$\left\{x \in U_i \;\middle|\; \phi_1|_{U_i}(x) = \phi_2|_{U_i}(x)\right\}$$

which is by definition closed. ∎

The following shows that the separated property is analogous to the Hausdorff property in the classical case (4.1.98).

Proposition 6.2.64
Let X be a variety. Then X is separated if and only if the diagonal

$$\Delta_X \coloneqq \{(x, x) \mid x \in X\}$$

is a closed subvariety of $X \times X$.

Proof. Suppose that Δ_X is closed. Then the equaliser of $\phi_1, \phi_2 : Z \rightrightarrows X$ is equal to $(\phi_1 \times \phi_2)^{-1}(\Delta_X)$. By (6.2.61) $\phi_1 \times \phi_2$ is regular and in particular continuous. We conclude the equaliser is closed.

Conversely Δ_X is the equaliser of $\pi_1, \pi_2 : X \times X \rightrightarrows X$ and so by (6.2.14) is closed. ∎

Proposition 6.2.65
Let X be a separated variety and $Z \subset X$ a closed (resp. open) subvariety. Then Z is separated.

Proof. Then $\Delta_Z = \Delta_X \cap (Z \times Z)$. By (6.2.61) $Z \times Z$ is a closed (resp. open) subvariety of $X \times X$. Then the result follows by (6.2.64). ∎

Lemma 6.2.66
Let $\phi : X \to Y$ be a regular map of varieties, with Y separated. Then the graph

$$\Gamma_\phi = \{(x, \phi(x)) \mid x \in X\}$$

is a closed subvariety of $X \times Y$. Further there is a canonical isomorphism of varieties

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$$X \xrightarrow{\sim} \Gamma_\phi$$
$$x \to (x, \phi(x))$$

Proof. Consider the regular map

$$(\phi \circ \pi_X) \times \pi_Y : X \times Y \to Y \times Y$$
$$(x, y) \to (\phi(x), y)$$

Then

$$\Gamma_\phi = ((\phi \circ \pi_X) \times \pi_Y)^{-1}(\Delta_Y)$$

which is closed. The map $X \to \Gamma_\phi$ is evidentally regular. Further $\Gamma_\phi \hookrightarrow X \times Y \to X$ is regular by (6.2.42) and (4.3.2). These are mutual inverses and so the last statement follows immediately. ∎

Proposition 6.2.67 (Intersection of Affines is Affine)
Let X be a separated variety and U, V open subsets. Then there is a regular isomorphism followed by a closed immersion

$$U \cap V \xrightarrow{\sim} (U \times V) \cap \Delta_X \hookrightarrow U \times V$$
$$x \to \quad (x, x)$$

In particular when U, V are affine then so is $U \cap V$.

Proof. By (6.2.61) $U \times V$ is an open subvariety of $X \times X$.

$(U \times V) \cap \Delta_X = ((U \cap V) \times V) \cap \Delta_X$ is the graph of the inclusion map $U \cap V \hookrightarrow V$. By (6.2.66) it is a closed subvariety of $(U \cap V) \times V$ and therefore of $U \times V$. The same result shows that the first map is a regular isomorphism.

When U, V are affine, so is $U \times V$. By (6.2.42) so is $(U \times V) \cap \Delta_X$, and therefore $U \cap V$. ∎

Proposition 6.2.68 (Criteria for Separated Variety)
Let X be a variety with an open cover $\bigcup_{i=1}^n U_i$. Then X is separated if and only if $(U_i \times U_j) \cap \Delta_X$ is a closed subvariety of $U_i \times U_j$ for all $i, j = 1...n$.

Proof. Suppose X is separated then $(U_i \times U_j) \cap \Delta_X$ is a closed subvariety by (6.2.67).

Conversely if $(U_i \times U_j) \cap \Delta_X$ is closed for all i, j then by (4.1.22)

$$\Delta_X = \bigcup_{i,j} (U_i \times U_j) \cap \Delta_X$$

is closed, and therefore X is separated by (6.2.64). ∎

Proposition 6.2.69 (Criteria for Separated Variety II)
Let X be a variety with an affine open cover $\bigcup_{i=1}^n U_i$. Then X is separated if and only if $U_i \cap U_j$ is affine and the k-algebra homomorphism

$$\mathcal{O}_X(U_i) \otimes_k \mathcal{O}_X(U_j) \longrightarrow \mathcal{O}_X(U_i \cap U_j)$$
$$f \otimes g \longrightarrow f|_{U_i \cap U_j} \cdot g|_{U_i \cap U_j}$$

is surjective for all $i, j = 1...n$

Proof. Suppose X is separated, then by (6.2.67) $U_i \cap U_j$ is affine and there is a closed immersion

$$U_i \cap U_j \xrightarrow{\sim} (U_i \times U_j) \cap \Delta_X \hookrightarrow U_i \times U_j$$

By (6.2.46) the corresponding map of global sections

$$\mathcal{O}_{U_i \times U_j}(U_i \times U_j) \to \mathcal{O}_{U_i \cap U_j}(U_i \cap U_j) \quad (*)$$
$$f \to u \to f(u, u)$$

is surjective. Recall from (6.2.61) there is a surjective k-algebra homomorphism

$$\mathcal{O}_{U_i}(U_i) \otimes_k \mathcal{O}_{U_i}(U_j) \to \mathcal{O}_{U_i \times U_j}(U_i \times U_j)$$
$$f \otimes g \to u \to f(u) \cdot g(u)$$

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which composes to give the required map.

Conversely suppose that $U_i \cap U_j$ are affine and the given map is surjective then evidentally the map in $(*)$ is surjective. By (6.2.46) this shows that the map $U_i \cap U_j \hookrightarrow U_i \times U_j$ is a closed immersion and in particular $\big(U_i \times U_j\big) \cap \Delta_X$ is a closed subvariety of $\big(U_i \times U_j\big)$. Then we are done by (6.2.68). ∎

6.2.13 Rational Maps

Definition 6.2.70 (Rational Maps)
*Let X, Y be varieties. A **rational map** $\phi : X \longrightarrow Y$ is an equivalence class of regular morphisms*

$$\phi : U \to Y$$

where U is a dense open subset of X. We say $(U, \phi) \sim (U', \phi')$ if there exists a dense open subset $W \subset U \cap U'$ such that $\phi|_W = \phi'|_W$.

*We say the **domain of definition** of a rational map ϕ is the set of points $x \in X$ for which there exists a neighbourhood U of x and a representative (U, ϕ). This is a dense open subset denoted by $\mathrm{dom}(\phi)$.*

Proposition 6.2.71
Let $\phi : X \longrightarrow Y$ be a rational map with Y separated. Then there exists a representative regular morphism $\phi : \mathrm{dom}(\phi) \to Y$.

Proof. It is enough to show that for any two representatives (U, ϕ) and (U', ϕ') we have $\phi|_{U \cap U'} = \phi'|_{U \cap U'}$. By (6.2.63) the equaliser is closed in $U \cap U'$ and by definition contains a dense open subset W. Therefore it equals $U \cap U'$ as required. ∎

Proposition 6.2.72 (Rational Functions are Rational Maps)
Let X be an integral variety. Then there is a bijection

$$k(X) \xrightarrow{\sim} \{f : X \longrightarrow \mathbb{A}^1_k\}$$

and every such map is either dominant or has finite image.

Proof. By (6.2.4) this is well-defined and bijective. Suppose $U \subset X$ is dense then by (4.1.70) it is irreducible. Therefore a regular map $f : U \to \mathbb{A}^1_k\big(\overline{k}\big)$ has irreducible image (4.1.64) and $\overline{f(X)}$ is irreducible (4.1.65) and closed. Clearly $\dim \overline{f(X)} \leq \dim \mathbb{A}^1_k\big(\overline{k}\big) = 1$. If equality holds then by (4.1.83).f) $\overline{f(X)} = \mathbb{A}^1_k$ and by definition f is dominant.

Otherwise $\dim \overline{f(X)} = 0$ whence by (6.2.56) $\overline{f(X)} = \overline{\{y\}}$ for some $y \in \mathbb{A}^1_k\big(\overline{k}\big)$. ∎

Proposition 6.2.73
Let X be an integral variety. Then as subrings of $k(X)$ (...) we have

$$\mathcal{O}_X(U) = \{f \in k(X) \mid U \subset \mathrm{dom}(f)\}$$

and

$$\mathcal{O}_{X,x} = \{f \in k(X) \mid x \in \mathrm{dom}(f)\}$$

Furthermore

$$\mathcal{O}_X(U) = \bigcap_{x \in U} \mathcal{O}_{X,x}$$

Proof. Recall (...) that $\mathbb{A}^1_k$ being affine is separated. Therefore every rational map $f \in k(X)$ has a regular representative in $\tilde{f} \in \mathcal{O}_X(\mathrm{dom}(f))$ by (6.2.71). So if $U \subset \mathrm{dom}(f)$ then $\tilde{f} \sim \tilde{f}|_U$ and so $f \in \mathcal{O}_X(U)$. Conversely if $\tilde{f} \in \mathcal{O}_X(U)$ then $\big(U, \tilde{f}\big)$ is a rational map whose domain contains U.

The second statement is similar, and then the final statement follows trivially. ∎

Proposition 6.2.74 (Dominant Rational Map)
Let X be an integral variety and $[(U, \phi)] : X \longrightarrow Y$ a rational map. Then the following are equivalent

a) $\phi(U)$ is dense in Y
b) $(U', \phi') \sim (U, \phi) \Longrightarrow \phi'(U')$ is dense in Y

*In this case we say that $[(U, \phi)]$ is a **dominant rational map***

Proof. Clearly $b) \Longrightarrow a)$. Conversely suppose $\emptyset \neq W \subset U$ is open, then it is dense by (4.1.59) so $\mathtt{cl}_U(W) = U$ by (4.1.21). Therefore by (4.1.30)

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$$\phi(U) = \phi(\mathtt{cl}_U(W)) \subset \overline{\phi(W)}$$

which means

$$X = \overline{\phi(U)} \subset \overline{\phi(W)}$$

and therefore $\phi(W)$ is dense. In particular by definition we have $W \subset U \cap U'$ such that $\phi(W) = \phi'(W)$. So we conclude that $\phi'(W)$ is dense and so a-fortiori is $\phi'(U')$. ∎

Proposition 6.2.75
Let X, Y, Z be integral varieties and $\phi : X \longrightarrow Y$, $\psi : Y \longrightarrow Z$ rational maps with ϕ dominant. For every equivalence class (U, ϕ) and (V, ψ) we have $U \cap \phi^{-1}(V) \neq \emptyset$ and the map

$$\psi \circ \phi : U \cap \phi^{-1}(V) \to Z$$

is a regular morphism determining a rational map

$$\psi \circ \phi : X \longrightarrow Z$$

which is dominant provided ψ is.

Further this is independent of the choice of equivalence class.

Proof. By definition $\phi(U)$ is dense which means that $\phi(U) \cap V \neq \emptyset \Longrightarrow U \cap \phi^{-1}(V) \neq \emptyset$, which is therefore also dense in X (4.1.59). ∎

Proposition 6.2.76
Let X, Y be integral varieties. Then there is a bijective map

$$\begin{aligned} \{\phi : X \longrightarrow Y \text{ dominant rational maps}\} &\xrightarrow{\sim} \mathrm{AlgHom}_k(k(Y), k(X)) \\ \phi &\to \ \phi_\star \end{aligned}$$

Further $\phi_\star$ is injective and the existence of ϕ implies that $\dim Y \leq \dim X$. Then following are equivalent

a) $\dim Y = \dim X$
b) $k(X)/\phi_\star(k(Y))$ *is algebraic*
c) $k(X)/\phi_\star(k(Y))$ *is finite*

In this case we denote the ***degree*** *by* $\deg(\phi) := [k(X) : \phi_\star(k(Y))]$.

Proof. The map is well-defined by (6.2.72) and (6.2.75), after we verify that the image is a k-algebra homomorphism.

Further it is injective by (6.2.29) and surjective by (6.2.30).

As $k(Y)$ is a field $\phi_\star$ is injective. We claim that $\mathrm{trdeg}_k(k(Y)) \leq \mathrm{trdeg}_k(k(X))$. For a transcendence base for $k(Y)$ is a-fortiori algebraicly independent as a subset of $k(X)$. Therefore the inequality follows from (3.18.140) and (6.2.54).

a) $\Longrightarrow$ *b)* Suppose $\dim X = \dim Y$ and let $\mathcal{B}$ be a transcendence base for Y, then by (3.18.141) $\phi_\star(\mathcal{B})$ is a transcendence base for X. Therefore $k(X)/\phi_\star(k(\mathcal{B}))$ is algebraic

b) $\Longleftrightarrow$ *c)* A-fortiori $k(X)/\phi_\star(k(Y))$ is finitely generated. Therefore the equivalence follows from (3.18.44).

b) $\Longrightarrow$ *a)* We have a-fortiori $k(X)/\phi_\star(k(\mathcal{B}))$ is algebraic whence by (3.18.140) $\dim Y \geq \dim X$, and so $\dim Y = \dim X$. ∎

6.2.14 Tangent Space

Definition 6.2.77
Let X be a variety and $W \subset X$ an irreducible subset. Define the ***tangent space***

$$T_W X := \mathrm{Der}\big(\mathcal{O}_{X,W}, k(\mathfrak{m}_{X,W})\big)$$

and the ***cotangent space***

$$T^\star_W X := \mathrm{Hom}\big(\mathfrak{m}_{X,W}/\mathfrak{m}^2_{X,W}, k(\mathfrak{m}_{X,W})\big)$$

Definition 6.2.78 (Non-Singular Point)
Let X be a variety and $W \subset X$ an irreducible subset. Then we say that X is ***non-singular*** *at W if*

a) W is contained in a unique irreducible component X_α (equivalently $\mathcal{O}_{X,W}$ is an integral domain), and
b) $\dim T_W X = \dim \mathrm{Der}(k(X_\alpha))$

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We say that X is non-singular if it is integral and non-singular at all points.

Definition 6.2.79 (Regular Point)
Let X be a variety and $W \subset X$ an irreducible subset.

*We say X is **regular** at W if*
a) W is contained in a unique irreducible component X_α, and
b) $\dim \mathcal{O}_{X,W} = \dim T^\star_x X$

We say that X is regular if it is integral and regular at all points.

Proposition 6.2.80 (Regularity / Non-Singularity is a local property)
Let X be a variety, $W \subset X$ an irreducible subset and $U \subset X$ an open subset such that $U \cap W \neq \emptyset$. Then there is a canonical isomorphism

$$T_W X \xrightarrow{\sim} T_{W \cap U} U.$$

X is regular (resp. non-singular) at W if and only if U is regular (resp. non-singular) at $U \cap W$.

Proposition 6.2.81
Let X be an integral variety. Then it is regular (resp. non-singular) if it has a regular (resp. non-singular) open cover.

Proposition 6.2.82
*Let X be a variety and $W \subset X$ an irreducible subset such that $k\big(\mathfrak{m}_{X,W}\big)/k$ is **separably generated** (e.g. k perfect). Then there is a canonical isomorphism*

$$T_W X \xrightarrow{\sim} T^\star_W W$$

*In particular when W is contained in a unique irreducible component we have **regular** $\iff$ **non-singular**.*

6.2.15 Completeness

Completeness is analogous to compactness in the Hausdorff case, and many results remain true when complete is replaced with compact and the standard product topology is used (in place of the topology induced by the product variety structure). We show in the next section that every projective variety is complete, so it is quite a natural condition.

Definition 6.2.83
Let X be a separated variety. Then it is

- ***universally closed** if for every variety Y the projection map*

$$\pi_Y : X \times Y \to Y$$

  is closed (i.e. the image of a closed subset is closed).
- ***complete** if for every irreducible closed subvariety $Z \subset X$ and valuation ring $k \subset R \subset k(Z)$ there is a point $z \in Z$ such that $\mathcal{O}_{Z,z} \subset R$*

Proposition 6.2.84 (Image of Complete Variety is Closed)
Let $\phi : X \to Y$ be a morphism of varieties where X is universally closed and Y is separated. Then $\phi(X)$ is a closed subvariety of Y.

Proof. By (6.2.66) the graph Γ_ϕ is a closed subset of $X \times Y$. Then $\phi(X)$ is the projection of Γ_ϕ and therefore closed by definition. ∎

Proposition 6.2.85 (Closed Subvariety of Complete Variety is also Complete)
A closed subvariety of a universally closed (resp. complete) variety is universally closed (resp. complete)

Proof. Suppose $Z \subset X$ is closed then by (6.2.61) $Z \times Y$ is a closed subset of $X \times Y$. Further T closed in $Z \times Y$ implies T closed in $X \times Y$ and the result follows.

The complete case is obvious. ∎

Proposition 6.2.86
A separated variety X is universally closed (resp. complete) iff all its irreducible components are universally closed (resp. complete).

Proof. By (4.1.66) every irreducible component is closed, and therefore complete by (6.2.85).

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Conversely suppose $X = X_1 \cup ... \cup X_n$ is the irreducible decomposition (6.2.8). If $T \subset X \times Y$ is closed then $T \cap (X_i \times Y)$ is closed in $X_i \times Y$ and therefore by assumption $\pi_Y(T \cap (X_i \times Y)) = \pi_Y(T) \cap X_i$ is closed. Therefore $\pi_Y(T) = \bigcup_{i=1}^n \pi_Y(T) \cap X_i$ is also closed.

The complete case is straightforward. ∎

Proposition 6.2.87
Let X be a separated variety. To demonstrate universal closedness it is sufficient to show that

$$\pi_Y : X \times Y \to Y$$

is closed for Y affine (or even $Y := \mathbb{A}^n_k$).

*Further it is sufficient to show that $\pi_Y(T)$ is closed for all **irreducible** closed subsets $T \subset X \times Y$.*

Proof. Suppose $Y = \bigcup_{i=1}^n U_i$ for U_i open affines. Then by (6.2.61) $X \times U_i$ is an open cover of $X \times Y$. Let $T \subset X \times Y$ closed then $T \cap (X \times U_i)$ is closed in $X \times U_i$. By assumption $\pi_Y(T \cap (X \times U_i)) = \pi_Y(T) \cap U_i$ is closed in U_i. Therefore by (4.1.22) $\pi_Y(T)$ is closed in Y as required.

Clearly the closed property is preserved under isomorphism of Y, therefore we may assume that $Y = V(\mathfrak{a}) \subset \mathbb{A}^n_k$ is a closed subset for some n and ideal $\mathfrak{a} \lhd k[X_1, ..., X_n]$.

By (6.2.61) $X \times Y$ is a closed subset of $X \times \mathbb{A}^n_k$. There $T \subset X \times Y$ is also closed in $X \times \mathbb{A}^n_k$ and the result follows.

Finally if T is closed then by (...) $T = T_1 \cup ... \cup T_m$ for T_j irreducible closed subsets of T (and therefore of $X \times Y$). Therefore

$$\pi_Y(T) = \pi_Y\left(\bigcup_{j=1}^m T_j\right) = \bigcup_{j=1}^m \pi_Y(T_j)$$

and it suffices to show that $\pi_Y(T_j)$ is closed. ∎

The following is not strictly needed in application to projective case as it's already known however we include it for completeness of the argument.

Lemma 6.2.88
Let X be an integral variety and $R \subset k(X)$ is a valuation ring with local homomorphism $\Phi : R \to \overline{k}$. Suppose that there exists $x \in X$ such that $\mathcal{O}_{X,x} \subset R$, then x may be chosen such that $\Phi|_{\mathcal{O}_{X,x}} \to \overline{k}$ is evaluation at x.

Proof. By (6.2.19) $\Phi|_{\mathcal{O}_{X,x}}$ is local and by (6.2.50) we may choose $x' \in \overline{\{x\}}$ such that $\Phi|_{\mathcal{O}_{X,x}}$ is precisely evaluation at z'. ∎

The following proof is adapted from [Bum98] (Prop. 7.17).

Proposition 6.2.89 (Complete $\implies$ Universally Closed)
A complete variety is universally closed.

Proof. We may consider the projection $\pi_Y : X \times Y \to Y$ for Y irreducible affine by (6.2.87), and demonstrate that $\pi_Y(T)$ is closed for every irreducible closed subset $T \subset X \times Y$.

Consider the closed subvariety $Z := \mathrm{cl}_X(\pi_X(T))$ of X and the irreducible and affine closed subvariety $Y' := \mathrm{cl}_Y(\pi_Y(T))$ of Y. Then evidentally $T = T \cap (Z \times Y')$ and so

$$\pi_Y(T) = \pi_Y|_{Z \times Y'}\ (T \cap (Z \times Y'))$$

Further by (...) $Z \times Y'$ is a closed subvariety of $X \times Y$ and T is an irreducible closed subset of $Z \times Y'$. Therefore for fixed T we may assume without loss of generality that $\pi_Y(T)$ is dense in Y. The composite maps

$$\pi_Y \circ i : T \hookrightarrow Z \times Y \to Y$$

$$\pi_X \circ i : T \hookrightarrow Z \times Y \to Z$$

are regular (6.2.42) and dominant, and we require to prove the first is surjective. Consider the injective algebra homomorphism (6.2.17)

$$(\pi_Y \circ i)_\star : k(Y) \hookrightarrow k(T)$$

For any $y \in Y$ there is a corresponding evaluation homomorphism $\phi : \mathcal{O}_Y(Y) \to \overline{k}$ (6.2.52). By (3.24.11) there is a valuation ring $(\pi_Y \circ i)_\star(\mathcal{O}_Y(Y)) \subset R \subset k(T)$ and a local homomorphism $\Phi : R \to \overline{k}$ such that $\Phi \circ (\pi_Y \circ i)_\star$ extends ϕ.

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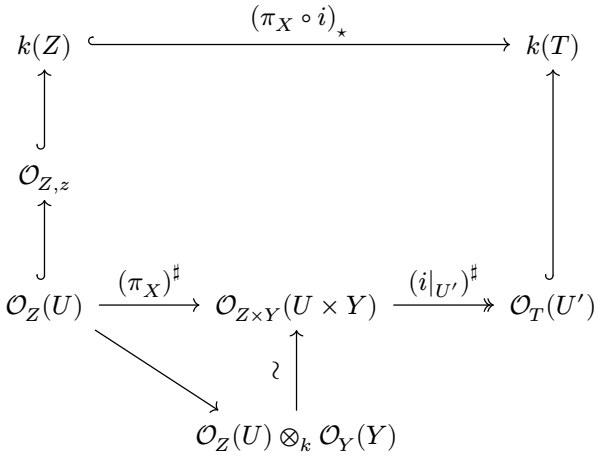
Consider further the injective algebra homomorphism

$$(\pi_X \circ i)_\star : k(Z) \hookrightarrow k(T)$$

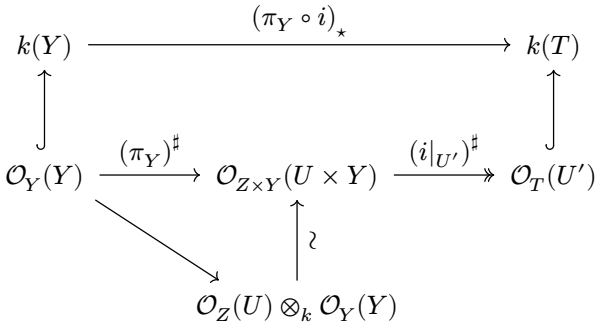
Then $R' \coloneqq (\pi_X \circ i)_\star^{-1}(R)$ is a valuation ring of $k(Z)$ together with local homomorphism $\Phi' : R' \to \overline{k}$ such that $\Phi' = \Phi \circ (\pi_X \circ i)_\star$. By hypothesis there is some $z \in Z$ such that $\mathcal{O}_{Z,z} \subset R'$ and by (6.2.88) we may assume that $\Phi'|_{\mathcal{O}_{Z,z}}$ is precisely evaluation at z.

Let U be an open affine neighbourhood of z in Z, then by (6.2.61) $U \times Y$ is an affine open subset of $Z \times Y$. As $\pi_X(T)$ is dense in Z then $U \cap \pi_X(T)$ is non-empty. This shows that $U' \coloneqq (U \times Y) \cap T$ is a non-empty open subset of T. Evidently U' is a closed subset of $U \times Y$. Therefore it is also affine (6.2.47) (6.2.42).

There is a commutative diagram (6.2.18)



By construction of R we see that $(i|_{U'})^\sharp(\mathcal{O}_Z(U) \otimes 1) \subset R$ (after suitable identifications). Similarly we have a commutative diagram



and by construction we have $(i|_{U'})^\sharp(1 \otimes \mathcal{O}_Y(Y)) \subset R$ (after suitable idenitifications). By properties of the tensor product this means

$$(i|_{U'})^\sharp(\mathcal{O}_Z(U) \otimes \mathcal{O}_Y(Y)) \subset R$$

By (6.2.42) the map $(i|_{U'})^\sharp$ is surjective, therefore we conclude that $\mathcal{O}_T(U') \subset R$. Then by (6.2.52) the restriction of Φ to $\mathcal{O}_T(U')$ corresponds to a point (z', y') of U'. Further $\Phi \circ (\pi_Y \circ i|_{U'})^\sharp = \left(\Phi \circ (\pi_Y \circ i)_\star\right)|_{\mathcal{O}_Y(Y)} = \phi$ and so by (6.2.52) we have $(\pi_Y \circ i)(z', y') = y$ that is to say $y' = y$. This shows that $\pi_Y \circ i$ is surjective as required. ∎

Proposition 6.2.90

Let X, Y be varieties with Y complete such that

a) $U \subset X$ is an irreducible open neighbourhood
b) $x \in \overline{U}$
c) $\mathcal{O}_{X,x}$ is a valuation ring
d) $\phi : U \to Y$ is a regular morphism

Then there exists an open subset $U' \supset U$ for which $\overline{\{x\}} \subset U'$ and ϕ lifts to U'.

Proof. We claim it's enough to find an open neighbourhood V of x such that $U \cap V \neq \emptyset$ and a regular function $\psi : V \to Y$ such that $\psi|_{U\cap V} = \phi|_{U\cap V}$. For then this extends to a regular function $U \cup V$ as required.

Replacing $X = \overline{U}$ we may assume that U is dense in X and X is irreducible. Observe $Z \coloneqq \mathrm{cl}_Y(\phi(U))$ is an irreducible closed subset of Y (4.1.70) (4.1.65). Therefore there is an injective k-algebra homomorphism (4.3.16)

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$$\phi_\star : k(Z) \hookrightarrow k(U) \xrightarrow{\sim} k(X)$$

By assumption $\mathcal{O}_{X,x}$ is a valuation ring, whence so is the inverse image R. By assumption there is $z \in Z$ such that $\mathcal{O}_{Z,z} \subset R$. Therefore we have a k-algebra homomorphism $\mathcal{O}_{Z,z} \to \mathcal{O}_{X,x}$. By (6.2.32) this lifts to a regular morphism $\psi : V \to Z \subset Y$ for which $x \in V$. As $x \in \overline{U}$ we have $U \cap V \neq \emptyset$ by (4.1.19). As Y is separated (6.2.63) the equaliser of ϕ, ψ is a closed subset of $U \cap V$. We have by construction a commutative diagram

$$\begin{array}{ccc} k(Z) & \xrightarrow{\phi_\star} & k(X) \\ \uparrow & & \uparrow \\ \mathcal{O}_{Z,z} & \xrightarrow{\psi_x^\sharp} & \mathcal{O}_{X,x} \\ \uparrow & & \uparrow \\ \mathcal{O}_Z(Z) & \xrightarrow{\psi_Z^\sharp} & \mathcal{O}_X(V) \end{array}$$

Regarding $\psi : X \longrightarrow Y$ as a rational map we find by (6.2.76) that $\psi_\star = \phi_\star$ whence $\psi = \phi$ as rational maps. This means that the equaliser contains a dense open subset of $U \cap V$ and being closed is therefore the whole set. That is $\phi|_{U\cap V} = \psi|_{U\cap V}$ as required. ∎

6.2.16 Projective Space

We introduce the notion of projective varieties by showing that projective space can naturally be given the structure of an abstract variety. A more low-tech approach is taken in the Projective Varieties section, where the constructions are shown to be equivalent.

Definition 6.2.91 (Projective Space)
Let K be a field extension define

$$\mathbb{P}^n(K) := K^{n+1} \setminus \{(0,...,0)\}/\sim$$

where

$$(x_0,...,x_n) \sim (x_{0'},...,x_{n'}) \iff x_i = \lambda x_{i'} \text{ some } \lambda \in K^\star$$

and write $(x_0 : ... : x_n)$ for the corresponding equivalence class in $\mathbb{P}^n(K)$.

Lemma 6.2.92 (Comparing projective points)
Consider two projective points $x = (x_0 : ... : x_n)$ and $y = (y_0 : ... : y_n)$.

a) $x = y \implies \forall i (x_i = 0 \iff y_i = 0)$

Suppose that $x_i \neq 0$ for some i. Then $x = y$ iff

a) $y_i \neq 0$
b) $\frac{x_j}{x_i} = \frac{y_j}{y_i}$ for all j

Proposition 6.2.93 (Projective Space)
Consider the set $X = \mathbb{P}^n\left(\overline{k}\right)$ and for every $i = 0,...,n$ define the subset

$$U_i := \left\{(x_0 : ... : x_n) \;\middle|\; x_0,...,x_n \in \overline{k}, x_i \neq 0\right\}.$$

There are canonical bijections

$$\begin{aligned} \theta_i : U_i &\xrightarrow{\sim} \mathbb{A}_k^n\left(\overline{k}\right) \quad i = 0...n \\ (x_0 : ... : x_n) &\to \left(\frac{x_0}{x_i},...,\frac{\widehat{x_i}}{x_i},...,\frac{x_n}{x_i}\right) \\ (z_1 : ... : 1 : ... : z_n) &\leftarrow (z_1,...,z_n) \end{aligned}$$

and there is a unique topology on X and space of functions $\mathcal{O}_X$ such that U_i is open in X and θ_i is a regular isomorphism for all $i = 0...n$.

Denote this by $\mathbb{P}_k^n$. With this structure $\mathbb{P}_k^n$ is an irreducible, separated algebraic variety of dimension n.

Further for any homogenous polynomial $F \in k[X_0,...,X_n]$ of degree d the expression

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$$(x_0 : ... : x_n) \to \frac{F(x_0, ..., x_n)}{x_i^d}$$

is a well-defined regular function on U_i. In particular on U_i we may define the ***homogenous coordinate functions*** $\frac{X_0}{X_i}, ..., \frac{X_n}{X_i}$.

Proof. It is straightforward to show that θ_i is a bijection. Define the cocycle maps

$$\theta_{ij} : \theta_i(U_i \cap U_j) \to \theta_j(U_i \cap U_j)$$

by $\theta_{ij} = \theta_j \circ \theta_i^{-1}$. We show that $\theta_i(U_i \cap U_j)$ is open in $\mathbb{A}_k^n\big(\overline{k}\big)$ and the maps θ_{ij} are regular. First observe that

$$\theta_i(U_i \cap U_j) = \begin{cases} D(Z_j) & i < j \\ D(Z_{j+1}) & j < i \end{cases}$$

Then the cocycle map is given pointwise by

$$\theta_{ij} : D(Z_j) \to D(Z_{i+1})$$
$$(z_1, ..., z_n) \to \left(\frac{z_1}{z_j}, ..., \frac{z_i}{z_j}, \frac{1}{z_j}, ..., \frac{z_n}{z_j}\right)$$

Using (6.2.45) and (6.1.39) we may show that θ_{ij} is a regular morphism of abstract varieties. Therefore the structure on X is uniquely determined by (6.2.38).

By construction U_i is affine and $U_i \cap U_j = D\left(\frac{X_j}{X_i}\right)$ where $\frac{X_j}{X_i} \in \mathcal{O}_X(U_i)$. Therefore $U_i \cap U_j$ is affine by (6.2.21) and there is a canonical isomorphism

$$\mathcal{O}_{U_i}(U_i)_{\frac{X_j}{X_i}} \xrightarrow{\sim} \mathcal{O}_{U_i}(U_i \cap U_j)$$

As $\frac{X_i}{X_j} \in \mathcal{O}_{U_j}(U_j)$ satisfies $\frac{X_i}{X_j}|_{U_i \cap U_j} = \left(\frac{X_j}{X_i}|_{U_i \cap U_j}\right)^{-1}$ we find the canonical map

$$\mathcal{O}_{U_i}(U_i) \otimes \mathcal{O}_{U_j}(U_j) \to \mathcal{O}_{U_i}(U_i \cap U_j)$$

is surjective. Therefore X is separated by (6.2.69).

By (6.1.21) the affine space $\mathbb{A}_k^n$ is irreducible whence so is U_i and $U_i \cap U_j$ (4.1.70). Therefore by (4.1.74) $\mathbb{P}_k^n$ is irreducible and has dimension n by (4.1.88). ∎

Proposition 6.2.94 (Riemann Sphere)
There is a bijection

$$\mathbb{P}_k^1 \xrightarrow{\sim} \mathbb{A}_k^1 \cup \{\infty\}$$

which sends $(x : 1)$ to x and $(1 : 0)$ to ∞. This gives $\mathbb{A}_k^1 \cup \{\infty\}$ the structure of a variety under which $\mathbb{A}_k^1$ is an open subvariety and x^{-1} is a regular automorphism interchanging 0 and ∞.

Consider $x = \frac{X_1}{X_0} \in \mathcal{O}_{\mathbb{P}_k^1}(U_0)$ as an element of the function field $k(\mathbb{P}_k^1)$ then x is transcendental and $k(\mathbb{P}_k^1) = k(x)$.

Proof. Evidently x is transcendental because $\mathcal{O}(U_0) = k[x]$ (...). Let $f \in k(\mathbb{P}_k^1)$ and suppose $\mathrm{dom}(f) \cap U_0 \neq \emptyset$. There is some $g(x) \in \mathcal{O}(U_0)$ such that $D(g(x)) \subset \mathrm{dom}(f)$. Then by (...) $\mathcal{O}(D(g)) = k[x]_{g(x)}$ and so $f|_{D(g(x))} = h(x)/g(x)^r$ for some polynomial $h \in k[X]$ and integer $r > 0$. Therefore $f \in k(x)$. Similarly if $\mathrm{dom}(f) \cap U_1 \neq \emptyset$ then $f \in k(x^{-1}) = k(x)$ as required, where $x^{-1} = \frac{X_0}{X_1} \in \mathcal{O}_{\mathbb{P}_k^1}(U_1)$. ∎

We show later that all projective closed subvarieties are complete, but here we demonstrate a simpler case more directly.

Proposition 6.2.95
$\mathbb{P}_k^1$ is complete.

Proof. Let $Z \subset \mathbb{P}_k^1$ be an irreducible closed subvariety. Then by (4.1.83) (6.2.56) it is either the (closure of) a single point or the whole space. In the former case the function field is algebraic over k and has no non-trivial valuation rings. Therefore we may assume wlog that $Z = \mathbb{P}_k^1$ and consider a valuation ring $R \subset k(\mathbb{P}_k^1) = k(x)$ where $x = \frac{X_1}{X_0}$. By (6.2.117) R is equal to either $k[x]_{p(x)}$ for some irreducible polynomial or $k\big[x^{-1}\big]_{x^{-1}}$. Let α be a root of $p(x)$ in $\overline{k}$, then this corresponds to the point $(1 : \alpha)$, or in the latter case just $(0 : 1)$. ∎

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6.2.17 Affine and Finite Morphisms

Definition 6.2.96
*Let $\phi : X \to Y$ be a regular morphism of varieties. We say ϕ is **affine** if for every affine open $V \subset Y$, $\phi^{-1}(V)$ is affine.*

*We say in addition that ϕ is **finite** if the ring map $\mathcal{O}_Y(V) \to \mathcal{O}_X(\phi^{-1}(V))$ is finite (and therefore integral ...).*

Other than the definition we do not require this section in the sequel. But it may be convenient for other applications to show that it's sufficient to demonstrate a morphism is affine (resp. finite) only over a single affine cover.

Lemma 6.2.97
Let X be a variety with U, V open affine subsets. Then there exists $g_1, ..., g_r \in \mathcal{O}_X(U)$ and $g'_1, ..., g'_r \in \mathcal{O}_X(V)$ such that

a) $W_i := D(g_i) = D(g'_i)$ for $i = 1...r$
b) $U \cap V = \bigcup_{i=1}^r W_i$

Proof. Given $x \in U \cap V$, then as $U \cap V$ is open in U by (6.2.10) we may find $f_x \in \mathcal{O}_X(U)$ such that $x \in D(f_x) \subset U \cap V$. Similarly we may find $f'_x \in \mathcal{O}_X(V)$ with $x \in D(f'_x) \subset D(f_x)$. By (6.2.21).c) there is a canonical isomorphism $\mathcal{O}_X(U)_{f_x} \xrightarrow{\sim} \mathcal{O}_X(D(f_x))$. Therefore $f'_x|_{D(f_x)} = \frac{g}{(f_x)^r}$ for some $g \in \mathcal{O}_X(U)$. Observe that $D(f'_x) = D(g) \cap D(f_x) = D(gf_x)$. This gives the required pair of sections for an arbitrary $x \in U \cap V$. Clearly these open sets cover $U \cap V$, and as $U \cap V$ is quasi-compact (6.2.7) we are done. ∎

Proposition 6.2.98
Let $\phi : X \to Y$ be a regular morphism of varieties, and suppose there is an open cover $Y = \bigcup_{i=1}^n V_i$ for which V_i and $\phi^{-1}(V_i)$ are affine. Then ϕ is affine.

If in addition $\mathcal{O}_Y(V_i) \to \mathcal{O}_X(\phi^{-1}(V_i))$ is finite for all $i = 1...n$ then ϕ is finite.

Proof. By assumption we have affine open sets $U_i := \phi^{-1}(V_i)$ for $i = 1...n$. Let $V \subset Y$ be an open affine set, then by (6.2.97) for each i we have an open cover

$$V \cap V_i = \bigcup_{j=1}^{n_i} W_{ij}$$

with $W_{ij} := D(f_{ij}) = D(f'_{ij})$ for $f'_{ij} \in \mathcal{O}_Y(V_i)$ and $f_{ij} \in \mathcal{O}_Y(V)$. Evidently

$$V = \bigcup_{i=1}^{n} \bigcup_{j=1}^{n_i} D(f_{ij}) \quad (*)$$

and so by (6.2.26) the f_{ij} generate $\mathcal{O}_Y(V)$, and consequentally $g_{ij} := \phi^\sharp(f_{ij})$ generate $\mathcal{O}_X(U)$ (as both ideals contain 1). Further $\phi^{-1}(W_{ij}) = \phi^{-1}(D(f'_{ij})) = D(\phi^\sharp(f'_{ij})) \subset U_i$ is affine (6.2.21).c). Therefore from (6.2.28) U is affine, because we also have the representation $\phi^{-1}(W_{ij}) = \phi^{-1}(D(f_{ij})) = D(g_{ij})$.

Suppose ϕ is finite then by definition this means the ring maps $\mathcal{O}_Y(V_i) \to \mathcal{O}_X(U_i)$ are finite. By (3.5.48).a) this means the localised maps are finite

$$\begin{array}{ccc}
\mathcal{O}_Y(V_i)_{f'_{ij}} & \longrightarrow & \mathcal{O}_X(U_i)_{g'_{ij}} \\
\downarrow \sim & & \downarrow \sim \\
\mathcal{O}_Y(W_{ij}) & \xrightarrow{\phi^\sharp} & \mathcal{O}_X(\phi^{-1}(W_{ij})) \\
\uparrow \sim & & \uparrow \sim \\
\mathcal{O}_Y(V)_{f_{ij}} & \dashrightarrow & \mathcal{O}_X(U)_{g_{ij}}
\end{array}$$

Using equation $(*)$ and (6.2.26) we see that f_{ij} generate the unit ideal of $\mathcal{O}_Y(V)$. Therefore by (3.5.48).b) $\phi^\sharp_V$ is finite. ∎

6.2.18 Normal Varieties

Definition 6.2.99
*Let A be an integral domain. We say that the ring A is **normal** if it is integrally closed in its field of fractions.*

Recall that A factorial $\implies$ A is normal (3.23.12).

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Proposition 6.2.100
The polynomial ring $A[X_1, ..., X_n]$ is normal for every unique factorisation domain A.

Proof. See (3.23.12) and (3.16.46). ∎

Proposition 6.2.101 (Normal Variety)
Let X be an integral variety. Then the following are equivalent

a) $\mathcal{O}_X(U)$ is normal for every open subset $U \subset X$
b) $\mathcal{O}_{X,W}$ is normal for every irreducible subset $W \subset X$
c) $\mathcal{O}_{X,x}$ is normal for every point $x \in X$

*In this case we say that X is a **normal variety**. If X is affine then this is equivalent to $\mathcal{O}_X(X)$ being normal.*

Proof. *a)* $\implies$ *b)* Let $U \subset X$ be any affine open subset such that $U \cap W \neq \emptyset$, then by (...) $\mathcal{O}_{X,W} \xrightarrow{\sim} \mathcal{O}_{U,W\cap U} \xrightarrow{\sim} \mathcal{O}_X(U)_{\mathfrak{m}_{U,W\cap U}}$, and the result follows from (3.23.10).

b) $\implies$ *c)* is straightforward because $\mathcal{O}_{X,x} = \mathcal{O}_{X,\{x\}}$.

c) $\implies$ *a)* We first show this for the case that U is affine. By (...) $\mathcal{O}_{X,x} \xrightarrow{\sim} \mathcal{O}_{U,x} \xrightarrow{\sim} \mathcal{O}_X(U)_{\mathfrak{m}_{U,x}}$ is normal for every $x \in U$. Therefore by (6.2.21).b) and (3.23.10) $\mathcal{O}_X(U)$ is normal.

Now let U be an arbitrary open set and $V \subset U$ affine. Then by (6.2.15) we have inclusions

$$\mathcal{O}_X(V) \hookrightarrow \mathcal{O}_X(U) \hookrightarrow \operatorname{Frac}(\mathcal{O}_X(U)) \hookrightarrow k(X)$$

and $k(X)$ is a fraction field for $\mathcal{O}_X(V)$. If $\alpha \in \operatorname{Frac}(\mathcal{O}_X(U))$ is integral over $\mathcal{O}_X(U)$, then it is a-fortiori integral over $\mathcal{O}_X(V)$. By the affine case then $\alpha \in \mathcal{O}_X(V)$ and we conclude that $\mathcal{O}_X(U)$ is normal as required.

For the case X is affine we may use (3.23.10). ∎

Corollary 6.2.102
The standard affine space $\mathbb{A}^n_k\left(\overline{k}\right)$ is normal.

Proof. The coordinate ring is a UFD (3.16.46) and therefore normal (3.23.12). ∎

Proposition 6.2.103
Let X be an integral variety with open cover $X = \bigcup_{i=1}^n U_i$. Then X is normal iff U_i are normal.

Proof. Straightforward, since normality is a local condition. ∎

6.2.19 Weil Divisors

The notion of regular curves can be generalised in one direction.

Definition 6.2.104 (Prime Divisors of an Integral Variety)
*Let X be an integral, separated variety. We say it is **regular in codimension one** if for every irreducible closed subset $Y \subset X$ for which $\operatorname{codim}(Y, X) = 1$ the local ring $\mathcal{O}_{X,Y}$ is regular (equivalently integrally closed or a DVR (3.30.2)).*

*In this case, we say an irreducible closed subset of codimension 1 is a **prime divisor**.*

We denote by $\operatorname{Div} X$ the free abelian group generated by the prime divisors. We write a prime divisor D by a formal sum

$$D = \sum_{Y \subset X} n_Y Y$$

*where $n_Y \in \mathbb{Z}$ and all but finitely many are zero. We define the **degree** of a divisor D by*

$$\deg D = \sum_{Y \subset X} n_Y \left[k\left(\mathfrak{m}_{X,Y}\right) : k\right]$$

Proposition 6.2.105
An integral, separated variety X is regular in codimension one for either of the following special cases

a) X is normal
b) X is non-singular and k is perfect.

Proof. See (3.30.2) and (6.2.82). ∎

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Proposition 6.2.106 (Valuation Associated to Prime Divisor)
Let X be a variety satisfying (6.2.104) and $Y \subset X$ a prime divisor. Then $\mathcal{O}_{X,Y}$ is a discrete valuation ring.

We denote by $v_Y : k(X)^\star \to \mathbb{Z}$ the corresponding valuation.

Proof. By (6.2.55) $\dim \mathcal{O}_{X,Y} = \operatorname{codim}(Y, X) = 1$ and is integral, so the result follows from (3.30.2). ∎

Proposition 6.2.107 (Prime Divisors on a Curve)
Let X be a regular curve. Then X satisfies (6.2.104) and the prime divisors are precisely irreducible closed subsets of the form $\overline{\{x\}}$. These are in bijective correspondence with the valuation rings of $k(X)$, via the map $x \to \mathcal{O}_{X,x}$.

Proof. By the codimension formula (6.2.55) the irreducible closed subsets of codimension 1 are precisely those of dimension 0. Therefore the result follows from (6.2.124). ∎

Proposition 6.2.108
Let X be a variety satisfying (6.2.104) and $f \in k(X)^\star$ a rational function. Then $v_Y(f) = 0$ for all but finitely many prime divisors Y.

Proof. By definition f is regular on some open subset U of X, which we may assume to be affine. Define $Z \coloneqq X \setminus U$. Then by (...) $Z = Z_1 \cup ... \cup Z_n$ is a decomposition into irreducible components. By (4.1.83).f) $\operatorname{codim}(Z, X) > 0$ whence $\operatorname{codim}(Z_i, X) > 0$ for all $i = 1...n$. Suppose $Y \subset Z$ is a prime divisor then by definition $\operatorname{codim}(Y, X) = 1$ and by (4.1.66) $Y \subset Z_i$ for some i. Therefore by (4.1.83).c) $\operatorname{codim}(Y, Z_i) \leq \operatorname{codim}(Y, X) - \operatorname{codim}(Z_i, X) \leq 0$ and therefore by (4.1.83).f) $Y = Z_i$.

We conclude all but finitely many Y intersect U, so we may without loss of generality assume $Y \cap U \neq \emptyset$. Evidently U also satisfies (6.2.104) and by (4.1.79) there is a bijection between prime divisors of X meeting U, and prime divisors of U.

Therefore we may assume that X is affine and $0 \neq f \in \mathcal{O}_X(X)$. Then $v_Y(f) > 0 \iff f \in \mathfrak{M}_{X,Y} \iff f \in I(Y) \iff Y \subset V(f)$. We claim $V(f)$ is a proper subset of X (for $V(f) = X \iff \sqrt{(0)} = \sqrt{f} \iff f = 0$). Therefore $\operatorname{codim}(V(f), X) > 0$ and

$$\operatorname{codim}(Y, V(f)) \leq \operatorname{codim}(Y, X) - \operatorname{codim}(V(f), X) \leq 0$$

so Y is an irreducible component of $V(f)$, of which there are only finitely many (4.1.66). ∎

Definition 6.2.109 (Principal Divisor)
Let X be a variety satisfying (6.2.104) then define the homomorphism

$$\begin{aligned} k(X)^\star &\longrightarrow \operatorname{Div}(X) \\ f &\longrightarrow (f) \end{aligned}$$

where we define a formal sum over all prime divisors

$$(f) \coloneqq \sum_{Y \subset X} v_Y(f) Y.$$

*The co-kernel of this map is known as the **divisor class group** of X, and denoted $\operatorname{Cl}(X)$. We say that two divisors are equivalent $D \sim D'$ if they become equal as elements of the divisor class group. Equivalently if $D - D' = (f)$ for some $f \in k(X)^\star$.*

6.2.20 Algebraic Function Fields

We develop the basic theory of **algebraic function fields** of one variable, the field-theoretic counterpart to a regular algebraic curve.

Proposition 6.2.110
Let K/k be a field extension. Then the following are equivalent

a) K/k is finitely generated with transcendence degree 1.
b) There exists a transcendental element $x \in K/k$ such that $K/k(x)$ is finite.

*In this case we say that K/k is an **algebraic function field**. We typically assume that k is perfect (either a number field or a finite field).*

Proof. *a)* $\implies$ *b)* By (3.18.139) a transcendence base consists of a single element x. By assumption $K/k(x)$ is algebraic and finitely generated, hence also finite (3.18.44).

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$b) \implies a)$ Evidently $\{x\}$ is a transcendence base. By assumption $K/k(x)$ is finite and therefore certainly finitely generated, whence so is K/k. ∎

Definition 6.2.111 (Algebraic Function Field)
*Let K/k be an algebraic function field. Denote by $\tilde{k}$ the **field of constants**, the relative algebraic closure of k in K:*

$$\tilde{k} := \{x \in K \mid x \text{ algebraic over } k\}$$

*(a subfield of K by (3.18.45)). By (3.18.46), $K/\tilde{k}$ is relatively algebraically closed. We say that K/k is **geometrically integral** if $k = \tilde{k}$ (c.f. (3.35.41)).*

Definition 6.2.112 (Place of an Algebraic Function Field)
*Let K/k be an algebraic function field. By a **place** of K/k we refer to a valuation ring $\mathcal{O}_P$ such that $k \subsetneq \mathcal{O}_P \subsetneq K$. By (3.24.2) this is a local ring with unique maximal ideal $\mathfrak{m}_P$.*

From (3.3.13) we verify immediately that $k \cap \mathfrak{m}_P = (0)$.

*For every $x \in K$ we say that x is **regular** at a place P if $x \in \mathcal{O}_P$.*

Proposition 6.2.113
Let K/k be a function field and $k \subsetneq \mathcal{O}_P \subsetneq K$ a valuation ring. Then $\tilde{k} \subset \mathcal{O}_P$ and $\tilde{k} \cap \mathfrak{m}_P = (0)$

Proof. By definition $\tilde{k}$ is the integral closure of k in K, so $\tilde{k} \subset \mathcal{O}_P$ by (3.24.14). Suppose $0 \neq x \in \tilde{k}$ then x is invertible in $\mathcal{O}_P$ and therefore x is not contained in any proper ideal, and in particular $x \notin \mathfrak{m}_P$. ∎

In order to use the theory of Dedekind domains and affine algebras we establish a connection between valuation rings and a certain "affine" subring.

Proposition 6.2.114
Let K/k be an algebraic function field and $f \in K \setminus \tilde{k}$ transcendental. Then the integral closure $\overline{k[f]}$ in K is a Dedekind domain with quotient field K and a finite $k[f]$-module. There is a correspondence

$$\left\{\mathfrak{B} \lhd \overline{k[f]} \;\middle|\; \mathfrak{B} \text{ non-zero prime}\right\} \longleftrightarrow \{\mathcal{O}_P \subsetneq K \mid f \in \mathcal{O}_P\}$$
$$\mathfrak{B} \longrightarrow \overline{k[f]}_{\mathfrak{B}}$$
$$\mathfrak{m}_P \cap \overline{k[f]} \longleftarrow \mathcal{O}_P$$

In particular every such $\mathcal{O}_P$ is a discrete valuation ring, and $k(\mathfrak{m}_P) = k(\mathfrak{B})$ is a finite extension of k. More precisely, writing $\mathfrak{p} := \mathfrak{B} \cap k[f]$ (a maximal ideal of $k[f]$) and $k(\mathfrak{p}) := k[f]/\mathfrak{p}$, which is finite over k, the residue degree satisfies

$$[k(\mathfrak{m}_P) : k(\mathfrak{p})] = [k(\mathfrak{B}) : k(\mathfrak{p})] \leq [K : k(f)]$$

Moreover every place P of K/k (whether or not $f \in \mathcal{O}_P$) is a discrete valuation ring whose residue field $k(\mathfrak{m}_P)$ is a finite extension of k. Therefore we may associate to each place P a discrete valuation

$$v_P : K \longrightarrow \mathbb{Z} \cup \{\infty\}$$

which is zero on k and $v_P(f) < 0 \iff f \notin \mathcal{O}_P \iff f^{-1} \in \mathfrak{m}_P$.

Proof. By (3.31.2) $k[f] \subset K$ is a Dedekind domain and so by (3.34.61) $B := \overline{k[f]}^K$ is a Dedekind domain and finite $k[f]$-module. As $\mathcal{O}_P$ is a valuation ring, it is integrally closed (3.24.2) and as it contains $k[f]$, also contains B. Then we may deduce this correspondence from (3.31.4) and the inequality from (3.31.19) applied to $k[f] \subset \overline{k[f]}$ and $\mathfrak{p}$, which bounds the inertial degree $[k(\mathfrak{B}) : k(\mathfrak{p})]$. Here $\mathfrak{p}$ is maximal, as $\mathfrak{B}$ is maximal (a non-zero prime of the Dedekind domain B) and B is integral over $k[f]$ (3.23.16), and so non-zero as $k[f]$ is not a field. As f is transcendental, $k[f]$ is a polynomial ring and so a PID, whence $\mathfrak{p} = (g(f))$ for an irreducible polynomial g. Thus $k(\mathfrak{p}) = k[f]/(g(f))$ has k-basis $1, f, ..., f^{\deg g - 1}$ and is finite over k, and hence so is $k(\mathfrak{B})$ as $[k(\mathfrak{B}) : k] = [k(\mathfrak{B}) : k(\mathfrak{p})][k(\mathfrak{p}) : k]$.

This covers every place P with $f \in \mathcal{O}_P$. For a place P with $f \notin \mathcal{O}_P$ instead, $f^{-1} \in \mathfrak{m}_P \subset \mathcal{O}_P$ (3.24.2), and f^{-1} is transcendental too (else $f = (f^{-1})^{-1}$ would be algebraic); applying the above with f^{-1} in place of f shows $\mathcal{O}_P$ is again a discrete valuation ring, with residue field $k(\mathfrak{m}_P)$ finite over k. So every place P of K/k is a discrete valuation ring with residue field finite over k, giving the associated valuation v_P. ∎

Corollary 6.2.115
Let K/k be an algebraic function field. Then the field of constants $\tilde{k}/k$ has finite degree.

Proof. For any place P we have an embedding $\tilde{k}/k \hookrightarrow k(\mathfrak{m}_P)/k$ which is finite (6.2.114). ∎

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We prove a number of facts in analogy with algebraic curves.

Lemma 6.2.116 (Every function has a zero)
Let K/k be an algebraic function field and $0 \neq f \in K$. If $f \in \tilde{k}$ (i.e. f is algebraic), then $v_P(f) = 0$ for every place P, so f has no zeros or poles; otherwise (f transcendental) f has between 1 and $[K : k(f)]$ zeros (resp. poles).

Proof. If $f \in \tilde{k}$, then $f \neq 0$ gives $f \in \tilde{k} \setminus \{0\}$, so $f \notin \mathfrak{m}_P$ for every place P (6.2.113), that is $v_P(f) = 0$.

Otherwise, clearly $k[f]$ is a polynomial ring with maximal ideal (f). Therefore by (3.24.12) there is a valuation ring $\mathcal{O}_P \supset k[f]$ such that $\mathfrak{m}_P \cap k[f] = (f)$, which means f has at least one zero.

Recall by (6.2.114) that zeros of f correspond to the maximal ideals of $\overline{k[f]}$ containing f. These are precisely the prime divisors of (f) (see (3.31.13)), so the bound follows from (3.31.20).

The statement about poles of f follows immediately because these are precisely zeros of f^{-1} and $k(f) = k(f^{-1})$. ∎

Proposition 6.2.117 (Places of the Rational Function Field)
Let $K = k(x)$ with x transcendental over k. Then the places of K/k, that is the valuation rings $k \subsetneq \mathcal{O} \subsetneq K$, are exactly the following pairwise distinct discrete valuation rings

a) $\mathcal{O}_{p(x)} := k[x]_{(p(x))}$ where $p \in k[X]$ is an irreducible monic polynomial, with uniformiser $p(x)$ and residue field $k[X]/(p)$, and
b) $\mathcal{O}_\infty := \left\{ \frac{f(x)}{g(x)} \;\middle|\; g \neq 0, \deg f \leq \deg g \right\}$, with uniformiser x^{-1} and residue field k.

If k is algebraically closed then the irreducible monic polynomials are the $X - \theta$ with $\theta \in k$, so every place is rational, with uniformiser $x - \theta$ or x^{-1}.

Proof. Let $t := x^{-1}$, which is again transcendental over k with $k(t) = K$.

The listed rings are discrete valuation rings. As $k[x] \cong k[X]$ is a PID (3.15.4), $\mathcal{O}_{p(x)}$ is a discrete valuation ring (3.30.4), with maximal ideal $p(x)\mathcal{O}_{p(x)}$ and residue field $k[x]/(p(x))$ (3.5.27).

Next $\mathcal{O}_\infty = k[t]_{(t)}$: if $f, g \in k[X]$ with $g(0) \neq 0$ and $N := \max(\deg f, \deg g)$ then $f(t)/g(t) = F(x)/G(x)$ with $F(x) := x^N f(1/x)$, $G(x) := x^N g(1/x)$, $\deg F \leq N = \deg G$; conversely $F(x)/G(x)$ with $\deg F \leq \deg G =: N$ equals $f(t)/g(t)$ with $f(t) := t^N F(1/t)$ and $g(t) := t^N G(1/t)$, where $g(0)$ is the leading coefficient of G. So $\mathcal{O}_\infty$ is a discrete valuation ring with uniformiser t and residue field $k[t]/(t) = k$, as above.

Every place is listed. Let $\mathcal{O}$ be a place with maximal ideal $\mathfrak{m}$. Suppose firstly that $x \in \mathcal{O}$. Then $k[x] \subset \mathcal{O}$ and the ideal $k[x] \cap \mathfrak{m} =: (p(x))$ is principal. There is an injective k-algebra homomorphism $k[x]/(p(x)) \hookrightarrow k(\mathfrak{m})$ whence $(p(x))$ is prime (3.3.59). As $k(\mathfrak{m})/k$ is algebraic (6.2.114) and x is transcendental, we must have $0 \neq p$, which we take monic, and therefore p is irreducible (3.16.11). Suppose $g(x) \notin (p(x))$ then $g(x) \notin \mathfrak{m} \Longrightarrow g(x)^{-1} \in \mathcal{O}$ (3.24.2). Therefore $\mathcal{O}_{p(x)} \subset \mathcal{O}$, and as $\mathcal{O}_{p(x)}$ is a discrete valuation ring and $\mathcal{O} \neq K$ we conclude that they are equal (3.30.5).

Alternatively suppose that $x \notin \mathcal{O}$ then by (3.24.2) $t = x^{-1} \in \mathfrak{m} \subset \mathcal{O}$. Applying the previous part with t in place of x gives $\mathcal{O} = \mathcal{O}_{p(t)}$ where $\mathfrak{m} \cap k[t] = (p(t))$. As $t \in \mathfrak{m}$ we conclude that $p(t) = t$, and $\mathcal{O} = k[t]_{(t)} = \mathcal{O}_\infty$.

Distinctness. $x \in \mathcal{O}_{p(x)}$ but $x \notin \mathcal{O}_\infty$, as $\deg x > \deg 1$. For distinct monic irreducible $p \neq p'$, $p \notin (p')$, so $p(x)$ is a unit of $\mathcal{O}_{p'(x)}$ but not of $\mathcal{O}_{p(x)}$.

Finally if k is algebraically closed the irreducible monic polynomials are the $X - \theta$ (3.18.58), with residue field $k[X]/(X - \theta) = k$. ∎

Lemma 6.2.118 (Restricting a Place is Proper)
Let L/k' be an algebraic function field, P' a place of L/k', $k \subseteq k'$ with k'/k algebraic, and $K \subseteq L$ a subfield with $\operatorname{trdeg}_k(K) \geq 1$ (e.g. K/k itself an algebraic function field). Then $\mathcal{O}_{P'} \cap K$ is a proper valuation ring of K (and therefore a place.)

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

$\mathcal{O}_{P'} \cap K$ is a valuation ring of K (3.24.3). Suppose it equals K, so $K \subset \mathcal{O}_{P'}$. Nonzero elements of K are units of $\mathcal{O}_{P'}$, hence not in $\mathfrak{m}_{P'}$, so $K \to \mathcal{O}_{P'}/\mathfrak{m}_{P'} = k'(\mathfrak{m}_{P'})$ is injective, and $\operatorname{trdeg}_k(K) \leq \operatorname{trdeg}_k(k'(\mathfrak{m}_{P'}))$ (3.18.136). But $k'(\mathfrak{m}_{P'})/k'$ is algebraic (6.2.114) and k'/k is algebraic, so the right-hand side is 0, contradicting $\operatorname{trdeg}_k(K) \geq 1$. ∎

Lemma 6.2.119 (Places Restrict Correctly)
Let K/k be an algebraic function field, P a place of K/k, and L/K a field extension. Let $k \subseteq k'' \subset L$ with k''/k algebraic and L/k'' itself an algebraic function field, and P' a place of L/k'' with $\mathcal{O}_P \subset \mathcal{O}_{P'}$. Then $\mathcal{O}_{P'} \cap K = \mathcal{O}_P$; in particular $\mathfrak{m}_P \subset \mathfrak{m}_{P'}$, and $\pi_P \notin k''$.

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Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

$\mathcal{O}_{P'} \cap K$ is a valuation ring of K containing $\mathcal{O}_P$ (3.24.3), and proper (as K/k is an algebraic function field, so $\operatorname{trdeg}_k(K) = 1 \geq 1$) (6.2.118). By (3.30.5) ($\mathcal{O}_P$ being a DVR, (6.2.114)), a valuation ring of K containing the DVR $\mathcal{O}_P$ is either $\mathcal{O}_P$ itself or all of K; as $\mathcal{O}_{P'} \cap K \neq K$, this forces $\mathcal{O}_{P'} \cap K = \mathcal{O}_P$.

For $0 \neq x \in \mathfrak{m}_P$ we have $x^{-1} \in K \setminus \mathcal{O}_P = K \setminus \mathcal{O}_{P'}$, so $x \in \mathfrak{m}_{P'}$ by (3.24.2): thus $\mathfrak{m}_P \subset \mathfrak{m}_{P'}$. In particular $\pi_P \in \mathfrak{m}_{P'}$ is a non-unit of $\mathcal{O}_{P'}$, so $\pi_P \notin k''$ (nonzero elements of the field $k'' \subset \mathcal{O}_{P'}$ are units). ∎

6.2.21 Algebraic Curves

Definition 6.2.120
We say an integral, separated algebraic variety of dimension 1 *is an* ***algebraic curve*** *or just a* ***curve****.*

Proposition 6.2.121 (Characterisation Regular Point on a curve)
Let X be an algebraic curve and $x \in X$. Then the following are equivalent

a) x is regular
b) $\mathcal{O}_{X,x}$ is a discrete valuation ring
c) $\mathcal{O}_{X,x}$ is an integrally closed domain

and when k is perfect this is equivalent to x being non-singular.

In particular X is regular if and only if it is normal.

Proof. Follows exactly the same lines as (6.1.100). ∎

Example 6.2.122
Both $\mathbb{A}^1_k$ and $\mathbb{P}^1_k$ are regular algebraic curves.

Proposition 6.2.123 (Curve has Cofinite Topology)
Let X be an algebraic curve (or just an algebraic variety of dimension 1*) then X_0 has the cofinite topology.*

Proof. Follows from (4.1.92). ∎

Proposition 6.2.124 (Points on Complete Regular Curve = Valuation Rings on Function Field)
Let X be a complete regular algebraic curve. Then there is a bijection

$$\{W \subset X \mid \dim W = 0 \text{ closed and irreducible}\} \longleftrightarrow X_0 \longleftrightarrow \{k \subset R \subsetneq k(X) \mid R \text{ a valuation ring}\}$$

Further the corresponding valuation rings are all discrete valuation rings.

For an affine open subset $U \subset X$ this restricts to a bijection

$$U_0 \longleftrightarrow \{\mathcal{O}_X(U) \subset R \subsetneq k(X) \mid R \text{ a valuation ring}\}$$

where $\mathcal{O}_X(U)$ is identified as a subring of $k(X)$ (6.2.18).

Proof. The first map is (4.1.94) and the second map is well-defined by (6.2.121) and injective by (6.2.20).

Let $R \subset k(X)$ be a valuation ring then by definition of completeness there is a point $x \in X$ such that $\mathcal{O}_{X,x} \subset R$. By (6.2.55) we have $\dim \mathcal{O}_{X,x} = \dim X = 1$. By (6.1.92) and (6.1.91) $1 = \dim \mathcal{O}_{X,x} \leq \dim R \leq \operatorname{trdeg}_k k(X) \overset{(6.2.54)}{=} \dim X = 1$. Therefore $\dim R = 1$ and by the same result $R = \mathcal{O}_{X,x}$. This shows that the map is surjective.

For the final statement we may use the affine case (6.1.101) and the commutative diagram

$$\begin{array}{ccc} \mathcal{O}_X(U) & \hookrightarrow & k(X) \\ \| & & \downarrow \sim \\ \mathcal{O}_U(U) & \hookrightarrow & k(U) \end{array}$$

∎

Proposition 6.2.125
Let X be a regular algebraic curve and Y a complete variety. Then the rational maps are precisely the regular maps. In otherwords the canonical map

$$\operatorname{Mor}(X, Y) \xrightarrow{\sim} \operatorname{Rat}(X, Y)$$

is a bijection. Further this restricts to dominant maps.

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Proof. Clearly the map is well-defined. Suppose that two regular maps ϕ, ψ are equal as rational maps. Then by definition they agree on a dense open subset. By definition Y is separated so by (6.2.63) the equaliser is a closed subset, and therefore the whole of Y. This means the map is injective.

Consider a regular map $\phi : U \to Y$ for U a (dense) open subset of X. Then $Z \coloneqq X \setminus U$ is a proper closed subset of X and therefore $\dim Z = 0$ by (4.1.84). By (6.2.12) we have $Z = \overline{\{z_1\}} \cup ... \cup \overline{\{z_n\}}$, and we may apply (6.2.90) finitely many times to lift to a regular map on an open set containing U and $z_1, ..., z_n$. By (4.1.41) this is the whole of X. Therefore the given map is surjective. ∎

Proposition 6.2.126 (Image of Complete Curve either Constant or Surjective)
Let X, Y be algebraic curves with X complete and $\phi : X \to Y$ a regular morphism. Then ϕ is either surjective or has image of the form $\overline{\{x\}}$ (which we call constant).

Proof. By (6.2.89) ϕ is universally closed, so by (6.2.84) $\phi(X)$ is an irreducible closed subvariety of Y. Evidently $\dim \phi(X) = 0$ or 1. Then we may use (6.2.56) or (4.1.84). ∎

Corollary 6.2.127 (Equivalence between nice curves and function fields)
Let X, Y be complete, regular algebraic curves. Then there are bijections

$$\{\phi : X \to Y \text{ non-constant}\} \longleftrightarrow \{\phi : X \longrightarrow Y \text{ dominant}\} \longleftrightarrow \{\phi_\star : k(Y) \hookrightarrow k(X)\}$$

All such morphisms are surjective and the degree

$$\deg(\phi) \coloneqq [k(X) : \phi_\star(k(Y))]$$

is always finite.

Finally the fibres are finite (i.e. ϕ is "quasi-finite").

Proof. This follows from (6.2.126), (6.2.125) and (6.2.76).

The fibres are finite because $\phi^{-1}(\{y\}) \subset \phi^{-1}\left(\overline{\{y\}}\right)$. By assumption this is a proper closed subset of X and therefore has dimension 0. So we are done by (6.2.56). ∎

Proposition 6.2.128
Let X be a regular algebraic curve. Then there is a bijection

$$k(X)^\star \longleftrightarrow \{\phi : X \to \mathbb{P}^1_k \text{ non-constant}\}$$

More precisely given $f \in k(X)^\star$, then for $x \in X$ the function

$$\phi_f(x) = \begin{cases} (f(x) : 1) & f \in \mathcal{O}_{X,x} \\ (1 : f^{-1}(x)) & f^{-1} \in \mathcal{O}_{X,x} \end{cases}$$

is well-defined and regular.

Proof. Recall (6.2.121) (6.2.15) that $\mathcal{O}_{X,x}$ is a (discrete) valuation ring with field of fractions $k(X)$.

Define the open subsets $U \coloneqq \mathrm{dom}(f)$ and $V \coloneqq \mathrm{dom}(f^{-1})$. We claim that $X = U \cup V$. For $x \in \mathrm{dom}(f) \iff f \in \mathcal{O}_{X,x}$ and as $\mathcal{O}_{X,x}$ is a valuation ring for $k(X)$ then we have either $x \in U$ or $x \in V$.

Define $\phi : U \to \mathbb{P}^1_k$ by $\phi(x) = (f(x) : 1)$, and similarly $\psi : V \to \mathbb{P}^1_k$ by $\psi(x) = (1 : f^{-1}(x))$. By definition of projective structure sheaf (6.2.93) these are regular. Then for $x \in U \cap V$ we must have $f \notin \mathfrak{m}_{X,x}$ that is $f(x) \neq 0$ and $f^{-1}(x) = f(x)^{-1}$. This shows that $\phi|_{U\cap V} = \psi|_{U\cap V}$. Therefore they glue to a well-defined regular map $X \to \mathbb{P}^1_k$ as required.

We note that $\phi_f^{-1}(U_1) = \mathrm{dom}(f)$ where U_1 is the second affine piece, and we conclude the map is injective.

Conversely given a morphism $\phi : X \to \mathbb{P}^1_k$ then $U \coloneqq \phi^{-1}(U_1)$ is a non-empty open subset of X and $f : X_0/X_1 \circ \phi|_U : U \to \overline{k}$ is a regular function. Evidently $\phi|_U = \phi_f|_U$, and we conclude $\phi = \phi_f$ by (6.2.125). Therefore the map is surjective. ∎

Proposition 6.2.129 (Regular Morphisms of Complete Curves are Finite)
Let $\phi : X \to Y$ be a non-constant morphism of complete regular algebraic curves. Suppose $V \subset Y$ is an affine open subset, then

$$\mathcal{O}_X(\phi^{-1}(V)) = \overline{\phi_\star(\mathcal{O}_Y(V))}$$

where $\bar{\cdot}$ denotes the integral closure in $k(X)$.

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Further ϕ is a finite (and therefore affine, and integral) morphism.

Proof. Define $A \coloneqq \mathcal{O}_Y(V)$ and $B \coloneqq \overline{\phi_\star(A)}$. Then

$$\begin{aligned} B &= \bigcap_{\phi_\star(A)\subset R} R && (3.24.14)\\ &= \bigcap_{A\subset\phi_\star^{-1}(\mathcal{O}_{X,x})} \mathcal{O}_{X,x} && (6.2.124)\\ &= \bigcap_{A\subset\mathcal{O}_{Y,\phi(x)}} \mathcal{O}_{X,x} && (6.2.18)\\ &= \bigcap_{\phi(x)\in V} \mathcal{O}_{X,x} && (6.2.124)\\ &= \mathcal{O}_X\big(\phi^{-1}(V)\big) && (6.2.73) \end{aligned}$$

By (6.2.54) $\dim V = \dim Y = 1$ and therefore $\dim \mathcal{O}_Y(V) = 1$ (6.1.50). By (3.34.61) B is a Dedekind domain and a finite A-module with quotient field $k(X)$.

Let $U \coloneqq \phi^{-1}(V)$ and observe that as $\mathcal{O}_{X,x}$ is integrally closed

$$B \subset \mathcal{O}_{X,x} \iff \phi_\star(A) \subset \mathcal{O}_{X,x} \iff A \subset \mathcal{O}_{Y,\phi(x)} \iff \phi(x) \in V \iff x \in U$$

where we have used the fact V is affine (6.2.124). This shows that there is a bijection between U_0 and the valuation rings of $k(X)$ containing $\mathcal{O}_X(U)$. It remains to show that U is affine.

Let Z denote an integral affine curve with coordinate ring (isomorphic to) $B = \mathcal{O}_X(U)$. By the Affine Mapping Property (6.2.22) (applicable to U without yet knowing it is affine, since only Z needs to be) there is a corresponding regular morphism

$$\psi : U \to Z$$

such that $\psi^\sharp_Z : \mathcal{O}_Z(Z) \to \mathcal{O}_U(U)$ is an isomorphism. Consider the commutative

$$\begin{array}{ccc} \mathcal{O}_Z(Z) & \xrightarrow[\sim]{\psi^\sharp_Z} & \mathcal{O}_U(U) \\ \big\downarrow & & \big\downarrow \\ k(Z) & \xrightarrow[\sim]{\psi_\star} & k(U) \end{array}$$

as $k(Z)$ is the field of fractions of $\mathcal{O}_Z(Z)$ and similarly for $\mathcal{O}_U(U)$ (6.2.15), we see that $\psi_\star$ is an isomorphism. In light of the correspondence of valuation rings to points we see that $\psi_0 : U_0 \to Z_0$ is a bijection. As these have the cofinite topology (6.2.123) then ψ_0 is necessarily a homeomorphism.

For $g \in \mathcal{O}_Z(Z)$ we have a commutative diagram

$$\begin{array}{ccc} \mathcal{O}_Z(Z) & \xrightarrow{\sim} & \mathcal{O}_U(U) \\ \big\downarrow & & \big\downarrow \\ \mathcal{O}_Z(D(g)) & \xrightarrow{\psi^\sharp_{D(g)}} & \mathcal{O}_U\big(D(\psi^\sharp(g))\big) \\ \sim\big\uparrow & & \big\uparrow\sim \\ \mathcal{O}_Z(Z)_g & \xrightarrow{\sim} & \mathcal{O}_U(U)_{\psi^\sharp(g)} \end{array}$$

where the bottom isomorphisms follow from (6.2.27). This shows that $\psi^\sharp_{D(g)}$ is an isomorphism, and in particular (as the $D(g)$ form a basis of Z) that ψ is continuous. We claim that the morphism of stalks

$$\psi_x : \mathcal{O}_{Z,\psi(x)} \to \mathcal{O}_{U,x}$$

is an isomorphism for all $x \in U$. Suppose we have $(W,\sigma) \in \mathcal{O}_{U,x}$ and let $g \in \mathcal{O}_Z(Z)$ be such that $\psi(x) \in D(g) \subset \psi(W)$ (6.1.15). Then as $\psi^\sharp_{D(g)}$ is an isomorphism there exists $\tau \in \mathcal{O}_Z(D(g))$ such that $\psi^\sharp(\tau) = \sigma|_{\psi^{-1}(D(g))}$ which has the required image in $\mathcal{O}_{U,x}$. Similarly suppose that $(T,\tau) \in \mathcal{O}_{Z,\psi(x)}$ maps to zero. Then there exists some neighbourhood $x \in W \subset \psi^{-1}(T)$ such that $\psi^\sharp(\tau)|_W = 0$. We may choose $g \in \mathcal{O}_Z(Z)$ such that $D(g) \subset \psi(W)$. Then $0 = \psi^\sharp(\tau)|_{\psi^{-1}(D(g))} = \psi^\sharp\big(\tau|_{D(g)}\big)$ whence $\tau|_{D(g)} = 0$ which shows that (T,τ) is zero.

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By (6.2.50) (applied to U and Z), together with ψ_0 already bijective and each stalk map ψ_x already an isomorphism, ψ is bijective. As ψ is a continuous bijection inducing the homeomorphism ψ_0 on Kolmogorov quotients, (4.1.38) shows ψ is a homeomorphism.

As $\psi^\sharp$ is an isomorphism on a basis of open sets, it is an isomorphism of sheaves. Therefore U is isomorphic to Z and affine as required. ∎

Definition 6.2.130 (Order of Zero / Poles)
Let X be a regular algebraic curve. For every point $x \in X$ there is a well-defined and unique discrete valuation

$$v_x : k(X)^\star \to \mathbb{Z}$$

*because $\mathcal{O}_{x,X}$ is a discrete valuation ring of $k(X)$ (6.2.124). For r a positive integer we say that $f \in k(X)^\star$ has a **zero** (resp. **pole**) of order r at x if $v_x(f) = r$ (resp. $v_x(f) = -r$).*

*As before define the **principal divisor***

$$(f) = \sum_{x \in X_0} v_x(f)x$$

on the free abelian group indexed by X_0.

Definition 6.2.131
*Let $\phi : X \to Y$ be a morphism of regular curves and $\phi_\star : k(Y) \hookrightarrow k(X)$ the correspond morphism of fields. Recall that for $x \in X$ that $\phi_\star$ restricts to a **local** homomorphism*

$$\mathcal{O}_{Y,\phi(x)} \to \mathcal{O}_{X,x}$$

*Therefore for $x \in X$ we may define the **ramification index** at x as follows*

$$e_x := v_x(\phi_\star(t)) \geq 1$$

*where t is a uniformiser for the discrete valuation ring $\mathcal{O}_{Y,\phi(x)}$. Similarly the **residue degree** is given by the degree of the field extension*

$$f_x := \left[k\left(\mathfrak{m}_{X,x}\right) : \phi_\star k\left(\mathfrak{m}_{Y,\phi(x)}\right)\right]$$

Observe that if x and x' are topologically indistinguishable (equivalently Galois conjugate, or equal in X_0) then $e_x = e_{x'}$ and $f_x = f_{x'}$.

Note when k is algebraically closed then $f_x = 1$ since $k(x), k(y)$ are finite and therefore trivial extensions of k.

Proposition 6.2.132 (Ramification Formula)
Let $\phi : X \to Y$ be a morphism of regular curves which is finite (e.g. X and Y are complete (6.2.129)). For $y \in Y$ then

$$\deg(\phi) = \sum_{x_0 \in \phi_0^{-1}(\{y_0\})} e_x f_x$$

Proof. Let V be an affine neighbourhood of y, and define $U = \phi^{-1}(V)$. Then by assumption U is affine and $\phi^\sharp : \mathcal{O}_Y(V) \to \mathcal{O}_X(U)$ is finite (and integral (3.23.4)).

Define $L := k(Y)$, $B := \mathcal{O}_Y(V)$, $K := \phi_\star(k(X))$ and $A := \phi_\star(\mathcal{O}_X(U))$, where $\phi_\star : k(X) \hookrightarrow k(Y)$ is the canonical embedding

Then by (6.2.15) L is the quotient field of B, and K is the quotient field of A. By (6.2.101) (6.2.54) both B and A are Dedekind domains. B is finite and integral over A, and therefore B is the integral closure of A, being integrally closed. Therefore we are in the situation of (3.31.12).

By (6.2.21) there is a bijection between the maximal ideals of B (resp. A) and the points of U_0 (resp. V_0). And by (...) $\phi_0(x_0) = y_0 \iff \mathfrak{M}_{U,x} \cap A = \mathfrak{M}_{V,y}$.

By (3.31.19) the notion of ramification index and residue degree coincide with the same notions for Dedekind domains. Therefore the result follows from the corresponding result for finite morphisms of Dedekind domains (3.31.20). ∎

Proposition 6.2.133 (Degree of Principal Divisor is Zero)
Let X be a complete regular curve and $f \in k(X)$ a non-constant rational function, then

$$\deg((f)) = 0$$

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i.e. the degree of a principal divisor is zero.

Proof. By (6.2.128) f corresponds to a non-constant and therefore surjective regular morphism $\phi_f : X \to \mathbb{P}^1_k \xrightarrow{\sim} \mathbb{A}^1_k \cup \{\infty\}$. From the construction we see that $v_x(f) > 0$ if and only if $\phi_f(x) = 0$ and $v_x(f) < 0$ if and only if $\phi_f(x) = \infty$. Therefore by definition

$$(f) = \sum_{\substack{x_0 \in X_0 \\ \phi(x_0)=0}} v_{x_0}(f) x_0 + \sum_{\substack{x_0 \in X_0 \\ \phi(x_0)=\infty}} v_{x_0}(f) x_0$$

We have canonical isomorphism $\mathcal{O}_{\mathbb{P}^1_k,0} \xrightarrow{\sim} k[X]_{(X)}$ and when $\phi_f(x) = 0$ the corresponding morphisms of stalks is given by $X \to f_x$, therefore the ramification index is $e_x = v_x(f)$. Similarly when $\phi_f(x) = \infty$ we have $e_x = v_x(f^{-1}) = -v_x(f)$. It's evident that $k\left(\mathfrak{m}_{\mathbb{P}^1_k,0}\right) = k$, whence $f_{x_0} = [k(x_0) : k]$. Therefore by definition of degree

$$\deg((f)) = \sum_{\substack{x_0 \in X_0 \\ \phi(x_0)=0}} e_{x_0} f_{x_0} - \sum_{\substack{x_0 \in X_0 \\ \phi(x_0)=\infty}} e_{x_0} f_{x_0}$$

As X is complete, and so is $\mathbb{P}^1_k$ (6.2.95) then ϕ_f is finite by (6.2.129) and we are done by (6.2.132) since both sums are equal to $\deg(\phi_f)$. ∎

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6.3 Function Fields and the Riemann-Roch Theorem

We develop the theory of algebraic function fields K/k, which by (6.2.127) are the same thing as complete regular algebraic curves over k. Our goal is the Riemann-Roch theorem for such curves over a perfect base field k, the algebraic counterpart of the classical theorem for compact Riemann surfaces. Via the correspondence it suffices to prove it for function fields, where it was proved by Weil and Chevalley [Che51] using adèles (répartitions). That proof needs the least machinery, but in it the genus g and the index of speciality $i(D)$ appear only as extremal or dimension-count invariants, with no direct geometric meaning. As in the classical case, the remaining work is to relate them to rational differentials: for a non-zero rational differential ω with divisor (ω) we show $i(D) = \ell((\omega) - D)$ and $g = \ell((\omega))$. This rests on the Residue Theorem $\sum_P \mathrm{Res}_P(\omega) = 0$ (6.3.54), which we need over a general perfect field and not only for $k = \overline{k}$. The closed points of a curve over k have residue fields $k(P)$ of arbitrary finite degree over k. At such a place we expand in $k(P)((X))$ and set $\mathrm{Res}_P := \mathrm{Tr}_{k(P)/k} \circ \mathrm{Res}_P^{k(P)}$ (6.3.25) equivalently Res_P is the sum of the residues at the conjugate rational places over $\overline{k}$ (6.3.53). A rational differential ω defines a Weil differential by $\alpha \mapsto \sum_P \mathrm{Res}_P(\alpha_P \omega)$, whose vanishing on K is exactly the Residue Theorem. We deduce it from the algebraically closed case (6.3.34) by base change to $\overline{k}$: the places of K correspond to Galois orbits of places of $K_{\overline{k}}$, and the residue at P is the sum of the residues over its orbit.

6.3.1 Relation to the Literature

These treatments fall into two methodological groups:

- Weil's répartition/adele method, in the tradition initiated by Chevalley [Che51]: used by Chevalley himself, by Stichtenoth [Sti09], by Lang [Lan82] (all three over a general constant field), and, at its computational core, even by Serre [Ser59] – whose own bibliographic note attributes the proof to Weil, though he additionally erects the formalism of sheaf cohomology around it, identifying $H^1(X, \mathcal{O}_X)$ with the classes of repartitions. Lang's book, unlike the others in this group, has a different overall aim: a bridge to the classical complex-analytic theory of abelian and theta functions, which the algebraic function-field theory here does not pursue.
- the classical algebraic-geometric (scheme-free, ringed-space) approach shared by Hartshorne [Har13] and Bump [Bum98], the latter specialised to curves with a dedicated treatment of differentials, residues and ramification.

I have interpolated between these, guided by four principles.

Dedekind domains over bare valuation theory. I use Dedekind domain theory for the key results on function fields and the Weil form of Riemann-Roch, rather than the purely valuation-theoretic development of [Che51] and [Sti09]. In the same spirit, finite extensions of $k((t))$ (3.32.34) are analysed directly via the integral closure in a Dedekind domain and a direct power-series computation, rather than via the P-adic completion machinery both of those references build on; this also yields the result in every characteristic, where [Che51]'s analogous Puiseux-expansion treatment (Ch. II, §6) is restricted to the tame case, as noted in the remark opening the Extensions of Power Series Rings subsection. We also use the theory of affine algebras to simplify the proof of some further technical results.

A perfect constant field, reduced to the algebraically closed case only once. Riemann-Roch is proved over a general constant field, matching [Sti09], but going beyond [Ser59], [Har13], [Bum98] and [Lan82], whose treatments of Riemann-Roch all assume an algebraically closed constant field throughout. The Residue Theorem, on which the identification of the genus and the index of speciality with rational differentials rests, is likewise stated in the literature only over an algebraically closed field ([Ser59] (Ch. II), [Lan82] (Ch. I, §5), [Sti09] and [Bum98]). Our proof in that case (6.3.34) follows [Ser59] (Ch. II, nos. 12–13) and [Lan82] (Ch. I, §§4–5): partial fractions on the rational function field, the trace formula for residues to reduce the general case to it, and a local trace formula for power series fields proved by a universal identity over $\mathbb{Z}$. The local theory rests on the structure of finite extensions of $k((t))$ from the Extensions of Power Series Rings subsection (3.32.34), (3.32.35) and the Weierstrass division lemma (3.32.30), as [Lan82] (Ch. I, §4, Prop. 4.3) does. We need the Residue Theorem over a perfect field (6.3.54), and deduce it from the algebraically closed case by relating the residue symbol over k to its base change to $\overline{k}$: the residue at a place P is the sum of the residues at the conjugate places of $K_{\overline{k}}$ above it (6.3.53).

The perfect case can also be assembled using only results of [Sti09]. There the residue is defined only at places of degree one (Def. 4.2.10), and the Residue Theorem (Cor. 4.3.3) is stated only over an algebraically closed field, where every place has degree one. But Thm. 4.3.2 identifies differentials with Weil differentials over any perfect constant field K, and for a place P of degree f, with the f places $\overline{P}_1, \ldots, \overline{P}_f$ of the constant field extension $F\overline{K}$ above it (Cor. 3.6.5), Rem. 4.3.7(d) gives the formula (4.38)

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$$\omega_P(u) = \sum_{i=1}^{f} \operatorname{res}_{\overline{P}_i}(u\omega),$$

that is, the local component of a differential at P is the sum of the residues at the conjugate rational places. This is exactly our residue symbol at P (6.3.53). The vanishing of the sum is then Prop. 1.7.2, $\sum_P \omega_P(1) = \omega(1) = 0$, valid over any constant field since a Weil differential vanishes on the principal adele 1. So over a perfect field the Residue Theorem is literally $\sum_P \omega_P(1) = 0$ once residues are read as local components; its algebraically closed statement is the special case where every place is rational. The compatibility of local components with constant field extension used there is Prop. 4.3.6, (4.33), which rests on the cotrace formula (4.21) – the cotrace of a Weil differential ω of F in a finite separable extension F' being the Weil differential ω' of F' characterised by $\omega_P\big(\operatorname{Tr}_{F'/F}(y)\big) = \sum_{P'|P} \omega'_{P'}(y)$ for $y \in F'$ – and the construction of the Weil differential $\delta(x)$ as a cotrace; our approach replaces it by the direct comparison of Laurent expansions (6.3.51).

Concrete constructions over general machinery, wherever one suffices. Like [Che51], [Sti09] and [Lan82], I work with Chevalley and Weil's répartition/adele-style methods directly, and connect this to cohomology only after the fact, rather than building up the general machinery of coherent sheaf cohomology as [Ser59] does around the same underlying répartition computation. Residues are defined directly from the Laurent expansion at a place, via the uniformiser derivation, without ever constructing the module of Kähler differentials $\Omega_{K/k}$, in contrast to [Ser59] and [Sti09], which both identify their (Weil-differential-based) residues with genuine differentials in this sense; at a non-rational place the residue is the trace of the residue in $k(P)((X))$, rather than a local component of a Weil differential defined through cotraces as in [Sti09]. As in [Lan82], the local analysis uses only explicit Laurent series fields and the structure of their finite extensions (3.32.34), rather than the abstract P-adic completion on which [Ser59] and [Sti09] depend, and the rational function field is handled by a short, self-contained proof by partial fractions rather than as a special case of the general machinery. The Trace Formula for Residues (6.3.42) likewise rests on this Laurent-series finite-extension lemma, and not on the general Complete Discrete Valuation Rings section developed alongside it, of which the lemma is the special case $\mathcal{O}_K \cong k[[t]]$.

Characteristic p handled uniformly, via universal $\mathbb{Z}$-polynomial identities. Invariance of the residue of a differential under change of uniformiser – notoriously tricky in characteristic p, where the naive derivative-of-a-power argument breaks down whenever p divides the relevant exponent – is settled once, universally, over $\mathbb{Z}[a_1, a_1^{-1}, a_2, a_3, \ldots]$ and specialised (6.3.26), the same device used by [Ser59] and [Sti09], rather than the Cartier-operator route of [Bum98] unlike [Ser59] and [Sti09], this is done over a general constant field rather than an algebraically closed one. The same device is reused for the Trace Formula for Residues, which holds without any tameness hypothesis (6.3.40) by the identical method [Ser59] uses (there attributed to Hasse) for the analogous local identity $\operatorname{Res}_t(f\,du) = \operatorname{Res}_u(\operatorname{Tr}(f)\,du)$ underlying its own residue formula; it is exactly this universal-identity device that lets the wildly ramified case go through in both results.

Two further points of departure concern scope and connections to the classical picture, rather than proof technique: the dictionary between the function field and the regular algebraic curve – established even over a non-algebraically closed field, generalising the algebraically-closed-field statement of [Har13] – is established earlier, in (6.2.127) of the Algebraic Geometry chapter's treatment of regular algebraic curves; and, in the classical case of a compact Riemann surface (6.3.19), Weil differentials are shown to coincide canonically with classical meromorphic 1-forms, reconnecting the algebraic formalism with the analytic theory it generalises.

6.3.2 Riemann-Roch Theorem

In what follows we generally assume that $\tilde{k} = k$ (i.e. K is geometrically integral).

Definition 6.3.1 (Divisors)
Let K/k be an algebraic function field. Denote by $\operatorname{Div}(K)$ *the free abelian group generated by the places of K/k (**divisor group**). The elements are written as a formal sum*

$$D = \sum_P n_P P$$

*with all but finitely many coefficients $n_P = 0$. Define the **degree** homomorphism*

$$\begin{aligned} \operatorname{Div}(K) &\longrightarrow \mathbb{Z} \\ \sum_P n_P P &\longrightarrow \sum_P n_P \cdot [k(\mathfrak{m}_P) : k] \end{aligned}$$

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We may also define $v_P(D) := n_P$*, and the* ***support*** *of* D *by* $\mathrm{Supp}(D) := \{P \mid v_P(D) \neq 0\}$*, a finite set. Finally we may define a partial order*

$$D \leq D' \iff v_P(D) \leq v_P(D') \quad \forall P$$

Definition 6.3.2 (Riemann-Roch Space)
Let K/k *be an algebraic function field and* D *a divisor. Define the* ***Riemann-Roch Space*** *associated to* D *by*

$$L(D) := \{x \in K^\star \mid (x) \geq -D\} \cup \{0\}$$

and define

$$\ell(D) := \dim_k L(D)$$

(we shortly show it is a k*-vector subspace of* K*)*

Definition 6.3.3 (Principal Divisors)
Let K/k *be an algebraic function field and* $x \in K^\star$*. Then define the* ***zero divisor*** *(resp.* ***pole divisor****,* ***principal divisor****) of* x *as follows*

$$\begin{aligned}(x)_0 &:= \sum_{P : v_P(x) > 0} v_P(x) P,\\ (x)_\infty &:= \sum_{P : v_P(x) < 0} (-v_P(x)) P,\\ (x) &:= (x)_0 - (x)_\infty = \sum_P v_P(x) P.\end{aligned}$$

which is well-defined by (6.2.116).

Observe similarly $x \in k^\star \iff (x) = 0$*. Define the subgroup of principal divisors by*

$$\mathrm{Princ}(K) := \{(x) \mid x \in K\}$$

and the quotient group $\mathrm{Cl}(K) := \mathrm{Div}(K)/\mathrm{Princ}(K)$ *is called the* ***divisor class group****. We say two divisors* D *and* D' *are* ***linearly equivalent*** *if there exists* $x \in K$ *such that* $D - D' = (x)$*, equivalently if they become equal in the divisor class group. This is written* $D \sim D'$*.*

We presently prove that these concepts are well-defined, namely every function $x \in K^\star$ has only finitely many zeros. More precisely we can show

Proposition 6.3.4 (Principal Divisors have degree zero)
Let K/k *be an algebraic function field and* $x \in K \setminus \tilde{k}$ *transcendental. Then*

$$\begin{aligned}\deg((x)_0) = \deg((x)_\infty) &= [K : k(x)]\\ \deg((x)) &= 0\end{aligned}$$

In particular there is a well-defined homomorphism

$$\deg : \mathrm{Cl}(K) \longrightarrow \mathbb{Z}$$

Proof. Recall (6.2.114) $k[x]$ is a Dedekind domain, $\overline{k[x]}$ is a finite $k[x]$-module and every place $\mathcal{O}_P$ containing x corresponds to prime ideal $\mathfrak{B}$ of $\overline{k[x]}$. By (3.31.19) then $v_P(x)$ is equal to the ramification index $e_{\mathfrak{B}/(x)}$, and the inertial degree is $f_{\mathfrak{B}/(x)} = [k(\mathfrak{m}_P) : k]$ where (x) is a prime ideal of $k[x]$. From (3.34.61) we have the ramification formula

$$[K : k(x)] = \sum_{\mathfrak{B} \mid (x)} e_{\mathfrak{B}/(x)} f_{\mathfrak{B}/(x)}$$

Therefore $\deg((x)_0) = [K : k(x)]$ and $\deg((x)_\infty) = \deg\left((x^{-1})_0\right) = [K : k(x^{-1})] = [K : k(x)]$. ∎

Proposition 6.3.5
Let K/k *be an algebraic function field and* D *a divisor. Then*

a) $L(D)$ *is a* k*-vector subspace of* K
b) If $D \sim D'$ *then* $L(D) = L(D')$
c) $L(0) = \tilde{k}$ *so under our standing assumption* $\ell(0) = 1$*.*
d) If $D < 0$ *then* $L(D) = \{0\}$

Proof.

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a) Given $x, y \in L(D)$ then by the strict triangle inequality for valuations $v_P(x+y) \geq \min(v_P(x), v_P(y)) \geq v_P(D)$. Similarly for $\lambda \in k$ we have $v_P(\lambda x) = v_P(\lambda) + v_P(x) = v_P(x)$ (6.2.114).
b) By definition $D' = D + (x)$ we may define a mapping

$$\begin{aligned} L(D) &\longrightarrow L(D') \\ y &\longrightarrow yx \end{aligned}$$

Then

$$v_P(yx) = v_P(x) + v_P(y) \geq v_P(x) + v_P(D) = v_P(D')$$

so this is well-defined. As x^{-1} defines the inverse map it is a bijection.
c) By (6.2.114).a) $\tilde{k} \subset L(0)$. Suppose $x \in K \setminus \tilde{k}$ then by definition x is transcendental and so is x^{-1}. By (6.2.116) x^{-1} has a zero P, i.e. $v_P(x^{-1}) > 0 \implies v_P(x) < 0$. Therefore by definition $x \notin L(0)$ as required.
d) If $x \in \tilde{k}^\star$ then $(x) = 0$ whence $x \notin L(D)$. If $x \in K \setminus \tilde{k}$ then as before x has a pole therefore $x \notin L(D)$ as required.

■

Lemma 6.3.6
Let K/k be an algebraic function field and $D \leq D'$ divisors. Then $L(D) \subset L(D')$ and

$$\dim_k L(D')/L(D) \leq \deg(D') - \deg(D)$$

In particular $\ell(D)$ is finite and we have

$$\deg(D) - \ell(D) \leq \deg(D') - \ell(D')$$

Proof. Consider a divisor D and suppose $D' \coloneqq D + P$ for some P possibly in the support of D. Let π be a uniformiser for $\mathcal{O}_P$ and define $t = \pi^{v_P(D')}$. We then have a well-defined map

$$\begin{aligned} \psi : L(D') &\longrightarrow k(\mathfrak{m}_P) \\ x &\longrightarrow xt + \mathfrak{m}_P \end{aligned}$$

Observe that $\psi(x) = 0 \iff xt \in \mathfrak{m}_P \iff v_P(xt) > 0 \iff v_P(x) + v_P(D') > 0 \iff v_P(x) \geq -v_P(D) \iff x \in L(D)$. Therefore $\ker(\psi) = L(D)$ and so

$$\dim_k(L(D')/L(D)) \leq [k(\mathfrak{m}_P) : k] = \deg(D') - \deg(D)$$

Now suppose $D'' = D' + Q$, then by the isomorphism theorem for vector spaces (...)

$$\begin{aligned} \dim_k(L(D'')/L(D)) &= \dim_k(L(D'')/L(D')) + \dim_k(L(D')/L(D)) \\ &\leq \deg(D'') - \deg(D') + \deg(D') - \deg(D) \\ &= \deg(D'') - \deg(D) \end{aligned}$$

from which it follows the general case. ■

Proposition 6.3.7 (Dimension Upper Bound)
Let K/k be an algebraic function field and D a divisor. Then

$$\ell(D) \leq \deg(D_+) + \dim_k \tilde{k}$$

where D_+ is the positive part of D.

Proof. $L(D) \subset L(D_+)$ so it is enough to show the result for positive divisors. As $L(0) = \tilde{k}$ (...) the result follows from (6.3.6). ■

We say that the function $\deg(D) - \ell(D)$ is increasing, we also show it is bounded above.

Lemma 6.3.8 (Dimension Lower Bound I)
Let K/k be an algebraic function field. There exists a positive divisor B and integer $\gamma \in \mathbb{Z}$ such that

$$\deg(mB) - \ell(mB) \leq \gamma$$

for all positive integers m.

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Proof. Let $x \in K$ be transcendental and define $B \coloneqq (x)_\infty$ with $\deg(B) = n$. It's positive by (...). Let $u_1, ..., u_n$ be a $k(x)$-basis for K and a divisor $C \geq 0$ such that $(u_i) \geq -C$. Then we claim that for all integers m the elements $\left\{x^i u_j \;\middle|\; 0 \leq i \leq m, 1 \leq j \leq n\right\}$ are linearly independent over k. But this follows immediately because suppose

$$\sum_{i,j} \lambda_{ij} x^i u_j = 0$$

then

$$\sum_{j=1}^{n} \left(\sum_{i=0}^{m} \lambda_{ij} x^i \right) u_j = 0$$

and by linear independence we have $\lambda_{ij} = 0$.

Further $x^i \in L(iB) \implies x^i \in L(mB)$ and $u_j \in L(C) \implies x^i u_j \in L(mB + C)$. Therefore $\ell(mB + C) \geq n(m+1) = \deg(mB) + n$ for all positive integers m.

On the other hand by (6.3.6) we have $\ell(mB + C) \leq \ell(mB) + \deg(C)$. Therefore

$$\deg(mB) - \ell(mB) \leq \deg(C) - n$$

■

We may extend this result to all divisors by way of the following lemma.

Lemma 6.3.9
Let K/k be an algebraic function field, B a divisor satisfying the conclusion of (6.3.8) and D an arbitrary divisor. Then there exists divisors D', D'' such that $D \leq D'$, $D' \sim D''$ and $D'' \leq mB$ for some $m \geq 0$.

Proof. Choose a positive divisor $D' \geq D$, then

$$\deg(mB - D') - \ell(mB - D') \leq \deg(mB) - \ell(mB) \leq \gamma$$

for all positive integers m. This means

$$\ell(mB - D') \geq m \deg(B) - \deg(D') - \gamma$$

which is positive for sufficiently large m. Therefore we may choose $0 \neq z \in L(mB - D')$. Let $D'' = D' - (z)$. By definition

$$(z) \geq D' - mB \implies D'' \leq mB$$

as required. ■

Proposition 6.3.10 (Genus is well-defined)
Let K/k be an algebraic function field. Then there exists a positive integer γ such that

$$\deg(D) - \ell(D) \leq \gamma$$

for all divisors D.

Proof. Then using the notation of (6.3.9) we have

$$\begin{aligned}
\deg(D) - \ell(D) &\overset{(6.3.6)}{\leq} \deg(D') - \ell(D') \\
&\overset{(6.3.4)\,(6.3.5)}{=} \deg(D'') - \ell(D'') \\
&\overset{(6.3.6)}{\leq} \deg(mB) - \ell(mB) \\
&\overset{(6.3.8)}{\leq} \gamma
\end{aligned}$$

■

Definition 6.3.11 (Genus of a Function Field)
*Let K/k be an algebraic function field with $\tilde{k} = k$. Define the **genus** by*

$$g \coloneqq \max\{\deg(D) - \ell(D) + 1 \mid D \in \mathrm{Div}(K)\}$$

*Note as $\ell(0) = 1$ the genus g is non-negative. For any divisor $D \in \mathrm{Div}(K)$ define the **index of specialty** as follows*

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$$i(D) := \ell(D) - \deg(D) + g - 1$$

which by definition is a non-negative integer. This satisfies a few key properties (6.3.6)

a) $i(0) = g$
b) $i(D) \geq 0$
c) $D \sim D' \implies i(D) = i(D')$
d) $D' \geq D \implies i(D') \leq i(D)$

The Riemann-Roch theorem then takes the following form

$$\ell(D) - i(D) = \deg(D) - g + 1$$

We say that a divisor D is ***non-special*** *if $i(D) = 0$ (equivalently $\ell(D) = \deg(D) - g + 1$).*

The full Riemann-Roch amounts to an explicit form of $i(D)$. In order to do this we identify it as the dimension of a certain vector space.

Definition 6.3.12 (Repartitions)
Let K/k be an algebraic function field. Define the K-algebra of ***repartitions*** *of K by*

$$\mathcal{A}_K := \left\{ (\alpha_P)_{P \in \mathbb{P}_K} \;\middle|\; \alpha_P \in K \text{ and } \alpha_P \in \mathcal{O}_P \text{ for all but finitely many places } P \right\}$$

For every divisor $D \in \mathrm{Div}(K)$ define the K-subspace

$$\mathcal{A}_K(D) := \{\alpha \in \mathcal{A}_K \mid v_P(\alpha_P) \geq -v_P(D) \text{ for all } P \in \mathbb{P}_K\}$$

There is a canonical K-algebra homomorphism

$$K \hookrightarrow \mathcal{A}_K$$

which is well-defined by (...). Then clearly we have $K \cap \mathcal{A}_K(D) = L(D)$.

We prove a number of results

Lemma 6.3.13
Let K/k be an algebraic function field. For divisors D, D' such that $D \leq D'$ we have
a) $\dim_k \mathcal{A}_K(D')/\mathcal{A}_K(D) = \deg(D') - \deg(D)$
b) $\dim_k((\mathcal{A}_K(D') + K)/(\mathcal{A}_K(D) + K)) = i(D) - i(D')$

Proof. We prove each in turn.

a) By the isomorphism theorem for k-modules it's enough to demonstrate this for the case $D' = D + P$. Let π be a uniformiser for $\mathcal{O}_P$ and let $r := v_P(D) + 1 = v_P(D')$. Then define the k-linear map

$$\begin{aligned} \mathcal{A}_K(D') &\longrightarrow k(\mathfrak{m}_P) \\ \alpha &\longrightarrow (\pi^r \alpha_P)(P) \end{aligned}$$

By definition we have $v_P(\alpha_P) \geq -v_P(D') \implies v_P(\pi^r \alpha_P) \geq 0 \implies \pi^r \alpha_P \in \mathcal{O}_P$ so that this map is well-defined. Further $\pi^r \alpha_P \in \mathfrak{m}_P \iff v_P(\pi^r \alpha_P) > 0 \iff v_P(D) + 1 + v_P(\alpha_P) > 0 \iff v_P(\alpha_P) \geq -v_P(D)$ whence the kernel is precisely $\mathcal{A}_K(D)$. It is clearly surjective because given $f \in \mathcal{O}_P$ we may consider the repartition given by

$$(1, ..., 1, \pi^{-r} f, 1, ...)$$

which clearly maps to f under this map. Therefore by the isomorphism theorem we have

$$\dim_k \mathcal{A}_K(D')/\mathcal{A}_K(D) = [k(\mathfrak{m}_P) : k] = \deg(D') - \deg(D)$$

b) We claim there is an exact sequence of k-modules

$$0 \longrightarrow L(D')/L(D) \xrightarrow{\sigma_1} \mathcal{A}_K(D')/\mathcal{A}_K(D) \xrightarrow{\sigma_2} (\mathcal{A}_K(D') + K)/(\mathcal{A}_K(D) + K) \longrightarrow 0$$

from which the result follows immediately by comparing dimensions. The first map is given by

$$\sigma_1([f]) := (v_P(f))_{P \in \mathbb{P}_K}$$

which is obviously well-defined and injective. The map σ_2 is clearly well-defined and surjective. Further it is evident that $\sigma_2 \circ \sigma_1 = 0$. Therefore it only remains to show that $\ker(\sigma_2) \subset \mathrm{im}(\sigma_1)$. Suppose $\alpha \in \mathcal{A}_K(D')$ then $\sigma_2(\alpha) = 0 \implies$

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$\alpha - (f) \in \mathcal{A}_K(D) \implies (f) \in \mathcal{A}_K(D')$ for some $f \in K$. Then by definition this means $f \in L(D')$ and $\sigma_1(f) = \alpha$ as required. ∎

Proposition 6.3.14 (Riemann's Theorem)
*Let K/k be an algebraic function field. Suppose D_0 is **non-special** (there always exists such a divisor) then*
a) $\deg(D) > \ell(D_0) \implies D$ *is non-special*
b) $\mathcal{A}_K = \mathcal{A}_K(D_0) + K$

Proof. We prove each in turn
a) For an arbitrary divisor D we have

$$\begin{aligned}\ell(D - D_0) &= i(D - D_0) + \deg(D - D_0) - g + 1 \\ &\geq 0 + \deg(D) - \ell(D_0)\end{aligned}$$

So when $\deg(D) > \ell(D_0)$ we have $\ell(D - D_0) > 0$. Choose $0 \neq f \in L(D - D_0)$. Then we may define $D' \coloneqq D + (f) \geq D_0$. Then by (...) $i(D) = i(D') \leq i(D_0) = 0$ whence $i(D) = 0$.
b) Given $\alpha \in \mathcal{A}_K$ choose $D \geq D_0$ such that $\alpha \in \mathcal{A}_K(D)$. Then by (6.3.11) $i(D) \leq i(D_0) = 0$ and therefore $i(D) = 0$. By (6.3.13).b) $\mathcal{A}_K(D) + K = \mathcal{A}_K(D_0) + K \implies \alpha \in \mathcal{A}_K(D_0) + K$ as required.

∎

Proposition 6.3.15 (Explicit form of $i(D)$)
Let K/k be an algebraic function field and $D \in \mathrm{Div}(K)$. Then

$$i(D) = \dim_k(\mathcal{A}_K/(\mathcal{A}_K(D) + K))$$

Proof. By (6.3.14) there exists some non-special divisor $D' \geq D$ such that $i(D') = 0$ and $\mathcal{A}_K = \mathcal{A}_K(D') + K$. The formula then follows from (6.3.13).b). ∎

Definition 6.3.16 (Weil Divisors)
Let K/k be an algebraic function field and $D \in \mathrm{Div}(K)$. Define

$$\Omega_K(D) \coloneqq \{\omega \in \mathrm{Hom}_k(\mathcal{A}_K, k) \mid \mathcal{A}_K(D) + K \subset \ker(\omega)\}$$

*As divisors form a directed system we may form the union of **Weil Divisors***

$$\Omega_K \coloneqq \bigcup_{D \in \mathrm{Div}(K)} \Omega_K(D) \subset \mathrm{Hom}_k(\mathcal{A}_K, k)$$

Proposition 6.3.17
Let K/k be an algebraic function field. Then there is a canonical isomorphism

$$\mathrm{Hom}_k(\mathcal{A}_K/(\mathcal{A}_K(D) + K), k) \xrightarrow{\sim} \Omega_K(D)$$

In particular $i(D) = \dim_k \Omega_K(D)$. Taking $D = 0$, since $i(0) = g$ (6.3.11) we get

$$\dim_k \Omega_K(0) = g$$

that is, the genus equals the dimension of the space of everywhere-regular Weil differentials.

Proof. This is just the isomorphism theorem for vector spaces. The statement on dimensions follows from (6.3.15) and (3.3.109). ∎

6.3.3 Meromorphic Differentials in the case $k = \mathbb{C}$

In this section we aim to provide a post-hoc motivation for the introduction of Weil Differentials, by showing that they are canonically isomorphic to the classical space of Meromorphic Differentials in the case $k = \mathbb{C}$. This relies on some techniques which are specific to the complex-analytic structure of Riemann Surfaces.

Proposition 6.3.18 (Residue Theorem for $k = \mathbb{C}$)
Let p be a point of a Riemann surface and π a local holomorphic coordinate centered at p. For f meromorphic near p, define

$$\mathrm{Res}_\pi(f) \coloneqq \mathrm{Res}_p(f\,\mathrm{d}\pi) \coloneqq \frac{1}{2\pi i}\oint_\gamma f\,\mathrm{d}\pi$$

for a small loop γ around p – the classical residue of the 1-form $f\,\mathrm{d}\pi$. Writing $g \coloneqq f \circ \psi$ for the re-expansion of f relative to $\pi' = \psi^{-1}(\pi)$, and $f = \sum_n a_n \pi^n$ for the Laurent series expansion of f at p, then

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a) Res_π *is* k*-linear in* f*;*
b) $\mathrm{Res}_\pi(f) = 0$ *whenever* f *is holomorphic at* p *(i.e. has no pole there);*
c) $\mathrm{Res}_\pi(f) = a_{-1}$*;*
d) $\mathrm{Res}_{\pi'}(g \cdot \psi') = \mathrm{Res}_\pi(f)$*;*
e) if p *ranges over a* compact *Riemann surface* X *and* ω *is a meromorphic* 1*-form on* X *(so has only finitely many poles), then, writing* ω_π *for the coefficient of* ω *relative to a local coordinate* π *at* p *(so* $\omega = \omega_\pi \,\mathrm{d}\pi$ *near* p*, and* $\mathrm{Res}_\pi(\omega_\pi) = \mathrm{Res}_p(\omega)$ *by definition – independent of the choice of* π *at each* p*, by d)),*

$$\sum_p \mathrm{Res}_\pi(\omega_\pi) = 0$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

a). Immediate from linearity of the contour integral, $\oint_\gamma (\alpha f + \beta h)\,\mathrm{d}\pi = \alpha \oint_\gamma f\,\mathrm{d}\pi + \beta \oint_\gamma h\,\mathrm{d}\pi$ for $\alpha, \beta \in k$.

b). Immediate from c) below: f holomorphic at p means $a_n = 0$ for every $n < 0$, in particular $a_{-1} = 0$, so $\mathrm{Res}_\pi(f) = a_{-1} = 0$.

c). Term-by-term integration (valid by uniform convergence of the Laurent series on γ) gives $\oint_\gamma f\,\mathrm{d}\pi = \sum_n a_n \oint_\gamma \pi^n\,\mathrm{d}\pi$, and $\oint_\gamma \pi^n\,\mathrm{d}\pi = 2\pi i \delta_{n,-1}$ ($n \neq -1$: exact, $\pi^n\,\mathrm{d}\pi = \mathrm{d}\left(\frac{\pi^{n+1}}{n+1}\right)$, so integrates to 0 around a closed loop; $n = -1$: winding number 1 around $\pi = 0$), so $\mathrm{Res}_\pi(f) = \frac{1}{2\pi i}\oint_\gamma f\,\mathrm{d}\pi = a_{-1}$.

d). By the chain rule, $f\,\mathrm{d}\pi = f(\psi(\pi'))\psi'(\pi')\,\mathrm{d}\pi' = (g \cdot \psi')\,\mathrm{d}\pi'$: these are the same 1-form ω, written in the two coordinates π and π'. The contour integral $\oint_\gamma \omega$ depends only on ω and γ, not on the coordinate used to write it down – reparametrising γ by π' instead of π changes nothing about which loop is being integrated over – so $\mathrm{Res}_p(\omega)$ computed from either expression gives the same number:

$$\mathrm{Res}_{\pi'}(g \cdot \psi') = \mathrm{Res}_p((g \cdot \psi')\,\mathrm{d}\pi') = \mathrm{Res}_p(\omega) = \mathrm{Res}_p(f\,\mathrm{d}\pi) = \mathrm{Res}_\pi(f)$$

e). This uses two facts special to the compact setting, neither purely local (unlike a)–d) above): ω is **closed** away from its poles ($\omega = f\,\mathrm{d}z$ locally with f holomorphic, so $\mathrm{d}\omega = \frac{\partial f}{\partial \overline{z}}\,\mathrm{d}\overline{z} \wedge \mathrm{d}z = 0$ by the Cauchy–Riemann equations), and X is **compact**. Removing a small disk D_p around each pole leaves $X_0 := X \setminus \bigcup_p D_p$, on which ω is holomorphic, so $\mathrm{d}\omega = 0$ there; Stokes' theorem gives

$$0 = \int_{X_0} \mathrm{d}\omega = \int_{\partial X_0} \omega = -\sum_p \oint_{\partial D_p} \omega = -\sum_p 2\pi i\,\mathrm{Res}_\pi(\omega_\pi)$$

(the sign, as ∂X_0 traverses each ∂D_p oppositely to the usual counterclockwise orientation used to define Res_π, and $\oint_{\partial D_p} \omega = 2\pi i\,\mathrm{Res}_\pi(\omega_\pi)$ by definition), giving $\sum_p \mathrm{Res}_\pi(\omega_\pi) = 0$. ∎

Proposition 6.3.19 (Classical Differentials are Weil Differentials)
Let X *be a compact Riemann surface with function field* K *(so* $k = \mathbb{C}$*), and let* $\mathcal{M}^1(X)$ *denote the* $\mathbb{C}$*-vector space of meromorphic* 1*-forms on* X*. For* $\omega \in \mathcal{M}^1(X)$*, define* $\varphi_\omega \in \mathrm{Hom}_k(\mathcal{A}_K, k)$ *by*

$$\varphi_\omega(\alpha) := \sum_P \mathrm{Res}_\pi(\alpha_P \cdot \omega_\pi), \quad \alpha = (\alpha_P)_P \in \mathcal{A}_K$$

(a finite sum, by b): $\alpha_P \omega_\pi$ *is holomorphic, hence contributes* 0*, at all but finitely many* P*). Then* $\varphi_\omega \in \Omega_K$*, and*

$$\mathcal{M}^1(X) \longrightarrow \Omega_K, \quad \omega \mapsto \varphi_\omega$$

is a canonical $\mathbb{C}$*-linear isomorphism.*

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

$\boldsymbol{\varphi_\omega \in \Omega_K}$**.** Let $D := (\omega)$, the divisor of ω (well-defined: $v_P(\omega_\pi)$ is independent of the choice of π at P, as $\omega_{\pi'} = (\omega_\pi \circ \psi) \cdot \psi'$ by the chain rule – the same identity underlying d) – and ψ' is a unit, so composition with ψ and multiplication by ψ' both preserve order of vanishing). We show $\mathcal{A}_K(D) + K \subset \ker(\varphi_\omega)$. For $\alpha \in \mathcal{A}_K(D)$, $v_P(\alpha_P) \geq -v_P(D) = -v_P(\omega_\pi)$ for every P, so $v_P(\alpha_P \omega_\pi) \geq 0$: each $\alpha_P \omega_\pi$ is holomorphic at P, so $\mathrm{Res}_\pi(\alpha_P \omega_\pi) = 0$ by b), giving $\varphi_\omega(\alpha) = 0$. For $f \in K$ (embedded diagonally, $\alpha_P = f$ for every P), $\varphi_\omega(f, f, ...) = \sum_P \mathrm{Res}_\pi(f \cdot \omega_\pi) = \sum_P \mathrm{Res}_\pi((f\omega)_\pi) = 0$ by e) (the Residue Theorem), applied to the meromorphic 1-form $f\omega \in \mathcal{M}^1(X)$. So φ_ω vanishes on $\mathcal{A}_K(D) + K$, that is $\varphi_\omega \in \Omega_K(D) \subset \Omega_K$.

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$\mathbb{C}$-linearity. Immediate from a): $\varphi_{\lambda\omega_1+\mu\omega_2}(\alpha) = \sum_P \mathrm{Res}_\pi\big(\alpha_P(\lambda\omega_1+\mu\omega_2)_\pi\big) = \lambda\varphi_{\omega_1}(\alpha) + \mu\varphi_{\omega_2}(\alpha)$ for $\lambda,\mu\in\mathbb{C}$, using that $(\lambda\omega_1+\mu\omega_2)_\pi = \lambda(\omega_1)_\pi + \mu(\omega_2)_\pi$ (coefficients relative to a fixed π add linearly) and linearity of Res_π in its argument.

Injectivity. Suppose $\varphi_\omega = 0$; we show $\omega = 0$. Fix P and, for $n\in\mathbb{Z}$, let $\alpha^{(n)}\in\mathcal{A}_K$ be the repartition with $\alpha^{(n)}_P := \pi^{-n}$ and $\alpha^{(n)}_Q := 0$ for $Q\neq P$ (a valid repartition: it is holomorphic away from the single place P). Writing $\omega_\pi = \sum_m a_m\pi^m$, $0 = \varphi_\omega(\alpha^{(n)}) = \mathrm{Res}_\pi(\pi^{-n}\omega_\pi) = a_{n-1}$ by c). As n was arbitrary, every coefficient of ω_π vanishes, so ω is 0 near P; as P was arbitrary, $\omega\equiv 0$ on X.

Surjectivity. *Sketch.* This needs the classical Mittag-Leffler theorem for meromorphic differentials: given principal parts (polar Laurent tails) prescribed at finitely many points of X, satisfying the residue-sum-zero constraint of e), there exists a global meromorphic 1-form on X realising them – solved via the $\overline{\partial}$-equation (Hodge theory / finite-dimensionality of $H^1(X,\mathcal{O})$), genuinely global analytic content with no counterpart among a)–e) above. Given $\psi\in\Omega_K(D)$, read off, at each pole P of D, the local principal part ψ demands by testing against the single-place repartitions $\alpha^{(n)}_P = \pi^{-n}$ used in the injectivity argument above; since ψ vanishes on K by construction, these prescribed principal parts automatically satisfy the residue-sum-zero compatibility condition Mittag-Leffler requires, and it produces a global ω realising them, with $\varphi_\omega = \psi$. ∎

6.3.4 Derivations on the Function Field

Proposition 6.3.20 (A Distinguished Basis for $\mathrm{Der}_k(K,K)$)
Let $x\in K$ be transcendental over k and assume k is perfect. Then there is a unique $\partial_x\in\mathrm{Der}_k(K,K)$ with $\partial_x(x) = 1$, and $D = D(x)\cdot\partial_x$ for every $D\in\mathrm{Der}_k(K,K)$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

By (3.34.44), $\dim_K \mathrm{Der}_k(K) = 1$. So there is at least one non-zero D_0. If $D_0(x) = 0$ then it vanishes on $k(x)$ by (3.25.9), (3.25.10) and therefore on K as $K/k(x)$ is separable (3.18.103) (3.34.42), a contradiction. Therefore $D_0(x)\neq 0$ for any non-zero derivation D_0. Therefore define $\partial_x := D_0(x)^{-1}D$ and by monogenicity we are done. ∎

Proposition 6.3.21 (Uniformisers are Transcendental)
Let P be a place of K/k, π a uniformiser. Then π is transcendental over k.

Proof. By (...) $k\cap\mathfrak{m}_P = (0)$ whence $\pi\notin k$. By assumption $\tilde{k} = k$ whence π is transcendental. ∎

Proposition 6.3.22 (Chain Rule for Uniformiser Derivations)
Let $x,x'\in K$ be transcendental. Then

$$\partial_{x'}(t) = \partial_{x'}(x)\cdot\partial_x(t) \quad \text{for every } t\in K.$$

Proof. This is immediate from (6.3.20). ∎

6.3.5 Residue Symbol at a Place

The residue symbol $\mathrm{Res}_P(z\,\mathrm{d}t)$ built in this subsection is defined via the Laurent coefficients of $z\partial_\pi(t)$ relative to a choice of uniformiser π, and shown independent of that choice (6.3.31) – exactly the invariance a genuine differential-form residue ought to have. This invariance is not straightforward in characteristic p: it reduces (6.3.31) to computing $(\psi^n\psi')_{-1}$ for the change-of-uniformiser series ψ, an identity which the naive derivative-of-a-power argument only settles when $n+1$ is invertible – automatic in characteristic 0, but not when $p\mid(n+1)$. We settle it uniformly in every characteristic by proving the identity once, universally, over $\mathbb{Z}[a_1,a_1^{-1},a_2,a_3,\ldots]$ (indeterminate coefficients, torsion-free over $\mathbb{Z}$) and specialising (6.3.26) – the same universal-$\mathbb{Z}$-polynomial philosophy used, e.g., in [Sti09] (Chap. 4, Prop. 4.2.9); another route entirely avoiding a characteristic-p detour is via finite-potent operators [Tat68] (§§1–2).

Throughout this subsection k is perfect, so the residue field $k(P)$ of every place P of K/k is a finite separable extension of k (3.18.103) and the coefficient expansion (3.32.17) embeds K in $k(P)((X))$. The residue of $z\,\mathrm{d}t$ at P is first a $k(P)$-valued Laurent coefficient, and the residue symbol $\mathrm{Res}_P(z\,\mathrm{d}t)\in k$ is its image under the trace $\mathrm{Tr}_{k(P)/k}$. At a *rational* place, that is $k(P) = k$, the trace is the identity and this is the residue of the classical case, a k-valued Laurent coefficient.

Proposition 6.3.23 (Laurent Coefficients at a Place)
Let P be a place of K/k, π a uniformiser of $\mathcal{O}_P$ and $k(P) := \mathcal{O}_P/\mathfrak{m}_P$ (a finite separable extension of k, as k is perfect). There is a unique k-algebra homomorphism

$$L_\pi : K\to k(P)((X))$$

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*with $L_\pi(\pi)=X$ which preserves valuation, $v_X(L_\pi(z))=v_P(z)$ for every $z\in K^\star$ (v_X being the valuation of the DVR $k(P)[[X]]$ (3.32.16)), and induces the identity on the residue field: $L_\pi(u)\equiv\overline{u}\pmod X$ for $u\in\mathcal{O}_P$. Moreover L_π is injective and $z\in\mathcal{O}_P\iff L_\pi(z)\in k(P)[[X]]$ for every $z\in K$. Writing $L_\pi(z)=\sum_j a_jX^j$ for $z\in K^\star$ (the **Laurent coefficients** $a_j\in k(P)$ of z at P relative to π): $a_j=0$ for $j<v_P(z)$ and $a_{v_P(z)}\neq 0$.*

The Laurent coefficients are compatible with the k-algebra sections $s_N:k(P)\to\mathcal{O}_P/\mathfrak{m}_P^{N+1}$ of (3.25.18), which do not depend on π. If $u\in\mathcal{O}_P$ and $L_\pi(u)=\sum_i c_iX^i$, then for every $N\geq 0$

$$u\equiv\sum_{i\leq N}s_N(c_i)\pi^i\pmod{\mathfrak{m}_P^{N+1}},$$

and for $c\in k(P)$ and every $c'\in\mathcal{O}_P$ with $c'\equiv s_N(c)\pmod{\mathfrak{m}_P^{N+1}}$ we have $L_\pi(c')\equiv c\pmod{X^{N+1}}$, for every uniformiser π.

If P is rational then $k(P)=k$, the s_N are induced by $k\subseteq\mathcal{O}_P$ and $L_\pi:K\to k((X))$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

As $k(P)/k$ is finite (6.2.114) and k is perfect, $k(P)/k$ is separable algebraic (3.18.103), so (3.32.17) applies to $A:=\mathcal{O}_P$, with the sections s_N of (3.25.18). It gives an injective k-algebra homomorphism $L_\pi:K\to k(P)((X))$ with $L_\pi(\pi)=X$ which on $\mathcal{O}_P$ is given by the digits $c_i(u)$, $u\equiv\sum_{i\leq N}s_N(c_i(u))\pi^i\pmod{\mathfrak{m}_P^{N+1}}$.

Valuation and residue. $v_X(L_\pi(u))=v_P(u)$ for $0\neq u\in\mathcal{O}_P$ is the valuation formula of that proposition, and for $z=a/s\in K^\star$ ($a,s\in\mathcal{O}_P$), $L_\pi(z)=L_\pi(a)L_\pi(s)^{-1}$, so $v_X(L_\pi(z))=v_P(a)-v_P(s)=v_P(z)$ (multiplicativity of v_X (3.32.15)). In particular $z\in\mathcal{O}_P\iff v_P(z)\geq 0\iff L_\pi(z)\in k(P)[[X]]$, and $a_j=0$ for $j<v_P(z)$ with $a_{v_P(z)}\neq 0$. For $u\in\mathcal{O}_P$ the case $N=0$ of the displayed congruence reads $u\equiv s_0(c_0(u))=c_0(u)\pmod{\mathfrak{m}_P}$ (s_0 is the identity of $k(P)$), that is $c_0(u)=\overline{u}$: L_π induces the identity on residue fields.

Compatibility with the sections. Let $c\in k(P)$ and $c'\in\mathcal{O}_P$ with $c'\equiv s_N(c)\pmod{\mathfrak{m}_P^{N+1}}$. Then $(c,0,\dots,0)$ satisfies the defining congruence for c' at level N, so by uniqueness of the digits $c_0(c')=c$ and $c_i(c')=0$ for $1\leq i\leq N$, that is $L_\pi(c')\equiv c\pmod{X^{N+1}}$. The s_N are unique, hence independent of π.

Uniqueness. Let $\phi:K\to k(P)((X))$ be a valuation-preserving k-algebra homomorphism with $\phi(\pi)=X$ inducing the identity on the residue field. For $N\geq 0$ it maps $\mathcal{O}_P$ into $k(P)[[X]]$ and $\mathfrak{m}_P^{N+1}$ into $X^{N+1}k(P)[[X]]$, so induces a k-algebra homomorphism $\phi_N:\mathcal{O}_P/\mathfrak{m}_P^{N+1}\to k(P)[X]/(X^{N+1})$. Then $\phi_N\circ s_N:k(P)\to k(P)[X]/(X^{N+1})$ is a k-algebra homomorphism which reduces to the identity of $k(P)$ modulo X (as ϕ induces the identity on residue fields), as does the inclusion of the constants. By the uniqueness of such sections (3.25.18) (applied to the local k-algebra $k(P)[X]/(X^{N+1})$) they agree: $\phi_N(s_N(c))=c$. So for $u\in\mathcal{O}_P$ with digits c_i,

$$\phi(u)\equiv\sum_{i\leq N}\phi_N(s_N(c_i))X^i=\sum_{i\leq N}c_iX^i\equiv L_\pi(u)\pmod{X^{N+1}}$$

for every N, so $\phi=L_\pi$ on $\mathcal{O}_P$ and hence on $K=\mathrm{Frac}(\mathcal{O}_P)$. ∎

Definition 6.3.24 (Residue Symbol of a Laurent Series)
Let A be a commutative ring, π an indeterminate, and $z,t\in A((\pi))$. Write ∂_π for the formal derivative, $\partial_\pi\left(\sum_n a_n\pi^n\right):=\sum_n na_n\pi^{n-1}$, and define

$$\mathrm{Res}_\pi(z\,\mathrm{d}t):=(z\partial_\pi(t))_{-1}\in A,$$

the coefficient of π^{-1} in $z\partial_\pi(t)$.

Definition 6.3.25 (Residue Symbol at a Place)
Let P be a place of K/k, π a uniformiser, and $t,z\in K$. Let $L_\pi:K\hookrightarrow k(P)((X))$ be the Laurent coefficient embedding (6.3.23). Define the $k(P)$-valued residue

$$\mathrm{Res}_P^{k(P)}(z\,\mathrm{d}t):=\mathrm{Res}_\pi(L_\pi(z)\,\mathrm{d}L_\pi(t))\in k(P),$$

*where Res_π is the residue of (6.3.24) with $A:=k(P)$, and $\mathrm{d}L_\pi(t):=\partial_\pi(L_\pi(t))\,\mathrm{d}\pi$, and the **residue symbol***

$$\mathrm{Res}_P(z\,\mathrm{d}t):=\mathrm{Tr}_{k(P)/k}\left(\mathrm{Res}_P^{k(P)}(z\,\mathrm{d}t)\right)\in k.$$

Since L_π commutes with the uniformiser derivation, $L_\pi(\partial_\pi(t))=\partial_\pi(L_\pi(t))$ (6.3.29), one has explicitly

$$\mathrm{Res}_P^{k(P)}(z\,\mathrm{d}t)=L_\pi(z\partial_\pi(t))_{-1}=(L_\pi(z)\partial_\pi(L_\pi(t)))_{-1}.$$

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If P is rational then $k(P) = k$, the trace is the identity and $\mathrm{Res}_P = \mathrm{Res}_P^k$ is the k-valued Laurent coefficient. We show that both are independent of the choice of uniformiser π below (6.3.31).

Lemma 6.3.26 (Residue of $\psi^n\psi'$)
Let F be a field and $\psi \in XF[[X]]$ with $\psi'(0) \neq 0$ (equivalently, ψ has order exactly 1). For every $n \in \mathbb{Z}$,

$$(\psi^n\psi')_{-1} = \delta_{n,-1}$$

Remark 6.3.27
The simplest proof using the identity $(n+1)\psi^n\psi' = (\psi^{n+1})'$ fails when $\mathrm{char}(k) = p \mid n+1$, so some additional work is required.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

$n \geq 0$. $\psi \in Xk[[X]]$ has no negative powers, so neither do ψ^n or ψ'; their product has no X^{-1} term, giving 0, matching $\delta_{n,-1} = 0$.

$n = -1$. Write $\psi(X) = Xv(X)$ with $v \in k[[X]]^\star$ a unit ($v(0) = \psi'(0) \neq 0$). Then $\psi' = v + Xv'$, so

$$\psi^{-1}\psi' = \frac{v + Xv'}{Xv} = \frac{1}{X} + \frac{v'}{v}$$

The second term is a genuine power series (product of v' and the unit v^{-1}, both in $k[[X]]$), so has zero -1-coefficient; the first contributes $(X^{-1})_{-1} = 1$. Total: 1, matching $\delta_{-1,-1}$.

$n \leq -2$: a universal identity. Fix $n \leq -2$; we prove the identity over a universal coefficient ring first, then specialise (see the remark above for why). Let $R := \mathbb{Z}[a_1, a_2, a_3, \ldots][a_1^{-1}]$, the polynomial ring over $\mathbb{Z}$ in countably many indeterminates $a_1, a_2, \ldots$, localised to invert a_1 – an integral domain of characteristic 0 (a localisation of the domain $\mathbb{Z}[a_1, a_2, \ldots]$, itself a domain as a polynomial ring over the domain $\mathbb{Z}$, (3.7.3), applied to finitely many variables at a time since any product only ever involves finitely many of them). Let $\Psi := \sum_{i\geq 1} a_i X^i \in XR[[X]]$, so $\Psi = a_1 X \cdot U$ for $U := 1 + \sum_{i\geq 2}(a_i/a_1)X^{i-1} \in 1 + XR[[X]]$; as U has constant coefficient $1 \in R^\star$, (3.32.13) gives $U \in R[[X]]^\star$, and hence $\Psi \in R((X))^\star$ too ($a_1 \in R^\star$, and X a unit of $R((X))$ directly by construction of the formal Laurent series ring, (3.32.1)). So $\Psi^n\Psi' \in R((X))$ is well-defined, with a well-defined coefficient of X^{-1}, which we still write $(\Psi^n\Psi')_{-1} \in R$.

As before (the power rule and vanishing of a derivative's X^{-1}-coefficient being valid over any commutative ring, not just k), $(n+1)\Psi^n\Psi' = (\Psi^{n+1})'$ ((3.32.19), Ψ being a unit) has vanishing X^{-1}-coefficient:

$$(n+1)\cdot(\Psi^n\Psi')_{-1} = \left((\Psi^{n+1})'\right)_{-1} = 0 \quad \text{in } R$$

As $n \leq -2$, $n+1$ is a nonzero integer; its image in R is therefore nonzero, R having characteristic 0. So $(n+1)\cdot(\Psi^n\Psi')_{-1} = 0$ forces $(\Psi^n\Psi')_{-1} = 0$ in R, by cancellation in the domain R.

Specialising. Write $\psi = \sum_{i\geq 1} c_i X^i$, so $c_1 = \psi'(0) \neq 0$ by hypothesis. As $c_1 \in k^\star$, the assignment $a_1 \mapsto c_1$, $a_i \mapsto c_i$ ($i \geq 2$) extends, by the universal property of localisation (3.5.5) (applied to $\mathbb{Z}[a_1, a_2, \ldots] \to k$, $a_1 \mapsto c_1$ a unit), to a ring homomorphism $\phi : R \to k$. Applying ϕ coefficientwise gives a ring homomorphism $\hat{\phi} : R((X)) \to k((X))$ (directly from the convolution formula for products, (3.32.1)) with $\hat{\phi}(\Psi) = \psi$, compatible with the formal derivative ($\hat{\phi}(f)' = \hat{\phi}(f')$, immediate from the coefficientwise formulas $f' = \sum_m mc_m X^{m-1}$ and $\phi(mc_m) = m\phi(c_m)$) and with the coefficient of X^{-1} (literally the same index, carried along by ϕ). So

$$(\psi^n\psi')_{-1} = \hat{\phi}(\Psi^n\Psi')_{-1} = \phi\big((\Psi^n\Psi')_{-1}\big) = \phi(0) = 0$$

using $(\Psi^n\Psi')_{-1} = 0$ in R (just shown), matching $\delta_{n,-1} = 0$ for $n \leq -2$. ∎

Lemma 6.3.28 (Laurent Expansions Compose Across Uniformisers)
Let P be a place of K/k, π, π' uniformisers at P, and $\psi := L_{\pi'}(\pi) \in Xk(P)[[X]]$.

Then for every $w \in K$,

$$L_{\pi'}(w) = L_\pi(w) \circ \psi$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

We may assume $w \neq 0$. Write $f := L_\pi(w) = \sum_j a_j X^j$ and $m := v_P(w)$, so $a_j = 0$ for $j < m$. As $v(\psi) = 1$ and the order function is multiplicative on the field $k(P)((X))$ (3.32.15), $v(\psi^j) = j$ for every j. So the family $h_i := a_{m+i}\psi^{m+i}$ ($i \geq 0$) is summable (3.32.9), as $v(h_i) \geq m + i$, and by definition of the substitution (3.32.20) $f \circ \psi = \sum_{j\geq m} a_j\psi^j = \sum_{i\geq 0} h_i$. By the Convergence Criterion (3.32.12) it suffices to show that for every $n \geq m$

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$$L_{\pi'}(w) - \sum_{j=m}^{n} a_j \psi^j \in X^{n+1} k(P)[[X]], \qquad (\star)$$

since then the partial sums $\sum_{i \le n-m} h_i$ converge to $L_{\pi'}(w)$, and so $L_{\pi'}(w) = \sum_i h_i = f \circ \psi$.

Proof of ($\star$) for a rational place. Here $k(P) = k \subseteq \mathcal{O}_P$, so $a_j \in k \subseteq \mathcal{O}_P$. Write $w = \pi^m u$ with $u \in \mathcal{O}_P^\star$; the digits of u are $c_i = a_{m+i}$ ((6.3.23)), so for $n \ge m$

$$w - \sum_{j=m}^{n} a_j \pi^j = \pi^m \left(u - \sum_{i \le n-m} a_{m+i} \pi^i \right) \in \pi^m \mathfrak{m}_P^{n-m+1} = \mathfrak{m}_P^{n+1}.$$

Apply $L_{\pi'}$: it preserves valuation, so the left side maps into $X^{n+1}k[[X]]$ (6.3.23) it is a k-algebra homomorphism of fields with $L_{\pi'}(\pi) = \psi$, so it sends $\pi^j \mapsto \psi^j$ for every $j \in \mathbb{Z}$ (3.2.6) and fixes each $a_j \in k$. Hence $L_{\pi'}(w) - \sum_{j=m}^{n} a_j \psi^j \in X^{n+1}k[[X]]$, which is ($\star$).

Proof of ($\star$) for a general place. The coefficients $a_j \in k(P)$ need not lie in $\mathcal{O}_P$, so we replace them by lifts through the sections s_N of (6.3.23). Fix $n \ge m$ and put $N := n - m$. Write $w = \pi^m u$ with $u \in \mathcal{O}_P^\star$, with digits $c_i = a_{m+i}$, and choose $\tilde{a}_j \in \mathcal{O}_P$ with $\tilde{a}_j \equiv s_N(a_j) \pmod{\mathfrak{m}_P^{N+1}}$ for $m \le j \le n$. Then $u \equiv \sum_{i \le N} s_N(c_i)\pi^i \pmod{\mathfrak{m}_P^{N+1}}$ gives, as before,

$$w - \sum_{j=m}^{n} \tilde{a}_j \pi^j = \pi^m \left(u - \sum_{i \le N} \tilde{a}_{m+i} \pi^i \right) \in \pi^m \mathfrak{m}_P^{N+1} = \mathfrak{m}_P^{n+1}.$$

Applying $L_{\pi'}$ exactly as in the rational case,

$$L_{\pi'}(w) - \sum_{j=m}^{n} L_{\pi'}(\tilde{a}_j) \psi^j \in X^{n+1} k(P)[[X]].$$

It remains to replace $L_{\pi'}(\tilde{a}_j)$ by a_j. By the compatibility with the sections (6.3.23) (valid for every uniformiser, in particular π') we have $L_{\pi'}(\tilde{a}_j) \equiv a_j \pmod{X^{N+1}}$, and $v(\psi^j) = j \ge m$, so $(L_{\pi'}(\tilde{a}_j) - a_j)\psi^j \in X^{N+1+j}k(P)[[X]] \subseteq X^{n+1}k(P)[[X]]$. This gives ($\star$). ∎

Corollary 6.3.29 (Laurent Coefficients of the Uniformiser Derivation)
Let P be a place of K/k, π a uniformiser. For every $w \in K$,

$$L_\pi(\partial_\pi(w)) = \frac{\mathrm{d}}{\mathrm{d}X} L_\pi(w)$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Let $M := k(P)((X))$, the target of $L_\pi : K \hookrightarrow k(P)((X))$ (6.3.23), viewed as a K-module through L_π, a k-algebra homomorphism (6.3.23). By (3.34.44) (with $n = 1$, transcendence basis π (6.3.21)), $\mathrm{Der}_k(K, M) \cong M$ via $D \mapsto D(\pi)$, so a k-derivation $K \to M$ is uniquely determined by its value at π.

Both $D_1 := L_\pi \circ \partial_\pi$ and $D_2 := \frac{\mathrm{d}}{\mathrm{d}X} \circ L_\pi$ are k-derivations $K \to M$: for D_1, $L_\pi(\partial_\pi(xy)) = L_\pi(x\partial_\pi(y) + y\partial_\pi(x)) = x \cdot D_1(y) + y \cdot D_1(x)$, using additivity and multiplicativity of L_π; for D_2, the same identity holds by the Leibniz rule for formal differentiation together with $L_\pi(xy) = L_\pi(x)L_\pi(y)$; and $D_1(\lambda) = L_\pi(\partial_\pi(\lambda)) = 0$, $D_2(\lambda) = 0$ for $\lambda \in k$, as $L_\pi(\lambda)$ is a constant of $k(P)((X))$. Both send $\pi \mapsto 1$ ($L_\pi(\partial_\pi(\pi)) = L_\pi(1) = 1$, $\frac{\mathrm{d}}{\mathrm{d}X}(L_\pi(\pi)) = \frac{\mathrm{d}}{\mathrm{d}X}(X) = 1$), so $D_1 = D_2$, that is $L_\pi(\partial_\pi(w)) = \frac{\mathrm{d}}{\mathrm{d}X} L_\pi(w)$ for every $w \in K$. ∎

Proposition 6.3.30 (The Uniformiser Derivation is Regular)
Let P be a place of K/k, π a uniformiser. Then $\partial_\pi(\mathcal{O}_P) \subseteq \mathcal{O}_P$.

Proof. This essentially follows from (6.3.23) and the fact that $\frac{\mathrm{d}}{\mathrm{d}X}$ preserves $k(P)[[X]]$. ∎

Corollary 6.3.31 (Well-Definedness of the Residue Symbol at a Place)
Let P be a place of K/k, π, π' uniformisers at P, and $z, t \in K$. Then

$$L_\pi(z\partial_\pi(t))_{-1} = L_{\pi'}(z\partial_{\pi'}(t))_{-1}$$

an equality in $k(P)$; that is, $\mathrm{Res}_P^{k(P)}(z\,\mathrm{d}t)$, and hence $\mathrm{Res}_P(z\,\mathrm{d}t)$, of (6.3.25) does not depend on the choice of uniformiser π used to compute it.

Proof. Write $w := z\partial_\pi(t)$ and $\psi := L_{\pi'}(\pi) \in Xk(P)[[X]]$.

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$\psi'(0) \neq 0$. Writing $\psi = \sum_j a_j X^j$ ($a_j \in k(P)$), this is the Laurent expansion of $\pi \in K^\star$ relative to π' (6.3.23), so $a_j = 0$ for $j < v_P(\pi)$ and $a_{v_P(\pi)} \neq 0$; as $v_P(\pi) = 1$ (π is itself a uniformiser), $a_0 = 0$ and $a_1 \neq 0$. The constant term of $\psi' = \sum_j j a_j X^{j-1}$ is its $j = 1$ term, $1 \cdot a_1 = a_1 \neq 0$, that is $\psi'(0) \neq 0$.

$L_\pi(w)_{-1} = (L_{\pi'}(w) \cdot \psi')_{-1}$. Write $f := L_\pi(w)$ and $g := f \circ \psi$ (well-defined, (3.32.20), as ψ has invertible linear term, just shown). Both sides of $f_{-1} = (g \cdot \psi')_{-1}$ are $k(P)$-linear in f (composition with ψ, multiplication by ψ', and $(-)_{-1}$ are all $k(P)$-linear), so it suffices to check $f = X^n$, $n \in \mathbb{Z}$, that is $(\psi^n \psi')_{-1} = \delta_{n,-1}$ – exactly (6.3.26). By (6.3.28), $g = L_\pi(w) \circ \psi = L_{\pi'}(w)$, giving $L_\pi(w)_{-1} = f_{-1} = (g \cdot \psi')_{-1} = (L_{\pi'}(w) \cdot \psi')_{-1}$.

By (6.3.29) (applied relative to π'), $L_{\pi'}(\partial_{\pi'}(\pi)) = \frac{\mathrm{d}}{\mathrm{d}X} L_{\pi'}(\pi) = \psi'$. As $L_{\pi'}$ is a ring homomorphism (6.3.23),

$$L_\pi(z\partial_\pi(t))_{-1} = L_\pi(w)_{-1} = (L_{\pi'}(w) \cdot L_{\pi'}(\partial_{\pi'}(\pi)))_{-1} = (L_{\pi'}(w\partial_{\pi'}(\pi)))_{-1}.$$

By (6.3.22), $\partial_\pi(t)\partial_{\pi'}(\pi) = \partial_{\pi'}(t)$, so $w\partial_{\pi'}(\pi) = z\partial_\pi(t)\partial_{\pi'}(\pi) = z\partial_{\pi'}(t)$, giving $L_\pi(z\partial_\pi(t))_{-1} = L_{\pi'}(z\partial_{\pi'}(t))_{-1}$. ∎

Lemma 6.3.32 (Residues are Invariant under Isomorphism)
Let $\varphi : K \to K'$ be an isomorphism of k-algebras between algebraic function fields K/k, K'/k, P a rational place of K/k and P' the place of K'/k with $\mathcal{O}_{P'} = \varphi(\mathcal{O}_P)$. Then $\mathrm{Res}_{P'}(\varphi(z)\,\mathrm{d}\varphi(t)) = \mathrm{Res}_P(z\,\mathrm{d}t)$ for all $z, t \in K$.

Proposition 6.3.33 (Elementary Properties of the Residue Symbol at a Place)
Let P be a place of K/k, π a uniformiser, $t \in K$. Then

a) $z \mapsto \mathrm{Res}_P^{k(P)}(z\,\mathrm{d}t)$ is k-linear $K \to k(P)$, and $z \mapsto \mathrm{Res}_P(z\,\mathrm{d}t)$ is k-linear $K \to k$;

b) for $x \in K$ transcendental over k and $\partial_x \in \mathrm{Der}_k(K, K)$ the distinguished derivation of (6.3.20), $\mathrm{Res}_P^{k(P)}(z\,\mathrm{d}t) = \mathrm{Res}_P^{k(P)}((z \cdot \partial_x(t))\,\mathrm{d}x)$ and $\mathrm{Res}_P(z\,\mathrm{d}t) = \mathrm{Res}_P((z \cdot \partial_x(t))\,\mathrm{d}x)$ for every $z \in K$;

c) if $z\partial_\pi(t) \in \mathcal{O}_P$ then $\mathrm{Res}_P^{k(P)}(z\,\mathrm{d}t) = 0$ and $\mathrm{Res}_P(z\,\mathrm{d}t) = 0$; in particular this holds if $z, t \in \mathcal{O}_P$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

a) By (6.3.25), $\mathrm{Res}_P^{k(P)}(z\,\mathrm{d}t) = L_\pi(z\partial_\pi(t))_{-1}$; k-linearity in z follows from that of ∂_π, of L_π (6.3.23), and of coefficient extraction (6.3.24). The trace is k-linear.

b) By (6.3.20) (applied to $D := \partial_\pi$), $\partial_\pi = \partial_\pi(x) \cdot \partial_x$, so $z\partial_\pi(t) = (z \cdot \partial_x(t))\partial_\pi(x)$; applying $L_\pi(-)_{-1}$ to both sides ((6.3.25), once with t and once with x) gives the first identity, and applying $\mathrm{Tr}_{k(P)/k}$ the second.

c) If $z\partial_\pi(t) \in \mathcal{O}_P$ then $L_\pi(z\partial_\pi(t)) \in k(P)[[X]]$ (6.3.23), so its X^{-1}-coefficient vanishes, that is $\mathrm{Res}_P^{k(P)}(z\,\mathrm{d}t) = 0$, and so is its trace. If $z, t \in \mathcal{O}_P$ then $\partial_\pi(t) \in \mathcal{O}_P$ (6.3.30), so $z\partial_\pi(t) \in \mathcal{O}_P$. ∎

6.3.6 The Algebraic Residue Theorem over an Algebraically Closed Field

Assume throughout this subsection that k is algebraically closed. Then every place of K/k is rational (6.2.114), so the entire theory of (6.3.25) applies directly, with no need for the base-change-to-$\overline{k}$ machinery of the following subsections (which is needed only once k is no longer assumed algebraically closed).

Proposition 6.3.34 (Algebraic Residue Theorem, over an Algebraically Closed Field)
Assume k is algebraically closed. For $t, f \in K$,

$$\sum_P \mathrm{Res}_P(f\,\mathrm{d}t) = 0,$$

summed over all places P of K/k (finitely many terms nonzero, as $f\,\mathrm{d}t$ has only finitely many poles).

Roadmap. The proof, given immediately below, proceeds in the following steps.

- *Reduce to a differential $f\,\mathrm{d}x$.* Choose a separating transcendental $x \in K$; then $f\,\mathrm{d}t = f\partial_x(t)\,\mathrm{d}x$ at every place (6.3.33).b).
- *The case $K = k(x)$.* Partial fractions over $k = \overline{k}$ (6.3.35) write a rational function as a sum of terms $c_{i,j}(x - \theta_i)^{-j}$; the finite residues are the $c_{i,1}$, and the place at infinity has residue $-\sum_i c_{i,1}$ (6.3.39).
- *Trace formula.* For a finite separable $K/k(x)$ and a place Q of $k(x)$, $\sum_{P|Q} \mathrm{Res}_P(f\,\mathrm{d}x) = \mathrm{Res}_Q\big(\mathrm{Tr}_{K/k(x)}(f)\,\mathrm{d}x\big)$ (6.3.42). This is proved locally, after completing at Q: the completed field splits as $\prod_i k((\pi_i))$ over the places above Q, reducing the identity to the local trace formula at each place, which is computed monomial by monomial via the roots of unity in k, following Lang.
- *Assemble.* Group the places P of K by the place Q of $k(x)$ below them. The trace formula turns each inner sum into $\mathrm{Res}_Q\big(\mathrm{Tr}_{K/k(x)}(f')\,\mathrm{d}x\big)$, and summing over Q gives 0 by the case $K = k(x)$.

Proof of (6.3.34). Drafted by Claude Sonnet, reviewed by David Rufino.

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Fix any transcendental $x \in K$ with $K/k(x)$ finite and separable (exists as K/k is a function field, separably generated), and let $\partial_x \in \mathrm{Der}_k(K,K)$ be the distinguished element of (6.3.20) with $\partial_x(x) = 1$. By (6.3.33).b), for every place P,

$$\mathrm{Res}_P(f\,\mathrm{d}t) = \mathrm{Res}_P((f \cdot \partial_x(t))\,\mathrm{d}x)$$

this handles $t \in k$ automatically too, as $\partial_x(k) = 0$ then makes both sides 0, matching $\mathrm{d}t = 0$. So it suffices to prove $\sum_P \mathrm{Res}_P(f'\,\mathrm{d}x) = 0$ for $f' := f \cdot \partial_x(t) \in K$.

Every place P of K/k lies over a unique place Q of $k(x)/k$, so (finitely many terms being nonzero) the sum may be regrouped:

$$\sum_{P \text{ of } K/k} \mathrm{Res}_P(f'\,\mathrm{d}x) = \sum_{Q \text{ of } k(x)/k} \;\sum_{P|Q} \mathrm{Res}_P(f'\,\mathrm{d}x).$$

By (6.3.42), the inner sum is $\mathrm{Res}_Q\big(\mathrm{Tr}_{K/k(x)}(f')\,\mathrm{d}x\big)$ for every Q, so

$$\sum_P \mathrm{Res}_P(f'\,\mathrm{d}x) = \sum_Q \mathrm{Res}_Q\big(\mathrm{Tr}_{K/k(x)}(f')\,\mathrm{d}x\big) = 0$$

by (6.3.39), applied to $\mathrm{Tr}_{K/k(x)}(f') \in k(x)$. ∎

This proof uses only (6.3.20) (already available, with no need for the module of differentials $\Omega_{K/k}$ to have been formally constructed).

Lemma 6.3.35 (Partial Fraction Decomposition over an Algebraically Closed Field)
Let k be algebraically closed and $r, s \in k[X]$ polynomials with $s \neq 0$. Write $s(X) = c\prod_{i=1}^m (X-\theta_i)^{e_i}$ for distinct $\theta_i \in k$, integers $e_i \geq 1$, and $c \in k^\times$. Then there exist a polynomial $Q \in k[X]$ and constants $c_{i,j} \in k$ $(1 \leq i \leq m,\ 1 \leq j \leq e_i)$ with

$$\frac{r(X)}{s(X)} = Q(X) + \sum_{i=1}^m \sum_{j=1}^{e_i} \frac{c_{i,j}}{(X-\theta_i)^j}$$

as an identity in $k(X)$. If $\deg r < \deg s$ then $Q = 0$.

Proof. Absorbing c into r, we may take $c = 1$. We argue by induction on m, the number of distinct roots of s. If $m = 0$ then $s = 1$ and we take $Q := r$.

Let $m \geq 1$ and set $s_1 = (X-\theta_1)^{e_1}$ and $s_2(X) := \prod_{i=2}^m (X-\theta_i)^{e_i}$, so $s = s_1 s_2$. By (3.18.10) s_1 and s_2 are co-prime as they have no common roots in k, so $gs_1 + hs_2 = 1$ for some $g, h \in k[X]$ and

$$\frac{r(X)}{s(X)} = \frac{r(X)h(X)}{s_1}(X) + \frac{r(X)g(X)}{s_2}(X)$$

By the division algorithm $rh = Q_1 s_1 + r_1$ with $\deg(r_1) < e_1$. By (3.7.12) $r_1 = \sum_{l=0}^{e_1-1} d_l (X-\theta_1)^l$ for some $d_l \in k$, so with $c_{1,j} := d_{e_1-j}$

$$\frac{rh}{s_1} = Q_1 + \sum_{j=1}^{e_1} \frac{c_{1,j}}{(X-\theta_1)^j}$$

The induction hypothesis applies to rg/s_2, which has $m-1$ distinct roots in its denominator, and gives the terms with $i \geq 2$ together with a polynomial Q_2. Adding, the claim holds with $Q := Q_1 + Q_2$.

Finally suppose $\deg r < \deg s$. Multiplying the decomposition by s gives $Qs = r - \sum_{i,j} c_{i,j} s/(X-\theta_i)^j$, where each $s/(X-\theta_i)^j$ is a polynomial of degree $\deg s - j \leq \deg s - 1$. So the right hand side has degree less than $\deg s$, whereas $\deg(Qs) \geq \deg s$ if $Q \neq 0$; hence $Q = 0$. ∎

Remark 6.3.36 (A Complex-Analytic Heuristic)
Over $k = \mathbb{C}$, this also follows from Liouville's theorem. At each θ_i, choose $c_{i,1}, \ldots, c_{i,e_i}$ to match the principal part of the Laurent expansion of r/s at θ_i (possible and unique, exactly as the induction above extracts them); then $r/s - \sum_{i,j} c_{i,j}/(X-\theta_i)^j$ is a rational function with no poles anywhere in $\mathbb{C}$, hence an entire function, and one that grows at most polynomially at infinity (being itself a rational function) – so by (the polynomial-growth form of) Liouville's theorem it is a polynomial, namely Q.

Lemma 6.3.37 (Residue at Infinity of a Proper Rational Function)
Let $f, g \in k[X]$ with $g \neq 0 \neq f$ and $\deg(f) < \deg(g)$. Then, for the place at infinity of $k(X)/k$ (6.2.117),

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$$\operatorname{Res}_\infty\left(\frac{f}{g}\,\mathrm{d}X\right)=\begin{cases}-\frac{\ell(f)}{\ell(g)} & \deg(f)=\deg(g)-1\\ 0 & \deg(f)<\deg(g)-1.\end{cases}$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Let $p:=\deg(f)$, $q:=\deg(g)$, and $m:=q-p\geq 1$. Set $A(U):=U^p f(1/U)$, $B(U):=U^q g(1/U)\in k[X]$, polynomials with $A(0)=\ell(f)$, $B(0)=\ell(g)\neq 0$, so $w:=A/B\in k[[U]]$ is a unit with $w(0)=\ell(f)/\ell(g)$. As $f(1/U)=U^{-p}A(U)$ and $g(1/U)=U^{-q}B(U)$,

$$\frac{f}{g}(1/U)=U^{q-p}\frac{A(U)}{B(U)}=U^m w(U).$$

Let U be an indeterminate and $\varphi:k(X)\to k(U)$ the isomorphism with $\varphi(X)=U^{-1}$; it maps the place at infinity to the place 0 of $k(U)$, with uniformiser U (6.2.117). By (6.3.32) and (6.3.33).b) (with $x:=U$, as $\partial_U(U^{-1})=-U^{-2}$),

$$\operatorname{Res}_\infty\left(\frac{f}{g}\,\mathrm{d}X\right)=\operatorname{Res}_0\left(\varphi(f/g)\,\mathrm{d}(U^{-1})\right)=\operatorname{Res}_0\left(-U^{m-2}w(U)\,\mathrm{d}U\right).$$

As $\partial_U(U)=1$, this is the coefficient of U^{-1} in $-U^{m-2}w(U)$, that is − the coefficient of U^{1-m} in $w(U)$. If $m=1$ this is $-w(0)=-\ell(f)/\ell(g)$. If $m\geq 2$ then $1-m<0$, and as $w\in k[[U]]$ has only non-negative powers, this coefficient is 0. ∎

Remark 6.3.38 (A Complex-Analytic Heuristic)
Over $k=\mathbb{C}$, this is controlled entirely by the leading-order behaviour of f/g at infinity. Writing $U=1/X$, the residue at infinity is $-\operatorname{Res}_0$ of the pulled-back differential, and $f/g(1/U)\cdot\partial_U(U^{-1})=-U^{m-2}w(U)$ has leading term $-\left(\frac{\ell(f)}{\ell(g)}\right)U^{m-2}$ ($w(0)=\ell(f)/\ell(g)\neq 0$); classically $\oint U^n\,\mathrm{d}U=0$ for every $n\neq -1$, so the residue vanishes unless this leading *term is itself the U^{-1} term, i.e. $m-2=-1$, or $\deg f=\deg g-1$ – exactly the borderline decay rate $f/g\sim(\ell(f)/\ell(g))X^{-1}$ at infinity – in which case the residue is read off directly from the leading coefficients, with no correction from the rest of the expansion of w.*

Lemma 6.3.39 (Residue Theorem for the Rational Function Field)
Assume k is algebraically closed. For $\theta\in k$, $n\in\mathbb{Z}$ and $g:=(X-\theta)^n\in k(X)$, the residues at the places of $k(X)/k$ (the $\mathcal{O}_{X-\theta'}$, $\theta'\in k$, written θ', and the place at infinity (6.2.117)) are

$$\operatorname{Res}_{\theta'}(g\,\mathrm{d}X)=\begin{cases}1 & \theta'=\theta\text{ and }n=-1\\ 0 & \text{otherwise}\end{cases}\qquad\operatorname{Res}_\infty(g\,\mathrm{d}X)=\begin{cases}-1 & n=-1\\ 0 & n\neq -1.\end{cases}$$

Consequently, for every $g\in k(X)$, where $k(X)/k$ is the rational function field,

$$\sum_Q\operatorname{Res}_Q(g\,\mathrm{d}X)=0,$$

summed over all places Q of $k(X)/k$ (finitely many terms nonzero).

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Write T for the variable of the Laurent series field, to avoid a clash with $X\in k(X)$, so that $L_\pi:k(X)\to k((T))$ is a ring homomorphism with $L_\pi(\pi)=T$ (6.3.23).

Finite places. The place θ' has uniformiser $\pi:=X-\theta'$ with $\partial_\pi(X)=1$. If $\theta'\neq\theta$ then g and X lie in $\mathcal{O}_{X-\theta'}$, so $\operatorname{Res}_{\theta'}(g\,\mathrm{d}X)=0$ (6.3.33).c). If $\theta'=\theta$ then by (6.3.25)

$$\operatorname{Res}_\theta(g\,\mathrm{d}X)=L_\pi(\pi^n)_{-1}=(T^n)_{-1}=\begin{cases}1 & n=-1\\ 0 & n\neq -1\end{cases}$$

The place at infinity. If $n\geq 0$ then g is a polynomial, so $g(1/U)$ is a k-linear combination of $U^0,\dots,U^{-n}$, and $g(1/U)\partial_U(X)=-U^{-2}g(1/U)$ a linear combination of $U^{-2},\dots,U^{-n-2}$; as every exponent is ≤ -2, $\operatorname{Res}_\infty(g\,\mathrm{d}X)=0$. If $n\leq -1$ then $g=1/(X-\theta)^{-n}$ is a proper fraction with numerator 1 and denominator $(X-\theta)^{-n}$ of degree $-n$, so by (6.3.37)

$$\operatorname{Res}_\infty(g\,\mathrm{d}X)=\begin{cases}-1 & n=-1\\ 0 & n\leq -2.\end{cases}$$

The sum vanishes for every $g\in k(X)$. By (6.3.35) every $g\in k(X)$ is the sum of a polynomial and a k-linear combination of the functions $(X-\theta)^{-n}$ with $\theta\in k$, $n\geq 1$; and by (3.7.12) every polynomial is a k-linear combination

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of the $(X-\theta)^n$, $n \geq 0$, for any fixed $\theta \in k$. The residue being k-linear, this reduces to the case $g = (X-\theta)^n$ above, where the residues at all places other than θ and ∞ vanish, and $\operatorname{Res}_\theta(g\,\mathrm{d}X) + \operatorname{Res}_\infty(g\,\mathrm{d}X)$ is $1-1=0$ if $n=-1$ and 0 otherwise. ∎

Lemma 6.3.40 (Local Trace Formula for Residues)
Let k be algebraically closed, $F := k((t))$, and let $E = k((\pi))$ be a finite separable extension of F with $v_\pi(t) = e$, so $[E:F] = e$ and $t = \pi^e w$ for some $w \in k[[\pi]]^\star$ (3.32.34). For $g \in E$ write $\operatorname{Res}_\pi(g\,\mathrm{d}t) := (g\partial_\pi(t))_{-1}$. Then, in every characteristic,

$$\operatorname{Res}_t\big(\operatorname{Tr}_{E/F}(g)\,\mathrm{d}t\big) = \operatorname{Res}_\pi(g\,\mathrm{d}t) \qquad (g \in E).$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Basis and matrices. By (3.32.30) (with $R := k$, $X := \pi$, $m := e$, $\tau = t$), every element of $k[[\pi]]$ is uniquely $\sum_{i<e} f_i(t)\pi^i$ with $f_i \in k[[Y]]$; multiplying by powers of t (which is π^e up to a unit) gives the same for E. So $1, \pi, \ldots, \pi^{e-1}$ is an F-basis of E. For $g \in E$ write $g\pi^i = \sum_{j<e} M_{ji}(g)\pi^j$ with $M_{ji}(g) \in F$, and $\operatorname{Tr}_{E/F}(g) = \sum_{i<e} M_{ii}(g)$.

Universal model. Let $R := \mathbb{Z}\big[b_0^{\pm 1}, b_1, b_2, \ldots\big]$, a domain of characteristic 0, and put $w_R := \sum_{i\geq 0} b_i\pi^i \in R[[\pi]]^\star$, $t_R := \pi^e w_R$. The construction above works verbatim over R (only b_0 needs inverting), with matrix entries $M_{ji}(g) \in R((t_R))$ given by universal formulas in $+$, $\times$ and w_R^{-1}, whose coefficients lie in R. For $r \in \mathbb{Z}$ set

$$\Psi(\pi^r) := \operatorname{Res}_{t_R}(\operatorname{Tr}_R(\pi^r)\,\mathrm{d}t_R) - \operatorname{Res}_\pi(\pi^r\,\mathrm{d}t_R) \in R,$$

both terms being coefficients of elements of $R((\pi))$ or $R((t_R))$.

Claim: $\Psi(\pi^r) = 0$ for all r. Let Ω be an algebraic closure of $\operatorname{Frac}(R)$; it has characteristic 0, so e is invertible in Ω and $R \subset \Omega$. Write $w_R = b_0 u$ with $u \in 1 + \pi R[[\pi]]$. The binomial series $\varepsilon := \sum_{i\geq 0}\binom{1/e}{i}(u-1)^i \in \Omega[[\pi]]$ satisfies $\varepsilon^e = u$, and for $\beta \in \Omega$ with $\beta^e = b_0$ the element $\varpi := \beta\pi\varepsilon$ satisfies $\varpi^e = t_R$. As $\varpi = \pi \cdot (\text{unit})$, it is a uniformiser of $\Omega((\pi)) = \Omega((\varpi))$, and $\Omega((t_R)) \subset \Omega((\pi))$ has basis $1, \varpi, \ldots, \varpi^{e-1}$ (the argument above, with ϖ in place of π).

Take $g = \varpi^r$ with $r = es + j$, $0 \leq j < e$. Then $\varpi^r = t_R^s\varpi^j$, and multiplication by ϖ^j sends the basis vector ϖ^i to ϖ^{i+j}, which is a basis vector times t_R^0 or t_R. So the matrix of ϖ^r has nonzero entries only in positions shifted by r modulo e: its diagonal vanishes unless $e \mid r$, and if $r = es$ it is the scalar t_R^s. Hence

$$\operatorname{Tr}(\varpi^r) = \begin{cases} e t_R^s & r = es \\ 0 & e \nmid r. \end{cases}$$

Therefore $\operatorname{Res}_{t_R}(\operatorname{Tr}(\varpi^r)\,\mathrm{d}t_R) = e$ if $r = -e$ and 0 otherwise. On the other side $\mathrm{d}t_R = e\varpi^{e-1}\,\mathrm{d}\varpi$, so $\varpi^r\,\mathrm{d}t_R = e\varpi^{r+e-1}\,\mathrm{d}\varpi$ and $\operatorname{Res}_\varpi(\varpi^r\,\mathrm{d}t_R)$ is again e for $r=-e$ and 0 otherwise. Residues are independent of the choice of uniformiser (6.3.31), so $\operatorname{Res}_\varpi(g\,\mathrm{d}t_R) = \operatorname{Res}_\pi(g\,\mathrm{d}t_R)$. Thus the identity holds for $g = \varpi^r$ for every r. Both sides are Ω-linear and continuous in g, and the ϖ^r span $\Omega((\pi))$ topologically, so it holds for every $g \in \Omega((\pi))$; in particular $\Psi(\pi^r) \in R$ maps to 0 in Ω. As $R \subset \Omega$, $\Psi(\pi^r) = 0$.

Specialisation. Given $E = k((\pi))$ and $t = \pi^e w$ with $w = \sum_{i\geq 0} c_i\pi^i$, $c_0 \in k^\times$, let $\phi : R \to k$ be the ring homomorphism with $b_i \mapsto c_i$ (it exists by the universal property of localisation, as c_0 is a unit). Applying ϕ coefficientwise gives ring homomorphisms $R((\pi)) \to k((\pi))$ and $R((t_R)) \to k((t))$ (with $t_R \mapsto t$, since $\phi(w_R) = w$). These commute with the formation of matrices M_{ji}, hence with traces, and with the coefficient extraction defining both residues. So $\phi(\Psi(\pi^r))$ is the defect $\operatorname{Res}_t\big(\operatorname{Tr}_{E/F}(\pi^r)\,\mathrm{d}t\big) - \operatorname{Res}_\pi(\pi^r\,\mathrm{d}t)$ over k, and it equals 0. Finally, for $g = \sum_r a_r\pi^r \in k((\pi))$ both sides depend on only finitely many a_r, since $\operatorname{Tr}(\pi^r)$ has t-order tending to ∞ with r and only finitely many terms contribute to a given coefficient of t^{-1} or π^{-1}. The identity for each monomial therefore gives it for g. ∎

Lemma 6.3.41 (Places over P and Irreducible Factors)
Let k be algebraically closed, $K/k(x)$ a finite separable extension of degree n, P the place (x) of $k(x)$, $K_P := k((x))$, and $y \in K$ a primitive element with minimal polynomial g over $k(x)$. Factor $g = g_1 \cdots g_r$ into distinct monic irreducibles over K_P, with $d_j := \deg g_j$, and let $E_j := K_P[Y]/(g_j)$ with $\iota_j : K \hookrightarrow E_j$ sending $y \mapsto Y$. Then $j \mapsto Q_j$, defined below, is a bijection from $\{1, \ldots, r\}$ onto the places of K/k over P, and $e_{Q_j} = d_j$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

The place Q_j. E_j/K_P is finite separable (as g_j is separable), so by (3.32.34) there is an isomorphism $\Psi_j : E_j \xrightarrow{\sim} k\big((\pi_j)\big)$ over K_P with $\Psi_j(x)$ of order d_j. Pulling back the valuation of $k\big((\pi_j)\big)$ along $\Psi_j \circ \iota_j$ gives a discrete valuation of K extending v_P with ramification index d_j; let Q_j be the corresponding place.

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Every place is some Q_j. Let Q be a place of K/k over P. As k is algebraically closed, Q is rational, so (6.3.23) gives a valuation-preserving embedding $L_Q : K \hookrightarrow k\big((\pi_Q)\big)$. Since $g(y) = 0$, the element $L_Q(y)$ is a root of g in $k\big((\pi_Q)\big)$. The g_j are pairwise coprime in $K_P[Y]$, so $L_Q(y)$ is a root of exactly one g_j (a common root of $g_j, g_{j'}$ would be a root of 1, by Bézout). Call this $j = j(Q)$. Then L_Q factors through E_j, and $L_Q(x)$ has ramification index e_Q over K_P; the valuation induced on E_j extends v_P, so by the uniqueness of extension ((3.32.35), applied to E_j/K_P) it equals the valuation of Q_j. Hence $Q = Q_{j(Q)}$.

Injectivity. If $Q_j = Q_{j'}$ then $j\big(Q_j\big) = j$ and $j\big(Q_{j'}\big) = j'$ are determined by the same valuation, and the minimal factor of $L(y)$ is unique, so $j = j'$. ∎

Proposition 6.3.42 (Trace Formula for Residues)
Assume k is algebraically closed. Let $\mathcal{O} := k[x]$, $\mathcal{B} := \overline{\mathcal{O}}^K$ the integral closure of $\mathcal{O}$ in a finite separable extension $K/k(x)$, P a place of $k(x)/k$ corresponding to a maximal ideal $\mathfrak{p} \lhd \mathcal{O}$, and $f \in K$. Then

$$\sum_{Q|P} \mathrm{Res}_Q(f\,\mathrm{d}x) = \mathrm{Res}_P\Big(\mathrm{Tr}_{K/k(x)}(f)\,\mathrm{d}x\Big),$$

with no hypothesis on $\mathrm{char}(k)$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

As k is algebraically closed every place is rational, so $\mathfrak{p} = (x - \theta)$ for some $\theta \in k$; replacing x by $x - \theta$ we may assume $\mathfrak{p} = (x)$.

By (3.31.19) the places $Q_1, \dots, Q_r$ of K/k over P have ramification indices $e_1, \dots, e_r$ with $\sum_i e_i = n := [K : k(x)]$ (residue degree 1, as k is algebraically closed). Let $K_P := k((x))$, with $k(x) \hookrightarrow K_P$ the expansion at P (6.3.23).

Factoring the minimal polynomial. Let $y \in \mathcal{B}$ be a primitive element with minimal polynomial $g \in \mathcal{O}[Y]$, monic of degree n. Factor $g = g_1 \cdots g_r$ over K_P as in (6.3.41), with $E_j = K_P[Y]/(g_j)$, $\iota_j : K \hookrightarrow E_j$, and $\Psi_j : E_j \xrightarrow{\sim} k\big((\pi_j)\big)$ the isomorphisms given there, so that $\Psi_j(x)$ has order $d_j = \deg g_j$. Put $\lambda_j := \Psi_j \circ \iota_j : K \hookrightarrow k\big((\pi_j)\big)$ and $L_{\pi_j} := \lambda_j$. By (6.3.41), the places Q_j attached to the g_j are exactly the places of K/k over P, with $e_{Q_j} = d_j$, so after relabelling Q_i is the place attached to g_i and $e_i = d_i$.

Trace additivity. By the Chinese remainder theorem applied to $g = \prod_j g_j$,

$$K \otimes_{k(x)} K_P \xrightarrow{\sim} \prod_{i=1}^{r} E_i, \quad f \otimes 1 \mapsto \big(\iota_i(f)\big)_i,$$

and composing with the isomorphisms Ψ_i identifies the right side with $\prod_i k((\pi_i))$, with $f \otimes 1 \mapsto \big(L_{\pi_i}(f)\big)_i$. The trace is unchanged by base change and additive over a finite product, so

$$\mathrm{Tr}_{K/k(x)}(f) = \sum_{i=1}^{r} \mathrm{Tr}_{k((\pi_i))/K_P}\big(L_{\pi_i}(f)\big) \qquad (f \in K).$$

Local identities. The residue at Q_i is the residue symbol of (6.3.25), computed in $k((\pi_i))$ through L_{π_i}, so $\mathrm{Res}_{Q_i}(f\,\mathrm{d}x) = \mathrm{Res}_{\pi_i}\big(L_{\pi_i}(f)\,\mathrm{d}x\big)$ (Laurent residue of (6.3.24)), and $\mathrm{Res}_P(z\,\mathrm{d}x) = \mathrm{Res}_x(z\,\mathrm{d}x)$ for $z \in K_P = k((x))$. Applying (6.3.40) to $E = k((\pi_i))$ over $F = K_P$ (with $t = x$ and $\Psi_i(x) = \tau_i$) gives, for every $f \in K$,

$$\mathrm{Res}_{Q_i}(f\,\mathrm{d}x) = \mathrm{Res}_P\Big(\mathrm{Tr}_{k((\pi_i))/K_P}\big(L_{\pi_i}(f)\big)\,\mathrm{d}x\Big).$$

Assembly. Summing over i and using k-linearity of Res_P and trace additivity,

$$\sum_{Q|P} \mathrm{Res}_Q(f\,\mathrm{d}x) = \sum_{i=1}^{r} \mathrm{Res}_P\Big(\mathrm{Tr}_{k((\pi_i))/K_P}\big(L_{\pi_i}(f)\big)\,\mathrm{d}x\Big) = \mathrm{Res}_P\Big(\mathrm{Tr}_{K/k(x)}(f)\,\mathrm{d}x\Big).$$

∎

6.3.7 Base Extension of the Constant Field

Throughout this section, in addition to the standing hypothesis that K/k is geometrically integral ($\tilde{k} = k$), assume also that k is **perfect**. Let k'/k be an algebraic field extension, and write $K_{k'} := K \otimes_k k'$.

Proposition 6.3.43 (The Base-Changed Function Field)
$K_{k'}$ is a field, and an algebraic function field over k', geometrically integral; if k'/k is finite, $[K_{k'} : K] = [k' : k]$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

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$K_{k'}$ is a field. As K/k is geometrically integral and k'/k is algebraic, $K_{k'} = K \otimes_k k'$ is a field (3.35.45).

$K_{k'}/k'$ is geometrically integral. By associativity of tensor products (3.4.13) and $\overline{k'} = \overline{k}$ (k'/k algebraic), $K_{k'} \otimes_{k'} \overline{k'} \cong K \otimes_k \left(k' \otimes_{k'} \overline{k}\right) = K \otimes_k \overline{k}$, a field, as K/k is geometrically integral.

$K_{k'}/k'$ is an algebraic function field. Fix a transcendental $x \in K$ with $K/k(x)$ finite (exists, as K/k is an algebraic function field). As $k[x] \otimes_k k' \cong k'[x]$ (tensoring a polynomial ring with a field extension), the image of x in $K_{k'}$ remains transcendental over k'. The image of $k(x) \otimes_k k'$ in $K_{k'}$ (well-defined and injective, as $k(x) \subset K$ is a k-subalgebra, (3.35.2)) is then a subring of the field $K_{k'}$, hence a domain; as k'/k is algebraic, it is thus a field (3.34.57), and – being the subfield of $K_{k'}$ generated by k' and the transcendental x – equals $k'(x)$. By associativity of tensor products (3.4.13) (as K is a $k(x)$-module),

$$K_{k'} = K \otimes_k k' \cong K \otimes_{k(x)} (k(x) \otimes_k k') = K \otimes_{k(x)} k'(x)$$

As $K/k(x)$ is finite, K is a finite-dimensional $k(x)$-vector space, hence free (3.3.126) with a basis of size $n := [K : k(x)]$; extension of scalars (3.4.26) then makes $K_{k'} \cong K \otimes_{k(x)} k'(x)$ a free $k'(x)$-module with a basis of the same size n, in particular finite over $k'(x)$. So $K_{k'}$ is a finitely generated field extension of k' (generated by x together with a finite $k'(x)$-basis) of transcendence degree 1: an algebraic function field.

Degree. If k'/k is finite, k' is a free k-module with a basis of size $[k' : k]$, so extension of scalars along $k \hookrightarrow K$ (3.4.26) makes $K_{k'} = K \otimes_k k'$ a free K-module with a basis of the same size, giving $[K_{k'} : K] = [k' : k]$. ∎

Proposition 6.3.44 (Constant Field Extension is Unramified)
Let P be a place of K/k and k'/k algebraic. Every place P' of $K_{k'}/k'$ lying over P is unramified.

There are exactly r such places, where r is the number of maximal ideals of $A := k(\mathfrak{m}_P) \otimes_k k'$ (finite, even when k'/k is not). Moreover each such P' satisfies $\mathcal{O}_{P'} \cap K = \mathcal{O}_P$ and $\mathfrak{m}_{P'} \cap K = \mathfrak{m}_P$.

More precisely, writing $\mathcal{O}_{P,k'} := \mathcal{O}_P \otimes_k k'$, $A \cong \mathcal{O}_{P,k'}/\mathfrak{m}_P\mathcal{O}_{P,k'}$ as k'-algebras (3.4.42) there is a bijection

$$\begin{aligned} \operatorname{Specm}(k(\mathfrak{m}_P) \otimes_k k') &\longleftrightarrow \{\text{places of } K_{k'}/k' \text{ lying over } P\} \\ \mathfrak{p}'/\mathfrak{m}_P\mathcal{O}_{P,k'} &\longrightarrow \left(\mathcal{O}_{P,k'}\right)_{\mathfrak{p}'} \\ \left(\mathfrak{m}_{P'} \cap \mathcal{O}_{P,k'}\right)/\mathfrak{m}_P\mathcal{O}_{P,k'} &\longleftarrow \mathcal{O}_{P'} \end{aligned}$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

We claim that we are in the situation of (3.35.47), applied with $\mathcal{O} := \mathcal{O}_P$, $F := K$:

- $\mathcal{O}_P$ is a discrete valuation ring containing k (a place of K/k is by definition a valuation ring $k \subsetneq \mathcal{O}_P \subsetneq K$ (6.2.112), automatically discrete (6.2.114))
- with fraction field K which is geometrically integral (the standing hypothesis of this section)
- $k(\mathfrak{m}_P)$ is finite over k (6.2.114)
- every valuation ring R of $K_{k'}/k'$ with $k' \subset R$ and $\mathcal{O}_P \subset R$ satisfies $R \cap K = \mathcal{O}_P$: such an R is exactly a place of $K_{k'}/k'$, and this is (6.2.119) (applied with $L := K_{k'}$, $k'' := k'$)

Therefore the result follows from there: writing $\mathcal{O}_{P,k'} := \mathcal{O}_P \otimes_k k'$, its $\mathfrak{b} := \mathfrak{m}_P\mathcal{O}_{P,k'}$ has quotient $\mathcal{O}_{P,k'}/\mathfrak{b} \cong A$ (above), so a) gives exactly r maximal ideals $\mathfrak{p}_1, ..., \mathfrak{p}_r$ of $\mathcal{O}_{P,k'}$; by b), each $\left(\mathcal{O}_{P,k'}\right)_{\mathfrak{p}_j}$ is a place P'_j of $K_{k'}/k'$ (with $\mathcal{O}_{P'_j} := \left(\mathcal{O}_{P,k'}\right)_{\mathfrak{p}_j}$), unramified over $\mathcal{O}_P$, and, as in the statement, corresponds to $\mathfrak{p}_j/\mathfrak{b} \in \operatorname{Specm}(A)$; and, by c) together with the last point above, *every* place P' of $K_{k'}/k'$ over P has $\mathcal{O}_{P'} = \left(\mathcal{O}_{P,k'}\right)_{\mathfrak{p}_j}$ for some j, that is $P' = P'_j$, giving the stated bijection. As $K_{k'}/k'$ is itself an algebraic function field (6.3.43), every place of $K_{k'}/k'$ – in particular P'_j – is automatically a discrete valuation ring (6.2.114), so $e(P'_j/P) = v_{P'_j}(\pi_P) = 1$: unramified, with π_P a uniformiser of $\mathcal{O}_P$ (3.30.2). Finally $\mathcal{O}_{P'_j} \cap K = \mathcal{O}_P$ and $\mathfrak{m}_{P'_j} \cap K = \mathfrak{m}_P$ by e). ∎

Lemma 6.3.45 (A Rational Place Extends Uniquely Under Any Algebraic Base Change)
Let P be a rational place of K/k. Then for any algebraic k'/k, there is a unique place Q of $K_{k'}$ over P, and $e(Q/P) = 1$, with Q again rational over k'.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

This is the case $k(\mathfrak{m}_P) = k$ of (6.3.44). ∎

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Proposition 6.3.46 (Complete Splitting)
Let K/k be an algebraic function field, P a place of K/k, and k'/k an algebraic extension. Suppose further that k' contains a normal closure of $k(\mathfrak{m}_P)$.

*Then P **splits completely** in $K_{k'}/k'$: that is there are exactly $n = [k(\mathfrak{m}_P) : k]$ places above P, each rational over k'.*

This holds in particular when $k' = \overline{k}$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Write $k_P := k(\mathfrak{m}_P)$; k_P/k is finite (6.2.114) and separable, as k is perfect (3.18.103). As k' contains a normal closure of k_P/k, (3.35.15) gives $k_P \otimes_k k' \cong (k')^n$ as k'-algebras, with exactly n maximal ideals, each with residue field k' (the coordinate kernels). By (6.3.44) (with $A := k_P \otimes_k k'$) this implies that P splits completely into n places, each rational (residue field k'). ∎

Proposition 6.3.47 (Automorphisms Act Transitively on Places over $\overline{k}$)
Let P be a place of K/k, $d := [k(\mathfrak{m}_P) : k]$, and $P'_1, ..., P'_d$ the places of $K_{\overline{k}}/\overline{k}$ over P ((6.3.46), with $k' := \overline{k}$). For $\sigma \in \mathrm{Aut}\left(\overline{k}/k\right)$, the map

$$\sigma_* : K_{\overline{k}} \to K_{\overline{k}}, \qquad x \otimes c \mapsto x \otimes \sigma(c)$$

is a well-defined ring automorphism of $K_{\overline{k}}$ fixing K pointwise, and $\sigma \mapsto \sigma_*$ is a group action of $\mathrm{Aut}\left(\overline{k}/k\right)$ on $K_{\overline{k}}$.

For any place P' of $K_{\overline{k}}/\overline{k}$ lying over P then $\sigma_*(\mathcal{O}_{P'})$ is a place lying over P and

$$v_{\sigma(P')}(\sigma_*(z)) = v_{P'}(z)$$

for all $z \in K_{\overline{k}}^\star$; this defines a transitive group action of $\mathrm{Aut}\left(\overline{k}/k\right)$ on $\{P'_1, ..., P'_d\}$.

Moreover, for P' among $P'_1, ..., P'_d$ (rational over $\overline{k}$), write $\iota_{P'} : k(\mathfrak{m}_P) \hookrightarrow k(\mathfrak{m}_{P'}) = \overline{k}$ for the residue embedding induced by $\mathcal{O}_P \subset \mathcal{O}_{P'}$ (well-defined as $\mathfrak{m}_P \subset \mathfrak{m}_{P'} \cap \mathcal{O}_P$). Then $\iota_{\sigma(P')} = \sigma \circ \iota_{P'}$.

This yields a bijection

$$\begin{aligned} \{\text{places of } K/k\} &\longleftrightarrow \left\{\mathrm{Aut}\left(\overline{k}/k\right)\text{-orbits of places of } K_{\overline{k}}/\overline{k}\right\} \\ P &\longrightarrow \{P'_1, ..., P'_d\} \end{aligned}$$

Proof. As $\sigma : \overline{k} \to \overline{k}$ is a k-algebra homomorphism, $1_K \otimes \sigma$ is a well-defined k-algebra endomorphism of $K \otimes_k \overline{k} = K_{\overline{k}}$ (3.4.36).

As $(1_K \otimes \sigma) \circ (1_K \otimes \tau) = 1_K \otimes (\sigma\tau)$ then σ_* is an automorphism of $K_{\overline{k}}/K$ and $\sigma \mapsto \sigma_*$ is a group homomorphism.

As σ_* is a field automorphism of $K_{\overline{k}}$ then $\sigma_*(\mathcal{O}_{P'})$ is a place of $K_{\overline{k}}$ which lies over the same place as $\mathcal{O}_{P'}$, as σ_* fixes $K \supset \mathcal{O}_P$.

The valuation identity follows because σ_* restricts to an isomorphism on the valuation rings.

As k is perfect, $k(\mathfrak{m}_P)/k$ is separable (3.18.103), so (3.35.15) gives a canonical isomorphism of $\overline{k}$-algebras

$$k(\mathfrak{m}_P) \otimes_k \overline{k} \cong \overline{k}^d, \quad p(\theta) \otimes c \mapsto (c\,p(\theta_1), ..., c\,p(\theta_d)) \qquad \left(p \in k[X], c \in \overline{k}\right)$$

identifying the d maximal ideals of this fibre with $\mathfrak{p}_j$, the kernel of the j-th coordinate.

Let $\mathfrak{p}'_1, ..., \mathfrak{p}'_d$ denote the corresponding maximal ideals of $\mathcal{O}_{P,\overline{k}} = \mathcal{O}_P \otimes_k \overline{k}$. Define $\mathcal{O}_{P'_i} := \left(\mathcal{O}_{P,\overline{k}}\right)_{\mathfrak{p}'_i}$ then by (6.3.44) these are precisely the places $P'_1, ..., P'_d$ above P, and $\mathcal{O}_{P'_i} \cap K = \mathcal{O}_P$, which shows that the given map is injective.

By (3.18.66) $\mathrm{Aut}\left(\overline{k}/k\right)$ permutes the roots θ_i transitively. We may show that if $\sigma(\theta_j) = \theta_{j'}$ then $\sigma_*\left(\mathcal{O}_{P'_j}\right) = \mathcal{O}_{P'_{j'}}$ and the action on places is transitive and $\{P'_1, ..., P'_d\}$ is precisely an $\mathrm{Aut}\left(\overline{k}/k\right)$-orbit.

Given a place $\overline{P}$ of $K_{\overline{k}}/\overline{k}$, $\mathcal{O} := \mathcal{O}_{\overline{P}} \cap K$ is a (proper) valuation ring of K by (6.2.118), giving a place P of K/k with $\mathcal{O}_P = \mathcal{O} \subset \mathcal{O}_{\overline{P}}$, that is $\overline{P}$ lies over P. As $P'_1, ..., P'_d$ are all the places of $K_{\overline{k}}/\overline{k}$ above P (above), $\overline{P}$ is one of them. So the given map is a surjection. ∎

Lemma 6.3.48 (Residue Embeddings are Exactly the Embeddings into $\overline{k}$)
Let P be a place of K/k, $d := [k(\mathfrak{m}_P) : k]$, and $P'_1, ..., P'_d$ the places of $K_{\overline{k}}/\overline{k}$ over P ((6.3.46), with $k' := \overline{k}$).

Denote by $\iota_j : k(\mathfrak{m}_P) \hookrightarrow k\left(\mathfrak{m}_{P'_j}\right) = \overline{k}$ the corresponding embeddings, then these exhaust all the k-algebra embeddings $k(\mathfrak{m}_P)/k \hookrightarrow \overline{k}/k$.

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Proof. By (6.3.47) there are $d = [k(\mathfrak{m}_P) : k]$ places and, as k is perfect so $k(\mathfrak{m}_P)/k$ is separable (3.18.103), there are $\#\operatorname{Mor}_k\big(k(\mathfrak{m}_P), \overline{k}\big) = [k(\mathfrak{m}_P) : k]_s = d$ k-algebra embeddings (3.18.83).

We show that $j \mapsto \iota_j$ is injective, i.e. that $\iota_j = \iota_{j'}$ implies $\mathcal{O}_{P'_j} = \mathcal{O}_{P'_{j'}}$ (and so $j = j'$). Consider the composite $\rho_j : \mathcal{O}_{P,\overline{k}} \to \big(\mathcal{O}_{P,\overline{k}}\big)_{\mathfrak{p}'_j} = \mathcal{O}_{P'_j} \to \mathcal{O}_{P'_j}/\mathfrak{m}_{P'_j} = \overline{k}$. It is $\overline{k}$-linear and is uniquely determined by

$$\rho_j(a \otimes c) = c\,\iota_j(\overline{a}) \qquad \big(a \in \mathcal{O}_P, c \in \overline{k}\big),$$

so $\iota_j = \iota_{j'} \Longrightarrow \rho_j = \rho_{j'}$, and their kernels agree, that is $\mathfrak{p}'_j = \mathfrak{p}'_{j'}$ (as $\ker \rho_j = \mathfrak{m}_{P'_j} \cap \mathcal{O}_{P,\overline{k}} = \Big(\mathfrak{p}'_j\big(\mathcal{O}_{P,\overline{k}}\big)_{\mathfrak{p}'_j}\Big)^c = \mathfrak{p}'_j$ (3.5.42)). By (6.3.44) $P'_j = P'_{j'}$ and we are done by counting. ∎

6.3.8 Residues and Base Change

We compare the residue symbol at a place P of K/k with the residues at the rational places of $K_{\overline{k}}/\overline{k}$ above P. The place P splits completely into $d := [k(P) : k]$ *rational* places of $K_{\overline{k}}/\overline{k}$, each carrying its own $\overline{k}$-valued Laurent coefficient theory, and these residues are the images $\iota_j\Big(\operatorname{Res}_P^{k(P)}(z\,\mathrm{d}t)\Big)$ of the single element $\operatorname{Res}_P^{k(P)}(z\,\mathrm{d}t) \in k(P)$ under the d embeddings of $k(P)$ in $\overline{k}$.

Definition 6.3.49 (Conjugate Places)
Let P be a place of K/k, $d := [k(\mathfrak{m}_P) : k]$, and $\iota_1, \dots, \iota_d : k(\mathfrak{m}_P) \hookrightarrow \overline{k}$ the embeddings, with P'_j the corresponding place of $K_{\overline{k}}/\overline{k}$ over P, $\iota_j = \iota_{P'_j}$ (6.3.48). As each P'_j/P is unramified, every uniformiser π at P remains a uniformiser at P'_j (6.3.44), so the residue symbol $\operatorname{Res}_{P'_j}(w\,\mathrm{d}s)$ at the rational place P'_j of $K_{\overline{k}}/\overline{k}$ (6.3.25) is defined for $w, s \in K_{\overline{k}}$.

Lemma 6.3.50 (The Uniformiser Derivation Extends to the Base Change)
Let P be a place of K/k, π a uniformiser, and $P'_1, \dots, P'_d$ the conjugate rational places of $K_{\overline{k}}/\overline{k}$ over P (6.3.49), at each of which $\pi \otimes 1$ remains a uniformiser.

Then $\partial_{\pi\otimes 1} \in \operatorname{Der}_{\overline{k}}(K_{\overline{k}}, K_{\overline{k}})$ is the unique D satisfying

$$D|_K = \partial_\pi$$

and is in particular regular at every place P'_j.

Lemma 6.3.51 (Laurent Coefficients and Base Change)
Let P be a place of K/k, π a uniformiser, and P'_j, ι_j $(1 \le j \le d)$ as in (6.3.49). Write $L_j : K_{\overline{k}} \hookrightarrow \overline{k}((X))$ for the Laurent coefficient embedding at the rational place P'_j relative to the uniformiser $\pi \otimes 1$, and $(\iota_j)_*$ for the map $k(P)((X)) \to \overline{k}((X))$ applying ι_j to the coefficients. Then for every $z \in K$

$$L_j(z \otimes 1) = (\iota_j)_* L_\pi(z).$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

The map $\Phi := (\iota_j)_* \circ L_\pi : K \to \overline{k}((X))$ is a k-algebra homomorphism, so $\tilde{\Phi}(z \otimes c) := c\,\Phi(z)$ defines an $\overline{k}$-algebra homomorphism $\tilde{\Phi} : K_{\overline{k}} = K \otimes_k \overline{k} \to \overline{k}((X))$. Recall $\mathcal{O}_{P'_j} = \big(\mathcal{O}_P \otimes_k \overline{k}\big)_{\mathfrak{p}'_j}$ with $\mathfrak{p}'_j = \ker \rho_j$ for $\rho_j(a \otimes c) = c\,\iota_j(\overline{a})$ (6.3.48). For $a \in \mathcal{O}_P$, $c \in \overline{k}$ the constant term of $\tilde{\Phi}(a \otimes c) = c\,(\iota_j)_* L_\pi(a)$ is $c\,\iota_j(\overline{a}) = \rho_j(a \otimes c)$, by (6.3.23) (the identity on residue fields). So $\tilde{\Phi}$ maps $\mathcal{O}_P \otimes_k \overline{k}$ into $\overline{k}[[X]]$, an element of $\mathfrak{p}'_j$ to $X\overline{k}[[X]]$ and an element outside $\mathfrak{p}'_j$ to a unit. Hence $\tilde{\Phi}$ extends to a local homomorphism $\mathcal{O}_{P'_j} \to \overline{k}[[X]]$ sending the uniformiser $\pi \otimes 1$ to X, and therefore $v_X\big(\tilde{\Phi}(z')\big) = v_{P'_j}(z')$ for $z' \in K_{\overline{k}}^\star$ (write $z' = (\pi \otimes 1)^n u$ with u a unit of $\mathcal{O}_{P'_j}$), and its constant term on $\mathcal{O}_{P'_j}$ is the residue class in $\overline{k} = k\big(P'_j\big)$. The uniqueness in (6.3.23), applied to the place P'_j of the function field $K_{\overline{k}}/\overline{k}$ (6.3.43) and the uniformiser $\pi \otimes 1$, gives $\tilde{\Phi} = L_j$. Restricting to $K \otimes 1$ gives the claim. ∎

Proposition 6.3.52 (Conjugate Residues)
Let P be a place of K/k and $t, z \in K$. With the notation of (6.3.49), for every $j = 1, \dots, d$

$$\operatorname{Res}_{P'_j}((z \otimes 1)\,\mathrm{d}(t \otimes 1)) = \iota_j\Big(\operatorname{Res}_P^{k(P)}(z\,\mathrm{d}t)\Big),$$

and consequently

$$\sum_{j=1}^{d} \operatorname{Res}_{P'_j}((z \otimes 1)\,\mathrm{d}(t \otimes 1)) = \operatorname{Tr}_{k(P)/k}\Big(\operatorname{Res}_P^{k(P)}(z\,\mathrm{d}t)\Big) = \operatorname{Res}_P(z\,\mathrm{d}t) \in k.$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

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Fix a uniformiser π at P; then $\pi\otimes 1$ is a uniformiser at P'_j (6.3.49). Applying the ring homomorphism $(\iota_j)_*$, which commutes with products, formal derivatives and coefficient extraction, and using $L_j(z\otimes 1)=(\iota_j)_*L_\pi(z)$ for z and for t (6.3.51),

$$\operatorname{Res}_{P'_j}((z\otimes 1)\,\mathrm{d}(t\otimes 1))=\operatorname{Res}_X\big(L_j(z\otimes 1)\,\mathrm{d}L_j(t\otimes 1)\big)=\operatorname{Res}_X\Big((\iota_j)_*L_\pi(z)\,\mathrm{d}(\iota_j)_*L_\pi(t)\Big)=\iota_j(\operatorname{Res}_X(L_\pi(z)\,\mathrm{d}L_\pi(t)))=\iota_j\Big(\operatorname{Res}_P^{k(P)}(z\,\mathrm{d}t)\Big).$$

By (6.3.48), $\operatorname{Mor}_k\big(k(P),\overline{k}\big)=\{\iota_1,\dots,\iota_d\}$; as k is perfect, $k(P)/k$ is separable (3.18.103), so $[k(P):k]_i=1$ (3.18.93). So (3.18.129) (trace is the sum of the embeddings in $\overline{k}$) gives, with $\alpha:=\operatorname{Res}_P^{k(P)}(z\,\mathrm{d}t)$,

$$\operatorname{Tr}_{k(P)/k}(\alpha)=\sum_{j=1}^{d}\iota_j(\alpha)=\sum_{j=1}^{d}\operatorname{Res}_{P'_j}((z\otimes 1)\,\mathrm{d}(t\otimes 1)).$$

■

Corollary 6.3.53 (Residue Symbol as a Sum over Conjugate Places)
Let P be a place of K/k and $t,z\in K$. Then

$$\operatorname{Res}_P(z\,\mathrm{d}t)=\sum_{j=1}^{d}\operatorname{Res}_{P'_j}((z\otimes 1)\,\mathrm{d}(t\otimes 1)),$$

the sum of the residues at the conjugate rational places $P'_1,\dots,P'_d$ of $K_{\overline{k}}/\overline{k}$ over P (6.3.49).

Proof. This is the second assertion of (6.3.52). ■

Once the module of differentials $\Omega_{K/k}$ is available, it is a 1-dimensional K-vector space dual to $\operatorname{Der}_k(K,K)$ (both being $\operatorname{trdeg}_k(K)=1$-dimensional, (3.34.44)), and the pairing of $z\,\mathrm{d}t\in\Omega_{K/k}$ with ∂_π is exactly $z\,\partial_\pi(t)$; $\operatorname{Res}_P(z\,\mathrm{d}t)$ is then the residue, in this sense, of the differential $z\,\mathrm{d}t$.

6.3.9 The Algebraic Residue Theorem

This subsection proves the (algebraic) Residue Theorem for a general base field k, by reduction to the algebraically closed case of (6.3.34).

Proposition 6.3.54 (Algebraic Residue Theorem)
For $t,f\in K$,

$$\sum_P\operatorname{Res}_P(f\,\mathrm{d}t)=0,$$

summed over all places P of K/k (finitely many terms nonzero, as $f\,\mathrm{d}t$ has only finitely many poles).

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

By (6.3.47) (applied to K/k), restriction $\overline{P}\mapsto\overline{P}\cap K$ is a bijection $\{\text{places of } K/k\}\leftrightarrow\big\{\operatorname{Aut}\big(\overline{k}/k\big)\text{-orbits of places of } K_{\overline{k}}/\overline{k}\big\}$ (where $K_{\overline{k}}:=K\otimes_k\overline{k}$, a field by (3.35.45) as in (6.3.43)); the fibre over P is $\big\{P'_1,\dots,P'_{d_P}\big\}$, $d_P:=\deg P$. By (6.3.53), for every place P of K/k and $z\in K$,

$$\operatorname{Res}_P(z\,\mathrm{d}t)=\sum_{j=1}^{d_P}\operatorname{Res}_{P'_j}((z\otimes 1)\,\mathrm{d}(t\otimes 1)).$$

Summing over all places P of K/k and using that the fibres $\big\{P'_1,\dots,P'_{d_P}\big\}$ partition the places of $K_{\overline{k}}/\overline{k}$ (restriction being a surjection),

$$\sum_{P\text{ of }K/k}\operatorname{Res}_P(f\,\mathrm{d}t)=\sum_{\overline{P}\text{ of }K_{\overline{k}}/\overline{k}}\operatorname{Res}_{\overline{P}}((f\otimes 1)\,\mathrm{d}(t\otimes 1))=0$$

by (6.3.34) (applicable as $K_{\overline{k}}/\overline{k}$ is itself an algebraic function field, over the algebraically closed field $\overline{k}$, (6.3.43)), applied to $K_{\overline{k}}/\overline{k}$, $t\otimes 1$, $f\otimes 1$. ■

6.3.10 Differentials are Weil Differentials, for General k

Fix a transcendental $t\in K$. By (6.3.20), every $D\in\operatorname{Der}_k(K,K)$ is $D(t)\cdot\partial_t$, so a differential $z\,\mathrm{d}t$ ($z\in K$) already carries all the information of an element of the (not yet formally constructed) module $\Omega_{K/k}$, and $\operatorname{Res}_P(z\,\mathrm{d}t)$ is available at every place (6.3.25). We show this recovers Ω_K exactly, generalising (6.3.19) beyond $k=\mathbb{C}$: the complex-analytic ingredients there (contour integration, Stokes' theorem, Mittag-Leffler) are replaced by the algebraic Residue Theorem, the trace formalism of (6.3.25), and the following fact, taken as given in this subsection.

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Proposition 6.3.55 (Ω_K is a K-Vector Space)
Let K/k be an algebraic function field. For $x \in K$ and $\omega \in \Omega_K(D)$ ($D \in \mathrm{Div}(K)$), define $x\omega \in \mathrm{Hom}_k(\mathcal{A}_K, k)$ by $(x\omega)(\alpha) := \omega(x\alpha)$ (using that $\mathcal{A}_K$ is a K-algebra). Then $x\omega \in \Omega_K(D-(x)) \subset \Omega_K$, and this action makes Ω_K a K-vector space.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

$x\omega \in \Omega_K(D-(x))$. For $\alpha \in \mathcal{A}_K(D-(x))$, $v_P(\alpha_P) \geq -v_P(D-(x)) = v_P(x) - v_P(D)$ for every P, so $v_P(x\alpha_P) \geq -v_P(D)$, that is $x\alpha \in \mathcal{A}_K(D)$; and $xK = K$. As $\omega \in \Omega_K(D)$ vanishes on $\mathcal{A}_K(D) + K$, it follows that $(x\omega)(\alpha) = \omega(x\alpha) = 0$ for every $\alpha \in \mathcal{A}_K(D-(x)) + K$, that is $x\omega \in \Omega_K(D-(x))$.

Module axioms. For $x, y \in K$, $\lambda \in k$, $\alpha, \beta \in \mathcal{A}_K$: $(x\omega)(\alpha+\beta) = \omega(x\alpha + x\beta) = (x\omega)(\alpha) + (x\omega)(\beta)$ and $(x\omega)(\lambda\alpha) = \omega(\lambda x\alpha) = \lambda(x\omega)(\alpha)$, so $x\omega \in \mathrm{Hom}_k(\mathcal{A}_K, k)$ is again k-linear; $(x(y\omega))(\alpha) = (y\omega)(x\alpha) = \omega(yx\alpha) = ((xy)\omega)(\alpha)$ and $1\omega = \omega$; $((x+y)\omega)(\alpha) = \omega((x+y)\alpha) = \omega(x\alpha) + \omega(y\alpha) = (x\omega)(\alpha) + (y\omega)(\alpha)$; and $(\omega \mapsto x\omega)$ is k-linear by the same computation applied to $\omega_1 + \omega_2$. These are exactly the axioms making Ω_K a K-module, that is a K-vector space (being already a k-vector space). ∎

Proposition 6.3.56 (Ω_K is 1-Dimensional over K)
Let K/k be an algebraic function field. Then $\dim_K \Omega_K = 1$ (for the K-vector space structure of (6.3.55)).

Proposition 6.3.57 (Differentials are Weil Differentials)
Let K/k be an algebraic function field and $t \in K$ transcendental. For $z \in K$, define $\varphi_{z\,\mathrm{d}t} \in \mathrm{Hom}_k(\mathcal{A}_K, k)$ by

$$\varphi_{z\,\mathrm{d}t}(\alpha) := \sum_P \mathrm{Res}_P(\alpha_P z\,\mathrm{d}t), \qquad \alpha = (\alpha_P)_P \in \mathcal{A}_K$$

(a finite sum: $\alpha_P \in \mathcal{O}_P$ for all but finitely many P, and $z\,\mathrm{d}t$ is regular at all but finitely many P, so $\mathrm{Res}_P(\alpha_P z\,\mathrm{d}t) = 0$ for almost all P (6.3.33).c)). Then $\varphi_{z\,\mathrm{d}t} \in \Omega_K$, and

$$K \longrightarrow \Omega_K, \qquad z \longmapsto \varphi_{z\,\mathrm{d}t}$$

is a canonical K-linear isomorphism (for the K-vector space structures of (6.3.55) and K itself).

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

$\varphi_{z\,\mathrm{d}t} \in \Omega_K$. Let D be the divisor with $v_P(D) := v_P(z\,\partial_\pi(t))$ for a uniformiser π at P (independent of the choice of π: for another uniformiser π', $\partial_{\pi'}(t) = \partial_{\pi'}(\pi)\,\partial_\pi(t)$ (6.3.22), with $\partial_{\pi'}(\pi)$ a unit at P, so $v_P(z\partial_{\pi'}(t)) = v_P(z\partial_\pi(t))$; and this vanishes for all but finitely many P, as z has finitely many poles and t finitely many poles). For $\alpha \in \mathcal{A}_K(D)$, $v_P(\alpha_P) \geq -v_P(D) = -v_P(z\partial_\pi(t))$ for every P, so $v_P(\alpha_P z\partial_\pi(t)) \geq 0$, that is $\alpha_P z\,\partial_\pi(t) \in \mathcal{O}_P$, giving $\mathrm{Res}_P(\alpha_P z\,\mathrm{d}t) = 0$ (6.3.33).c) so $\varphi_{z\,\mathrm{d}t}(\alpha) = 0$. For $f \in K$ embedded diagonally, $\varphi_{z\,\mathrm{d}t}(f, f, \ldots) = \sum_P \mathrm{Res}_P(fz\,\mathrm{d}t) = 0$ by (6.3.54) (the Residue Theorem), applied to $fz \in K$. So $\varphi_{z\,\mathrm{d}t}$ vanishes on $\mathcal{A}_K(D) + K$, that is $\varphi_{z\,\mathrm{d}t} \in \Omega_K(D) \subset \Omega_K$.

K-linearity. k-linearity in z is immediate from (6.3.33).a), applied place by place and summed. For $x \in K$, using $x\alpha_P \cdot z = \alpha_P \cdot (xz)$ in the field K,

$$(x\,\varphi_{z\,\mathrm{d}t})(\alpha) = \varphi_{z\,\mathrm{d}t}(x\alpha) = \sum_P \mathrm{Res}_P((x\alpha_P)z\,\mathrm{d}t) = \sum_P \mathrm{Res}_P(\alpha_P(xz)\,\mathrm{d}t) = \varphi_{(xz)\,\mathrm{d}t}(\alpha),$$

so $x\,\varphi_{z\,\mathrm{d}t} = \varphi_{(xz)\,\mathrm{d}t}$: the map is K-linear (6.3.55).

Injectivity. Suppose $\varphi_{z\,\mathrm{d}t} = 0$ and $z \neq 0$; we derive a contradiction. Fix a place P and a uniformiser π at P. As ∂_π is a nonzero derivation and t is transcendental, $\partial_\pi(t) \neq 0$ (6.3.20), so $w := z\,\partial_\pi(t) \neq 0$ in K. Let $m := v_P(w)$ and $c := \pi^{-m-1} \in K^\star$; then $L_\pi(cw)$ has valuation exactly -1 (6.3.23), so

$$\gamma := \mathrm{Res}_P^{k(P)}(cz\,\mathrm{d}t) = L_\pi(cw)_{-1} \in k(P)$$

is its leading coefficient, and $\gamma \neq 0$. As k is perfect, $k(P)/k$ is separable (3.18.103), so by (3.18.131) there is $\overline{e} \in k(P)$ with $\mathrm{Tr}_{k(P)/k}(\overline{e}\gamma) \neq 0$; lift it to a unit $e \in \mathcal{O}_P^\star$ (any lift of a nonzero residue class is a unit, $\mathcal{O}_P$ being local). Then $L_\pi(e)$ has constant term $\overline{e}$ (6.3.23), and $L_\pi(cw)$ has no terms below X^{-1}, so the coefficient of X^{-1} in $L_\pi(ecw) = L_\pi(e)L_\pi(cw)$ is $\overline{e}\gamma$. That is $\mathrm{Res}_P^{k(P)}(ecz\,\mathrm{d}t) = \overline{e}\gamma$, and

$$\mathrm{Res}_P(ecz\,\mathrm{d}t) = \mathrm{Tr}_{k(P)/k}(\overline{e}\gamma) \neq 0.$$

Testing $\varphi_{z\,\mathrm{d}t}$ against the repartition supported at P with value $ec \in K$ then gives $\varphi_{z\,\mathrm{d}t}(\ldots) = \mathrm{Res}_P(ecz\,\mathrm{d}t) \neq 0$, contradicting $\varphi_{z\,\mathrm{d}t} = 0$.

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Surjectivity. The map $K \to \Omega_K$ is K-linear and injective, and K is a 1-dimensional K-vector space over itself; so its image is a 1-dimensional K-subspace of Ω_K, which by (6.3.56) is itself 1-dimensional over K. So the image is all of Ω_K. ∎

6.3.11 Tate's Approach to Residues: A Sketch

The route taken above defines Res_P through a uniformiser and the formal derivative, and proves the Residue Theorem (6.3.54) by reducing to $K = k(x)$ along a finite separable extension. Tate [Tat68] gives a different, intrinsic route, which we only sketch here (nothing later depends on it): the residue is defined from linear algebra alone, and the Residue Theorem is deduced from the finiteness results of the Riemann-Roch theory.

a) *Traces of finite-rank operators.* The trace of a finite-rank endomorphism T of a k-vector space V is its trace on any finite-dimensional subspace containing $T(V)$. It is additive, $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$ whenever $A : V \to W$, $B : W \to V$ and one of them has finite rank, and $\mathrm{Tr}(T) = \mathrm{Tr}(T|_H) + \mathrm{Tr}(\overline{T})$ if T preserves a finite-dimensional H, with $\overline{T}$ the induced operator on V/H. In particular $\mathrm{Tr}([T, F]) = 0$ for F of finite rank and T arbitrary.

b) *The local residue.* Let π be a k-linear retraction of the local space onto the lattice $\mathcal{O}_P$ (in Tate's setting the completion K_P, with π the projection killing the principal parts). For f in the local space let $T_f := \pi \circ (f \cdot -)|_{\mathcal{O}_P}$ be the Toeplitz operator on $\mathcal{O}_P$. Then $[T_f, T_g]$ has finite rank, because $(1-\pi)(f\mathcal{O}_P)$ lies in the finite-dimensional image of $t^{-n}\mathcal{O}_P/\mathcal{O}_P$, and $\mathrm{Res}_P(f\,\mathrm{d}g) := \mathrm{Tr}\big([T_f, T_g]\big)$. Changing π changes each T_f by a finite-rank operator, which does not change the trace of the commutator. So no uniformiser, derivation or chain rule enters.

c) *Local properties.* It is k-linear, vanishes on differentials regular at P, has $\mathrm{Res}_P(\mathrm{d}t/t) = \deg P$ (the trace is taken over k) and is invariant under automorphisms of K_P; it should agree with the trace of the coefficient of t^{-1} used above.

d) *Global input.* The finiteness underlying Riemann-Roch: $H := K \cap \mathcal{A}_K(0) = L(0)$ and $\mathcal{A}_K/(K + \mathcal{A}_K(0))$ are finite dimensional (6.3.15) and $\mathcal{A}_K(D)/\mathcal{A}_K(0) = \bigoplus_P t_P^{-n_P}\mathcal{O}_P/\mathcal{O}_P$ is finite dimensional for every divisor $D = \sum_P n_P P$.

e) *The Residue Theorem as a trace identity.* Write $\mathcal{A} := \mathcal{A}_K$ and $L := \mathcal{A}(0) = \prod_P \mathcal{O}_P$, and for $f, g \in K$ let D be a divisor with $f\mathcal{A}(0), g\mathcal{A}(0) \subset \mathcal{A}(D)$.

   (a) For the retraction $\pi_L := \prod_P \pi_P : \mathcal{A} \to L$ and $T_f := \pi_L \circ (f \cdot -)|_L$, the commutator $[T_f, T_g]$ is the direct sum over P of the local commutators, of which only finitely many are nonzero (where f, g are regular the operators are multiplication by f, g and commute). So $\mathrm{Tr}\big([T_f, T_g]\big) = \sum_P \mathrm{Res}_P(f\,\mathrm{d}g)$.

   (b) As in b), $\mathrm{Tr}\big([T_f, T_g]\big)$ does not depend on the retraction $\pi : \mathcal{A} \to L$: another retraction changes T_f by $\psi \circ (1-\pi) f|_L$, which has finite rank since $(1-\pi)(\mathcal{A}(D))$ is finite dimensional.

   (c) Choose π adapted to K: write $K = H \oplus W_K$ and choose a finite-dimensional F with $\mathcal{A} = L \oplus W_K \oplus F$ (possible by d)), and let π be the projection onto L along $W_K \oplus F$. For $h \in H$, $fh \in K = H \oplus W_K$ so $T_f h \in H$: both T_f, T_g preserve H and act there as operators on a finite-dimensional space, whose commutator has trace 0.

   (d) On $L/H \cong (L+K)/K =: \Lambda \subset \mathcal{A}/K$ the induced operator $\overline{T}_f$ is the Toeplitz operator of multiplication by f on $\mathcal{A}/K$ (where f, g commute exactly), with respect to the retraction $\mathcal{A}/K = \Lambda \oplus \overline{F} \to \Lambda$. Writing $\overline{\pi} = 1 - p$ with p the projection onto $\overline{F}$, and using $fg = gf$ on $\mathcal{A}/K$, one finds $[\overline{T}_f, \overline{T}_g] = -\overline{\pi} f p g + \overline{\pi} g p f$ on Λ, whose trace equals $\mathrm{Tr}_{\overline{F}}([pg|_{\overline{F}}, pf|_{\overline{F}}]) = 0$, a commutator of operators on the finite-dimensional space $\overline{F}$.

   By (c), (d) and additivity of the trace, $\mathrm{Tr}\big([T_f, T_g]\big) = 0$, so $\sum_P \mathrm{Res}_P(f\,\mathrm{d}g) = 0$ by (a) and (b).

Nothing here uses the topology of K_P. The construction of b) uses only the subspace $\mathcal{O}_P \subset K$, a retraction $\pi : K \to \mathcal{O}_P$, and $\dim_k t^{-n}\mathcal{O}_P/\mathcal{O}_P < \infty$ (as $k(\mathfrak{m}_P)/k$ is finite): "lattices" and "continuity" reduce to commensurability of subspaces and to the pole order of f, so the local residue may be defined on K itself, and the global step on the repartitions $\mathcal{A}_K$ of this book (components in K, not in the K_P) together with the finiteness of d). Compared with the proof above, this avoids the reduction to $k(x)$, the trace formula and separability, and works in any characteristic. What is left to check is that this Res_P agrees with the Laurent-coefficient residue used above (for a place of degree d, $\mathrm{Res}_P(\mathrm{d}t/t) = d$ with the trace taken over k); the global argument of e) is then complete.

6.3.12 Sheaf-Theoretic Interpretation of Repartitions

Unlike the rest of this chapter, this section depends on sheaf theory developed later in this book (4.2.2), and so is not logically self-contained at this point in the text; nothing later in this chapter depends on it.

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Let $X \coloneqq \mathbb{P}_K$, the set of places of K/k, equipped with the **cofinite topology** (opens are $\emptyset$ and complements of finite subsets); as $\mathbb{P}_K$ is infinite, any two nonempty opens intersect, so X is irreducible. Let $\mathcal{K}_X$ be the constant sheaf with value K (4.2.50). For $D \in \mathrm{Div}(K)$ define the subsheaf

$$\mathcal{O}(D)(U) \coloneqq \{x \in K \mid v_P(x) \geq -v_P(D)\ \forall P \in U\} \subseteq \mathcal{K}_X(U)$$

generalising the structure sheaf ($\mathcal{O}(0) = \mathcal{O}_X$), with inclusion $\iota_D : \mathcal{O}(D) \hookrightarrow \mathcal{K}_X$.

Proposition 6.3.58 (Stalk of $\mathcal{O}(D)$)
For $P \in X$, the canonical map $\mathcal{O}(D)_P \to \mathcal{K}_{X,P} = K$ is injective, identifying $\mathcal{O}(D)_P$ with the fractional ideal

$$\{y \in K \mid v_P(y) \geq -v_P(D)\} =: \pi_P^{-v_P(D)}\mathcal{O}_P$$

of K.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Stalk of $\mathcal{K}_X$. Since X is irreducible, $\mathcal{K}_X(U) = K$ for every nonempty open $U \subset X$, with every restriction map the identity (4.2.50). We can therefore represent the stalk as $\mathcal{K}_{X,P} = K$ with the identity maps. Under this identification, the canonical map $\mathcal{O}(D)_P \to \mathcal{K}_{X,P} = K$ is given by $(U, x) \mapsto x$.

Injectivity. Two representatives $(U, x), (U', x')$ of elements of $\mathcal{O}(D)_P$ become identified in the stalk exactly when they agree on some common refinement; as restriction maps are the identity, this means $x = x'$ already, as elements of K. So $(U, x) \mapsto x$ is injective, and we identify $\mathcal{O}(D)_P$ with its image in K.

$\mathcal{O}(D)_P \subseteq \{y : v_P(y) \geq -v_P(D)\}$. Every element of $\mathcal{O}(D)_P$ is represented by a pair (U, x), $P \in U$, $x \in \mathcal{O}(D)(U)$, whose image in $\mathcal{O}(D)_P$ is x itself (viewed in K); taking the defining condition of $\mathcal{O}(D)(U)$ at the point $P \in U$ gives $v_P(x) \geq -v_P(D)$.

$\{y : v_P(y) \geq -v_P(D)\} \subseteq \mathcal{O}(D)_P$. Let $y \in K$ with $v_P(y) \geq -v_P(D)$. By (6.2.116), y has finitely many poles, so together with finiteness of $\mathrm{Supp}(D)$, the set $T \coloneqq \{Q \in X \mid v_Q(y) < -v_Q(D)\}$ is finite. As $P \notin T$, $U \coloneqq X \setminus T$ is an open neighbourhood of P, and $y \in \mathcal{O}(D)(U)$ by construction of T; so (U, y) represents an element of $\mathcal{O}(D)_P$ mapping to y.

Fractional ideal. By the previous two paragraphs $\mathcal{O}(D)_P = \{y \in K \mid v_P(y) \geq -v_P(D)\}$. Writing π_P for a uniformiser of $\mathcal{O}_P$, (3.30.2) represents every $y \in K^\star$ uniquely as $y = u\pi_P^{v_P(y)}$ for $u \in \mathcal{O}_P^\star$, so $v_P(y) \geq -v_P(D) \iff y \in \pi_P^{-v_P(D)}\mathcal{O}_P$; thus $\mathcal{O}(D)_P = \pi_P^{-v_P(D)}\mathcal{O}_P$. This is the image of $\mathcal{O}_P$ under multiplication by the fixed element $\pi_P^{-v_P(D)} \in K$, hence an $\mathcal{O}_P$-submodule of K and so a fractional ideal of $\mathcal{O}_P$ (3.29.1) (indeed $\mathcal{O}_P$ is a discrete valuation ring by (6.2.114), so this is the fractional ideal $\mathfrak{m}_P^{-v_P(D)}$ of (3.30.7)); in particular it is a k-vector subspace of K, since $k \subset \mathcal{O}_P$ (6.2.113). ∎

Proposition 6.3.59 (Stalk of the Cokernel)
Since ι_D is a monomorphism (6.3.58), write $\mathcal{K}_X/\mathcal{O}(D) \coloneqq \mathrm{coker}(\iota_D)$ (4.2.36). There is a unique morphism θ_P making

$$\begin{array}{ccc} & K & \\ \swarrow & & \searrow \\ (\mathcal{K}_X/\mathcal{O}(D))_P & \xrightarrow[\sim]{\theta_P} & K/\mathcal{O}(D)_P \end{array}$$

commute, and θ_P is (moreover) an isomorphism; in particular $(\mathcal{K}_X/\mathcal{O}(D))_P \cong K/\mathcal{O}(D)_P = K/\pi_P^{-v_P(D)}\mathcal{O}_P$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

This is (4.2.37), applied to $\iota_D : \mathcal{O}(D) \to \mathcal{K}_X$ (using $\mathcal{K}_{X,P} = K$), together with (6.3.58) identifying $\mathcal{O}(D)_P$ with $\pi_P^{-v_P(D)}\mathcal{O}_P$. ∎

Used freely below.

Proposition 6.3.60 (Repartitions compute the Cokernel Sheaf)
Writing $p_X : K = \Gamma(X, \mathcal{K}_X) \to \Gamma(X, \mathcal{K}_X/\mathcal{O}(D))$ for the global sections of the cokernel map (4.2.35), there is an exact sequence of k-vector spaces

$$0 \longrightarrow L(D) \longrightarrow K \xrightarrow{p_X} \Gamma(X, \mathcal{K}_X/\mathcal{O}(D)) \xrightarrow{\Phi} \mathcal{A}_K/(\mathcal{A}_K(D) + K) \longrightarrow 0$$

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where Φ sends $\tau \in \Gamma(X, \mathcal{K}_X/\mathcal{O}(D))$ – a function $\tau : X \to \coprod_{P\in X} K/\mathcal{O}(D)_P$ (4.2.35) – to the class of any repartition $(\alpha_P)_P \in \mathcal{A}_K$ with $\alpha_P \equiv \tau(P) \pmod{\mathcal{O}(D)_P}$ for every P; that is, sending a representative $x \in K$ to its corresponding residue vector. In particular $i(D)$ is (via (6.3.60)) the dimension of the cokernel of p_X – via sheaf cohomology, the group $H^1(X, \mathcal{O}(D))$, a cohomological invariant of X, not merely of the algebra of repartitions.

Sketch. $(\iota_D)_U$ is the inclusion $\mathcal{O}(D)(U) \hookrightarrow K$, injective.

By (4.2.35) a global section of $\mathcal{K}_X/\mathcal{O}(D)$ is represented by a cover of X together with representatives in K over each open, and since the cofinite topology is quasi-compact (4.1.93) this cover may be taken finite, say $X = U_1 \cup \cdots \cup U_n$ together with representatives $x_i \in K$, well-defined modulo $\mathcal{O}(D)(U_i)$, that **agree at every point of overlap**: for every place $P \in U_i \cap U_j$, the images of x_i, x_j in the stalk $(\mathcal{K}_X/\mathcal{O}(D))_P = K/\pi_P^{-v_P(D)}\mathcal{O}_P$ (6.3.59) coincide, that is

$$v_P(x_i - x_j) \geq -v_P(D)$$

Given such data, set $\alpha_P := x_i$ (a literal element of K, well-defined only up to the choice of i) for any i with $P \in U_i$. The class of α_P modulo $\pi_P^{-v_P(D)}\mathcal{O}_P$ is independent of the choice of i: if also $P \in U_j$, then $P \in U_i \cap U_j$, so the overlap condition above applies directly at this P and gives $v_P(x_i - x_j) \geq -v_P(D)$, that is $x_i \equiv x_j \pmod{\pi_P^{-v_P(D)}\mathcal{O}_P}$. Each x_i has finitely many poles (6.2.116), so $\{P \mid \alpha_P \notin \mathcal{O}_P\} \subseteq \bigcup_{i=1}^n \{P \in U_i \mid v_P(x_i) < 0\}$, a finite union of finite sets, is finite; so $\alpha_P \in \mathcal{O}_P$ for all but finitely many P, giving $(\alpha_P)_P \in \mathcal{A}_K$.

Independence of the choice of representing data. Let $(U_i, x_i)_{i=1}^n$ and $(U_i', x_i')_{i=1}^m$ be two choices representing the same τ, with resulting repartitions $(\alpha_P)_P, (\alpha_P')_P \in \mathcal{A}_K$ as above. For each P, choosing i, i' with $P \in U_i \cap U_i'$, both x_i and x_i' represent $\tau(P) \in K/\mathcal{O}(D)_P$ (4.2.35), that is $x_i \equiv x_i' \pmod{\mathcal{O}(D)_P}$, so $\alpha_P - \alpha_P' = x_i - x_i' \in \mathcal{O}(D)_P$. As this holds for every P, $(\alpha_P - \alpha_P')_P \in \mathcal{A}_K(D)$; so $(\alpha_P)_P$ and $(\alpha_P')_P$ have the same class in $\mathcal{A}_K/(\mathcal{A}_K(D) + K)$. This assignment therefore descends to a well-defined map

$$\Phi : \Gamma(X, \mathcal{K}_X/\mathcal{O}(D)) \longrightarrow \mathcal{A}_K/(\mathcal{A}_K(D) + K)$$

Exactness at K. For $c \in K$, $p_X(c)$ is represented by the trivial cover $\{X\}$ with $x_1 := c$, so $p_X(c) = 0$ if and only if $c \in \mathcal{O}(D)(X) = L(D)$; that is, $\ker(p_X) = L(D)$, matching exactness at K (the map $L(D) \hookrightarrow K$ being the inclusion).

Exactness at $\Gamma(X, \mathcal{K}_X/\mathcal{O}(D))$. ($\mathrm{Im}(p_X) \subseteq \ker(\Phi)$) For $c \in K$, $p_X(c)$ has $\alpha_P = c$ for every P, that is, the constant repartition $\delta(c)$ (diagonal embedding $\delta : K \hookrightarrow \mathcal{A}_K$), whose class in $\mathcal{A}_K/(\mathcal{A}_K(D) + K)$ is 0, since $\delta(c) \in K$; so $\Phi(p_X(c)) = 0$.

($\ker(\Phi) \subseteq \mathrm{Im}(p_X)$) Suppose $\Phi(\tau) = 0$, that is $(\alpha_P)_P \in \mathcal{A}_K(D) + K$, say $(\alpha_P)_P = \beta + \delta(c)$ with $\beta \in \mathcal{A}_K(D)$, $c \in K$. Then $x_i - c \in \mathcal{O}(D)(U_i)$ for every i (as $x_i - c = \alpha_P - c = \beta_P \in \mathcal{O}(D)_P$ for every $P \in U_i$), so τ and $p_X(c)$ are represented by the same local data, that is $\tau = p_X(c) \in \mathrm{Im}(p_X)$.

Exactness at $\mathcal{A}_K/(\mathcal{A}_K(D) + K)$. Given $(\alpha_Q)_Q \in \mathcal{A}_K$, let

$$S := \{Q \mid \alpha_Q \notin \mathcal{O}_Q\} \cup \mathrm{Supp}(D) = \{P_1, \ldots, P_n\}$$

finite, being a union of two finite sets; note $v_Q(D) = 0$ for every $Q \notin S$, so $\mathcal{O}(D)_Q = \mathcal{O}_Q$ there. Set $x_0 := 0$ and, for $i = 1, \ldots, n$, $x_i := \alpha_{P_i}$ and $Z_i := \{Q \neq P_i \mid \alpha_{P_i} \notin \mathcal{O}_Q\}$, the poles of α_{P_i} other than possibly at P_i itself, finite by (6.2.116). Since the cofinite topology has no "small" neighbourhoods – any nonempty open already excludes only finitely many points – each U_i ($i \geq 1$) must be taken to also avoid x_i's own poles, not merely to contain P_i: set

$$U_0 := X \setminus S, \qquad U_i := X \setminus ((S \cup Z_i) \setminus \{P_i\}) \quad (i = 1, \ldots, n)$$

These cover X: $U_0 \cup \cdots \cup U_n \supseteq (X \setminus S) \cup \{P_1, \ldots, P_n\} = X$, as $P_i \in U_i$ for each i.

Each U_i meets S in at most the single point P_i: $U_0 \cap S = \emptyset$, and $U_i \cap S = \{P_i\}$ for $i \geq 1$, since U_i excludes every other point of S. Consequently, for $i \neq j$, no point of S lies in both U_i and U_j (it would have to be the marked point of each, and the marked points – or absence thereof, for $i = 0$ – differ), so $U_i \cap U_j \subseteq X \setminus S$; in particular $Q \in U_i \cap U_j$ gives $v_Q(D) = 0$ and $Q \neq P_i, P_j$.

They agree at overlaps: for $i \neq j$ in $\{0, \ldots, n\}$ and $Q \in U_i \cap U_j$, $v_Q(x_i) \geq 0$ (trivially if $i = 0$, as $x_0 = 0$; else since $Q \in U_i$, $Q \neq P_i$ gives $Q \notin Z_i$) and likewise $v_Q(x_j) \geq 0$; so by the ultrametric inequality $v_Q(x_i - x_j) \geq \min(v_Q(x_i), v_Q(x_j)) \geq 0 = -v_Q(D)$.

This data therefore glues to a class $\tau \in \Gamma(X, \mathcal{K}_X/\mathcal{O}(D))$ with $\tau(Q) = \alpha_Q \pmod{\mathcal{O}(D)_Q}$ for every Q (on U_0 this matches $x_0 = 0$, since $Q \notin S$ gives $\alpha_Q \in \mathcal{O}_Q = \mathcal{O}(D)_Q$; at $Q = P_i$ this matches $x_i = \alpha_{P_i}$ directly), so $\Phi(\tau) = (\alpha_Q)_Q$ exactly; so Φ is surjective, matching exactness at the final term. ∎

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6.4 Projective Varieties

6.4.1 Projective Algebraic Sets

Proposition 6.4.1 (Projective Space is Functorial)
Let $\phi : K \to L$ be an injective ring homomorphism Then there is a well-defined injective map

$$\begin{aligned} \mathbb{P}^n(K) &\to \mathbb{P}^n(L) \\ (x_0 : ... : x_n) &\to (\phi(x_0) : ... : \phi(x_n)) \end{aligned}$$

Proof. Evidently the map is well-defined. Suppose that

$$(\phi(x_0) : ... : \phi(x_n)) \sim (\phi(x_0') : ... : \phi(x_n'))$$

Then for some i we have $x_i \neq 0$ and $\phi(x_i) = \lambda\phi(x_i')$ and $\lambda \in L^\star$. Therefore $\phi(x_i') \neq 0 \implies x_i' \neq 0$ and $\lambda = \phi(x_i/x') =: \phi(\mu)$. Similarly $\phi(x_j) = \phi(\mu)\phi(x_j') \implies \phi(x_j - \mu x_j') = 0 \implies x_j = \mu x_j'$. ∎

Proposition 6.4.2 (Zero Loci in Projective Space)
*Let $A = k[X_0, ..., X_n]$ be the graded polynomial ring in $(n+1)$-variables over a field k. For a homogenous ideal $\mathfrak{a} \subset k[X_0, ..., X_n]$ and field extension K/k define the **zero-locus***

$$V_+(\mathfrak{a})(K) \coloneqq \{(x_0 : ... : x_n) \in \mathbb{P}^n(K) \mid F(x_0, ..., x_n) = 0 \quad \forall F \in \mathfrak{a}\}$$

Similarly for a subset $Y \subset \mathbb{P}^n(K)$ define

$$I_+(Y) \coloneqq \{F \in k[X_0, ..., X_n] \mid F_d(x_0, ..., x_n) = 0 \quad \forall d \geq 0, (x_0 : ... : x_n) \in Y\}$$

Then

a) $I_+(Y)$ is a homogenous radical ideal and $\sqrt{\langle S \rangle} \subseteq I_+(V_+(S)(K))$
b) I_+ and V_+ are order-reversing
c) $S \subseteq I_+(V_+(S)(K))$ and $Y \subseteq V_+(I_+(Y))(K)$
d) $V_+ \circ I_+ \circ V_+ = V_+$ and $I_+ \circ V_+ \circ I_+ = I_+$
e) $V_+(S) = V_+\left(\sqrt{\langle S \rangle}\right)$ and $V_+(\mathfrak{a}) = V_+(\sqrt{\mathfrak{a}})$
f) $\bigcap_i V_+(S_i) = V_+\left(\bigcup_i S_i\right)$ and $\bigcap_i V_+(\mathfrak{a}_i) = V_+\left(\sum_i \mathfrak{a}_i\right)$
g) $\bigcap_i I_+(W_i) = I_+(\bigcup W_i)$
h) $V_+(\mathfrak{a}) \cup V_+(\mathfrak{b}) = V_+(\mathfrak{a} \cap \mathfrak{b})$
i) $V_+((0)) = \mathbb{P}^n(K)$

*The sets of the form $V_+(\mathfrak{a})(K)$ determine the **Zariski Topology** on $\mathbb{P}^n(K)$. We denote the topological space by $\mathbb{P}_k^n(K)$. The topological closure satisfies the following identities*

$$V_+(I_+(X)) = \overline{X}$$

and

$$I_+(X) = I_+\left(\overline{X}\right)$$

There is a form of the Weak Nullstellensatz

$$V_+(\mathfrak{a})\left(\overline{k}\right) = \emptyset \iff A_+ \not\subset \sqrt{\mathfrak{a}} \iff \mathfrak{a} \text{ irrelevant}$$

and a form of the Strong Nullstellensatz namely

$$\mathfrak{a} \text{ essential} \implies I_+\left(V_+(\mathfrak{a})\left(\overline{k}\right)\right) = \sqrt{\mathfrak{a}}$$

Further there is a dual isomorphism

$$\left\{\mathfrak{a} \lhd A \;\middle|\; \mathfrak{a} \text{ radical homogenous and } A_+ \not\subset \mathfrak{a}\right\} \underset{I_+}{\overset{V_+}{\rightleftarrows}} \left\{Z \in \mathbb{P}_k^n\left(\overline{k}\right) \;\middle|\; Z \neq \emptyset\right\}$$

Proof. Let $V_K(S) \coloneqq \left\{(x) \in K^{n+1} \;\middle|\; f(x) = 0 \quad \forall s \in S\right\}$ denote the affine zero-locus. Then $V_+(S)(K) = \pi(V_K(S) \setminus \{\mathbf{0}\})$, where $\mathbf{0} = (0, ..., 0)$ is the origin in K^{n+1}.

a) Homogeneity follows from definition. The relation is straightforward.
b) Follows by definition
c) The pair I, V form a Galois connection by (2.1.54) and the relations follow formally by (2.1.50).

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d) Follows formally by (2.1.52)
e) Follows from (2.1.53) by considering the closure operators $\sqrt{\langle - \rangle}$ and $\langle - \rangle$.
f) The first equality follows from (2.1.55). The second equality follows from (3.3.29).
g) This follows from (2.1.55).
h) Then using the affine case we may deduce

$$V_+(\mathfrak{a} \cap \mathfrak{b}) = \pi(V(\mathfrak{a}) \cup V(\mathfrak{b}) \setminus \mathbf{0}) = \pi(V(\mathfrak{a}) \setminus \mathbf{0}) \cup \pi(V(\mathfrak{b}) \setminus \mathbf{0}) = V_+(\mathfrak{a}) \cup V_+(\mathfrak{b})$$

i) Trivial

Weak Nullstellensatz. Observe that using the affine case $V_{\overline{k}}(A_+) = \{\mathbf{0}\}$ and $I_{\overline{k}}(\mathbf{0}) = A_+$. Therefore

$$V_+(\mathfrak{a})\left(\overline{k}\right) = \emptyset \iff V_{\overline{k}}(\mathfrak{a}) \subset \{\mathbf{0}\} \overset{(3.12.7)}{\iff} A_+ \not\subset \sqrt{\mathfrak{a}}$$

Strong Nullstellensatz. Assume $\mathfrak{a}$ is an essential homogenous ideal then $Y := V(\mathfrak{a}) \setminus \{\mathbf{0}\}$ is non-empty and $\overline{Y} = V(\mathfrak{a})$. Because $\mathfrak{a}$ is homogenous then Y is a cone – $x \in Y \implies \lambda x \in Y$ for $\lambda \in \overline{k}^\star$. We claim that $V_+(\mathfrak{a}) = \pi(Y)$ and $I(Y) = I_+(\pi(Y))$. Suppose $F \in I(Y)$ then $F(\lambda x) = 0$ for all $\lambda \in \overline{k}^\star$ and $x \in Y$. As $\overline{k}$ is infinite we deduce that $F_d \in I(Y)$ whence $F \in I_+(\pi(Y))$. The remaining claims are clear. Therefore we may deduce from (6.1.2) and (6.1.13)

$$I_+V_+(\mathfrak{a}) = I_+(\pi(Y)) = I(Y) = I\left(\overline{Y}\right) = I(V(\mathfrak{a})) = \sqrt{\mathfrak{a}}$$

Topological Closure. By (2.1.52) $V_+ \circ I_+$ is a closure operator with image precisely the closed sets. Therefore by (2.1.41) $V_+ \circ I_+$ is the topological closure (4.1.19). Finally $I_+\left(\overline{X}\right) = I_+(V_+(I_+(X))) = I_+(X)$ using d).

The dual isomorphism follows from the Strong Nullstellensatz. ∎

Definition 6.4.3 (Projective Variety)
For an essential homogenous radical ideal $\mathfrak{a} \lhd k[X_0, ..., X_n]$ we say the family

$$X(K) := V_+(\mathfrak{a})(K) \subset \mathbb{P}^n_k(K)$$

*is a **projective variety** and write it as $X(K) := V_+(\mathfrak{a})(K)$. We say X is **integral** if $X\left(\overline{k}\right)$ is irreducible in the subspace topology. We also define the **homogenous coordinate ring***

$$k[X]_{\text{hom}} := k[X_0, ..., X_n]/\mathfrak{a}$$

*which is naturally a positive graded ring by (3.11.12). Define the **irrelevant ideal***

$$k[X]_+ := \bigoplus_{d>0} k[X]_d$$

We may generalise (6.4.2) as follows.

Proposition 6.4.4 (Zero Loci in Projective Varieties)
*Let $X = V_+(\mathfrak{a}) \subset \mathbb{P}^n_k$ be a projective variety and $A = k[X]_{\text{hom}}$ the coordinate ring. For any homogenous ideal $\mathfrak{b} \lhd A$ and field extension K/k define the **zero-locus** by*

$$V_+(\mathfrak{b})(K) := \{(x_0 : ... : x_n) \in X(K) \mid f(x_0, ..., x_n) = 0 \quad \forall f \in \mathfrak{b}\}$$

Similarly for a subset $Y \subset X(K)$ define

$$I_+(Y) := \{f \in k[X] \mid f_d(x) = 0 \quad \forall d \geq 0, x \in Y\}$$

Suppose $\pi : k[X_0, ..., X_n] \to k[X]$ is the quotient map. Then we have the following properties

$$V_+(\mathfrak{b})(K) = V_+(\pi^{-1}(\mathfrak{b}))(K) \cap X(K)$$
$$I_+(Y \cap X(K)) = \pi(I_+(Y)) \quad Y \subset \mathbb{P}^n_k(K)$$
$$I_+(V_+(\mathfrak{b})(K)) = \pi(I_+(V_+(\pi^{-1}(\mathfrak{b}))))$$

The pair $(V_+(-)(K), I_+)$ satisfies the same properties as (6.4.2). The image of $V_+(-)(K)$ forms the closed sets of a topology on $X(K)$ which coincides with the subspace topology from $\mathbb{P}^n_k(K)$. Further the Weak and Strong Nullstellensatz also hold

$$V_+(\mathfrak{b})\left(\overline{k}\right) = \emptyset \iff k[X]_+ \not\subset \sqrt{\mathfrak{b}} \iff \mathfrak{b} \text{ is irrelevant}$$

and

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$$\mathfrak{b} \text{ essential} \implies I_+\left(V_+(\mathfrak{b})\left(\overline{k}\right)\right) = \sqrt{\mathfrak{b}}$$

Further there is a dual isomorphism

$$\left\{\mathfrak{b} \lhd k[X]_{\text{hom}} \;\middle|\; \mathfrak{b} \text{ radical, homogenous and } A_+ \not\subset \mathfrak{b}\right\} \underset{I_+}{\overset{V_+}{\rightleftharpoons}} \left\{Z \in X\left(\overline{k}\right) \;\middle|\; Z \neq \emptyset\right\}$$

Proposition 6.4.5 (Projective Subvariety)
Let X be a projective variety, $\mathfrak{b} \lhd k[X]_{\text{hom}}$ an essential radical homogenous ideal. Then $V_+(\mathfrak{b})(-)$ is a ***closed subvariety*** *which coincides with the projective variety $V_+(\pi^{-1}(\mathfrak{b}))$. There is a bijective correspondence between non-empty closed subsets of $X\left(\overline{k}\right)$ and closed subvarieties of X.*

Conversely if X', X are two projective varieties such that $X'\left(\overline{k}\right) \subset X\left(\overline{k}\right)$ then X' may be expressed as a closed subvariety.

Proof. There is a canonical graded surjective k-algebra homomorphism $\pi : k[X_0, ..., X_n] \to k[X]_{\text{hom}}$ by (3.11.12). Then $\pi^{-1}(\mathfrak{b})$ is graded (3.11.10) and radical (3.3.50) g). Also by definition of the quotient grading on π we find $\pi^{-1}(k[X]_+) = k[X_0, ..., X_n]_+$ so that $\mathfrak{b}$ essential $\implies \pi^{-1}(\mathfrak{b})$ is essential (3.12.7). The bijective correspondence exists by (6.4.4).

If $X' = V_+(\mathfrak{b}')$, $X = V_+(\mathfrak{b})$ are two projective varieties such that $X'\left(\overline{k}\right) \subset X\left(\overline{k}\right)$ then by the dual isomorphism we have $\mathfrak{b} \subset \mathfrak{b}'$. Therefore X' may be represented as the closed subvariety $V_+(\pi(\mathfrak{b}'))$ ∎

Proposition 6.4.6 (Principal Open Sets)
Let $X \subset \mathbb{P}^n_k$ be a projective variety. The Zariski topology on $X(K)$ has a base consisting of subsets of the form

$$D_+(f)(K) \coloneqq X(K) \setminus V_+((f)) = \{(x) \in X(K) \mid f_d(x) \neq 0 \text{ some } d > 0\}$$

where $f \in k[X]$ is a homogenous form. Note that $(x) = (y) \implies (f_d(x) = 0 \iff f_d(y) = 0)$ so the second form is well-defined.

Proof. A generic open set $U = X(K) \setminus V_+(\mathfrak{b})(K)$ where $\mathfrak{b}$ is a homogenous ideal of $k[X]$. Given $(x) \in U$ there is some $f \in \mathfrak{b}$ and $d > 0$ such that $f_d(x) \neq 0$. Therefore $D_+(f_d) \subset U$ is an open neighbourhood of (x). ∎

Proposition 6.4.7
Let $X = V_+(\mathfrak{b})$ be a projective variety and $f, g \in k[X]$ essential homogenous forms. Then

$$D_+(g) \subset D_+(f) \iff f \mid g^N \text{ some } N > 0$$

Proof.

$$D_+(g) \subset D_+(f) \iff V_+((f)) \subset V_+((g)) \iff \sqrt{(g)} \subset \sqrt{(f)} \iff f \mid g^N$$

∎

Proposition 6.4.8 (Irreducible Subsets)
Let $X = V_+(\mathfrak{a}) \subset \mathbb{P}^n_k$ be a projective variety and $\mathfrak{b} \lhd k[X]$ a homogenous radical ideal. Then a closed subvariety $V_+(\mathfrak{b}) \subset X$ is integral iff $\mathfrak{b}$ is prime.

In particular $\mathbb{P}^n_k\left(\overline{k}\right)$ is integral.

Proof. Let $Y \coloneqq V_+(\mathfrak{b})$. Suppose that Y is not irreducible then $Y \subseteq V_+(\mathfrak{c}) \cup V_+(\mathfrak{d})$ is a non-trivial decomposition into closed subsets for $\mathfrak{c}, \mathfrak{d}$ essential homogenous radical ideals for which $Y \not\subset V_+(\mathfrak{c}) \implies \mathfrak{c} \not\subset \mathfrak{b}$ and similarly $\mathfrak{d} \not\subset \mathfrak{b}$ (see (4.1.61)). Choose $f \in \mathfrak{c} \setminus \mathfrak{b}$ and $g \in \mathfrak{d} \setminus \mathfrak{b}$. Then fg is zero on Y whence $fg \in \mathfrak{b}$. This shows that $\mathfrak{b}$ is not prime.

Conversely suppose $Y \subset V_+(\mathfrak{c}) \cup V_+(\mathfrak{d}) = V_+(\mathfrak{c} \cap \mathfrak{d})$ then by (6.4.2) $\mathfrak{c}\mathfrak{d} \subset \mathfrak{c} \cap \mathfrak{d} \subset \mathfrak{b}$. By (3.3.39) we have $\mathfrak{c} \subset \mathfrak{b}$ or $\mathfrak{d} \subset \mathfrak{b}$, whence $Y \subset V_+(\mathfrak{c})$ or $Y \subset V_+(\mathfrak{d})$. Therefore we conclude that Y is irreducible by (4.1.61).

Therefore $\mathbb{P}^n_k\left(\overline{k}\right)$ is irreducible by the special case both $\mathfrak{a}$ and $\mathfrak{b}$ being the zero ideals. ∎

Proposition 6.4.9 (Dual Isomorphism between Closed Sets and Homogenous Ideals)
Let $X = V_+(\mathfrak{a}) \subset \mathbb{P}^n_k$ be a projective variety and $A = k[X]$ the coordinate ring. Under the dual isomorphism

$$\left\{\mathfrak{b} \lhd A \;\middle|\; \mathfrak{b} \text{ radical homogenous and } A_+ \not\subset \mathfrak{b}\right\} \underset{I_+}{\overset{V_+}{\rightleftharpoons}} \left\{Z \subset X\left(\overline{k}\right) \text{ closed} \;\middle|\; Z \neq \emptyset\right\}$$

we have the following correspondence

- *irreducible sets correspond to prime ideals*

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- *irreducible components (of $V_+(\mathfrak{b})$) correspond to minimal prime ideals (of $\mathfrak{b}$)*

Proposition 6.4.10 (Decomposition into Irreducible Components)
Let $X = V_+(\mathfrak{a})$ be a projective variety then the topological space $X\left(\overline{k}\right)$ is Noetherian. Furthermore every closed subset Z has finitely many irreducible components Z_i and the decomposition

$$Z = Z_1 \cup ... \cup Z_n$$

is the unique incomparable decomposition into irreducible closed subsets.

If $Z = V_+(\mathfrak{b})$ then the irreducible components correspond to precisely the minimal prime ideals over $\mathfrak{b}$.

Proof. $X\left(\overline{k}\right)$ is Noetherian by using the order isomorphism in (6.4.9) and the fact $k[X]$ is Noetherian. The result then follows from (4.1.76). ∎

6.4.2 Affine Charts

References:

- Algebraic Curves [Ful08] (Chap. 4.3)
- Algebraic Geometry [Har13] (Chap. I §2)

For every $i = 0, ..., n$ the subset of $\mathbb{P}^n_k(K)$ consisting of points of the form

$$\{(x_0 : ... : x_n) \mid x_i \neq 0\}$$

may be naturally identified with $\mathbb{A}^n_k(K)$. The purpose of this section is to make this notion precise.

Lemma 6.4.11
There is an isomorphism of rings

$$\theta_i^\sharp : k[X_0, ..., X_n]_{(X_i)} \longleftrightarrow k[x_0, ..., x_{i-1}, x_{i+1}, ..., x_n]$$
$$\frac{X_j}{X_i} \longleftrightarrow x_j$$

Proof. The algebra homomorphism $\theta_i^\sharp$ exists by (3.7.6) and (3.5.5), restricting to the subring $k[X_0, ..., X_n]_{(X_i)}$. The inverse algebra homomorphism exists by (3.7.6), sending $x_j \to \frac{X_j}{X_i}$. We may verify directly that these are mutually inverse. ∎

Lemma 6.4.12 ((De-)Homogenisation)
Fix n and $0 \leq i \leq n$. For $G \in k[X_0, ..., X_n]$ homogenous and $F \in k[x_0, ..., x_{i-1}, x_{i+1}, ..., x_n]$ define

$$G^a := G(x_0, ..., x_{i-1}, 1, x_{i+1}, ..., x_n)$$
$$F^h := X_i^{\deg(F)} F\left(\frac{X_0}{X_i}, ..., \frac{X_{i-1}}{X_i}, \frac{X_{i+1}}{X_i}, ..., \frac{X_n}{X_i}\right)$$
$$= \sum_{j=0}^{\deg F} X_i^{\deg(F)-j} F_j(X_0, ..., X_{i-1}, X_{i+1}, ..., X_n)$$

Then

a) $(FF')^h = F^h F'^h$ *and* $(GG')^a = G^a G'^a$
b) $F^{ha} = F$ *and* $X_i^k G^{ah} = G$ *where* $k := v_{X_i}(G)$ *is the smallest power of X_i appearing in G.*
c) For $\mathfrak{b} \triangleleft k[X_0, ..., X_n]$ *a homogenous ideal the set* $\mathfrak{b}^a := \{G^a \mid G \in \mathfrak{b}\}$ *is an ideal.*

For $\mathfrak{a} \triangleleft k[x_0, ..., x_{i-1}, x_{i+1}, ..., x_n]$ *define the homogenous ideal*

$$\mathfrak{a}^h := \left\langle \{F^h \mid F \in \mathfrak{a}\} \right\rangle$$

Then we also have the identities

d) $\mathfrak{b} \subset \mathfrak{b}^{ah}$ *with equality when* $\mathfrak{b}$ *is radical and* $\mathfrak{b} \subset \mathfrak{p}$ *minimal* $\implies X_i \notin \mathfrak{p}$
e) $\mathfrak{a} = \mathfrak{a}^{ha}$
f) $\mathfrak{b}$ *prime and* $X_i \notin \mathfrak{b} \implies \mathfrak{b}^a$ *prime.*
g) $\mathfrak{a}$ *prime* $\implies \mathfrak{a}^h$ *prime*
h) $\mathfrak{a}$ *radical* $\implies \mathfrak{a}^h$ *radical*

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Proof. Mostly tedious verification

a) Using the notation of (6.4.11) we see that $F^h = X_i^{\deg(F)}\left(\theta_i^\sharp\right)^{-1}(F)$ so evidently $(FF')^h = F^h F'^h$ because $\deg(FF') = \deg(F) + \deg(F')$.

   The second relation follows immediately from (6.4.11).

b) The first identity is clear. We may write

$$G = \sum_{j=0}^{\deg(G)-v_{X_i}(G)} X_i^{\deg(G)-j} G_j(X_0, ..., X_{i-1}, X_{i+1}, ..., X_n)$$

   where $\deg\left(G_j\right) = j$ and $v_{X_i}(G)$ is the smallest power of X_i appearing in G. So

$$G^a = \sum_{j=0}^{\deg(G)-v_{X_i}(G)} G_j(x_0, ..., x_{i-1}, x_{i+1}, ..., x_n)$$

$$G^{ah} = \sum_{j=0}^{\deg(G)-v_{X_i}(G)} X_i^{\deg(G)-v_{X_i}(G)-j} G_j(X_0, ..., X_{i-1}, X_{i+1}, ..., X_n)$$

   whence $G = X_i^{v_{X_i}(G)} G^{ah}$.

c) This is evident because $(-)^a$ is a ring homomorphism.

d) The first part is evident from b). Suppose $G \in \mathfrak{b}^{ah}$ then

$$G = \sum_{j=1}^{r} \lambda_j G_j^{ah} \Longrightarrow X_i^k G = \sum_{j=1}^{r} \lambda_j X_i^{k_j} G_j$$

   for $G_j \in \mathfrak{b}$. Then $X_i^k G \in \mathfrak{b} \Longrightarrow G \in \mathfrak{b}$ as we assumed $X_i \notin \mathfrak{b}$.

e) Trivially $F \in \mathfrak{a} \Longrightarrow F^h \in \mathfrak{a}^h \Longrightarrow F^{ha} = F \in \mathfrak{a}^{ha}$. Conversely any $G \in \mathfrak{a}^h$ is of the form

$$G = \sum_j \lambda_j F_j^h \Longrightarrow G^a = \sum_j \lambda_j^a F_j^{ha} = \sum_j \lambda_j^a F_j \in \mathfrak{a}^{ha}$$

   for some $\lambda_i \in k[X_0, ..., X_n]$.

f) Suppose $FF' \in \mathfrak{b}^a \Longrightarrow F^h F'^h \in \mathfrak{b}^{ah} = \mathfrak{b} \Longrightarrow F^h \in \mathfrak{b}$ or $F'^h \in \mathfrak{b} \Longrightarrow F \in \mathfrak{b}^a$ or $F' \in \mathfrak{b}^a$.

g) Suppose $GG' \in \mathfrak{a}^h$ then $GG' = \sum_j \lambda_j F_j^h$ for $F_j \in \mathfrak{a}$. Then $G^a G'^a = \sum_j \lambda_j^a F_j^{ha} = \sum_j \lambda_j^a F_j \in \mathfrak{a}$. Therefore say $G^a \in \mathfrak{a} \Longrightarrow G^{ah} \in \mathfrak{a}^h \Longrightarrow G = X_i^k G^{ah} \in \mathfrak{a}^h$.

h) Suppose $G^n \in \mathfrak{a}^h$, then $(G^a)^n \in \mathfrak{a}^{ha} = \mathfrak{a}$ whence $G^a \in \mathfrak{a}$ and $G^{ah} \in \mathfrak{a}^h \Longrightarrow G \in \mathfrak{a}^h$

■

Proposition 6.4.13 (Projective Space has Affine Covering)
Consider subsets $D_+(X_i)(K) = U_i(K) \subset \mathbb{P}_k^n(K)$ of the form

$$D_+(X_i)(K) = U_i(K) \coloneqq \{(x_0 : ... : x_n) \mid x_i \neq 0\} \quad i = 0, ..., n$$

These are open in the Zariski topology. Furthermore under the subspace topology U_i is homeomorphic to $\mathbb{A}_k^n(K)$, with an explicit homeomorphism given by

$$\theta_i : \mathbb{A}_k^n(K) \to U_i$$
$$(x_0, ..., x_{i-1}, x_{i+1}, ..., x_n) \to [x_0 : ... : 1 : ... : x_n]$$

For $F \in k[x_0, ..., x_{i-1}, x_{i+1}, ..., x_n]$ and $(x) \in U_i$ we have

$$F(x) = 0 \Longleftrightarrow F^h(\theta_i(x)) = 0$$

and similarly for $G \in k[X_0, ..., X_n]$ we have

$$G(\theta_i(x)) = 0 \Longleftrightarrow G^a(x) = 0$$

Further we have

a) $\theta_i\left(V^{\text{aff}}(\mathfrak{a})\right) = V_+\left(\mathfrak{a}^h\right) \cap U_i$ for $\mathfrak{a} \lhd k[x_0, ..., x_{i-1}, x_{i+1}, ..., x_n]$
b) $\theta_i\left(V^{\text{aff}}(\mathfrak{b}^a)\right) = V_+(\mathfrak{b}) \cap U_i = V_+\left(\mathfrak{b}^{ah}\right) \cap U_i$ for $\mathfrak{b} \lhd k[X_0, ..., X_n]$ homogenous
c) $I^{\text{aff}}(Y)^h = I_+(\theta_i(Y))$ for $Y \subset U_i$

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Proof. This follows largely from (6.4.12). The relations show that $\theta_i(D(F)) = D_+(F^h)$ and $\theta_i^{-1}(D_+(G)) = D(G^a)$, so that θ_i is a homeomorphism as required.

a) $x \in \theta_i(V^{\text{aff}}(\mathfrak{a})) \implies x \in U_i$ and $F(\theta_i^{-1}(x)) = 0 \quad \forall F \in \mathfrak{a} \implies x \in U_i$ and $F^h(x) = 0 \quad \forall F \in \mathfrak{a} \implies G(x) = 0 \quad \forall G \in \mathfrak{a}^h \implies x \in V_+(\mathfrak{a}^h) \cap U_i$. Conversely $x \in V_+(\mathfrak{a}^h) \cap U_i \implies G(x) = 0 \quad \forall G \in \mathfrak{a}^h \implies F(\theta_i^{-1}(x)) = 0 \quad \forall F \in \mathfrak{a}^{ha} = \mathfrak{a} \implies x \in \theta_i(V^{\text{aff}}(\mathfrak{a}))$

b) $\theta_i(V^{\text{aff}}(\mathfrak{b}^a)) = V_+(\mathfrak{b}^{ah}) \cap U_i$. It remains to show that $V_+(\mathfrak{b}) \cap U_i = V_+(\mathfrak{b}^{ah}) \cap U_i$. As (6.4.12) $\mathfrak{b} \subset \mathfrak{b}^{ah}$ then $V_+(\mathfrak{b}^{ah}) \cap U_i \subset V_+(\mathfrak{b}) \cap U_i$. Conversely if $x \in V_+(\mathfrak{b}) \cap U_i$ then $G(x) = 0 \implies x_i^k G^{ah}(x) = 0 \implies G^{ah}(x) = 0$ because $x_i \neq 0$. This shows $x \in V_+(\mathfrak{b}^{ah}) \cap U_i$

c) Suppose $G \in I^{\text{aff}}(Y)^h$ then $G = \sum_j \lambda_j F_j^h$ for $F_j \in I^{\text{aff}}(Y)$ and $\lambda_j \in k[X_0, ..., X_n]$. Then $F_j(x) = 0 \implies F_j^h(\theta_i(x)) = 0 \implies G(\theta_i(x)) = 0$. Therefore we conclude $G \in I_+(\theta_i(Y))$. Conversely if $G \in I_+(\theta_i(Y))$ then $G^a(x) = 0$ for all $x \in Y$ and $G^a \in I^{\text{aff}}(Y) \implies G^{ah} \in I^{\text{aff}}(Y)^h \implies G = X_i^k G^{ah} \in I^{\text{aff}}(Y)^h$.

■

Lemma 6.4.14
Let $X = V_+(\mathfrak{b})$ be a projective variety. Then $X \cap U_i \neq \emptyset \iff X_i \notin \mathfrak{b} \iff \overline{X_i} \neq 0$.

Proof. Observe $X \cap U_i = \emptyset \iff X \subset V_+(X_i) \iff X_i \in I_+(X) = \mathfrak{b}$. ■

Corollary 6.4.15
Let $X = V_+(\mathfrak{b}) \subset \mathbb{P}_k^n$ be a projective variety and for each affine chart $U_i \subset \mathbb{P}_k^n$ consider the affine variety $X^a := V^{\text{aff}}(\mathfrak{b}^a) \subset \mathbb{A}_k^n$. Then there is a natural homeomorphism

$$\theta_i : X^a(K) \xrightarrow{\sim} D_+(\overline{X_i}) = X(K) \cap U_i(K)$$

for all algebraic extensions K/k.

Proposition 6.4.16 (Projective Closure)
Identify $\mathbb{A}_k^n(\overline{k})$ with $U_i \subset \mathbb{P}_k^n(\overline{k})$. Let $Y \subset U_i$ be a subset. Then we have the following expression for topological closure

$$\overline{Y} = V_+\left(I^{\text{aff}}(Y)^h\right)$$
$$\overline{Y} \cap U_i = \text{cl}_{U_i}(Y)$$

*We call $\overline{Y}$ the **projective closure** of Y.*

Proof. By (6.4.13).c) $I_+(Y) = I^{\text{aff}}(Y)^h \implies \overline{Y} \overset{(6.4.4)}{=} V_+ I_+(Y) = V_+\left(I^{\text{aff}}(Y)^h\right)$.

Then $\text{cl}_{U_i}(Y) = \overline{Y} \cap U_i$ by (4.1.21) (or more directly $\overset{(6.1.2)}{=} V_+(I_+(Y)) \overset{(6.4.13).a)}{=} V_+\left(I^{\text{aff}}(Y)^h\right) \cap U_i = \overline{Y} \cap U_i$). ■

6.4.3 Galois Orbits

Proposition 6.4.17
Let $X = V_+(\mathfrak{a}) \subset \mathbb{P}_k^n$ be a projective variety and K/k a normal algebraic extension (e.g. $\overline{k}$). Then there is a well-defined group action

$$\text{Aut}(K/k) \times X(K) \to X(K)$$
$$(\sigma, (x_0 : ... : x_n)) \to (\sigma(x_0) : ... : \sigma(x_n))$$

which commutes with the affine charts

$$\sigma(\theta_i(x)) = \theta_i(\sigma(x)) \quad \forall x \in U_i(K) \cap X(K)$$

Write $X_0(K)$ for the set of equivalence classes under this action. The projection map

$$\pi : X(K) \to X_0(K)$$

is the Kolmogorov Quotient (4.1.36). In particular points $x, y \in X(K)$ are topologically indistinguishable iff they are Galois conjugate. Further closed (resp. open) subsets are Galois-equivariant.

Proof. First consider the case $X = \mathbb{P}_k^n$. If $(x_0 : ... : x_n) = (x_0' : ... : x_n')$ then $x_i = \lambda_i x_i'$ for $\lambda_i \in K^\star$. Then $\sigma(x_i) = \sigma(\lambda_i)\sigma(x_i')$ so $(\sigma(x_0) : ... : \sigma(x_n)) \sim (\sigma(x_0') : ... : \sigma(x_n'))$ and the action is well-defined. For $x \in X(K)$ write $\sigma(x)$. Then $\tau(\sigma(x)) = (\tau \circ \sigma)(x)$ so it is a group action.

For the general case if $F(x) = 0$ for $F \in k[X_0, ..., X_n]$ then $F(\sigma(x)) = 0$. Therefore $x \in V(\mathfrak{a})(K) \implies \sigma(x) \in V(\mathfrak{a})(K)$ and the action is well-defined on $X(K)$.

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We need to show that two points $x, y \in X(K)$ are conjugate by $\mathrm{Aut}(K/k)$ iff they are topologically indistinguishable. Suppose $x = \sigma(y)$ and $x \in V(\mathfrak{b})$. Then evidently $y \in V(\mathfrak{b})$. Conversely if they are topologically indistinguishable then we may suppose $x, y \in U_i$. Then $\theta_i(x), \theta_i(y)$ are also indistinguishable, so by (6.1.9) they are Galois conjugate, and so $\theta_i(x) = \theta_i(\sigma(y))$. As θ_i is bijective we deduce $x = \sigma(y)$ as required. ∎

Proposition 6.4.18
Let $X = V_+(\mathfrak{a}) \subset \mathbb{P}^n_k$ be a projective variety and $L/K/k$ a tower of normal extensions. Then there is an injective map

$$\begin{array}{ccc} X(K) & \xhookrightarrow{i_{KL}} & X(L) \\ \downarrow{\scriptstyle \pi} & & \downarrow{\scriptstyle \pi} \\ X_0(K) & \overset{i_0}{\dashrightarrow} & X_0(L) \end{array}$$

Proof. In order for i_0 to be well-defined, we require that $\pi(x) = \pi(y) \implies \pi(i_{KL}(x)) = \pi(i_{KL}(y))$. Recall $\pi(x) = \pi(y) \implies x = \sigma(y)$ for some $\sigma \in \mathrm{Aut}(K/k)$. Then $i_{KL} \circ \sigma \in \mathrm{Mor}_k(K, L)$ and by (3.18.69) there exists $\hat{\sigma} \in \mathrm{Aut}(L/k)$ such that $\hat{\sigma}|_K := \hat{\sigma} \circ i_{KL} = i_{KL} \circ \sigma$. Therefore $\hat{\sigma}(i_{KL}(x)) = i_{KL}(y) \implies \pi(i_{KL}(x)) = \pi(i_{KL}(y))$.

To show i_0 is injective we need to show the converse, which follows from (3.18.71). ∎

6.4.4 Projective Varieties are Abstract Varieties

Definition 6.4.19 (Regular Function)
Let $X = V_+(\mathfrak{a}) \subset \mathbb{P}^n_k$ be a projective variety and $U \subset X\left(\overline{k}\right)$ an open subset. We say a function

$$f : U \to \overline{k}$$

*is **regular** at $x \in U$ there is a neighbourhood V of x such that*

$$f(y) = \frac{g(y)}{h(y)} \quad \forall\, y \in V$$

*for $g, h \in k[X]$ homogenous polynomials of the same degree. We therefore define the **structure sheaf** as follows*

$$\mathcal{O}_X(U) := \left\{ f : U \to \overline{k} \;\middle|\; f \text{ regular at all } x \in U \right\}$$

The pair $(X, \mathcal{O}_X)$ is a space of functions.

Proposition 6.4.20 (Structure Sheaf of Regular Functions)
Let $X = V_+(\mathfrak{a}) \subset \mathbb{P}^n_k$ be a projective variety, $U \subset X\left(\overline{k}\right)$ open and $f : U \to \overline{k}$ a function. Consider the affine charts

$$\theta_i : X_i^a\left(\overline{k}\right) \xrightarrow{\sim} D_+\left(\overline{X_i}\right) = X\left(\overline{k}\right) \cap U_i \quad i = 0, ..., n$$

where $X_i^a = V^{\mathrm{aff}}(\mathfrak{a}^a)$ as in (6.4.15). Then the following are equivalent

a) f is regular (at x) in the sense of (6.4.19)
b) $f \circ \theta_i : \theta_i^{-1}(U) \to \overline{k}$ is regular (at $\theta_i^{-1}(x)$) in the sense of (6.1.25) whenever $U \cap U_i \neq \emptyset$ ($x \in U \cap U_i$)

In other words for each $i = 1, ..., n$ there is an isomorphism of spaces with functions

$$\theta_i : \left(X_i^a\left(\overline{k}\right), \mathcal{O}_{X_i^a}\right) \xrightarrow{\sim} \left(D_+\left(\overline{X_i}\right), \mathcal{O}_X\,|_{D_+\left(\overline{X_i}\right)}\right)$$

In particular $(X, \mathcal{O}_X)$ is an abstract variety.

Proof. *a)* $\implies$ *b)* Let $x := \theta(\hat{x})$. There exists an open nbhd V of x such that $f(y) = \frac{g(y)}{h(y)}$ for all $y \in V$. By definition $g = G + \mathfrak{a}$ and $h = H + \mathfrak{a}$ for $G, H \in k[X_0, ..., X_n]$ homogenous polynomials of the same degree, for which H doesn't vanish on V. Then $\hat{V} := \theta_i^{-1}(V)$ is a open nbhd of $\hat{x}$. For all $\hat{y} \in \hat{V}$ we have

$$(f \circ \theta_i)(\hat{y}) = f(\theta_i(\hat{y})) = \frac{G(\hat{y})}{H(\hat{y})} = \frac{G^a(y)}{H^a(y)}$$

b) $\implies$ *a)* Suppose $x \in U \cap U_i$. We have by definition an open nbhd $\hat{V}$ of $\theta_i^{-1}(x)$ and $G, H \in k[x_0, x_1, ..., x_{i-1}, x_{i+1}, ..., x_n]$ for which

$$(f \circ \theta_i)(\hat{y}) = \frac{G(\hat{y})}{H(\hat{y})} \quad \forall \hat{y} \in \hat{V}$$

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As θ_i is a homeomorphism, $V := \theta_i\left(\hat{V}_i\right)$ is an open nbhd of x. By (6.4.13) we have for $y := \theta_i(\hat{y})$

$$f(y) = \frac{G^h(y)}{H^h(y)}$$

which holds for all $y \in V$. ∎

We may generalise the dehomogenisation

Proposition 6.4.21 (Structure Sheaf on Basic Affines)
Let $X = V_+(\mathfrak{p})$ be an irreducible projective variety and suppose $X \cap U_i \neq \emptyset$. Then there is an isomorphism of k-algebras

$$\begin{array}{ccc} \mathcal{O}_X\left(D_+\left(\overline{X_i}\right)\right) & \xrightarrow[\sim]{(6.4.20)} & \mathcal{O}_{X\cap U_i}(X \cap U_i) \\ \vdots & & \sim \uparrow (6.1.26) \\ k[X]_{(\overline{X_i})} & \xrightarrow{\sim} & k[X \cap U_i] \end{array}$$

$$\frac{\overline{X_0}}{\overline{X_i}} \longleftrightarrow \overline{x_i}$$

and the dashed arrow is defined in the obvious way (and is therefore an isomorphism).

Proof. By (6.4.14) $X_i \notin \mathfrak{p}$. We first demonstrate the bottom map is well-defined and an isomorphism. Consider the following commutative diagram

$$\begin{array}{ccccc} & & k[X_0,...,X_n]_{(X_i)} & \xrightarrow[\sim]{\theta_i^\sharp} & k[U_i] \\ & \swarrow & \downarrow \pi_1 & & \downarrow \pi_2 \\ k[X_0,...,X_n]_{(X_i)}/\mathfrak{p}_{(X_i)} & \xrightarrow[(3.11.17)]{\sim} & k[X]_{(\overline{X_i})} & \xrightarrow{\sim} & k[X \cap U_i] \end{array}$$

where the isomorphism $\theta_i^\sharp$ is defined in (6.4.11).

It follows from (3.3.56) that π_1 is uniquely defined and has kernel $\mathfrak{p}_{(X_i)}$ which equals $\mathfrak{p}_{X_i} \cap k[X_0,...,X_n]_{(X_i)}$ by (3.11.15). Therefore we may regard π_1 as a quotient map.

We also claim that the diagonal map $\pi_2 \circ \theta_i^\sharp$ has kernel $\mathfrak{p}_{(X_i)}$. It is enough by (3.5.5).b) and the preceding comment to show that the composite map $\pi_2 \circ (-)^a : k[X_0,...,X_n] \to k[X \cap U_i]$ has kernel $\mathfrak{p}$. However this follows from (6.4.12) for $F \in \ker(\pi_2 \circ (-)^a) \implies F^a \in \mathfrak{p}^a \implies F^a = G^a \implies F^{ah} = G^{ah} \in \mathfrak{p}^{ah} = \mathfrak{p} \implies F^{ah} \in \mathfrak{p} \implies X_i^k F \in \mathfrak{p} \implies F \in \mathfrak{p}$.

The required map exists by (3.3.56), as π_1 is a quotient map. Further it is injective, surjective and therefore an isomorphism

The map $k[X]_{(\overline{X_i})} \to \mathcal{O}_X\left(D_+\left(\overline{X_i}\right)\right)$ is defined in the obvious way. It is clear the diagram commutes and so this map is an isomorphism. ∎

Proposition 6.4.22
Let X be a projective variety. Then it is separated.

6.4.5 Regular Morphisms

For the category of quasi-projective varieties we introduce a number of equivalent definitions of morphism. In light of (6.4.20) then b) shows a function is regular iff it is regular in all affine coordinates.

Proposition 6.4.23 (Regular Morphism)
Let X be a projective variety, $U \subset X\left(\overline{k}\right)$ an open subset, Y a quasi-projective variety and $\phi : U \to Y\left(\overline{k}\right)$ a function. Then the following are equivalent

*a) **pulls back regular functions** ϕ is continuous and determines a morphism of sheaves*

$$\begin{aligned} \mathcal{O}_Y &\to \phi_\star \mathcal{O}_U \\ f &\to f \circ \phi \end{aligned}$$

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b) ***affine projections are regular*** *There is an open cover* $U = \bigcup_{i\in I} U_i$ *such that* $\phi(U_i)$ *is a subset of an affine chart* $W_{j(i)}$ *and*

$$\pi_k \circ \theta_{j(i)}^{-1} \circ \phi|_{U_i} : U_i \to \overline{k}$$

is regular (6.4.19) for all $i \in I$ *and* $k = 1, ..., m$

c) ***locally homogenous*** *There is an open cover* $U = \bigcup_{i\in I} U_i$ *and families of homogenous polynomials* $\{f_{i0}, ..., f_{im}\} \subset k[X]_{\text{hom}}$ *of the same degree such that* $U_i \subset D_+(f_{i0}) \cup ... \cup D_+(f_{im})$ *and*

$$\phi(y) = (f_{i0}(y) : ... : f_{im}(y)) \quad \forall y \in U_i$$

In this case we say ϕ *is* ***regular*** *or a* ***regular morphism*** *of quasi-projective varieties.*

Proof. $b) \implies a)$ We may assume wlog that U_i is contained in an affine chart V_i. Therefore by definition and (6.4.20)

$$\theta_{j(i)}^{-1} \circ \phi \circ \theta_i \mid_{\theta_i^{-1}(U_i)} : \theta_i^{-1}(U_i) \to \mathbb{A}_k^m$$

is regular as a map of quasi-affine varieties. Therefore by (6.1.39) it is continuous. Whence $\phi|_{U_i}$ is continuous by (6.4.13), and ϕ is continuous by (4.1.27).

Suppose $f \in \mathcal{O}_Y(W)$ then it is enough to show for all $x \in \phi^{-1}(W)$ that $f \circ \phi$ is regular at x. Suppose $x \in U_i$ and replace W with $W \cap W_{j(i)}$. Then by definition $f \circ \theta_{j(i)} : \theta_{j(i)}^{-1}(W) \to \overline{k}$ is regular at $\theta_{j(i)}^{-1}(x)$. Whence by (6.1.40)

$$f \circ \phi \circ \theta_i = \left(f \circ \theta_{j(i)}\right) \circ \left(\theta_{j(i)}^{-1} \circ \phi \circ \theta_i\right)$$

is regular at $\theta_i^{-1}(x)$, and therefore by (6.4.20) $f \circ \phi$ is regular at x.

$a) \implies b)$ For $x \in U$ there is some affine W_j such that $\phi(x) \in W_j \implies x \in \phi^{-1}(W_j) =: V_x$. By (6.4.20) $\pi_k \circ \theta_j^{-1} : W_j \to \overline{k}$ is regular whence by assumption $\pi_k \circ \theta_j^{-1} \circ \phi|_{V_x}$ is regular. As every point has a nbhd V_x we may find the required open cover.

$b) \implies c)$ By definition there exists homogenous $f_{i0}, ..., f_{im}, g_{i0}, ..., g_{im} \in k[X]_{\text{hom}}$ such that

$$\phi(y) = \left(\frac{f_{i0}(y)}{g_{i0}(y)} : ... : 1 : ... : \frac{f_{im}(y)}{g_{im}(y)}\right) \quad \forall y \in U_i$$

Therefore we may define

$$h_{il} := f_{il} \prod_{k\neq l} g_{ik}$$

Then evidently

$$\phi(y) = (h_{i0}(y) : ... : h_{im}(y)) \quad \forall y \in U_i$$

$c) \implies b)$ For every $x \in U$ there exists some open nbhd V_x such that

$$\phi(y) = (f_0(y) : ... : f_m(y)) \quad \forall y \in V_x$$

Suppose that $f_j(x) \neq 0$ then replace V_x with $V_x \cap D_+(f_j)$. Then

$$\left(\pi_k \circ \theta_j^{-1} \circ \phi|_{V_x}\right)(y) = \frac{f_k(y)}{f_j(y)}$$

which is regular by definition. The family $\{V_x\}_{x\in U}$ is the corresponding open cover. ∎

Corollary 6.4.24
Let $\phi : X \to Y$ *and* $\psi : Y \to Z$ *be regular morphisms of quasi-projective varieties. Then so is* $\psi \circ \phi : X \to Z$.

Proof. This is immediate by (6.4.23).a). ∎

Corollary 6.4.25
A function $f : X\left(\overline{k}\right) \to \overline{k}$ *is regular iff it coincides with a regular morphism* $f : X\left(\overline{k}\right) \to D_+(X_0) \cong \mathbb{A}_k^1$. *In particular a regular function is continuous with respect to the co-finite topology on* $\overline{k}$.

Proof. This follows from (6.4.23).b). ∎

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Proposition 6.4.26
Let $X \subset \mathbb{P}^n_k$ be a projective variety, $U \subset X\left(\overline{k}\right)$ an open subset and $U = \bigcup_{i\in I} U_i$ an open cover. Suppose that we have $f_{i0}, ..., f_{im} \in k[X]_{\text{hom}}$ homogenous of the same degree such that

a) For every $i \in I$

$$U_i \subset D_+(f_{i0}) \cup ... \cup D_+(f_{im})$$

b) For every pair $U_i \cap U_{i'} \neq \emptyset$ we have

$$f_{ik'}(y)f_{i'k}(y) = f_{ik}(y)f_{i'k'}(y) \quad \forall y \in U_i \cap U_{i'}, \forall k, k' = 0, ..., m$$

Then there exists a unique regular morphism $f : U \to \mathbb{P}^m_k\left(\overline{k}\right)$ such that

$$f(x) = (f_{i0}(x) : ... : f_{im}(x)) \quad \forall x \in U_i$$

6.4.6 Local Ring

Proposition 6.4.27
Let $X = V_+(\mathfrak{b}) \subset \mathbb{P}^n_k$ be a projective variety and $W = V_+(\mathfrak{p})$ an irreducible subset. Then there is an isomorphism

$$\begin{aligned} k[X]_{(\mathfrak{p}/\mathfrak{b})} &\xrightarrow{\sim} \mathcal{O}_{X,W} \\ \frac{\overline{F}}{\overline{G}} &\to \left[\left(D\left(\overline{G}\right), \frac{F(-)}{G(-)}\right)\right] \end{aligned}$$

More precisely for any affine chart U_i such that $U_i \cap W \neq \emptyset$ there is a commutative diagram

$$\begin{array}{ccccc} k[U_i]_{\mathfrak{p}^a} & \xrightarrow[(3.5.45)]{\sim} & k[U_i \cap X]_{\mathfrak{p}^a/\mathfrak{b}^a} & \xrightarrow[(6.1.57)]{\sim} & \mathcal{O}_{U_i\cap X, U_i\cap W} \\ \wr\downarrow & & & & \downarrow\wr \\ & & & & \text{(4.3.17)} \\ k[X_0, ..., X_n]_{(\mathfrak{p})} & \xrightarrow[(3.11.18)]{\sim} & k[X]_{(\mathfrak{p}/\mathfrak{b})} & \dashrightarrow^{\sim} & \mathcal{O}_{X,W} \end{array}$$

Proof. We construct the left vertical map, and then the dashed arrow is uniquely defined.

As $U_i \cap W \neq \emptyset$ then $W \not\subset V(X_i) \implies X_i \notin \mathfrak{p}$. So by (6.4.12).d) $\mathfrak{p}^{ah} = \mathfrak{p}$. To complete the diagram we consider the mutually inverse maps

$$\begin{aligned} k[U_i] = k[x_0, ..., x_{i-1}, x_{i+1}, ..., x_n]_{\mathfrak{p}^a} &\to k[X_0, ..., X_n]_{(\mathfrak{p})} \\ \frac{F}{F'} &\to \frac{F\left(\frac{X_0}{X_i}, ..., \frac{X_n}{X_i}\right)}{F'\left(\frac{X_0}{X_i}, ..., \frac{X_n}{X_i}\right)} \\ \frac{G^a}{G'^a} &\leftarrow \frac{G}{G'} \end{aligned}$$

Note $F'^h \in \mathfrak{p} \implies F' = F'^{ha} \in \mathfrak{p}^a$. Therefore $F' \notin \mathfrak{p}^a \implies F'^h = X_i^{\deg(F)} F'\left(\frac{X_0}{X_1}, ..., \frac{X_n}{X_i}\right) \notin \mathfrak{p} \implies F'\left(\frac{X_0}{X_1}, ..., \frac{X_n}{X_i}\right) \notin \mathfrak{p}$. Similarly $G'^a \in \mathfrak{p}^a \implies G'^{ah} \in \mathfrak{p}^{ah} = \mathfrak{p} \implies G' = X_i^k G'^{ah} \in \mathfrak{p}$, so $G' \notin \mathfrak{p} \implies G'^a \notin \mathfrak{p}^a$. Finally

$$\frac{G^a\left(\frac{X_0}{X_i}, ..., \frac{X_n}{X_i}\right)}{G'^a\left(\frac{X_0}{X_i}, ..., \frac{X_n}{X_i}\right)} = \frac{X_i^{-\deg(G)} G}{X_i^{-\deg(G')} G'} = \frac{G}{G'}$$

so the maps are mutually inverse. ∎

Corollary 6.4.28
Let X be an irreducible projective variety with $W = V_+(\mathfrak{p})$ an irreducible subvariety. Then there is a commutative diagram

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$$\begin{array}{ccc} k[U_i]_{\mathfrak{p}^a} & \hookrightarrow & k[U_i]_{(0)} \\ \downarrow \sim & & \downarrow \sim \\ k[X]_{(\mathfrak{p})} & \hookrightarrow & k[X]_{((0))} \\ \downarrow \sim & & \downarrow \sim \\ \mathcal{O}_{X,W} & \hookrightarrow & k(X) \end{array}$$

In particular $k(X)$ is the field of fractions for $\mathcal{O}_{X,W}$.

Proof. The vertical maps are from (6.4.27) and the horizontal maps are the obvious ones. As the top map is injective and corresponds to field of fractions we deduce that $k(X)$ is field of fractions for $\mathcal{O}_{X,W}$. ∎

6.4.7 Dimension

Proposition 6.4.29
Projective space $\mathbb{P}^n_k$ has dimension n. Furthermore it is non-singular, normal and regular.

Proof. Projective space is integral (6.4.8) and has an open subset isomorphic to $\mathbb{A}^n_k$. So we are done by (6.2.54) and (6.1.50).

The conditions are local (6.2.81) (3.23.10). ∎

Proposition 6.4.30
Let $X = V_+(\mathfrak{p})$ be an irreducible projective variety then $\dim k[X] = \dim X + 1$.

Proof. We have $\overline{X_i} \neq 0$ for some i as X is non-empty. Therefore

$$\dim k[X] \overset{(3.34.31)}{=} \dim k[X]_{\overline{X_i}} \overset{(3.12.8)}{=} \dim k[X]_{(\overline{X_i})}[T, T^{-1}] \overset{(3.9.3)}{=} \dim k[X]_{(\overline{X_i})}[T]_T \overset{(3.34.31)}{=} \dim k[X]_{(\overline{X_i})}[T] \overset{(3.34.29)}{=} \dim k[X]_{(\overline{X_i})} + 1$$

By (6.4.21) $\dim k[X]_{(\overline{X_i})} = \dim k[X \cap U_i] = \dim X \cap U_i$.

Finally this equals $\dim X$ (4.1.88). ∎

Proposition 6.4.31
Let X be a projective variety and $x \in X\left(\overline{k}\right)$. Then

$$\overline{\{x\}} = \left\{\sigma(x) \;\middle|\; \sigma \in \operatorname{Aut}\left(\overline{k}/k\right)\right\}$$

is an irreducible set of order $\left[k\left(\mathfrak{m}_{X,x}\right) : k\right]_s$. Further X is a symmetric space.

Proof. We have $x \in U_i$ for some affine chart then we may use (6.4.20) to reduce to the affine case (6.1.18). ∎

Corollary 6.4.32
Let X be a projective variety and $Z \subset X$ a closed subset. Then the following are equivalent

a) $\dim Z = 0$
b) *Z is finite*

More precisely

$$Z = \overline{\{x_1\}} \cup ... \cup \overline{\{x_n\}}$$

for a finite number of points $x_1, ..., x_n \in Z$.

Proof. This follows from (4.1.91) and (6.4.31). ∎

Proposition 6.4.33
Let $X = \mathbb{P}^n_k\left(\overline{k}\right)$ and $G \in k[X_0, ..., X_n]$ an irreducible, homogenous polynomial of positive degree. Then $\dim V_+(G) = n - 1$.

Proof. Let $Y := V_+(G)$ then by (...) we have $Y \neq \emptyset$. Choose an affine patch U_i such that $U_i \cap Y \neq \emptyset$. Then $Y \cap U_i = V(G^a) \neq \emptyset$ which implies G^a has positive degree. By (...) $\dim Y \cap U_i = n - 1$. ∎

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6.4.8 Valuative Completeness

Lemma 6.4.34
Let R be a valuation ring for K. Suppose $x_0, ..., x_n \in K$ not all zero, then there exists $0 \leq j \leq n$ such that $\frac{x_i}{x_j} \in R$ for all $i = 0, ..., n$.

Proof. Recall (3.24.17) that the abelian group $K^\star/R^\star$ is totally ordered by the condition $x \leq y \iff yx^{-1} \in R^\star$. Choose j for which $x_j \neq 0$ such that x_j is minimal under this ordering. ∎

Proposition 6.4.35 (Projective Varieties are Complete)
Let $X = V_+(\mathfrak{p}) \subset \mathbb{P}^n_k$ be an irreducible projective variety. Let $k \subset R \subset k(X)$ be a valuation ring. Then there exists a point $x \in X\left(\overline{k}\right)$ such that $\mathcal{O}_{X,x} \subset R$ and $\mathfrak{m}_R \cap \mathcal{O}_{X,x} = \mathfrak{m}_{X,x}$.

More precisely there exists a local homomorphism $\Phi : R \to \overline{k}$ such that $\Phi|_{\mathcal{O}_{X,x}} = \mathrm{ev}_x$.

Proof. Denote the homogenous coordinate ring by simply $k[X]$. In light of (6.4.28) we may assume that $k(X) = k[X]_{((0))}$ and $\mathcal{O}_{X,x} = k[X]_{\mathfrak{p}_x}$ with $\mathfrak{m}_{X,x} = \left(\mathfrak{p}_x k[X]_{\mathfrak{p}_x}\right)_{(0)}$.

By (...) we may assume that $k(\mathfrak{m}_R)/k$ is algebraic and therefore by (3.18.61) there exists a local k-algebra homomorphisms

$$\Phi : R \to k(\mathfrak{m}_R) \hookrightarrow \overline{k}$$

We may assume wlog that $X \cap U_0 \neq \emptyset$ and therefore $X_0 \notin \mathfrak{p}$ where $X_0, ..., X_n$ are the homogenous coordinates. Define $x_i := \frac{X_i}{X_0} \in k[X]_{((0))}$ for $i = 0, ..., n$. By (6.4.34) there is $1 \leq j \leq n$ such that $t_i := \frac{x_i}{x_j} = \frac{\overline{X}_i}{\overline{X}_j} \in R$ for all $i = 0, ..., n$.

Note that $t_j = 1 \notin \mathfrak{m}_R$ so $\Phi\left(t_j\right) \neq 0$ we have a well-defined projective point $x = [\Phi(t_0) : ... : \Phi(t_n)] \in \mathbb{P}^n_k\left(\overline{k}\right)$. For any homogenous polynomial $F \in k[X_0, ..., X_n]$ of degree d we have

$$\begin{aligned} F(t_0, ..., t_n) &= \overline{X}_j^d F\left(\overline{X}_0, ..., \overline{X}_n\right) \\ &= \overline{X}_j^d \overline{F} \end{aligned}$$

regarding $k(X)$ as a subring of $\mathrm{Frac}(k[X])$ and $\bar{\cdot}$ denoting the reduction map $k[X_0, ..., X_n] \to k[X]$. Further

$$F(\Phi(t_0), ..., \Phi(t_n)) = \Phi(F(t_0, ..., t_n))$$

If $F \in \mathfrak{p}$ then $\overline{F} = 0$ and therefore $F(x) = 0$. This shows that $x \in X\left(\overline{k}\right)$.

We require to show that $\mathcal{O}_{X,x} \subset R$ and $\Phi|_{\mathcal{O}_{X,x}} = \mathrm{ev}_x$, from which it immediately follows that $\mathfrak{m}_R \cap \mathcal{O}_{X,x} = \mathfrak{m}_{X,x}$

Consider $g, h \in k[X]$ homogenous of the same degree with $h \notin \mathfrak{p}_x$. We claim that $g/h \in R$. For there exists $G, H \in k[X_0, ..., X_n]$ homogenous such that

$$\begin{aligned} g &= G\left(\overline{X_0}, ..., \overline{X_n}\right) \\ &= \overline{X}_j^d G(t_0, ..., t_n) \\ h &= H\left(\overline{X_0}, ..., \overline{X_n}\right) \\ &= \overline{X}_j^d H(t_0, ..., t_n) \\ \frac{g}{h} &= \frac{G(t_0, ..., t_n)}{H(t_0, ..., t_n)} \end{aligned}$$

Evidently $G(t_0, ..., t_n) \in R$ since R is a k-algebra. Further by assumption

$$\begin{aligned} 0 &\neq h(\Phi(t_0), ..., \Phi(t_n)) \\ &= H(\Phi(t_0), ..., \Phi(t_n)) \\ &= \Phi(H(t_0, ..., t_n)) \end{aligned}$$

which means $H(t_0, ..., t_n) \notin \mathfrak{m}_R \implies H(t_0, ..., t_n) \in R^\star$. Using the expression above for $\frac{g}{h}$ shows it lies in R as required. Further

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$$\begin{aligned}\Phi\left(\frac{g}{h}\right) &= \frac{\Phi(G(t_0,...,t_n))}{\Phi(H(t_0,...,t_n))} \\ &= \frac{G(\Phi(t_0),...,\Phi(t_n))}{H(\Phi(t_0),...,\Phi(t_n))} \\ &= \frac{G(x)}{H(x)} \\ &= \frac{g}{h}(x)\end{aligned}$$

as required. ■

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6.5 Arithmetic Varieties

Remark 6.5.1
This section is rough working notes rather than a polished treatment, kept here for personal reference. Omitted from the HTML editions.

Proposition 6.5.2
Let $\mathfrak{o}$ be a ring with quotient field k. For an ideal $\mathfrak{a} \lhd \mathfrak{o}[X_1, ..., X_n]$ define the zero-locus

$$|X| := V(\mathfrak{a}) := \left\{ x \in \overline{k}^n \;\middle|\; f(x_1, ..., x_n) = 0 \quad \forall f \in \mathfrak{a} \right\}$$

and the coordinate ring $\mathfrak{o}[X] := \mathfrak{o}[X_1, ..., X_n]/\mathfrak{a}$. Then

$$|X| = V(\mathfrak{a}_k) \subset \overline{k}^n$$

$$\mathfrak{a}_k := \left\{ \sum_{i=1}^{n} \lambda_i f_i \in k[X_1, ..., X_n] \;\middle|\; f_i \in \mathfrak{a}, \lambda_i \in k \right\}$$

where $\mathfrak{a}_k \lhd k[X_1, ..., X_n]$. For every $f \in \mathfrak{o}[X_1, ..., X_n]/\mathfrak{a}$ the sets of the form

$$D(f) := \{x \in |X| \mid f(x) \neq 0\}$$

form a basis for the usual Zariski topology. For every principal open set $D(f) \subset |X|$ define

$$\mathcal{O}_X(D(f)) := \left\{ \phi : D(f) \to \overline{k} \;\middle|\; \phi(x) = \frac{g(x)}{f^r(x)} \forall x \in D(f), \text{some } g \in \mathfrak{o}[X], r \geq 0 \right\}$$

then $\mathcal{O}_X$ extends uniquely to sheaf of $\overline{k}$-valued functions with the obvious $\mathfrak{o}$-algebra structure.

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6.6 Scheme Notes

Remark 6.6.1
This section is rough working notes rather than a polished treatment, kept here for personal reference. Omitted from the HTML editions.

6.6.1 Schemes over $\mathbb{Z}$

Let A be a finitely-generated $\mathbb{Z}$-algebra, it may be expressed in the form $A = \mathbb{Z}[X_1, ..., X_n]/\mathfrak{a}$. For any $\mathbb{Q}$-algebra R the rational points are given by $\mathrm{AlgHom}_{\mathbb{Z}}(A, [R]_{\mathbb{Z}}) = \mathrm{AlgHom}_{\mathbb{Q}}(A \otimes_{\mathbb{Z}} \mathbb{Q}, R)$.

6.6.2 Closed Points vs Maximal Ideals

In the case of affine schemes there is a straightforward relationship between closed points and maximal ideals.

Proposition 6.6.2
Let $X = \mathrm{Spec}(A)$ be an affine scheme. Then the closed points of X correspond precisely to maximal ideals of A.

For a general scheme a closed point corresponds to a maximal ideal of all affine neighbourhoods.

Proposition 6.6.3
Let X be a scheme, $x \in X$ and $U \subset X$ an affine neighbourhood, with $U \xrightarrow{\sim} \mathrm{Spec}(A)$. Then if x is a closed in X we have x is closed in U and corresponds to a maximal ideal of A.

Remark 6.6.4
However the converse may not hold – there may be maximal ideals of affines subsets which do not correspond to a closed point. As an extreme example it's possible to construct a scheme with no closed points at all. Consider a local ring A with the following ideal structure

$$0 \subsetneq \mathfrak{p}_0 \subsetneq ... \subsetneq \mathfrak{p}_n \subsetneq ... \subsetneq \mathfrak{m} \subsetneq A$$

Then $\mathrm{Spec}(A) \setminus \{[\mathfrak{m}]\}$ is an open subscheme with no closed points.

More simply consider the spectrum of a discrete valuation ring $\mathrm{Spec}(A) = \{(0), \mathfrak{m}\}$. Then $\{(0)\}$ is not closed in $X := \mathrm{Spec}(A)$, but it is open (being the complement of $V(\mathfrak{m})$) and even isomorphic to $\mathrm{Spec}\big(A_{(0)}\big)$, and so (0) is closed in an affine neighbourhood.

However we may obtain some partial converses

Proposition 6.6.5
Let X be a quasi-compact scheme, $U \subset X$ an affine open neighbourhood and $x \in U$ a closed point. Then there exists $y \in \overline{\{x\}}$ such that y is closed.

Proposition 6.6.6
Let X be a integral scheme locally of finite type over k. Then if $x \in U$ is closed for U affine, it is also closed in X.

Proof. By assumption we may write $X = \bigcup_{i \in I} V_i$ where V_i are affine (as finite type $\implies$ quasi-compact). Then by (4.1.21)

$$\mathrm{cl}_X(\{x\}) \cap V_i = \mathrm{cl}_{V_i}(\{x\})$$

whenever $x \in V_i$ (and it is empty otherwise). Therefore it is sufficient to show that $x \in V_i \implies x$ closed in V_i.

So we may consider the following situation: $U = \mathrm{Spec}(A)$ and $V = \mathrm{Spec}(B)$ are affine open neighbourhoods of x, for A, B finitely generated k-algebras, where additionally x is closed in U. Then by (6.6.2) x corresponds to a maximal ideal $\mathfrak{m}$ of A, and a prime ideal of $\mathfrak{p}$ of B. As these are both stalks of X we have a local k-algebra isomorphism $A_{\mathfrak{m}} \xrightarrow{\sim} B_{\mathfrak{p}}$, and therefore also of residue fields $k(\mathfrak{m}) \xrightarrow{\sim} k(\mathfrak{p})$. Therefore by following lemma we have x is closed in V as required. ∎

Lemma 6.6.7
Let A be a finitely-generated k-algebra and $\mathfrak{p}$ a prime ideal. Then the following are equivalent

a) $\mathfrak{p}$ is maximal
b) $[\mathfrak{p}]$ is a closed point of $\mathrm{Spec}(A)$.
c) $k(\mathfrak{p})/k$ is algebraic
d) $k(\mathfrak{p})/k$ is finite

In particular the closed points of $\mathrm{Spec}(A)$ correspond precisely to the maximal ideals of A.

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Proof. First we observe that $k(\mathfrak{p})/k$ is always finitely generated, so $c) \iff d)$ is simply (3.18.44). Further $a) \iff b)$ is simply (6.6.2).

$d) \implies a)$ We have an inclusion $A/\mathfrak{p} \hookrightarrow k(\mathfrak{p})/k$. We deduce that $A/\mathfrak{p}$ is a finite-dimensional, integral k-algebra and therefore a field (3.34.56).

$a) \implies d)$ As A is finitely generated, so is $k(\mathfrak{p})/k$. By Zariski's Lemma (3.34.23) it is finite. ■

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